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Gram's Law in the Theory of the Riemann Zeta-Function. Part 1

M. A. Korolev*

Steklov Institute of Mathematics, Russian Academy of Sciences, ul. Gubkina 8, Moscow, 119991 Russia

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To the blessed memory of my teacher
Anatolii Alekseevich Karatsuba

INTRODUCTION

The Riemann zeta-function and its zeros. Given a complex number $s = \sigma + it$ satisfying the condition $\operatorname{Re} s = \sigma > 1$, the value $\zeta(s)$ of the Riemann zeta-function at s is defined as the sum of the infinite series

$$\sum_{n=1}^{+\infty} \frac{1}{n^s}.$$

The zeta-function was introduced in mathematics by Leonhard Euler. In his reasoning, Euler first used the identity[1]

$$\sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1} \tag{1}$$

(for real $s > 1$), which now bears his name. The study of the zeta-function as a function of a complex variable was initiated by Bernhard Riemann.

On the right of the unit line $\operatorname{Re} s = 1$, the relation

$$\zeta(s) = \frac{s}{s-1} - s \int_1^{\infty} \frac{\{u\}}{u^{s+1}}\, du \tag{2}$$

holds, where $\{u\}$ denotes the fractional part of u. Since the improper integral in (2) converges for any s with positive real part, this relation can be used to define a meromorphic continuation of the zeta-function to the domain $0 < \operatorname{Re} s \leq 1$.

To continue $\zeta(s)$ over the entire complex plane, consider the function

$$\xi(s) = s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s),$$

which is called the Riemann xi-function. The xi-function, which is defined for $\operatorname{Re} s > 1$, can be represented in this domain as

$$\xi(s) = 1 + s(s-1)\int_1^{\infty} (x^{s/2-1} + x^{-(s+1)/2})\omega(x)\, dx, \tag{3}$$

*E-mail: korolevma@mi.ras.ru

[1] The (infinite) product on the right-hand side of identity (1) is taken over all consecutive primes.

where $\omega(x)$ denotes the sum of the series

$$\sum_{n=1}^{\infty} e^{-\pi n^2 x}.$$

Since the improper integral in (3) converges for any s, it follows that the xi-function and, hence, $\zeta(s)$ extend to the entire complex plane. Representation (3) implies

$$\xi(s) = \xi(1-s). \tag{4}$$

Relation (4) is called the functional equation of the Riemann zeta-function.

The Riemann xi-function is an entire function of order 1 and decomposes into an infinite Weierstrass product of the form

$$\xi(s) = e^{as} \prod_{\varrho} \left(1 - \frac{s}{\varrho}\right) e^{s/\varrho}, \tag{5}$$

where ϱ denotes the zeros of $\xi(s)$,

$$a = \log 2\sqrt{\pi} - 1 - \frac{1}{2}E = -0.023\,095\,708\ldots,$$

and E is the Euler constant.[2]

It follows from relation (5) and properties of the gamma-function that $\zeta(s)$ has infinitely many zeros. These are, first, the "nontrivial" zeros, which coincide with the zeros ϱ of the Riemann xi-function, and, secondly, the "trivial" zeros at the points $s = -2n$, $n = 1, 2, 3, \ldots$, which coincide with the poles of $s\Gamma(s/2)$. Setting $s = \varrho$ in (4) and noting that $\overline{\xi}(s) = \xi(\overline{s})$, we conclude that, together with each ϱ, the numbers $1 - \varrho$, $\overline{\varrho}$, and $1 - \overline{\varrho}$ are also zeros of $\zeta(s)$. In other words, the nontrivial zeros of the zeta-function are located symmetrically with respect to the real axis and the line $\operatorname{Re} s = 1/2$.

It is known that all nontrivial zeros of the Riemann zeta-function lie in the vertical strip $0 \le \operatorname{Re} s \le 1$, which is called the critical strip in the theory of $\zeta(s)$. It can be shown that the right-hand side of (2) is negative at $0 < s < 1$ and $\xi(0) = 1 \ne 0$. Therefore, all zeros of $\zeta(s)$ in the critical strip are complex.

The zeros ϱ in the upper half-plane are numbered in the order of increasing ordinates; zeros with equal ordinates are numbered in any order:[3]

$$\varrho_n = \beta_n + i\gamma_n, \quad 0 \le \beta_n \le 1, \quad 0 < \gamma_1 < \gamma_2 < \gamma_3 < \cdots \le \gamma_n \le \gamma_{n+1} \le \cdots, \qquad n = 1, 2, 3, \ldots.$$

The Riemann hypothesis is that all zeros ϱ_n lie on the line $\operatorname{Re} s = 1/2$, which is called the critical line. This hypothesis has been neither proved nor disproved. At the same time, it is known that the critical line contains a positive fraction of all nontrivial zeros of $\zeta(s)$. The positive ordinates of such zeros, also numbered in increasing order, are denoted[4] by $\mathfrak{c}_n$.

Riemann's celebrated memoir *On the Number of Primes Less than a Given Magnitude* [1] brought into being new directions in number theory and the theory of functions of a complex variable, including the research related to calculating the complex zeros ϱ_n of the zeta-function.

Although Riemann himself approximately determined the ordinates of the first three zeros by his own powerful method, this fact has become known only after Siegel published his 1932 paper [2], in which he reconstructed Riemann's line of reasoning from Riemann's drafts. Apparently, the first publication on this subject was the 1895 paper [3] by the Danish mathematician Jørgen Pedersen Gram, in which the following approximate values of the first three ordinates were given:

$$\gamma_1 = 14.135, \qquad \gamma_2 = 20.82, \qquad \gamma_3 = 25.1.$$

The method used by Gram is quite laborious and virtually useless for further research. Gram's subsequent many-year investigations have led to the discovery of a different, more perfect, approximate method for calculating zeros of $\zeta(s)$ in the autumn of 1902.

[2] The Euler constant is usually denoted by γ; in this paper, we depart from this tradition, because, in what follows, we use γ to denote the ordinates of zeros of the Riemann zeta-function, for which this notation is also conventional.

[3] Some of the inequalities are weak, because, although all zeros of $\zeta(s)$ whose approximate values are known at present are simple, it has not been proved that the zeta-function has no multiple zeros.

[4] This notation is not conventional. It is easy to see that $\gamma_n \le \mathfrak{c}_n$; if the Riemann hypothesis is true, then $\mathfrak{c}_n = \gamma_n$ for any n.

Gram points. For each positive t, by $\vartheta(t)$ we denote the increment of a continuous branch of the argument of $\pi^{-s/2}\Gamma(s/2)$ as s varies along the straight line segment joining the points $s=1/2$ and $s=1/2+it$. Equation (4) implies the relation

$$e^{i\vartheta(t)}\zeta\left(\frac{1}{2}+it\right)=e^{-i\vartheta(t)}\zeta\left(\frac{1}{2}-it\right),$$

which means that the function

$$Z(t)=e^{i\vartheta(t)}\zeta\left(\frac{1}{2}+it\right)$$

takes real values at real t and its real zeros coincide with the ordinates of the zeros of $\zeta(s)$ that lie on the critical line.

Gram's method used in [4] is based on the following simple consideration. Let $A(t)$ and $B(t)$ be, respectively, the real and imaginary parts of $\zeta(1/2+it)$. Then

$$\zeta\left(\frac{1}{2}+it\right)=e^{-i\vartheta(t)}Z(t)=Z(t)(\cos\vartheta(t)-i\sin\vartheta(t)),$$

whence

$$A(t)=Z(t)\cos\vartheta(t)\qquad\text{and}\qquad B(t)=-Z(t)\sin\vartheta(t).$$

Consider the values t at which $B(t)$ vanishes. These are, first, the real roots of the function $Z(t)$, which, obviously, coincide with the ordinates $\mathfrak{c}_n$ of those zeros of $\zeta(s)$ which lie on the critical line, and, secondly, the roots of the equation $\sin\vartheta(t)=0$, i.e., those t at which the values $\vartheta(t)$ are multiples of π (see Fig. 1).

It can be shown that the function $\vartheta(t)$ is not bounded above and monotonically increases at $t>t^*=6.289836\ldots$, and $\vartheta(t^*)=-3.530573\ldots$. Therefore, for any $n\ge 0$, the equation $\vartheta(t)=(n-1)\pi$ has a unique solution t_n in the interval $t>t^*$.

Thus, $\zeta(1/2+it_n)$ turns out to be a real number, and

$$\zeta\left(\frac{1}{2}+it_n\right)=A(t_n)=Z(t_n)\cos\pi(n-1)=(-1)^{n-1}Z(t_n).$$

Suppose that the function $A(t)$ takes values of the same sign at neighboring points t_{n-1} and t_n. Then $Z(t_{n-1})$ and $Z(t_n)$ are of opposite signs, which is possible only if $Z(t)$ has an odd number of zeros between t_{n-1} and t_n.

Having convinced himself by using the Euler–Maclaurin summation formula that $A(t_n)>0$ for all $n=1,2,\ldots,15$, Gram managed to prove that all zeros of $\zeta(s)$ with positive ordinates not exceeding 66 lie on the critical line and found very accurate approximations of $\gamma_1,\gamma_2,\ldots,\gamma_{15}$.

Thus, Gram discovered that each of the intervals

$$G_n=(t_{n-1},t_n],\qquad n=1,2,\ldots,15,$$

contains precisely one zero of $Z(t)$ and that $t_{n-1}<\mathfrak{c}_n<t_n$. In relation to this discovery, Gram made the following statement in [4]: "*The values*[5] $A(t_n)$ *are positive at all* t_n *in the interval from* 10 *to* 65. *Apparently, the function* $A(t)$ *itself remains positive on a major part of this interval. Without doubt, the reason for this phenomenon is that the first summand in* $\sum_1^n n^{-1/2}\cos(t\log n)$ *being equal to* 1 *leads to the prevalence of positive terms* (over negative—*M. K.*). *But if this is so, then this law governing the mutual arrangement of the numbers* $\mathfrak{c}_n$ *and* t_n *remains valid for roots* $\mathfrak{c}$ *close to* $\mathfrak{c}_{15}$ *and succeeding it, until the balance is restored little by little.*"

The numbers t_n, now called the Gram points, form a monotonically increasing unbounded sequence. The asymptotic expansion

$$\vartheta(t)\sim\frac{t}{2}\log\frac{t}{2\pi}-\frac{t}{2}-\frac{\pi}{8}+\sum_{n=1}^{\infty}\frac{2^{2n-1}-1}{2^{2n}2n(2n-1)}(-1)^{n+1}B_{2n}t^{-(2n-1)},$$

[5] To avoid confusion and redundant explanations, in the quotation of Gram's work, we use the notation of our paper.

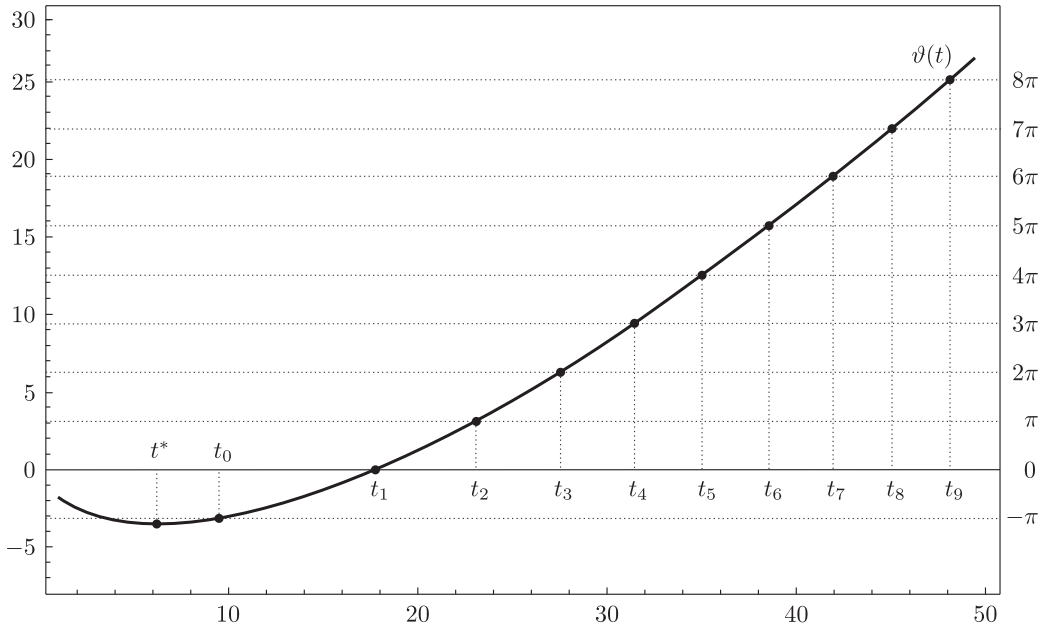


Fig. 1. The Gram points being the abscissas of the intersection points of the graph of the function $y = \vartheta(t)$ with the horizontal lines $y = m\pi$, $m = -1, 0, 1, \ldots, 9$.

where the B_{2n} are the Bernoulli numbers

$$B_2 = \frac{1}{6}, \quad B_4 = -\frac{1}{30}, \quad B_6 = \frac{1}{42}, \quad B_8 = -\frac{1}{30}, \quad B_{10} = \frac{5}{66}, \quad \ldots$$

(see [5]), makes it possible to calculate the numbers t_n with any given accuracy. In particular, it can be shown that

$$t_n = \frac{2\pi n}{\log n}\left(1 + \frac{\log\log n}{\log n}(1 + o(1))\right) \qquad \text{and} \qquad t_{n+1} - t_n = \frac{2\pi}{\log n}\left(1 + \frac{\log\log n}{\log n}(1 + o(1))\right)$$

as $n \to +\infty$. Since $\log n$ changes very slightly under the passage from n to $n + 1$, it follows from the last relation that the Gram points form an "almost uniform" grid on the real line.

The functions $S(t)$ and $N(t)$. Let us introduce several more notions necessary for what follows. If t does not equal the ordinate of a zero of $\zeta(s)$, then we set

$$S(t) = \frac{1}{\pi}\arg\zeta\left(\frac{1}{2} + it\right)$$

equal to the increment of a continuous branch of $\pi^{-1}\arg\zeta(s)$ along the polygonal chain joining the points 2, $2 + it$, and $1/2 + it$. Otherwise, we set

$$S(t) = \frac{1}{2}\lim_{h\to 0}(S(t + h) + S(t - h)).$$

The function $S(t)$ thus defined is called the argument of the Riemann zeta-function on the critical line.

Let $N(t)$ denote the number of zeros of $\zeta(s)$ in the rectangle $0 < \operatorname{Im} s \le t$, $0 \le \operatorname{Re} s \le 1$, except that, at the points of discontinuity (which, obviously, coincide with the ordinates γ_n), $N(t)$ is defined as the half-sum of its right and left limits:

$$N(t) = \frac{1}{2}\lim_{h\to 0}(N(t + h) + N(t - h)).$$

The functions $N(t)$, $\vartheta(t)$, and $S(t)$ are related by

$$N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t);$$

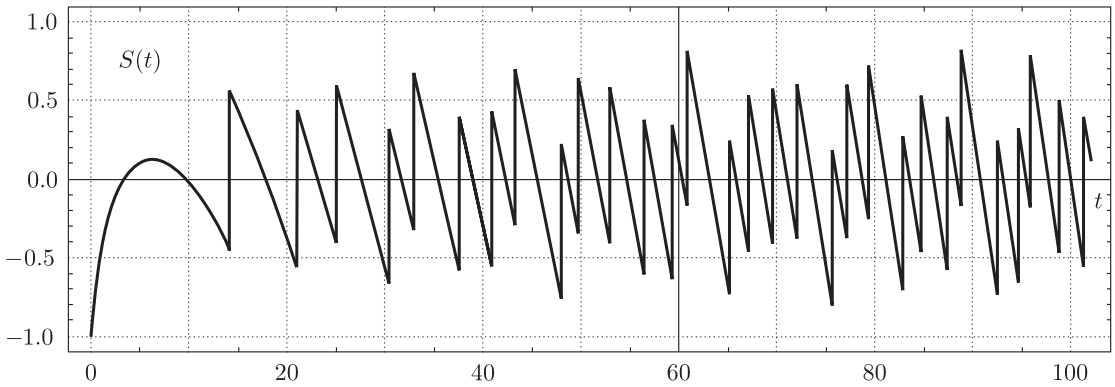


Fig. 2. The graph of the function $S(t)$. The vertical segments of the graph correspond to the jumps of $S(t)$ at the points of discontinuity, which coincide with the ordinates of the complex zeros of $\zeta(s)$.

this relation is called the Riemann–Mangoldt formula (see, e.g., [6]). The Riemann–Mangoldt formula and the classical estimate $S(t) = O(\log t)$, which is due to Littlewood [7], imply that the numbers of the Gram points t_n and the ordinates γ_n in the interval $(0, T]$ coincide, up to the addition of $O(\log T)$, with

$$\frac{1}{\pi}\vartheta(T) = \frac{T}{2\pi}\log\frac{T}{2\pi} - \frac{T}{2\pi} - \frac{1}{8} + O\left(\frac{1}{T}\right).$$

Thus, as n increases, the numbers t_n behave asymptotically as the ordinates of zeros of the Riemann zeta-function: $t_n \sim \gamma_n$.

The existence of such a relation between the sequence of Gram points, which has a simple structure, and the hard-to-study zeros of $\zeta(s)$ has generated interest in the law discovered by Gram.

Classification of Gram's laws. The term "Gram's law" was first used in Hutchinson's 1925 paper [8] for the property of neighboring zeros $\mathfrak{c}_n$ and $\mathfrak{c}_{n+1}$ of the Hardy function to be separated by the Gram point t_n. Wishing to confirm Gram's conjecture cited above that this law breaks down sooner or later, Hutchinson calculated zeros of $\zeta(s)$ in a broader range than Gram and found two values of n not obeying this law, namely, $n = 127$ and 135 (see Figs. 3(a), 3(b)), for which

$$t_{127} < \gamma_{127} < \gamma_{128} < t_{128} \qquad \text{and} \qquad t_{134} < \gamma_{135} < \gamma_{136} < t_{135}.$$

Ten years later, Hutchinson's calculations were continued by Titchmarsh and Comrie on the *Brunsviga*, *Hollerith*, and *National* tabulating machines. In the course of computation, whose results were presented in [9, 10], new exceptions to Gram's law were detected; their fraction among all studied values did not exceed[6] 4.5%.

In addition to numerical data, paper [9] contains the first theoretical results related to Gram's law. First, Titchmarsh proved that the sequence of n for which the inequality $A(t_n) > 0$ does not hold is unbounded. Titchmarsh also established that the sequence of fractions

$$\tau_n = \frac{\mathfrak{c}_n - t_n}{t_{n+1} - t_n} \tag{6}$$

is unbounded as well.

This immediately implies that, for infinitely many n, the zeros $\mathfrak{c}_n$ of the Hardy function do not fall in "their" Gram intervals $G_n = (t_{n-1}, t_n]$.

Strictly speaking, a rule with infinitely many exceptions cannot be called a law. Nevertheless, the term "Gram's law" has become used fairly extensively, although with different meanings. Thus, some authors use it for the property of the Gram interval $G_n = (t_{n-1}, t_n]$ to contain precisely one zero of the

[6] Titchmarsh himself mentioned 43 exceptions encountered in the calculation of 1041 zeros of $Z(t)$ in the interval $0 < t \le 1468$. In reality, between $t = 0$ and $t = 1468$, there are 1042 zeros of the Hardy functions, 1041 Gram points t_n, and 45 values of n for which $(-1)^{n-1}Z(t_n) < 0$.

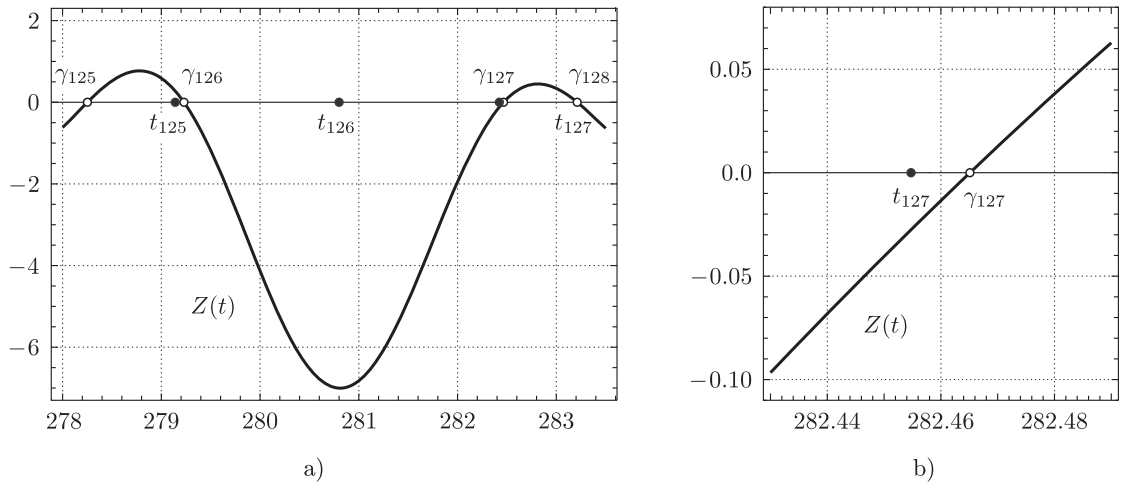


Fig. 3. The graph of the Hardy function $Z(t)$ near the point t_{127} corresponding to the first exception to Gram's law found by Hutchinson; the interval $G_{127} = (t_{126}, t_{127}]$ contains no ordinates.

Hardy function $Z(t)$. Together with Gram's law, the weak Gram law is also considered, which states that each interval G_n contains an odd number of zeros of $Z(t)$. As is known, each of these versions of Gram's law is violated for a positive fraction of the intervals G_n. The weak Gram's law holds in a positive fraction of cases, but it is not even known whether or not the number of Gram intervals containing precisely one zero of the function $Z(t)$ is finite.[7] The problem of describing the set of those zeros $\mathfrak{c}_n$ of the Hardy function each of which falls in "its own" Gram interval G_n turns out to be even more difficult.[8]

Since the formal descriptions of the law observed by Gram for a few numerical examples are so diverse, it is appropriate to use the term "Gram's law" for any statement concerning the mutual arrangement of the terms of infinite sequences of Gram points and ordinates of $\zeta(s)$ zeros.

Gram's law in Selberg's work. In his talk *The Zeta-Function and the Riemann Hypothesis* [12] given at the 10th Scandinavian Congress of Mathematicians in Copenhagen (1946), Selberg, desiring to demonstrate how wrong number-theoretic conjectures based only on numerical data can be, gave two examples.

The first, classical, example dealt with the inequality[9] $\pi(x) < \operatorname{li}(x)$, which has been supported by numerous examples but, nevertheless, is violated for infinitely many integers x.

The second example was concerned with the numerical verification of Gram's law. Let us explain this example. Consider any ordinate γ_n of a zero of the Riemann zeta-function. For its number n, we define the integer $m = m(n)$ so that

$$t_{m-1} < \gamma_n \le t_m$$

and set $\Delta_n = m - n$.

Calculations performed by Titchmarsh for the interval $1 \le n \le 1041$ show that $\Delta_n = 0$ for all ordinates γ_n in this range except 45 of them, for which $\Delta_n = \pm 1$. The development of more powerful computers has not strongly affected the picture in a certain sense. Thus, 2004 computations of Gordon and Demichel (see [13]) in the intervals

$$10^k \le n \le 10^k + 2 \cdot 10^9, \qquad k = 13, 14, \dots, 24,$$

have not revealed any n for which $|\Delta_n| > 3$.

[7] For a detailed discussion of this question, see [11].

[8] An obvious consequence of Titchmarsh's theorem on the unboundedness of fractions (6) is that the zeros $\mathfrak{c}_n$ do not fall in G_n for infinitely many n.

[9] Here, $\pi(x)$ is the number of primes not exceeding x and $\operatorname{li}(x) = \int_2^x (1/\log u)\, du$ is the integral logarithm of x.

These facts seem to indicate that the numbers $|\Delta_n|$ are bounded in absolute value. To disprove such a hasty conclusion, Selberg presented the following relations obtained by him in his talk:

$$\sum_{n\le N}\Delta_n^{2k}=\frac{(2k)!}{k!}\frac{N}{(2\pi)^{2k}}(\log\log N)^k+O(N(\log\log N)^{k-1/2}), \tag{7}$$

$$\sum_{n\le N}\Delta_n^{2k-1}=O(N(\log\log N)^{k-1}), \tag{8}$$

where $N\to+\infty$ and $k\ge 1$ is a fixed integer.

Relation (7) suggests that values Δ_n of order $\sqrt{\log\log n}$ must occur fairly often. Thus, in [12], Selberg stated the following conjecture: "*It is probably true, though I have not been able to prove it rigorously, that $\sqrt{\log\log n}$ is the 'normal' order of magnitude of Δ_n in the sense that if $\Phi(n)$ is a positive function of n which tends to infinity with n, the inequalities*

$$\frac{\sqrt{\log\log n}}{\Phi(n)}<|\Delta_n|\le\Phi(n)\sqrt{\log\log n} \tag{9}$$

hold true for almost all n. In particular, this should imply that γ_ν 'almost never' lies in the interval $(t_{\nu-1},t_\nu)$. It is the first inequality which is difficult point, and which I have not been able to prove completely."[10]

Afterward, Selberg had succeeded in proving this conjecture.[11] This can be concluded from the following remark on the text of the talk in the first volume of the collection of Selberg's papers: ... *It follows from these equations* (i.e., (7) and (8)—*M. K.*) *by standard theory that the quantity $\Delta_n/\sqrt{\log\log n}$ has a simple Gaussian distribution. In particular this answers in the affirmative question raised here. In 1946 I did not know that these results about the moments of Δ_n allow one to determine this distribution function*" (see [14, p. 355]).

Selberg's results mentioned above explain also the fact that all values Δ_n that has been calculated so far are very small. Indeed, for $N=10^{24}$, we have $|\Delta_n|\approx\sqrt{\log\log N}\approx 2.001$ for "most" numbers $n\le N$. In other words, the influence of the growing factor $\log\log N$ in (7) in the domain available for computation is imperceptible.

In [12], by Gram's law Selberg means the property of Δ_n to be equal to zero, or, equivalently, of γ_n, to fall in the Gram interval G_n with the same number.

This version of Gram's law differs from those listed above in that it imposes a constraint on the mutual arrangement of the Gram points t_n and arbitrary ordinates γ_m of zeros of $\zeta(s)$, while Gram, Hutchinson, and Titchmarsh considered the ordinates $\mathfrak{c}_m$ of only those zeros which lie on the critical line.[12]

This difference is of fundamental importance. The following fact shows how deep the gap is between the levels of our knowledge about the sequences of ordinates γ_n and $\mathfrak{c}_n$. It is known, e.g., that the distance between neighboring ordinates γ_n and γ_{n+1} tends to zero with increasing n. Moreover, if n is sufficiently large, then the difference between these ordinates satisfies the inequality

$$\gamma_{n+1}-\gamma_n\le\frac{32}{\log\log\log\gamma_n}.$$

However, no similar facts are known about $\mathfrak{c}_n$. The best that can be guaranteed is

$$\mathfrak{c}_{n+1}-\mathfrak{c}_n\le\mathfrak{c}_n^{\alpha+\varepsilon},\qquad\alpha=0.1559458\ldots$$

(proofs of this and the preceding result can be found in [15] and [16, p. 261], respectively).

[10] The words "almost never" in this context mean that if $\nu=\nu(\Phi;N)$ is the number of integers $n\le N$ not satisfying conditions (9), then $\nu/N\to 0$ as $N\to+\infty$. A similar meaning is attached to "almost all," "almost always," etc. here and in what follows.

[11] Proofs of relations (7) and (8) have never been published by Selberg or anybody else, so that the problem of substantiating them still remains open.

[12] Certainly, this difference disappears if the Riemann hypothesis is true.

To distinguish the statement of Gram's law proposed by Selberg from all other ones, we say that, in the case $\Delta_n = 0$ (or, equivalently, in the case $\gamma_n \in G_n$), the *Gram–Selberg phenomenon* occurs for the ordinate γ_n. Thus, Selberg's statement made in [14, p. 355] and quoted above means that, in reality, the Gram–Selberg phenomenon occurs very rarely, "almost never."

Object of study. The initial purpose of the author was to rigorously substantiate Selberg's relations (7) and (8). The search for proof has led to the consideration of a whole circle of questions related to the quantity Δ_n, which are the contents of this paper. They include:

(1) moments of trigonometric polynomials of a special form;

(2) the behavior of the argument of the Riemann zeta-function at the Gram points;

(3) Selberg's relations for the moments of Δ_n and their generalizations; the distribution of Δ_n;

(4) the order of growth of Δ_n as $n \to +\infty$; the set of values of Δ_n;

(5) distribution of pairs, triples, quadruples, etc. of neighboring ordinates γ_n for which Δ_n is different from zero;

(6) the small values of the function $\zeta(1/2 + it)$ at the Gram points t_n.

Let us explain each of these problems.

(1) The numbers Δ_n are closely related to the values of the function $S(t)$ at the Gram points t_n. In turn, the function $S(t)$ is approximated (in the metric of the space $L_a[T; T + H]$, $a > 0$) by trigonometric polynomials of the form

$$W_x(t) = W(t) = -\frac{1}{\pi} \sum_{p \le x} \frac{\sin(t \log p)}{\sqrt{p}},$$

where $x = x(t)$ and p ranges over the set of primes (see Fig. 4). For this reason, of special interest is the problem of finding asymptotic formulas for the "discrete" moments of $W(t)$, i.e., the quantities

$$\sum_{N < n \le N+M} |W(t_n)|^a, \qquad a > 0.$$

This problem is considered in Chapter 1.

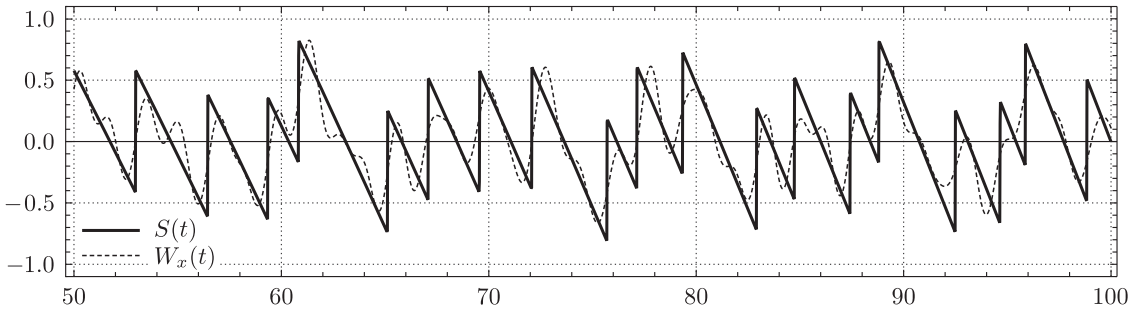


Fig. 4. The graphs of the functions $S(t)$ and $W_x(t)$ at $x = 100$.

The main results of Chapter 1 are new relations which give asymptotic expansions in powers of the parameter

$$\mathfrak{S} = \sum_{p \le x} \frac{1}{p}$$

for these moments. An example of such an expansion decomposition is the following expression for the first moment:

$$\sum_{N<n\le N+M}|W(t_n)|=\frac{M}{\pi\sqrt{\pi}}\left\{\mathfrak{S}_1^{1/2}+\sum_{n=1}^{+\infty}\frac{(-1)^{n+1}}{2^{2n-1}}\frac{(2n-2)!}{(n-1)!}\Phi_{n+1}\mathfrak{S}_1^{-1/2-n}\right\}+O\left(M\sqrt{\frac{\log x}{\log N}}\right),$$

where the Φ_n are coefficients which can be expressed explicitly as polynomials in

$$\mathfrak{S}_r=\sum_{p\le x}p^{-r},\qquad r=2,\dots,n.$$

The method used to derive such relations turns out to apply to the fairly large class of polynomials of the form

$$\sum_{p\le x}a_pf(t\log p),$$

where the a_p are real numbers and $f(u)$ is any of the functions $\sin u$, $\cos u$, and e^{iu}; it also applies to the "continuous" case, where summation over the Gram points t_n is replaced by integration over the interval $T<t\le T+H$.

(2) The theorems of Chapter 1 make it possible to significantly improve the existing estimates for the discrete moments of the argument of the Riemann zeta-function on the critical line, i.e., for sums of the form

$$\sum_{N<n\le N+M}|S(t_n+0)|^a,\qquad a>0.$$

For example, the estimate

$$r\ll\frac{\log\log\log\log N}{\log\log\log N}$$

of the remainder term in

$$\sum_{N<n\le N+M}|S(t_n+0)|=\frac{M}{\pi\sqrt{\pi}}\sqrt{\log\log N}\,(1+r)$$

can be replaced by

$$r\ll\sqrt{\frac{\log\log\log N}{\log\log N}}\,.$$

Proofs of this and other results are given in Chapter 2.

(3) Some characteristics of the sequences of Δ_n and $\Delta(n)=S(t_n+0)$ turn out to be similar in many respects. This refers, in particular, to the sums

$$\sum_{N<n\le N+M}|\Delta_n|^a\qquad\text{and}\qquad\sum_{N<n\le N+M}|\Delta(n)|^a.$$

This makes it possible to derive Selberg's relations (7) and (8) directly from theorems of Chapter 2. Moreover, the numbers Δ_n are well approximated by the values of $S(t)$ at the points of discontinuity (i.e., by the numbers $S(\gamma_n\pm0)$) and by the numbers

$$(\gamma_n-t_n)\frac{\log n}{2\pi}\,.$$

Therefore, a direct consequence of Selberg's relations is asymptotic expressions for the sums

$$\sum_{N<n\le N+M}|S(\gamma_n\pm0)|^a\qquad\text{and}\qquad\sum_{N<n\le N+M}|\gamma_n-t_n|^a.$$

The passage from the moments of $\Delta(n)$ to those of Δ_n can be performed by several methods, three of which are described in this paper. Each of them is of independent interest. Thus, one of these methods leads to the following result: for a positive proportion of the numbers n, the Gram interval G_n contains no ordinates of zeros of $\zeta(s)$, and, simultaneously, in a positive proportion of cases, G_n contains at least two such ordinates.[13] This indicates a significant irregularity of the distribution of zeros of $\zeta(s)$, because, "on average," one Gram interval contains precisely one ordinate γ_n.

Proofs of this and other results mentioned above can be found in Chapter 3.

(4) It is easy to conclude from Selberg's relations (7) and (8) that

$$\liminf_{n\to+\infty}\Delta_n=-\infty \qquad \text{and} \qquad \limsup_{n\to+\infty}\Delta_n=+\infty. \tag{10}$$

The relationship between the quantity Δ_n and the argument of the Riemann zeta-function $\zeta(s)$ makes it possible to use the most accurate results on the order of growth of the function $S(t)$ known at present. In this way, we obtain, e.g., the relations

$$\Delta_n=O\big(\log n\big) \qquad \text{and} \qquad \Delta_n=\Omega_{\pm}\left(\sqrt[3]{\frac{\log n}{\log\log n}}\right),$$

which sharpen (10).

Additional considerations based on Selberg's method for estimating the number of changes of the sign of $S(t)$ in a given interval make it possible to prove that, given any integer k between the maximum and minimum values of Δ_n in an interval $N<n\le N+M$, Δ_n takes the value k in this interval, and each of such values occurs fairly often. For example, in the case $k=0$, it turns out that the Gram–Selberg phenomenon occurs for at least $Me^{-c(\log\log\log N)^2}$ ordinates γ_n with $N<n\le N+M$. All these results are given in Chapter 4.

(5) The Gram–Selberg phenomenon turns out to occur fairly rarely: as follows from results of Chapter 3, $\Delta_n=0$ for at most $O(N(\log\log\log N)^{-0.5})=o(N)$ numbers $n\le N$. It is easy to see from here that the inequality

$$\Delta_n\Delta_{n+1}\ne 0 \tag{11}$$

holds for all integers $n\le N$ except $o(N)$ of them.

Theorems of Chapter 3 on the distribution of Δ_n imply that the numbers of positive and negative values among the first n members of the sequence of Δ_n is asymptotically equivalent to $0.5N$ as $N\to+\infty$. It is natural to ask how the signs of Δ_n and Δ_{n+1} satisfying condition (11) are distributed. This problem can be generalized to the cases of three, four, and any other fixed number s of consecutive values $\Delta_n,\Delta_{n+1},\dots$.

It turns out that, among all of the 2^s s-tuples of the symbols "+" and "−," there are two "special" s-tuples consisting of equal signs (i.e., only of pluses or only of minuses), each of which corresponds to $0.5N+o(N)$ s-tuples $\Delta_n,\Delta_{n+1},\dots,\Delta_{n+s-1}$, $n\le N$, with the corresponding distribution of signs. The remaining combinations of signs either are not realized at all or occur in no more than $o(N)$ cases. Roughly speaking, this means that changes of sign occur very rarely in the sequence of Δ_n.

Moreover, it becomes possible to investigate the distribution of the products $|\Delta_n\cdots\Delta_{n+s-1}|$ with $n\le N$ corresponding to s-tuples of numbers of the same sign. The distribution of these products, appropriately normalized, converges to the gamma-distribution with density

$$f(u)=\frac{e^{-u}}{\sqrt{\pi u}}, \qquad u>0,$$

as $N\to+\infty$.

[13] This result strengthens Trudgian's theorem [11], according to which a positive fraction of the Gram intervals G_n contain an "anomalous" (i.e., different from one) number of ordinates.

Using the results mentioned above, we can also prove the existence of an arbitrarily large number of consecutive anomalously large or anomalously small values Δ_n. Namely, for any $s \geq 2$, there exist functions $h(x)$ and $H(x)$ such that

$$\frac{h(x)}{\sqrt{\log\log x}} \to 0 \quad \text{and} \quad \frac{H(x)}{\sqrt{\log\log x}} \to +\infty \qquad \text{as} \quad x \to +\infty$$

and each of the inequalities

$$0 < \Delta_n, \Delta_{n+1}, \dots, \Delta_{n+s-1} \leq h(n) \qquad \text{and} \qquad \Delta_n, \Delta_{n+1}, \dots, \Delta_{n+s-1} > H(n)$$

has infinitely many solutions.

(6) Gram and some of the subsequent authors paid special attention to the fact that the points t_n with small n indeed separate the zeros $\mathfrak{c}_n$ of the Hardy function:

$$\mathfrak{c}_1 < t_1 < \mathfrak{c}_2 < t_2 < \mathfrak{c}_3 < t_3 < \cdots .$$

At the same time, above we considered "hits" of the zeros $\mathfrak{c}_n$ or the ordinates γ_n in the Gram intervals $G_n = (t_{n-1}, t_n]$. Thereby, we allowed a zero $\mathfrak{c}_n$ or an ordinate γ_n to coincide with the Gram point t_n. This suggests the question of whether such coincidences can indeed occur. Note that the same question arises from the following argument.

As mentioned above, the Gram points t_n on the real line form an almost uniform grid in the sense that, as n ranges in an interval of the form $N < n \leq N + M$, $M = o(N)$, all differences $t_{n+1} - t_n$ are approximately equal to the same number $2\pi(\log N)^{-1}$.

It is natural to expect that many special features of the behavior of the zeta-function $\zeta(1/2 + it)$ with t continuously varying along the real line are preserved in the case where t ranges over the discrete set of Gram points t_n.

In some cases, this expectation is realized (completely or partially). This refers, e.g., to the behavior of the quantities

$$\int_0^T \left|\zeta\left(\frac{1}{2} + it\right)\right|^{2k} dt \qquad \text{and} \qquad \sum_{n \leq N} \left|\zeta\left(\frac{1}{2} + it_n\right)\right|^{2k},$$

for which similar asymptotic relations or, at least, upper and lower bounds with close order of growth have been obtained at some k (see [17, 18], [19–21], [22–26], respectively). This also refers to the quantities

$$\max_{0<t\leq T} \left|\zeta\left(\frac{1}{2} + it\right)\right| \qquad \text{and} \qquad \max_{0<n\leq N} \left|\zeta\left(\frac{1}{2} + it_n\right)\right|,$$

for which lower bounds of the same order have been found (see [27–29]).

In the case where minimum values of the zeta-function over sets of points on the critical line are considered instead of its maximum values, the situation becomes less obvious. Since the zeta-function has infinitely many zeros on the critical line, it readily follows that

$$\min_{0<t\leq T} \left|\zeta\left(\frac{1}{2} + it\right)\right| = 0.$$

At the same time, the coincidence of a Gram point t_n with the ordinate of some zero of $\zeta(s)$ seems to be implausible.

This supposition can be verified in two ways. The first is to obtain maximally sharp lower bounds for the number of integers n, $n \leq N$, satisfying the condition

$$\zeta\left(\frac{1}{2} + it_n\right) \neq 0. \tag{12}$$

The second way is to study sequences of Gram points t_{n_k}, $k = 1, 2, \dots$, such that

$$\zeta\left(\frac{1}{2} + it_{n_k}\right) \to 0$$

as $k \to +\infty$.

It turns out, e.g., that at least $Ne^{-c(\log\log\log N)^2}$ of the first N Gram points satisfy condition (12); at the same time, given any $A>0$, for a positive proportion of numbers n in the same interval, we have

$$\left|\zeta\left(\frac{1}{2}+it_n\right)\right|<e^{-A\sqrt{\log\log N}}.$$

All these results are contained in Chapter 6.

Notation. We use the following notation:

- $N(t)$ is the number of zeros of $\zeta(s)$ in the rectangle $0<\operatorname{Re}s<1$, $0<\operatorname{Im}s\le t$;
- $N_0(t)$ is the number of zeros of $\zeta(s)$ in the interval $0<\operatorname{Im}s\le t$ of the critical line $\operatorname{Re}s=1/2$;
- $N_j(t)$ is the number of ordinates of zeros of $\zeta(s)$ of multiplicity $j\ge 1$ in the rectangle $0<\operatorname{Re}s<1$, $0<\operatorname{Im}s\le t$;
- $N(\sigma,t)$ is the number of zeros of $\zeta(s)$ in the rectangle $\sigma<\operatorname{Re}s<1$, $0<\operatorname{Im}s\le t$;
- $\vartheta(t)$ is the increment of a continuous branch of the argument of $\pi^{-s/2}\Gamma(s/2)$ along the interval with endpoints $s=1/2$ and $1/2+it$;
- $S(t)=\pi^{-1}\arg\zeta(1/2+it)$ is the argument of the Riemann zeta-function on the critical line;
- $0<\gamma_1<\gamma_2<\cdots\le\gamma_n\le\gamma_{n+1}\le\cdots$ are the positive ordinates of zeros of $\zeta(s)$ numbered in increasing order with multiplicity taken into account (coinciding ordinates are numbered in any order);
- $0<\mathfrak{c}_1<\mathfrak{c}_2<\cdots\le\mathfrak{c}_n\le\mathfrak{c}_{n+1}\le\cdots$ are the positive ordinates of zeros of $\zeta(s)$ on the critical line numbered in increasing order with multiplicity taken into account (coinciding ordinates are numbered in any order), or, equivalently, the real zeros of the Hardy function $Z(t)$;
- $J_0(z)$ is the Bessel function of the first kind;
- $G(u)$ is the normal distribution:

$$G(u)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{u}e^{-v^2/2}\,dv;$$

- $\sigma_{x,t}$ and $\Lambda_x(n)$ are the Selberg functions (see Section 2.1);
- B_n are the Bernoulli numbers:

$$B_0=1,\quad B_1=-\frac{1}{2},\quad B_2=\frac{1}{6},\quad B_4=\frac{1}{42},\quad\ldots,\qquad B_3=B_5=\cdots=0;$$

- t_n are the Gram points;
- $G_n=(t_{n-1},t_n]$ are the Gram intervals;
- $\Delta(n)=S(t_n+0)$;
- $r(n)=\Delta(n)-\Delta(n-1)$;
- $\#\mathcal{A}$ and $|\mathcal{A}|$ denote the number of elements in a set $\mathcal{A}$;
- $\mathsf{E}(\eta)$ is the mathematical expectation of a random variable η;
- $\mathsf{P}\{\xi\le u\}$ is the probability that a random variable ξ does not exceed u;

- ε is an arbitrarily small fixed number, $0 < \varepsilon < 10^{-3}$; $\varepsilon_1 = 0.9\varepsilon$;
- $\alpha = 27/82 = 1/3 - 1/246$;
- N and M are sufficiently large integers such that $N^{\alpha+\varepsilon_1} \le M \le N^{\alpha+\varepsilon}$;
- $\sum_n$ and $\sum_{n\in A}$ denote $\sum_{N<n\le N+M}$ and $\sum_{\substack{N<n\le N+M\\ n\in A}}$, respectively;
- $p, p_1, p_2, \dots, q, q_1, q_2, \dots$ are primes;
- $V_x(t) = \sum_{p\le x} \dfrac{\sin(t\log p)}{\sqrt{p}}, \quad W_x(t) = -\pi^{-1}V_x(t)$;
- $L = \log\log N$;
- $\mathcal{L}_k = \dfrac{(2k)!}{k!}\dfrac{ML^k}{(2\pi)^{2k}}$;
- $\varkappa(a) = \dfrac{2}{\sqrt{\pi}}\displaystyle\int_0^{+\infty}(2v)^a e^{-v^2}\,dv = \dfrac{2^a}{\sqrt{\pi}}\Gamma\biggl(\dfrac{a+1}{2}\biggr)$,

  in particular, $\varkappa(2k) = (2k)!/k!$ for integer $k \ge 1$;
- $\theta, \theta_1, \theta_2, \dots$ are complex numbers not exceeding 1 in absolute value; these numbers are generally different in different relations;
- $a, a_1, a_2, \dots, b, b_1, b_2, \dots$ are positive quantities depending on no other quantities except possibly ε;
- $c_0 = 2880\varepsilon^{-1}$ (Lemma 2.7);
- $c_1 = (1.75\pi^2)^{1/3}c_0^{-1}$ (Theorem 2.1);
- $c_2 = 11^{-2/3}c_0^{-1}$ (Theorem 2.3);
- $c_3 = (6c_0)^{-2}$ (Theorem 3.13);
- $c_4 = 8.7ec_0$, $c_5 = 9e^5c_0$ (Lemma 5.2);
- $c_6 = 5132.8\varepsilon^{-1}$ (Lemma 6.5).

We refer to Lemma n in Appendix I (Part 2 of this paper) as Lemma I.n. The remaining notation is introduced and explained in the course of exposition.

1. MOMENTS OF SPECIAL TRIGONOMETRIC POLYNOMIALS

The purpose of this chapter is to obtain asymptotic relations for the discrete moments of the polynomial

$$V(t) = V_x(t) = \sum_{p \le x} \frac{\sin(t \log p)}{\sqrt{p}},$$

where p ranges over consecutive primes, i.e., for sums of the form

$$V_a = \sum_n |V(t_n)|^a,$$

where a is any (not necessarily integer) positive number.

In Section 1.1, we prove an asymptotic formula for the even moments V_{2k}, whose main term has the form

$$\frac{(2k)!}{2^{2k}} M \left\{ \frac{\Phi_0}{k!} \mathfrak{S}_1^k + \frac{\Phi_1}{(k-1)!} \mathfrak{S}_1^{k-1} + \frac{\Phi_2}{(k-2)!} \mathfrak{S}_1^{k-2} + \cdots + \Phi_n \right\},$$

where

$$\mathfrak{S}_1 = \sum_{p \le x} \frac{1}{p} \sim \log\log x.$$

The coefficients Φ_n in this expression are, in turn, polynomials in the quantities

$$\mathfrak{S}_r = \sum_{p \le x} \frac{1}{p^r}, \qquad r = 2, 3, \ldots,$$

and have a fairly complicated form. Properties of the numbers Φ_n are studied in Sections 1.2 and 1.3.

Results of Sections 1.2 and 1.3 are applied in Section 1.4 to derive an expression for the moment V_a with noninteger a. The main term of such an expression can be represented by the asymptotic series

$$\frac{M}{\sqrt{\pi}} \Gamma\left(\frac{a+1}{2}\right) \sum_{n=0}^{+\infty} (-1)^n \Phi_n \frac{\Gamma(n - a/2)}{\Gamma(-a/2)} \mathfrak{S}_1^{a/2-n}. \tag{1.1}$$

For the main purpose of this paper, which is to study Gram's law, the results of Sections 1.1–1.4 are quite sufficient. Nevertheless, since asymptotic expansions of fractional moments may be of independent interest, we found it appropriate to include the additional Section 1.5 in this chapter, which is devoted to improving the remainder term in the expression for the first moment V_1.

All considerations of Sections 1.1–1.5 can be carried over, virtually without change, to the case of "integral" moments, i.e., quantities of the form

$$\int_T^{T+H} |V(t)|^a \, dt, \qquad T^{\alpha+\varepsilon} \le H \le T.$$

We also mention that many results of Chapter 1 (or their analogues) remain valid for polynomials of the forms

$$\sum_{p \le x} a_p \sin(t \log p), \qquad \sum_{p \le x} a_p \cos(t \log p), \qquad \text{and} \qquad \sum_{p \le x} a_p p^{it}$$

with real coefficients a_p. In this case, the role of the quantities $\mathfrak{S}_r$ is played by the sums $\sum_{p \le x} a_p^{2r}$. The problem of describing the class of such polynomials requires a separate study.

1.1. Even moments of $V(t_n)$. First, we need a number of auxiliary assertions about the behavior of the Gram points t_n and their generalizations to noninteger n as the parameter n unboundedly increases. All relations of this kind needed for what follows are contained in the following lemma.

Lemma 1.1. *Let $u \geq 0$ be any (not necessarily integer) number, and let t_u be the unique solution of the equation*

$$\vartheta(t_u) = (u-1)\pi \tag{1.2}$$

satisfying the condition $\vartheta'(t_u) > 0$. Then

(1) *as $u \to +\infty$,*

$$t_u = \frac{2\pi u}{\log u}\biggl(1 + (1+o(1))\frac{\log\log u}{\log u}\biggr), \tag{1.3}$$

$$\frac{dt_u}{du} = \frac{2\pi}{\log u}\biggl(1 + (1+o(1))\frac{\log\log u}{\log u}\biggr), \tag{1.4}$$

$$\frac{d^2t_u}{du^2} = -\frac{\pi}{u(\log u)^2}\biggl(1 + (2+o(1))\frac{\log\log u}{\log u}\biggr); \tag{1.5}$$

(2) *if $u \to +\infty$ and $|v| \leq u/5$, then*

$$t_{u+v} - t_u = \frac{2\pi v}{\log u}\biggl(1 + (1+o(1))\frac{\log\log u}{\log u}\biggr), \tag{1.6}$$

$$|t_{u+v} - t_u| > \frac{2\pi|v|}{\log u}, \tag{1.7}$$

$$t_{u+v} - t_u = \frac{\pi v}{\vartheta'(t_u)}\biggl(1 + \frac{\theta|v|}{u\log u}\biggr); \tag{1.8}$$

(3) *if $N < u \leq u+v \leq N+M$, then*

$$t_{u+v} - t_u = \frac{\pi v}{\vartheta'(t_N)} + \frac{2\pi\theta v M}{N(\log N)^2}, \tag{1.9}$$

$$\frac{2\pi}{\log N} < \frac{dt_u}{du} < \frac{2\pi}{\log N - \log\log N}. \tag{1.10}$$

Proof. Relation (1.3) is a corollary of Lemma 2 in [21]. Twice differentiating (1.2), we obtain the relations

$$t'_u = \frac{\pi}{\vartheta'(t_u)}, \qquad t''_u = -\frac{\pi^2\vartheta''(t_u)}{(\vartheta'(t_u))^3}, \tag{1.11}$$

from which we derive (1.4) and (1.5) by using (1.2) and the relations

$$\vartheta'(t) = \frac{1}{2}\log\frac{t}{2\pi} + O(t^{-2}) \qquad \text{and} \qquad \vartheta''(t) = \frac{1}{2t} + O(t^{-3}).$$

Let us prove (1.6)–(1.8). We assume that $0 \leq v \leq u/5$ (the case of negative u is considered in a similar way). The finite increment formula implies that, for some w, $0 \leq w \leq v$, we have

$$\pi v = \vartheta(t_{u+v}) - \vartheta(t_u) = (t_{u+v} - t_u)\vartheta'(t_{u+w}),$$

whence we conclude that, according to (1.4) and (1.11),

$$\begin{aligned} t_{u+v} - t_u = \frac{\pi v}{\vartheta'(t_{u+w})} = v t'_{u+w} &= \frac{2\pi v}{\log(u+w)}\biggl(1 + (1+o(1))\frac{\log\log(u+w)}{\log(u+w)}\biggr) \\ &= \frac{2\pi v}{\log u}\biggl(1 + (1+o(1))\frac{\log\log u}{\log u}\biggr). \end{aligned} \tag{1.12}$$

Next, Lagrange's mean value theorem also implies that, for some (other) w satisfying the condition $|w| \le u/5$, we have

$$\pi v = \vartheta(t_{u+v}) - \vartheta(t_u) = (t_{u+v} - t_u)\vartheta'(t_u) + \frac{1}{2}(t_{u+v} - t_u)^2\vartheta''(t_{u+w}),$$

so that

$$t_{u+v} - t_u = \frac{\pi v}{\vartheta'(t_u)}(1+r)^{-1}, \tag{1.13}$$

where, according to (1.11), r has the form

$$r = \frac{1}{2}(t_{u+v} - t_u)\frac{\vartheta''(t_{u+w})}{\vartheta'(t_u)} = -\frac{1}{2}(t_{u+v} - t_u)\frac{t'_u t''_{u+w}}{(t'_{u+w})^3}.$$

Using relation (1.12) and estimate (1.13), we successively find

$$\begin{aligned} r &= \frac{v}{4(\log u)^2}\frac{\log(u+w)}{u+w}\left(1 + (1+o(1))\frac{\log\log u}{\log u}\right) \\ &= \frac{v}{4u\log u}\left(1 - (1+o(1))\frac{w}{u}\right)\left(1 + (1+o(1))\frac{\log\log u}{\log u}\right), \\ |r| &\le \frac{|v|}{3u\log u}, \qquad (1+r)^{-1} = 1 + \frac{\theta|v|}{2u\log u}. \end{aligned}$$

Thus, (1.6) and (1.7) follow from (1.13) and the last relation.

If $N < u \le u+v \le N+M$, then, for some y, z, and w such that $u \le y, z \le w \le u+v$, we have

$$t_{u+v} - t_u = \frac{\pi v}{\vartheta'(t_w)} = \frac{\pi v}{\vartheta'(t_N)} - h,$$

where

$$\begin{aligned} 0 \le h &= \pi v\left(\frac{1}{\vartheta'(t_N)} - \frac{1}{\vartheta'(t_w)}\right) = \frac{\pi v(\vartheta'(t_w) - \vartheta'(t_N))}{\vartheta'(t_N)\vartheta'(t_w)} \le \frac{\pi v(t_w - t_N)\vartheta''(t_y)}{(\vartheta'(t_N))^2} \\ &= \frac{\pi v(w-N)}{\vartheta'(t_z)}\frac{\vartheta''(t_y)}{(\vartheta'(t_N))^2} \le \frac{\pi v\vartheta''(t_N)}{(\vartheta'(t_N))^3} = \frac{\pi v M}{N(\log N)^2}\left(1 + (2+o(1))\frac{\log\log N}{\log N}\right) < \frac{2\pi v M}{N(\log N)^2}, \end{aligned}$$

which proves (1.9).

Finally, the upper bound in (1.10) follows from relations (1.11) and (1.3) and the monotone decrease of the function dt_u/du in the interval $N \le u \le N+M$. The lower bound in (1.10) obviously follows from (1.4). This completes the proof of the lemma.

Lemma 1.2. *Let μ and ν be integers satisfying the conditions $1 \le \mu, \nu \le \sqrt{N/\log N}$, and $\mu \ne \nu$. Then the sum*

$$S = \sum_n \left(\frac{\mu}{\nu}\right)^{it_n}$$

does not exceed $\sqrt{\mu\nu}\log N$ in absolute value.

Proof. Suppose for definiteness that $\mu > \nu \ge 1$. Let us write S in the form

$$\sum_n e^{2\pi i f(n)}, \qquad \text{where} \quad f(u) = \frac{t_u}{2\pi}\log\frac{\mu}{\nu}.$$

Using bounds (1.10) of Lemma 1.1 and the assumptions of the lemma, we obtain

$$0 < f'(u) = \frac{1}{2\pi}\frac{dt_u}{du}\log\frac{\mu}{\nu} < \frac{1}{2\pi}\frac{2\pi\log\mu}{\log N - \log\log N} \le \frac{1}{2}.$$

By virtue of Lemma 1.1, we have

$$|S| < |j| + 3, \qquad j = \int_N^{N+M} e^{2\pi i f(u)}\, du.$$

Integrating by parts and using the monotonicity of $1/f'(u)$ and relations (1.10), we obtain

$$\begin{aligned} |j| &= \frac{1}{2\pi}\left|\frac{e^{2\pi i f(N+M)}}{f'(N+M)} - \frac{e^{2\pi i f(N)}}{f'(N)} - \int_N^{N+M} e^{2\pi i f(x)}\, d\frac{1}{f'(x)}\right| \\ &\le \frac{1}{2\pi}\left(\frac{1}{f'(N+M)} + \frac{1}{f'(N)} + \frac{1}{f'(N+M)} - \frac{1}{f'(N)}\right) = \frac{1}{\pi f'(N+M)} < \frac{1}{\pi}\frac{\log N}{\log(\mu/\nu)}. \end{aligned}$$

It is easy to see that

$$\log\frac{\mu}{\nu} \ge \log\frac{\nu+1}{\nu} = \frac{\lambda}{\nu}, \qquad \lambda = 1 - \frac{1}{2\nu} + \frac{1}{3\nu^2} - \frac{1}{4\nu^3} + \cdots.$$

Observing that $\lambda = \log 2$ for $\nu = 1$ and $\lambda \ge 3/4$ for $\nu \ge 2$, we arrive at

$$|j| < \frac{\nu \log N}{\pi \log 2} < \frac{\sqrt{\mu\nu}}{\pi \log 2}\log N \qquad \text{and} \qquad |S| < \frac{\sqrt{\mu\nu}}{\pi \log 2}\log N + 3 < \sqrt{\mu\nu}\log N,$$

which proves the lemma.

Lemma 1.3. *Let k, m, and r be integers such that $m \ge 0$, $r \ge 0$, and $m + r = k$, and let $e^2 < y < (N/\log N)^{1/(2k)}$. Suppose that $p_1, \dots, p_m$, $q_1, \dots, q_r$ range over the set of primes in the interval $(1, y]$ and satisfy the condition $p_1 \cdots p_m \ne q_1 \cdots q_r$. Finally, suppose that all members of the sequence $a(p)$ with $p \le y$ satisfy the condition $|a(p)| \le \delta$. Then the sum*

$$S = \sum_n \sum_{\substack{p_1,\dots,p_m \\ q_1,\dots,q_r}} \frac{a(p_1)\cdots a(p_m)\overline{a}(q_1)\cdots\overline{a}(q_r)}{(p_1\cdots p_m q_1\cdots q_r)^{1/2}}\left(\frac{q_1\cdots q_r}{p_1\cdots p_m}\right)^{it_n} \tag{1.14}$$

is estimated as

$$|S| < (\delta y)^k \log N.$$

Proof. Changing the order of summation in (1.14) and applying Lemma 1.2 with $\mu = q_1\cdots q_r$ and $\nu = p_1\cdots p_m$, we obtain

$$|S| < \delta^k \sum_{p_1,\dots,p_m \le x} \sum_{q_1,\dots,q_r \le x} (p_1\cdots q_r)^{-1/2}(p_1\cdots q_r)^{1/2}\log N = (\delta\pi(y))^k \log N < (\delta y)^k \log N,$$

as required.

Lemma 1.4. *Let $k \ge 1$ be an integer, and let $x_0 < x \le M^{1/(2k)}$. Then*

$$V_{2k} = \sum_n V^{2k}(t_n) = \binom{2k}{k}\frac{M}{2^{2k}}\,\mathfrak{C}_k + \theta x^{2k}\log N,$$

where

$$\mathfrak{C}_k = \sum_{\substack{p_1\cdots p_k = q_1\cdots q_k \\ p_1,\dots,p_k,q_1,\dots,q_k \le x}} (p_1\cdots p_k)^{-1}$$

(the summation in $\mathfrak{C}_k$ is over all sets of primes $p_1, \dots, q_k$ not exceeding x and satisfying the equation $p_1\cdots p_k = q_1\cdots q_k$).

Proof. Setting

$$U(t)=\sum_{p\le x}\frac{p^{it}}{\sqrt{p}},\qquad \text{so that}\quad V(t)=\frac{1}{2i}(U(t)-\overline{U}(t)),$$

we represent the sum V_{2k} in the form

$$\frac{(-1)^k}{2^{2k}}\sum_n(U(t_n)-\overline{U}(t_n))^{2k}=\frac{(-1)^k}{2^{2k}}\sum_{\nu=0}^{2k}(-1)^\nu\binom{2k}{\nu}v_\nu,$$

where

$$v_\nu=\sum_n U^\nu(t_n)\overline{U}^{2k-\nu}(t_n)=\sum_n\sum_{\substack{p_1,\dots,p_\nu\le x\\ q_1,\dots,q_{2k-\nu}\le x}}(p_1\cdots q_{2k-\nu})^{-1/2}\biggl(\frac{p_1\cdots p_\nu}{q_1\cdots q_{2k-\nu}}\biggr)^{it_n}.$$

According to Lemma 1.2, for $\nu\ne k$, we have

$$|v_\nu|<(\log N)\sum_{\substack{p_1,\dots,p_\nu\le x\\ q_1,\dots,q_{2k-\nu}\le x}}(p_1\cdots q_{2k-\nu})^{-1/2}(p_1\cdots q_{2k-\nu})^{1/2}<x^{2k}\log N.$$

The contribution to v_k of the summands satisfying the condition $p_1\cdots p_k\ne q_1\cdots q_k$ is estimated in a similar way. Therefore, setting

$$\mathfrak{C}_k=\sum_{\substack{p_1\cdots p_k=q_1\cdots q_k\\ p_1,\dots,p_k,q_1,\dots,q_k\le x}}(p_1\cdots q_k)^{-1/2}=\sum_{\substack{p_1\cdots p_k=q_1\cdots q_k\\ p_1,\dots,p_k,q_1,\dots,q_k\le x}}(p_1\cdots p_k)^{-1},$$

we arrive at

$$V_{2k}=\binom{2k}{k}\frac{M}{2^{2k}}\,\mathfrak{C}_k+\theta 2^{-2k}\sum_{\nu=0}^{2k}\binom{2k}{\nu}x^{2k}\log N,$$

which implies the required assertion.

Lemma 1.5. *For each $k\ge 1$, the quantity*

$$\mathfrak{C}_k=\sum_{\substack{p_1\cdots p_k=q_1\cdots q_k\\ p_1,\dots,p_k,q_1,\dots,q_k\le x}}(p_1\cdots p_k)^{-1}$$

satisfies the relation $\mathfrak{C}_k=k!H^{(k)}(0)$, where

$$H(z)=\prod_{p\le x}J_0\biggl(2i\sqrt{\frac{z}{p}}\biggr),$$

and $J_0(u)$ is the Bessel function of the first kind.

Proof. We transform the quantity $\mathfrak{C}_k$, following Radziwiłł's paper [30]. Let $n\ge 2$ be an integer all of whose prime divisors do not exceed x, and let $r_1^{\alpha_1}\cdots r_s^{\alpha_s}$ be its canonical factorization. If the number $\Omega(n)=\alpha_1+\cdots+\alpha_s$ of its prime divisors equals k, then the equation

$$n=p_1\cdots p_k=q_1\cdots q_k \tag{1.15}$$

can be solved in primes $p_1,\dots,q_k\le x$, and the number of its solutions equals

$$\biggl(\frac{k!}{\alpha_1!\cdots\alpha_s!}\biggr)^2.$$

If $\Omega(n)\ne k$ or at least one of the primes $r_1,\dots,r_s$ is greater than x, then, obviously, Eq. (1.15) is unsolvable.

Consider the multiplicative function $f(n)$ defined at prime powers by

$$f(r^\alpha) = \begin{cases} (\alpha!)^{-2} & \text{if } r \le x, \\ 0 & \text{if } r > x. \end{cases}$$

Then the number of solutions of Eq. (1.15) is $(k!)^2 f(n)$ for any n, and the sum $\mathfrak{C}_k$ equals $(k!)^2 h(k)$, where

$$h(k) = \sum_{\substack{n=1 \\ \Omega(n)=k}}^{+\infty} \frac{f(n)}{n}.$$

To calculate $h(k)$, consider the formal series

$$H(z) = \sum_{k=0}^{+\infty} h(k) z^k.$$

Changing the order of summation and using the multiplicativity (with respect to n) of the function $z^{\Omega(n)}$, we obtain

$$H(z) = \sum_{k=0}^{+\infty} \Biggl\{ \sum_{\substack{n=1 \\ \Omega(n)=k}}^{+\infty} \frac{f(n)}{n} \Biggr\} z^k = \sum_{n=1}^{+\infty} \frac{f(n)}{n} z^{\Omega(n)}$$

$$= \prod_p \Biggl\{ \sum_{m=0}^{+\infty} \frac{f(p^m)}{p^m} z^{\Omega(p^m)} \Biggr\} = \prod_{p \le x} \Biggl\{ \sum_{m=0}^{+\infty} \frac{1}{(m!)^2} \biggl(\frac{z}{p} \biggr)^m \Biggr\}. \tag{1.16}$$

Each multiplier in the last product is a series absolutely convergent on any finite part of the plane. Therefore, the series $H(z)$, which is the product of finitely many such series, converges absolutely for any z. Thus, the change of the order of summation in (1.16) is legitimate. Hence, $H(z)$ is an entire function, so that

$$h(k) = \frac{H^{(k)}(0)}{k!} \qquad \text{and} \qquad \mathfrak{C}_k = k! H^{(k)}(0).$$

Observing, finally, that

$$J_0(z) = \sum_{m=0}^{+\infty} \frac{(-1)^m}{(m!)^2} \biggl(\frac{z}{2} \biggr)^{2m},$$

we obtain the required result.

Lemma 1.6. *For any $k \ge 0$,*

$$H^{(k)}(0) = k! \sum_{n=0}^{k} \frac{\Phi_n}{(k-n)!} \mathfrak{S}_1^{k-n},$$

where

$$\mathfrak{S}_1 = \sum_{p \le x} \frac{1}{p}$$

and the absolute values of the coefficients Φ_n remain bounded above by a constant depending only on n as x unboundedly increases.

Proof. The first terms of the series

$$J_0 \biggl(2i \sqrt{\frac{z}{p}} \biggr) = \sum_{m=0}^{+\infty} \frac{1}{(m!)^2} \biggl(\frac{z}{p} \biggr)^m = 1 + \frac{z}{p} + \cdots$$

on the right-hand side of (1.16) coincide with the first terms of the expansion

$$e^{z/p} = 1 + \frac{z}{p} + \cdots .$$

This observation allows us to surmise that, for sufficiently small z, product (1.16) expressing $H(z)$ behaves similarly to the function

$$\prod_{p\le x} e^{z/p} = \exp\left\{ z \sum_{p\le x} \frac{1}{p} \right\} = e^{z\mathfrak{S}_1}.$$

Therefore, for what follows, it is convenient to represent $H(z)$ in the form

$$\prod_{p\le x} e^{z/p}\cdot e^{-z/p}\left\{ \sum_{m=0}^{+\infty} \frac{1}{(m!)^2}\left(\frac{z}{p}\right)^m \right\} = \prod_{p\le x} e^{z/p}\varphi\left(\frac{z}{p}\right) = e^{z\mathfrak{S}_1}\Phi(z),$$

where

$$\varphi(v) = e^{-v}\sum_{m=0}^{+\infty} \frac{v^m}{(m!)^2} \qquad \text{and} \qquad \Phi(z) = \prod_{p\le x} \varphi\left(\frac{z}{p}\right).$$

Differentiating the relation $H(z) = e^{z\mathfrak{S}_1}\Phi(z)$, we obtain

$$H^{(k)}(z) = \sum_{n=0}^{+\infty} \binom{k}{n} \Phi^{(n)}(z)(e^{z\mathfrak{S}_1})^{(k-n)} = \sum_{n=0}^{k} \binom{k}{n} \Phi^{(n)}(z)\mathfrak{S}_1^{k-n} e^{z\mathfrak{S}_1},$$

whence

$$H^{(k)}(0) = \sum_{n=0}^{k} \binom{k}{n} \Phi^{(n)}(0)\mathfrak{S}_1^{k-n}.$$

For the coefficients Φ_n in the expansion of $\Phi(z)$ in powers of z, we have

$$\Phi^{(n)}(0) = n!\,\Phi_n, \qquad \text{so that} \quad H^{(k)}(0) = \sum_{n=0}^{k} n! \binom{k}{n} \Phi_n \mathfrak{S}_1^{k-n}.$$

Note that the power series expansion of $\varphi(z/p)$ contains no term with z/p:

$$\varphi\left(\frac{z}{p}\right) = 1 - \frac{z^2}{4p^2} + \frac{z^3}{18p^3} - \cdots .$$

Therefore, in the product of such expansions over all $p \le x$, the coefficient of z^n is a finite linear combination of sums of products of the form

$$\frac{1}{p_1^{m_1}\cdots p_r^{m_r}}, \qquad r \le n, \quad p_1, \ldots, p_r \le x, \quad m_1 + \cdots + m_r = n,$$

in which all exponents $m_1, \ldots, m_r$ are greater than or equal to 2. Since the series $\sum p^{-r}$, $r = 2, 3, \ldots$, converge, it follows that each coefficient Φ_n is bounded in absolute value by a constant depending only on n and not depending on x. This completes the proof of the lemma.

Lemmas 1.4–1.6 proved above allow us to state the main result of this section as follows.

Theorem 1.1. *If $k \ge 1$ is an integer and $x_0 < x \le M^{1/(2k)}$, then*

$$V_{2k} = \sum_n V^{2k}(t_n) = \frac{M}{2^{2k}} \sum_{n=0}^{k} \frac{(2k)!}{(k-n)!}\,\Phi_n \mathfrak{S}_1^{k-n} + \theta x^{2k}\log N,$$

where each coefficient Φ_n is a polynomial in the quantities $\mathfrak{S}_2, \dots, \mathfrak{S}_n$ defined by

$$\mathfrak{S}_r = \sum_{p \le x} \frac{1}{p^r}.$$

The first part of this theorem follows directly from Lemmas 1.4–1.6. The proof of the second part is given in the next section, which is entirely devoted to the study of properties of the coefficients Φ_n.

1.2. Explicit expressions for the coefficients Φ_n. The expression for an even moment V_{2k} given in Theorem 1.1 contains quantities Φ_n, which are defined as the coefficients in the expansion of the function $\Phi(z)$ in powers of z. The further analysis of such expressions requires determining certain properties of these coefficients. This is the purpose of this and the next section.

First, we complete the proof of Theorem 1.1. Namely, we explicitly specify Φ_n for small n and describe a procedure for expressing Φ_n as a polynomial in $\mathfrak{S}_2, \dots, \mathfrak{S}_n$ in finitely many steps for any n.

Lemma 1.7. *The following relations hold:*

$$\Phi_0 \equiv 1, \qquad \Phi_1 \equiv 0, \qquad \Phi_2 = -\frac{1}{2^2}\,\mathfrak{S}_2, \qquad \Phi_3 = \frac{1}{3^2}\,\mathfrak{S}_3,$$
$$\Phi_4 = -\frac{11}{2^6 \cdot 3}\,\mathfrak{S}_4 + \frac{1}{2^5}\,\mathfrak{S}_2^2, \qquad \Phi_5 = \frac{19}{2^3 \cdot 3 \cdot 5^2}\,\mathfrak{S}_5 - \frac{1}{2^2 \cdot 3^2}\,\mathfrak{S}_2\mathfrak{S}_3,$$
$$\Phi_6 = -\frac{11 \cdot 43}{2^6 \cdot 3^4 \cdot 5}\,\mathfrak{S}_6 + \frac{11}{2^8 \cdot 3}\,\mathfrak{S}_2\mathfrak{S}_4 + \frac{1}{2 \cdot 3^4}\,\mathfrak{S}_3^2 - \frac{1}{2^7 \cdot 3}\,\mathfrak{S}_2^3,$$

and

$$\Phi_7 = \frac{229}{2^4 \cdot 3^3 \cdot 7^2}\,\mathfrak{S}_7 - \frac{19}{2^5 \cdot 3 \cdot 5^2}\,\mathfrak{S}_2\mathfrak{S}_5 - \frac{11}{2^6 \cdot 3^3}\,\mathfrak{S}_3\mathfrak{S}_4 + \frac{1}{2^5 \cdot 3^2}\,\mathfrak{S}_2^2\mathfrak{S}_3.$$

Proof. Let ϖ_n denote the Fourier coefficients of the function $\varphi(v)$, so that

$$\varphi(v) = e^{-v} \sum_{m=0}^{+\infty} \frac{v^m}{(m!)^2} = \sum_{n=0}^{+\infty} \varpi_n v^n.$$

It is easy to verify that

$$\varpi_0 = 1, \qquad \varpi_1 = 0, \qquad 2!\,\varpi_2 = -\frac{1}{2}, \qquad 3!\,\varpi_3 = \frac{2}{3}, \qquad 4!\,\varpi_4 = -\frac{5}{8},$$
$$5!\,\varpi_5 = \frac{7}{15}, \qquad 6!\,\varpi_6 = -\frac{37}{144}, \qquad 7!\,\varpi_7 = \frac{17}{420},$$

and, in general,

$$n!\,\varpi_n = (-1)^n \sum_{m=0}^{n} \frac{(-1)^m}{m!} \binom{n}{m}.$$

Multiplying out the expansions of $\varphi(z/p)$ over all $p \le x$, we obtain

$$\Phi(z) = \prod_{p \le x} \sum_{m=0}^{+\infty} \varpi_m \left(\frac{z}{p}\right)^m = \sum_{r \ge 0} \sum_{m_1, \dots, m_r \ge 0} \sum_{p_1 < \dots < p_r \le x} \frac{\varpi_{m_1} \cdots \varpi_{m_r}}{p_1^{m_1} \cdots p_r^{m_r}}\, z^{m_1 + \dots + m_r},$$

whence

$$\Phi_n = \sum_{0 \le r \le n} \sum_{\substack{m_1 + \dots + m_r = n \\ m_1, \dots, m_r \ge 0}} \varpi_{m_1} \cdots \varpi_{m_r} T_{m_1, \dots, m_r},$$

where

$$T_{m_1, \dots, m_r} = \sum_{p_1 < \dots < p_r \le x} (p_1^{m_1} \cdots p_r^{m_r})^{-1}.$$

We have $\varpi_1 = 0$; hence, a nonzero contribution to Φ_n is made only by the terms corresponding to the r-tuples $m_1, \dots, m_r$ not containing 1, in which $m_j = 0$ or $m_j \geq 2$ for all j. Let s be the number of nonzero components of such an r-tuple, and let $\alpha_1, \dots, \alpha_s$ be these nonzero components written in the order of their occurrence. Since $\alpha_1, \dots, \alpha_s \geq 2$, it follows that $n = \alpha_1 + \dots + \alpha_s \geq 2s$, which implies the inequality $1 \leq s \leq n/2$ as a necessary condition for $n \geq 2$. Therefore,

$$\Phi_n = \sum_{1 \leq s \leq n/2} \sum_{\substack{\alpha_1 + \dots + \alpha_s = n \\ \alpha_1, \dots, \alpha_s \geq 2}} \varpi_{\alpha_1} \dots \varpi_{\alpha_s} T_{\alpha_1, \dots, \alpha_s}. \tag{1.17}$$

Thus,

$$\Phi_2 = \varpi_2 T_2 = \varpi_2 \mathfrak{S}_2 = -\frac{1}{4}\mathfrak{S}_2 \qquad \text{and} \qquad \Phi_3 = \varpi_3 T_3 = \varpi_3 \mathfrak{S}_3 = \frac{1}{9}\mathfrak{S}_3.$$

Next,

$$\Phi_4 = \varpi_4 T_4 + \varpi_2^2 T_{2,2}, \qquad \Phi_5 = \varpi_5 T_5 + \varpi_2 \varpi_3 (T_{2,3} + T_{3,2}),$$
$$\Phi_6 = \varpi_6 T_6 + \varpi_3^2 T_{3,3} + \varpi_2 \varpi_4 (T_{2,4} + T_{4,2}) + \varpi_2^3 T_{2,2,2},$$

and

$$\Phi_7 = \varpi_7 T_7 + \varpi_2 \varpi_5 (T_{2,5} + T_{5,2}) + \varpi_3 \varpi_4 (T_{3,4} + T_{4,3}) + \varpi_2^2 \varpi_3 (T_{2,2,3} + T_{2,3,2} + T_{3,2,2}).$$

Let α and β be any positive integers. Then

$$\mathfrak{S}_\alpha \mathfrak{S}_\beta = \sum_{p,q \leq x} \frac{1}{p^\alpha q^\beta} = \sum_{p<q \leq x} \frac{1}{p^\alpha q^\beta} + \sum_{q<p \leq x} \frac{1}{q^\beta p^\alpha} + \sum_{p \leq x} \frac{1}{p^{\alpha+\beta}} = T_{\alpha,\beta} + T_{\beta,\alpha} + \mathfrak{S}_{\alpha+\beta},$$

whence

$$T_{\alpha,\beta} + T_{\beta,\alpha} = \mathfrak{S}_\alpha \mathfrak{S}_\beta - \mathfrak{S}_{\alpha+\beta}.$$

Similarly,

$$\begin{aligned}
\mathfrak{S}_\alpha^2 \mathfrak{S}_\beta &= \sum_{p,q,r \leq x} \frac{1}{p^\alpha q^\alpha r^\beta} = \biggl\{ \sum_{p<q \leq x} + \sum_{q<p \leq x} \biggr\} \frac{1}{p^\alpha q^\alpha} \sum_{r \leq x} \frac{1}{r^\beta} + \sum_{p,r \leq x} \frac{1}{p^{2\alpha} r^\beta} \\
&= 2 \sum_{p<q \leq x} \sum_{r \leq x} \frac{1}{p^\alpha q^\alpha r^\beta} + \sum_{p<r \leq x} \frac{1}{p^{2\alpha} r^\beta} + \sum_{r<p \leq x} \frac{1}{r^\beta p^{2\alpha}} + \sum_{p \leq x} \frac{1}{p^{2\alpha+\beta}} \\
&= 2 \biggl\{ \sum_{r<p<q \leq x} \frac{1}{r^\beta p^\alpha q^\alpha} + \sum_{p<q \leq x} \frac{1}{p^{\alpha+\beta} q^\alpha} + \sum_{p<r<q \leq x} \frac{1}{p^\alpha r^\beta q^\alpha} + \sum_{p<q \leq x} \frac{1}{p^\alpha q^{\alpha+\beta}} + \sum_{p<q<r \leq x} \frac{1}{p^\alpha q^\alpha r^\beta} \biggr\} \\
&\quad + T_{2\alpha,\beta} + T_{\beta,2\alpha} + \mathfrak{S}_{2\alpha+\beta} \\
&= 2(T_{\beta,\alpha,\alpha} + T_{\alpha+\beta,\alpha} + T_{\alpha,\beta,\alpha} + T_{\alpha,\alpha+\beta} + T_{\alpha,\alpha,\beta}) + T_{2\alpha,\beta} + T_{\beta,2\alpha} + \mathfrak{S}_{2\alpha+\beta},
\end{aligned}$$

so that

$$T_{\alpha,\alpha,\beta} + T_{\alpha,\beta,\alpha} + T_{\beta,\alpha,\alpha} = \frac{1}{2}(\mathfrak{S}_\alpha^2 \mathfrak{S}_\beta - \mathfrak{S}_{2\alpha} \mathfrak{S}_\beta) + \mathfrak{S}_{2\alpha+\beta} - \mathfrak{S}_\alpha \mathfrak{S}_{\alpha+\beta}.$$

Thus,

$$T_{2,2} = \frac{1}{2}(\mathfrak{S}_2^2 - \mathfrak{S}_4), \qquad T_{3,3} = \frac{1}{2}(\mathfrak{S}_3^2 - \mathfrak{S}_6),$$
$$T_{2,3} + T_{3,2} = \mathfrak{S}_2 \mathfrak{S}_3 - \mathfrak{S}_5, \qquad T_{2,4} + T_{4,2} = \mathfrak{S}_2 \mathfrak{S}_4 - \mathfrak{S}_6,$$
$$T_{2,5} + T_{5,2} = \mathfrak{S}_2 \mathfrak{S}_5 - \mathfrak{S}_7, \qquad T_{3,4} + T_{4,3} = \mathfrak{S}_3 \mathfrak{S}_4 - \mathfrak{S}_7,$$
$$T_{2,2,2} = \frac{1}{6}\mathfrak{S}_2^3 - \frac{1}{2}\mathfrak{S}_2 \mathfrak{S}_4 + \frac{1}{3}\mathfrak{S}_6,$$

and, finally,

$$T_{2,2,3}+T_{2,3,2}+T_{3,2,2}=\frac{1}{2}\,\mathfrak{S}_2^2\mathfrak{S}_3-\mathfrak{S}_2\mathfrak{S}_5-\frac{1}{2}\,\mathfrak{S}_3\mathfrak{S}_4+\mathfrak{S}_7.$$

Returning to the expressions for Φ_n, $n=4,\dots,7$, we obtain

$$\Phi_4=\varpi_4\mathfrak{S}_4+\frac{1}{2}\,\varpi_2^2(\mathfrak{S}_2^2-\mathfrak{S}_4)=\biggl(\varpi_4-\frac{1}{2}\,\varpi_2^2\biggr)\mathfrak{S}_4+\frac{1}{2}\,\varpi_2^2\mathfrak{S}_2=-\frac{11}{192}\,\mathfrak{S}_4+\frac{1}{32}\,\mathfrak{S}_2^2,$$

$$\Phi_5=\varpi_5\mathfrak{S}_5+\varpi_2\varpi_3(\mathfrak{S}_2\mathfrak{S}_3-\mathfrak{S}_5)=(\varpi_5-\varpi_2\varpi_3)\mathfrak{S}_5+\varpi_2\varpi_3\mathfrak{S}_2\mathfrak{S}_3=\frac{19}{600}\,\mathfrak{S}_5-\frac{1}{36}\,\mathfrak{S}_2\mathfrak{S}_3,$$

$$\begin{aligned}\Phi_6&=\varpi_6\mathfrak{S}_6+\frac{1}{2}\,\varpi_3^2(\mathfrak{S}_3^2-\mathfrak{S}_6)+\varpi_2\varpi_4(\mathfrak{S}_2\mathfrak{S}_4-\mathfrak{S}_6)+\varpi_2^3\biggl(\frac{1}{3}\,\mathfrak{S}_6-\frac{1}{2}\,\mathfrak{S}_2\mathfrak{S}_4+\frac{1}{6}\,\mathfrak{S}_2^3\biggr)\\&=\biggl(\varpi_6-\frac{1}{2}\,\varpi_3^2-\varpi_2\varpi_4+\frac{1}{3}\,\varpi_2^3\biggr)\mathfrak{S}_6+\biggl(\varpi_2\varpi_4-\frac{1}{2}\,\varpi_2^3\biggr)\mathfrak{S}_2\mathfrak{S}_4++\frac{1}{2}\,\varpi_3^2\mathfrak{S}_3^2+\frac{1}{6}\,\varpi_2^3S_2^3\\&=-\frac{473}{25\,920}\,\mathfrak{S}_6+\frac{11}{768}\,\mathfrak{S}_2\mathfrak{S}_4+\frac{1}{162}\,\mathfrak{S}_3^2-\frac{1}{384}\,\mathfrak{S}_2^3,\end{aligned}$$

and

$$\begin{aligned}\Phi_7&=\varpi_7\mathfrak{S}_7+\varpi_2\varpi_5(\mathfrak{S}_2\mathfrak{S}_5-\mathfrak{S}_7)+\varpi_3\varpi_4(\mathfrak{S}_3\mathfrak{S}_4-\mathfrak{S}_7)\\&\qquad+\varpi_2^2\varpi_3\biggl(\frac{1}{2}\,\mathfrak{S}_2^2\mathfrak{S}_3-\mathfrak{S}_2\mathfrak{S}_5+\mathfrak{S}_7-\frac{1}{2}\,\mathfrak{S}_3\mathfrak{S}_4\biggr)\\&=(\varpi_7-\varpi_2\varpi_5-\varpi_3\varpi_4+\varpi_2^2\varpi_3)\mathfrak{S}_7+(\varpi_2\varpi_5-\varpi_2^2\varpi_3)\mathfrak{S}_2\mathfrak{S}_5\\&\qquad+\biggl(\varpi_3\varpi_4-\frac{1}{2}\,\varpi_2^2\varpi_3\biggr)\mathfrak{S}_3\mathfrak{S}_4+\frac{1}{2}\,\varpi_2^2\varpi_3\mathfrak{S}_2^2\mathfrak{S}_3\\&=\frac{229}{21\,168}\,\mathfrak{S}_7-\frac{19}{2\,400}\,\mathfrak{S}_2\mathfrak{S}_5-\frac{11}{1\,728}\,\mathfrak{S}_3\mathfrak{S}_4+\frac{1}{288}\,\mathfrak{S}_2^2\mathfrak{S}_3.\end{aligned}$$

This completes the proof of the lemma.

The method for finding Φ_n, $n\le 7$, applied above turns out to be very laborious even for comparatively small n. To render the relations of Theorem 1.1 (as well as those of Theorems 1.2 and 1.3) usable in computations, we describe an algorithm which yields an explicit expression for Φ_n in the form of a polynomial in $\mathfrak{S}_2,\dots,\mathfrak{S}_n$ in finitely many steps.

We precede the description of the algorithm by auxiliary statements.

Let $\alpha_1,\dots,\alpha_s$ be an increasing s-tuple of any positive integers $\alpha_1\le\dots\le\alpha_s$. By $Q_{\alpha_1,\dots,\alpha_s}(x)$ (or, briefly, $Q_{\alpha_1,\dots,\alpha_s}$) we denote the sum

$$\sum_{p_1,\dots,p_s\le x}{}'\,(p_1^{\alpha_1}\cdots p_s^{\alpha_s})^{-1},\tag{1.18}$$

where the prime denotes summation over all s-tuples of primes $p_1,\dots,p_s$ without repetitions.

Lemma 1.8. *The relation*

$$Q_{\alpha_1,\dots,\alpha_s}=\nu_1!\cdots\nu_m!\sum_{\overline{\beta}}T_{\beta_1,\dots,\beta_s}$$

holds, where m is the number of different integers in the s-tuple $(\alpha_1,\dots,\alpha_s)$, $\nu_1,\dots,\nu_m$ are the numbers of occurrences of these integers, and the summation is over all different permutations $\overline{\beta}=(\beta_1,\dots,\beta_s)$ of $(\alpha_1,\dots,\alpha_s)$.

Proof. For each summand in (1.18) and any pair of primes p_i, p_j, one of the inequalities $p_i < p_j$ and $p_j < p_i$ holds. Therefore, the range of variation of the variables in (1.18) can be partitioned into $s!$ regions, each of which is determined by a relation of the form

$$p_{i_1} < p_{i_2} < \cdots < p_{i_s} \le x,$$

where $i_1, i_2, \dots, i_s$ is a permutation of the numbers $1, 2, \dots, s$. Accordingly, $Q_{\alpha_1,\dots,\alpha_s}$ decomposes into $s!$ sums of the form

$$\sum_{p_{i_1}<p_{i_2}<\cdots<p_{i_s}\le x} (p_1^{\alpha_1}\cdots p_s^{\alpha_s})^{-1} = \sum_{p_{i_1}<p_{i_2}<\cdots<p_{i_s}\le x} (p_1^{\alpha_{i_1}}\cdots p_s^{\alpha_{i_s}})^{-1} = T_{\alpha_{i_1},\dots,\alpha_{i_s}}.$$

Some of the numbers $\alpha_1, \dots, \alpha_s$ may be equal; hence, some of these $s!$ sums may coincide. This happens when the permutations $(\alpha_{i_1}, \dots, \alpha_{i_s})$ coincide. Interchanging equal elements in $(\alpha_{i_1}, \dots, \alpha_{i_s})$, we obtain $\nu_1! \cdots \nu_m!$ permutations coinciding with the initial one. Therefore, each of the sums $T_{\alpha_{i_1},\dots,\alpha_{i_s}}$ corresponding to different permutations $(\alpha_{i_1}, \dots, \alpha_{i_s})$, occurs in $Q_{\alpha_1,\dots,\alpha_s}$ with the coefficient $\nu_1! \cdots \nu_m!$. This completes the proof of the lemma.

Example 1.1. Let $(\alpha_1, \alpha_2, \alpha_3) = (2, 2, 3)$. Then

$$\begin{aligned}
Q_{2,2,3} &= {\sum_{p_1,p_2,p_3\le x}}' (p_1^2p_2^2p_3^3)^{-1} \\
&= \Bigg\{ \sum_{p_1<p_2<p_3} + \sum_{p_1<p_3<p_2} + \sum_{p_2<p_1<p_3} + \sum_{p_2<p_3<p_1} + \sum_{p_3<p_1<p_2} + \sum_{p_3<p_2<p_1} \Bigg\} (p_1^2p_2^2p_3^3)^{-1} \\
&= \sum_{q_1<q_2<q_3} (q_1^2q_2^2q_3^3)^{-1} + \sum_{q_1<q_2<q_3} (q_1^2q_2^3q_3^2)^{-1} + \sum_{q_1<q_2<q_3} (q_1^2q_2^2q_3^3)^{-1} \\
&\quad + \sum_{q_1<q_2<q_3} (q_1^2q_2^3q_3^2)^{-1} + \sum_{q_1<q_2<q_3} (q_1^3q_2^2q_3^2)^{-1} + \sum_{q_1<q_2<q_3} (q_1^3q_2^2q_3^2)^{-1} \\
&= T_{2,2,3} + T_{2,3,2} + T_{2,2,3} + T_{2,3,2} + T_{3,2,2} + T_{3,2,2} = 2(T_{2,2,3} + T_{2,3,2} + T_{3,2,2}).
\end{aligned}$$

Lemma 1.9. *The sum $Q_{\alpha_1,\dots,\alpha_s}$ is a polynomial in $\mathfrak{S}_2, \dots, \mathfrak{S}_n$, where $n = \alpha_1 + \cdots + \alpha_s$.*

Proof. We prove the lemma by induction on s. If $s = 1$, then

$$Q_\alpha = \sum_{p\le x} \frac{1}{p^\alpha} = \mathfrak{S}_\alpha,$$

and the required assertion is obvious.

Suppose that the assertion is true for all $s \le m$. Let us prove it for $s = m + 1$. Consider any $(m + 1)$-tuple $(\alpha_1, \dots, \alpha_{m+1})$ in which $\alpha_i \ge 2$ for $i = 1, \dots, m + 1$. Multiplying out the expressions

$$\mathfrak{S}_{\alpha_1} = \sum_{p_1\le x} p_1^{-\alpha_1}, \qquad \dots, \qquad \mathfrak{S}_{\alpha_{m+1}} = \sum_{p_{m+1}\le x} p_{m+1}^{-\alpha_{m+1}},$$

we obtain the sum

$$\sum_{p_1,\dots,p_{m+1}\le x} (p_1^{\alpha_1}\cdots p_{m+1}^{\alpha_{m+1}})^{-1}.$$

We divide all terms in this sum into two classes, the class of the terms corresponding to repetition-free $(m + 1)$-tuples $(p_1, \dots, p_{m+1})$ and the class of all remaining terms.

The terms in the first class form the sum $Q_{\alpha_1,\dots,\alpha_{m+1}}$.

Each term in the second class corresponds to an $(m + 1)$-tuple containing at least two equal primes and, therefore, can be written in the form $(q_1^{\beta_1}\cdots q_r^{\beta_r})^{-1}$, where the number r of factors corresponding to different primes $q_1, \dots, q_r$ is strictly less than s: $r \le s - 1 = m$.

Therefore, all terms in the second class can be distributed over the sums $Q_{\beta_1,\dots,\beta_r}$ with $1 \le r \le s-1 = m$. By the induction hypothesis, any such sum is a polynomial in $\mathfrak{S}_2, \dots, \mathfrak{S}_n$. Hence, the quantity $Q_{\alpha_1,\dots,\alpha_{m+1}}$, which is the difference between the monomial $\mathfrak{S}_{\alpha_1} \cdots \mathfrak{S}_{\alpha_{m+1}}$ and the terms in the second class, is a polynomial in $\mathfrak{S}_2, \dots, \mathfrak{S}_n$ as well. This proves the lemma and, thereby, Theorem 1.1.

Analyzing this proof, we see that the sums $Q_{\beta_1,\dots,\beta_r}$ obtained by grouping terms in the second class correspond to the r-tuples $\overline{\beta} = (\beta_1, \dots, \beta_r)$ obtained from $\overline{\alpha} = (\alpha_1, \dots, \alpha_s)$ by replacing some components $\alpha_{i_1}, \dots, \alpha_{i_k}$ by one component equal to the sum $\alpha_{i_1} + \cdots + \alpha_{i_k}$ (or by performing this operation several times). For any $\overline{\beta}$ obtained from $\overline{\alpha}$ in this way, we write $\overline{\beta} \prec \overline{\alpha}$.

The expression for the sum $Q_{\beta_1,\dots,\beta_r}$ contains only one monomial being the product of r of the quantities $\mathfrak{S}_2, \dots, \mathfrak{S}_n$, namely, the monomial $\mathfrak{S}_{\beta_1} \cdots \mathfrak{S}_{\beta_r}$, and this monomial is contained in $Q_{\beta_1,\dots,\beta_r}$ with coefficient 1. Therefore,

$$Q_{\alpha_1,\dots,\alpha_s} = \mathfrak{S}_{\alpha_1} \cdots \mathfrak{S}_{\alpha_s} + \sum_{r=1}^{s-1} \sum_{\overline{\beta}=(\beta_1,\dots,\beta_r) \prec \overline{\alpha}} A_{\overline{\beta}} \mathfrak{S}_{\beta_1} \cdots \mathfrak{S}_{\beta_r}, \tag{1.19}$$

where the $A_{\overline{\beta}}$ are numbers not depending on $\mathfrak{S}_2, \dots, \mathfrak{S}_n$ and x.

Now, we determine how the quantities Φ_n are related to the sums $Q_{\alpha_1,\dots,\alpha_s}$.

In the proof of Lemma 1.7, we obtained the expression

$$\Phi_n = \sum_{1 \le s \le n/2} \sum_{\substack{\beta_1+\cdots+\beta_s=n \\ \beta_1,\dots,\beta_s \ge 2}} \varpi_{\beta_1} \cdots \varpi_{\beta_s} T_{\beta_1,\dots,\beta_s},$$

in which the summation is over all s-tuples $\overline{\beta} = (\beta_1, \dots, \beta_s)$ being solutions of the equation

$$\beta_1 + \cdots + \beta_s = n \tag{1.20}$$

and satisfying the condition $\beta_1, \dots, \beta_s \ge 2$. In the inner sum, we collect together the terms corresponding to the permutations $\overline{\beta} = (\beta_1, \dots, \beta_s)$ of the same solution $\overline{\alpha} = (\alpha_1, \dots, \alpha_s)$ of Eq. (1.20) with increasing components: $\alpha_1 \le \cdots \le \alpha_s$.

For all such s-tuples $\overline{\beta}$, the product $\varpi_{\beta_1} \cdots \varpi_{\beta_s}$ equals the same number $\varpi_{\alpha_1} \cdots \varpi_{\alpha_s}$. By Lemma 1.8, the sum of $T_{\beta_1,\dots,\beta_s}$ coincides with $Q_{\alpha_1,\dots,\alpha_s}$ divided by $\nu_1! \cdots \nu_m!$, where m is the number of different components in $\overline{\alpha}$ and $\nu_1, \dots, \nu_m$ are the numbers of occurrences of these components. Thus,

$$\Phi_n = \sum_{1 \le s \le n/2} {\sum_{\overline{\alpha}=(\alpha_1,\dots,\alpha_s)}}' \frac{\varpi_{\alpha_1} \cdots \varpi_{\alpha_s}}{\nu_1! \cdots \nu_m!} Q_{\alpha_1,\dots,\alpha_s},$$

where the prime denotes summation over all partitions of n into s summands greater than 1; partitions which differ from each other only in the order of summands are considered equal.

Example 1.2. In the case $n = 8$, we have the partitions

$$n = 8 = 2+6 = 3+5 = 4+4 = 2+2+4 = 2+3+3 = 2+2+2+2.$$

Therefore,

$$\begin{aligned}\Phi_8 = {} & \varpi_8 T_8 + \varpi_2\varpi_6(T_{2,6} + T_{6,2}) + \varpi_3\varpi_5(T_{3,5} + T_{5,3}) + \varpi_4^2 T_{4,4} + \varpi_2^2\varpi_4(T_{2,2,4} + T_{2,4,2} + T_{4,2,2}) \\ & + \varpi_2\varpi_3^2(T_{2,3,3} + T_{3,2,3} + T_{3,3,2}) + \varpi_2^4 T_{2,2,2,2}.\end{aligned}$$

The results for this case are collected in Table 1, in which the notation introduced above is used.

Finally, we obtain

$$\begin{aligned}\Phi_8 = {} & \varpi_8 Q_8 + \varpi_2\varpi_6 Q_{2,6} + \varpi_3\varpi_5 Q_{3,5} \\ & + \frac{1}{2}\varpi_4^2 Q_{4,4} + \frac{1}{2}\varpi_2^2\varpi_4 Q_{2,2,4} + \frac{1}{2}\varpi_2\varpi_3^2 Q_{2,3,3} + \frac{1}{24}\varpi_2^4 Q_{2,2,2,2}.\end{aligned} \tag{1.21}$$

Table 1

$\overline{\beta}$	m	$\nu_1,\dots,\nu_m$	$\nu_1!\cdots\nu_m!$
(8)	1	$\nu_1=1$	1
$(2,6)$	2	$\nu_1=\nu_2=1$	1
$(3,5)$	2	$\nu_1=\nu_2=1$	1
$(4,4)$	1	$\nu_1=2$	2
$(2,2,4)$	2	$\nu_1=2,\nu_2=1$	2
$(2,3,3)$	2	$\nu_1=1,\nu_2=2$	2
$(2,2,2,2)$	1	$\nu_1=4$	24

Thus, to express a coefficient Φ_n in terms of $\mathfrak{S}_2,\dots,\mathfrak{S}_n$, it suffices to find a way to calculate the coefficients $A_{\overline{\beta}}$ in (1.19). For this purpose, we compose a system of linear equations with respect to the unknowns $A_{\overline{\beta}}$ that has a unique solution.

First, we arrange all monomials $\mathfrak{S}_{\beta_1}\cdots\mathfrak{S}_{\beta_r}$ in (1.19) so that

(1) the expression begins with the monomial $\mathfrak{S}_{\alpha_1}\cdots\mathfrak{S}_{\alpha_s}$;

(2) for any $\overline{\gamma}=(\gamma_1,\dots,\gamma_k)$ and $\overline{\beta}=(\beta_1,\dots,\beta_r)$ such that $\overline{\gamma}\prec\overline{\beta}$, the monomial $\mathfrak{S}_{\beta_1}\cdots\mathfrak{S}_{\beta_r}$ precedes $\mathfrak{S}_{\gamma_1}\cdots\mathfrak{S}_{\gamma_k}$.

Based on this order of monomials, we compose the matrix of a system with unknowns $A_{\overline{\beta}}$, calculating the coefficient of each fraction $\mathfrak{f}=\mathfrak{f}(\overline{\gamma})=(p_1^{\gamma_1}\cdots p_k^{\gamma_k})^{-1}$ in the decomposition $\mathfrak{S}_{\beta_1}\cdots\mathfrak{S}_{\beta_r}$.

Each of the fractions $\mathfrak{f}(\overline{\gamma})$ is absent from the sum $Q_{\alpha_1,\dots,\alpha_s}$ but occurs in the sum resulting from multiplying out the monomials $\mathfrak{S}_{\alpha_1},\dots,\mathfrak{S}_{\alpha_s}$.

The fraction $\mathfrak{f}(\overline{\gamma})$ occurs in the decomposition of the monomial $\mathfrak{S}_{\gamma_1}\cdots\mathfrak{S}_{\gamma_k}$ at least once and does not occur in the decompositions of the monomials $\mathfrak{S}_{\beta_1}\cdots\mathfrak{S}_{\beta_r}$ written afterward. Therefore, the matrix is lower triangular, and all elements on its main diagonal are nonzero. Obviously, the system of equations with this matrix has a unique solution.

Example 1.3. Let us express the quantity $Q=Q_{2,2,2,2}$, which occurs in the expression (1.21) for Φ_8, in terms of $\mathfrak{S}_2,\dots,\mathfrak{S}_8$. First, we have

$$(2,2,2,2)\succ(2,2,4)\succ(2,6),(4,4)\succ(8).$$

Therefore, $Q_{2,2,2,2}$ has the form

$$\mathfrak{S}_2^4+A\mathfrak{S}_2^2\mathfrak{S}_4+B\mathfrak{S}_2\mathfrak{S}_6+C\mathfrak{S}_4^2+D\mathfrak{S}_8$$

(for brevity, we write A, B, and so on instead of $A_{\overline{\beta}}$). In Table 2, the element in the intersection of the row corresponding to $\overline{\gamma}$ (and to the fraction $\mathfrak{f}(\overline{\gamma})$) and the column corresponding to a monomial $\mathfrak{M}=\mathfrak{S}_{\beta_1}\cdots\mathfrak{S}_{\beta_2}$ equals the coefficient multiplying $\mathfrak{f}(\overline{\gamma})$ in $\mathfrak{M}$.

Table 2

$\overline{\gamma}$	$\mathfrak{f}^{-1}(\overline{\gamma})$	$\mathfrak{S}_2^4$	$\mathfrak{S}_2^2\mathfrak{S}_4$	$\mathfrak{S}_2\mathfrak{S}_6$	$\mathfrak{S}_4^2$	$\mathfrak{S}_8$	Q
$(2,2,4)$	$p_1^2p_2^2p_3^4$	12	2	0	0	0	0
$(2,6)$	$p_1^2p_2^6$	4	2	1	0	0	0
$(4,4)$	$p_1^4p_2^4$	6	2	0	2	0	0
(8)	p_1^8	1	1	1	1	1	0

Indeed, the fraction $(p_1^2p_2^2p_3^4)^{-1}$ is absent from the decompositions of $\mathfrak{S}_2\mathfrak{S}_6$, $\mathfrak{S}_4^2$, and $\mathfrak{S}_8$, the fraction $(p_1^2p_2^6)^{-1}$ is absent from those of $\mathfrak{S}_4^2$ and $\mathfrak{S}_8$, and $(p_1^4p_2^4)^{-1}$ is absent from those of $\mathfrak{S}_2\mathfrak{S}_6$ and $\mathfrak{S}_8$; therefore, the corresponding cells of Table 2 are filled with zeros. Next, the sum $\mathfrak{S}_2^4$ has the form

$$\sum_{q_1,\dots,q_4\le x}(q_1^2q_2^2q_3^2q_4^2)^{-1}.$$

Fixing various primes p_1, p_2, and p_3, we calculate the number of quadruples q_1, q_2, q_3, q_4 for which the fraction $(q_1^2q_2^2q_3^2q_4^2)^{-1}$ coincides with $(p_1^2p_2^2p_3^4)^{-1}$. Clearly, this number equals the number of solutions of the equation

$$q_1^2q_2^2q_3^2q_4^2=p_1^2p_2^2p_3^4 \tag{1.22}$$

in primes q_1, q_2, q_3, and q_4. Obviously, two of the numbers q_i must coincide with p_3. Such a pair can be chosen in $\binom{4}{2}=6$ ways. One of the remaining numbers q_i must equal p_1 and the other must equal p_2; hence they can be chosen in two ways. Thus, the total number of solutions of Eq. (1.22) is $6\cdot 2=12$.

A similar argument shows that the coefficient multiplying the fraction $(p_1^2p_2^2p_3^4)^{-1}$ in $\mathfrak{S}_2^2\mathfrak{S}_4$ equals the number of solutions of the equation $q_1^2q_2^2q_3^4=p_1^2p_2^2p_3^4$. Since its only solutions are (p_1,p_2,p_3) and (p_2,p_1,p_3), it follows that the coefficient in question equals 2.

It is easy to verify that the equations

$$q_1^2q_2^2q_3^2q_4^2=p_1^2p_2^6,\qquad q_1^2q_2^2q_3^4=p_1^2p_2^6,\qquad\text{and}\qquad q_1^2q_2^6=p_1^2p_2^6$$

have four, two, and one solution, respectively. These numbers are placed in the second row of Table 2.

The third row contains the numbers of solutions of the equations

$$q_1^2q_2^2q_3^2q_4^2=p_1^4p_2^4,\qquad q_1^2q_2^2q_3^4=p_1^4p_2^4,\qquad\text{and}\qquad q_1^2q_2^6=p_1^4p_2^4,$$

i.e., six, two, and two.

Finally, observing that each of the equations

$$q_1^2q_2^2q_3^2q_4^2=p_1^8,\qquad q_1^2q_2^2q_3^4=p_1^8,\qquad\text{and}\qquad q_1^2q_2^6=p_1^8$$

has a unique solution, all of whose components coincide with p_1, we fill the last row with ones.

Since none of the fractions considered above occurs in the sum Q, it follows that the last column of Table 2 contain only zeros, and the system of equations has the form

$$\begin{cases}12+2A=0,\\ 4+2A+B=0,\\ 6+2A+2C=0,\\ 1+A+B+C+D=0.\end{cases}$$

Therefore, $A=-6$, $B=8$, $C=3$, $D=-6$, and

$$Q_{2,2,2,2}=\mathfrak{S}_2^4-6\mathfrak{S}_2^2\mathfrak{S}_4+8\mathfrak{S}_2\mathfrak{S}_6+3\mathfrak{S}_4^2-6\mathfrak{S}_8.$$

Using the last relation and expression (1.21), we can calculate Φ_8. Indeed, the relations in Lemma 1.7 imply

$$Q_{2,6}=\mathfrak{S}_2\mathfrak{S}_6-\mathfrak{S}_8,\qquad Q_{3,5}=\mathfrak{S}_3\mathfrak{S}_5-\mathfrak{S}_8,\qquad Q_{4,4}=\mathfrak{S}_4^2-\mathfrak{S}_8,$$
$$Q_{2,2,4}=2(T_{2,2,4}+T_{2,4,2}+T_{4,2,2})=\mathfrak{S}_2^2\mathfrak{S}_4-2\mathfrak{S}_2\mathfrak{S}_6-\mathfrak{S}_4^2+2\mathfrak{S}_8,$$

and

$$Q_{2,3,3}=2(T_{2,3,3}+T_{3,2,3}+T_{3,3,2})=\mathfrak{S}_3^2\mathfrak{S}_2-\mathfrak{S}_2\mathfrak{S}_6-2\mathfrak{S}_3\mathfrak{S}_5+2\mathfrak{S}_8.$$

Substituting these expressions into (1.21), we obtain

$$\begin{aligned}\Phi_8 &= \mathfrak{S}_8\biggl(\varpi_8 - \varpi_2\varpi_6 - \varpi_3\varpi_5 - \frac{1}{2}\,\varpi_4^2 + \varpi_2^2\varpi_4 + \varpi_2\varpi_3^2 - \frac{1}{4}\,\varpi_2^4\biggr)\\ &\quad + \mathfrak{S}_2\mathfrak{S}_6\biggl(\varpi_2\varpi_6 - \varpi_2^2\varpi_4 - \frac{1}{2}\,\varpi_2\varpi_3^2 + \frac{1}{3}\,\varpi_2^4\biggr) + \mathfrak{S}_3\mathfrak{S}_5(\varpi_3\varpi_5 - \varpi_2\varpi_3^2)\\ &\quad + \mathfrak{S}_4^2\biggl(\frac{1}{2}\,\varpi_4^2 - \frac{1}{2}\,\varpi_2^2\varpi_4 + \frac{1}{8}\varpi_2^4\biggr) + \mathfrak{S}_2^2\mathfrak{S}_4\biggl(\frac{1}{2}\,\varpi_2^2\varpi_4 - \frac{1}{4}\,\varpi_2^4\biggr) + \frac{1}{2}\,\varpi_2\varpi_3^2\,\mathfrak{S}_3^2\mathfrak{S}_2 + \frac{1}{24}\,\varpi_2^4\mathfrak{S}_2^4\\ &= -\frac{167\cdot 607}{2^{14}\cdot 3^3\cdot 5\cdot 7}\,\mathfrak{S}_8 + \frac{11\cdot 43}{2^8\cdot 3^4\cdot 5}\,\mathfrak{S}_2\mathfrak{S}_6 + \frac{19}{2^3\cdot 3^3\cdot 5^2}\,\mathfrak{S}_3\mathfrak{S}_5 + \frac{11^2}{2^{13}\cdot 3^2}\,\mathfrak{S}_4^2\\ &\quad - \frac{11}{2^{11}\cdot 3}\,\mathfrak{S}_2^2\mathfrak{S}_4 - \frac{1}{2^3\cdot 3^4}\,\mathfrak{S}_2\mathfrak{S}_3^2 + \frac{1}{2^{11}\cdot 3}\,\mathfrak{S}_2^4.\end{aligned}$$

To complete the description of the algorithm for expressing Φ_n in the general case, it remains to specify a procedure for calculating the coefficient multiplying the fraction $(q_1^{\beta_1}\cdots q_r^{\beta_r})^{-1}$ in the sum resulting from multiplying out $\mathfrak{S}_{\alpha_1},\dots,\mathfrak{S}_{\alpha_s}$, i.e., the sum

$$\sum_{p_1,\dots,p_s\le x}(p_1^{\alpha_1}\cdots p_s^{\alpha_s})^{-1}.$$

It is easy to see that this coefficient coincides with the number $\mathcal{N}(\overline{\alpha},\overline{\beta})$ of solutions of the equation

$$p_1^{\alpha_1}\cdots p_s^{\alpha_s} = q_1^{\beta_1}\cdots q_r^{\beta_r} \tag{1.23}$$

in primes $p_1,\dots,p_s$. To determine $\mathcal{N}(\overline{\alpha},\overline{\beta})$, we need more notions.

We define a partition of a vector $\overline{\beta}=(\beta_1,\dots,\beta_r)$ with positive integer components β_i as a decomposition of each component in a sum of positive integers:

$$\begin{cases}\beta_1 = \alpha_{u_1}+\cdots+\alpha_{v_1},\\ \dots\dots\dots\dots\dots\dots\\ \beta_r = \alpha_{u_r}+\cdots+\alpha_{v_r}.\end{cases} \tag{1.24}$$

Decompositions which differ only in the order of their terms α_i are considered equal.

Let $\overline{\alpha}$ be any vector whose components are all summands $\alpha_{u_1},\dots,\alpha_{v_r}$ on the right-hand sides of (1.24). Then (1.24) implies $\overline{\beta}\prec\overline{\alpha}$ (or $\overline{\beta}=\overline{\alpha}$). Given two vectors $\overline{\alpha}$ and $\overline{\beta}$ such that $\overline{\beta}\prec\overline{\alpha}$, we denote the set of all partitions of the vector $\overline{\beta}$ into the summands being the components of $\overline{\alpha}$ by $\mathcal{P}(\overline{\alpha},\overline{\beta})$.

Example 1.4. Let $\overline{\alpha}=(2,2,2,3,3)$, and let $\overline{\beta}=(6,6)$. Then $\mathcal{P}(\overline{\alpha},\overline{\beta})$ consists of the partitions π_1 and π_2 determined by

$$\pi_1\colon\ \begin{cases}\beta_1 = 6 = 2+2+2,\\ \beta_2 = 6 = 3+3\end{cases}\qquad\text{and}\qquad \pi_2\colon\ \begin{cases}\beta_1 = 6 = 3+3,\\ \beta_2 = 6 = 2+2+2.\end{cases}$$

Let $\mathcal{Q}(\overline{\alpha},\overline{\beta})$ denote the set of those partitions of $\overline{\beta}$ in $\mathcal{P}(\overline{\alpha},\overline{\beta})$ which cannot be obtained from each other by permuting the right-hand sides of (1.24).

Example 1.5. For the vectors $\overline{\alpha}$ and $\overline{\beta}$ in Example 1.4, the set $\mathcal{Q}(\overline{\alpha},\overline{\beta})$ consists of the single partition

$$\pi\colon\ \begin{cases}\beta_1 = 6 = 2+2+2,\\ \beta_2 = 6 = 3+3.\end{cases}$$

If $\overline{\alpha}=(2,2,2,3,3,4)$ and $\overline{\beta}=(4,6,6)$, then $\mathcal{Q}(\overline{\alpha},\overline{\beta})$ consists of the two partitions

$$\pi_1\colon\ \begin{cases}4 = 4,\\ 6 = 3+3,\\ 6 = 2+2+2\end{cases}\qquad\text{and}\qquad \pi_2\colon\ \begin{cases}4 = 2+2,\\ 6 = 2+4,\\ 6 = 3+3.\end{cases}$$

Let us fix any partition $\pi \in \mathcal{Q}(\overline{\alpha}, \overline{\beta})$ of a vector $\overline{\beta}$ and count different partitions π' which can be obtained from π by permuting the right-hand sides in (1.24).

Let β be any number among $\beta_1, \dots, \beta_r$. Consider all relations in (1.24) whose left-hand sides coincide with β. Let l be the number of different types of these relations (two relations are of different types if they cannot be obtained from each other by changing the order of terms), and let $m_1, \dots, m_l$ denote the numbers of their occurrences. We obtain new partitions in $\mathcal{P}(\overline{\alpha}, \overline{\beta})$ by permuting right-hand sides of relations of different types. The number of such permutations equals

$$\frac{(m_1 + \dots + m_l)!}{m_1! \cdots m_l!}.$$

The total number of partitions that can be obtained from π in this way is

$$\mathcal{N}_1(\pi) = \prod_{\beta} \frac{(m_1 + \dots + m_l)!}{m_1! \cdots m_l!},$$

where the product is over all different numbers among $\beta_1, \dots, \beta_r$.

Example 1.6. Consider the partitions π_1 and π_2 in Example 1.5. For the first partition and the element $\beta = 4$, we obviously have $l = 1$ and $m_1 = 1$; if $\beta = 6$, then $l = 2$ and $m_1 = m_2 = 1$. Therefore,

$$\mathcal{N}_1(\pi_1) = \frac{1!}{1!} \cdot \frac{(1+1)!}{1!\,1!} = 2.$$

Similarly, $\mathcal{N}_1(\pi_2) = 2$. The total number of partitions in $\mathcal{P}(\overline{\alpha}, \overline{\beta})$ equals $\mathcal{N}_1(\pi_1) + \mathcal{N}_1(\pi_2) = 4$.

We proceed to count the solutions of Eq. (1.23). Let $\pi \in \mathcal{P}(\overline{\alpha}, \overline{\beta})$ be any partition of a vector $\overline{\beta}$, and suppose that this partition corresponds to system (1.24). We represent the right-hand side in the form

$$q_1^{\alpha_{u_1} + \dots + \alpha_{v_1}} \cdots q_r^{\alpha_{u_r} + \dots + \alpha_{v_r}} = (q_1^{\alpha_{u_1}} \cdots q_1^{\alpha_{v_1}}) \cdots (q_r^{\alpha_{u_r}} \cdots q_r^{\alpha_{v_r}}).$$

Ordering accordingly the factors on the left-hand side of (1.21), we obtain

$$(p_{u_1}^{\alpha_{u_1}} \cdots p_{v_1}^{\alpha_{v_1}}) \cdots (p_{u_r}^{\alpha_{u_r}} \cdots p_{v_r}^{\alpha_{v_r}}) = (q_1^{\alpha_{u_1}} \cdots q_1^{\alpha_{v_1}}) \cdots (q_r^{\alpha_{u_r}} \cdots q_r^{\alpha_{v_r}}). \tag{1.25}$$

Obviously, the initial equation has the solution

$$\begin{cases} p_{u_1} = \dots = p_{v_1} = q_1, \\ \dots\dots\dots\dots\dots\dots \\ p_{u_r} = \dots = p_{v_r} = q_r. \end{cases} \tag{1.26}$$

This solution generates a whole series of new ones. Indeed, let us fix any number α among $\alpha_1, \dots, \alpha_s$ and consider all factors q_i^{α} within each pair of parentheses on the right-hand side of (1.25).

Interchanging any two of these factors but retaining the order of the factors p_i on the left-hand side, we obtain new sets $(p_1, \dots, p_s)$, which are again solutions of (1.23). Some of these sets coincide with solution (1.26). This happens if two equal factors q_i^{α} and q_i^{α} are interchanged. New solutions are obtained only under permutations that transpose factors q_i^{α} and q_j^{α} corresponding to different primes q_i and q_j.

Let us find the number of such permutations. Suppose that the exponent α occurs k_1 times within the first pair of parentheses, …, k_r times within the rth pair of parentheses (in other words, $k_j = k_j(\alpha)$ equals the number of summands equal to α on the right-hand side of the jth equation of system (1.24)). Then the sought number of permutations equals

$$\frac{(k_1 + \dots + k_r)!}{k_1! \cdots k_r!}.$$

The total number of different solutions which can be obtained from (1.26) in this way equals

$$\mathcal{N}_2(\pi) = \prod_{\alpha} \frac{(k_1 + \dots + k_r)!}{k_1! \cdots k_r!},$$

where the product is over all different numbers among $\alpha_1, \dots, \alpha_r$.

Note that the interchange of the factors q_i^α and q_j^α and the solution $p_1, \dots, p_s$ thus obtained do not affect the form of partition (1.24). Indeed, this operation is equivalent to the interchange of two equal terms $\alpha_u = \alpha_v = \alpha$ on the right-hand sides of different rows in system (1.24).

Note that the product $\mathcal{N}_2(\pi)$ does not depend on the order of relations in (1.24); therefore, this product is the same for all partitions $\pi' \in \mathcal{P}(\overline{\alpha}, \overline{\beta})$ obtained from a given partition $\pi \in \mathcal{Q}(\overline{\alpha}, \overline{\beta})$. Since the total number of partitions π' obtained from $\pi \in \mathcal{Q}(\overline{\alpha}, \overline{\beta})$ equals $\mathcal{N}_1(\pi)$, it follows from the above considerations that we obtain

$$\sum_{\pi \in \mathcal{Q}(\overline{\alpha}, \overline{\beta})} \mathcal{N}_1(\pi)\mathcal{N}_2(\pi)$$

solutions of Eq. (1.23).

Let us show that all solutions of Eq. (1.23) can be obtained by the procedure described above.

Recall that the procedure begins with the exhaustive search for all partitions π of a vector $\overline{\beta}$ in the set $\mathcal{Q}(\overline{\alpha}, \overline{\beta})$. For each of them, $\mathcal{N}_1(\pi)$ different partitions in $\mathcal{P}(\overline{\alpha}, \overline{\beta})$ are constructed by permuting relations in system (1.24). Finally, for each of the $\mathcal{N}_1(\pi)$ solutions corresponding to these partitions, $\mathcal{N}_2(\pi)$ new solutions are obtained by permuting factors q_i^α and q_j^α with $i \neq j$.

Now, let $p_1, \dots, p_s$ be any solution of (1.23). By virtue of the fundamental theorem of arithmetic, the unknowns p_i cannot take values different from $q_1, \dots, q_r$. Hence, collecting together the factors p_i in $p_1^{\alpha_1} \cdots p_s^{\alpha_s}$ which equal q_1, then the factors p_i which equal q_2, and so on, up to the factors which equal q_r, we obtain

$$(p_{i_1}^{\alpha_{i_1}} \cdots p_{j_1}^{\alpha_{j_1}}) \cdots (p_{i_r}^{\alpha_{i_r}} \cdots p_{j_r}^{\alpha_{j_r}}) = q_1^{\beta_1} \cdots q_r^{\beta_r},$$

where

$$\begin{cases} p_{i_1} = \cdots = p_{j_1} = q_1, \\ \dots\dots\dots\dots\dots\dots \\ p_{i_r} = \cdots = p_{j_r} = q_r \end{cases}$$

and, therefore,

$$\begin{cases} \beta_1 = \alpha_{i_1} + \cdots + \alpha_{j_1}, \\ \dots\dots\dots\dots\dots\dots \\ \beta_r = \alpha_{i_r} + \cdots + \alpha_{j_r}. \end{cases} \tag{1.27}$$

System (1.27) determines a partition $\pi \in \mathcal{P}(\overline{\alpha}, \overline{\beta})$. This partition must be among those treated in the course of the above procedure for constructing solutions. The system of type (1.24) corresponding to a partition π arising in the course of the procedure differs from system (1.27) only by a permutation of equal numbers α_i with different indices which occur in different rows of the system. But the procedure treats all such permutations, including system (1.27), and it must construct the corresponding solution $p_1, \dots, p_s$. Thus, all solutions of (1.23) are obtained by the method described above.

It only remains to note that all of the constructed solutions are different. Obviously, the solutions corresponding to different partitions $\pi \in \mathcal{P}(\overline{\alpha}, \overline{\beta})$ are different, because equal solutions generate the same system (1.24) and, therefore, the same partition. Next, the solutions corresponding to the same partition $\pi \in \mathcal{P}(\overline{\alpha}, \overline{\beta})$ and obtained by interchanging factors q_i^α and q_j^α are different as well, because any such interchange entails the interchange of unequal components in the set $p_1, \dots, p_s$.

Thus, the number of solutions of Eq. (1.23) is

$$\mathcal{N}(\overline{\alpha}, \overline{\beta}) = \sum_{\pi \in \mathcal{Q}(\overline{\alpha}, \overline{\beta})} \mathcal{N}_1(\pi)\mathcal{N}_2(\pi) = \sum_{\pi \in \mathcal{Q}(\overline{\alpha}, \overline{\beta})} \prod_{\alpha} \frac{(k_1 + \cdots + k_r)!}{k_1! \cdots k_r!} \prod_{\beta} \frac{(m_1 + \cdots + m_l)!}{m_1! \cdots m_l!}.$$

Example 1.7. Let us find the number of solutions of the equation

$$p_1^2p_2^2p_3^2p_4^3p_5^3p_6^4 = 729\,000\,000 = 5^4 \cdot 3^6 \cdot 2^6.$$

Consider $\overline{\alpha} = (2,2,2,3,3,4)$ and $\overline{\beta} = (4,6,6)$. In Examples 1.5 and 1.6, we found that the set $\mathcal{Q}(\overline{\alpha},\overline{\beta})$ consists of the two partitions

$$\pi_1\colon \begin{cases} 4=4, \\ 6=3+3, \\ 6=2+2+2 \end{cases} \qquad \text{and} \qquad \pi_2\colon \begin{cases} 4=2+2, \\ 6=2+4, \\ 6=3+3 \end{cases}$$

and that $\mathcal{N}_1(\pi_1) = \mathcal{N}_1(\pi_2) = 2$. Consider the partition π_1. Searching through the different components α of the vector $\overline{\alpha}$, we obtain

$$\begin{aligned} &\alpha = 2, \qquad k_1 = k_2 = 0, \quad k_3 = 3, \\ &\alpha = 3, \qquad k_1 = 0, \quad k_2 = 2, \quad k_3 = 0, \end{aligned}$$

and

$$\alpha = 4, \qquad k_1 = 1, \quad k_2 = k_3 = 0.$$

Therefore,

$$\mathcal{N}_2(\pi_1) = \frac{(0+3+0)!}{0!\,3!\,0!}\frac{(0+0+2)!}{0!\,0!\,2!}\frac{(1+0+0)!}{1!\,0!\,0!} = 1.$$

Passing to the partition π_2, we obtain

$$\begin{aligned} &\alpha = 2, \qquad k_1 = 2, \quad k_2 = 1, \quad k_3 = 0, \\ &\alpha = 3, \qquad k_1 = k_2 = 0, \quad k_3 = 2, \end{aligned}$$

and

$$\alpha = 4, \qquad k_1 = 0, \quad k_2 = 1, \quad k_3 = 0,$$

whence

$$\mathcal{N}_2(\pi_2) = \frac{(2+1+0)!}{2!\,1!\,0!}\frac{(0+0+2)!}{0!\,0!\,2!}\frac{(0+1+0)!}{0!\,1!\,0!} = 3.$$

As a result, we have

$$\mathcal{N}(\overline{\alpha},\overline{\beta}) = \mathcal{N}_1(\pi_1)\mathcal{N}_2(\pi_1) + \mathcal{N}_1(\pi_2)\mathcal{N}_2(\pi_2) = 8.$$

Thus, the initial equation has eight solutions.

This completes the description of the algorithm for calculating Φ_n. A description of methods for finding all possible partitions $\pi \in \mathcal{P}(\overline{\alpha},\overline{\beta})$ of a vector $\overline{\beta}$ is beyond the scope of this paper. For comparatively small

$$n = \sum_{i=1}^{s}\alpha_i = \sum_{j=1}^{r}\beta_j,$$

this problem can be successfully solved by a "manual" exhaustive search.

Example 1.8. Calculating the coefficients Φ_n with $n = 9, 10$ by the above algorithm, we successively obtain

$$\begin{aligned} Q_{2,3,4} &= \mathfrak{S}_2\mathfrak{S}_3\mathfrak{S}_4 - \mathfrak{S}_2\mathfrak{S}_7 - \mathfrak{S}_3\mathfrak{S}_6 - \mathfrak{S}_4\mathfrak{S}_5 + 2\mathfrak{S}_9, \\ Q_{2,2,2,3} &= \mathfrak{S}_2^3\mathfrak{S}_3 - 3\mathfrak{S}_2^2\mathfrak{S}_5 - 3\mathfrak{S}_2\mathfrak{S}_3\mathfrak{S}_4 + 3\mathfrak{S}_4\mathfrak{S}_5 + 2\mathfrak{S}_3\mathfrak{S}_6 + 6\mathfrak{S}_2\mathfrak{S}_7 - 6\mathfrak{S}_9, \end{aligned}$$

and

$$\begin{aligned}\Phi_9 &= \varpi_9 Q_9 + \varpi_2\varpi_7 Q_{2,7} + \varpi_3\varpi_6 Q_{3,6} + \varpi_4\varpi_5 Q_{4,5} + \frac{1}{2}\varpi_2^2\varpi_5 Q_{2,2,5} + \varpi_2\varpi_3\varpi_4 Q_{2,3,4}\\ &\quad + \frac{1}{6}\varpi_3^3 Q_{3,3,3} + \frac{1}{6}\varpi_2^3\varpi_3 Q_{2,2,2,3}\\ &= \mathfrak{S}_9\biggl(\varpi_9 - \varpi_2\varpi_7 - \varpi_3\varpi_6 - \varpi_4\varpi_5 + \varpi_2^2\varpi_5 + 2\varpi_2\varpi_3\varpi_4 + \frac{1}{3}\varpi_3^3 - \varpi_2^3\varpi_3\biggr)\\ &\quad + \mathfrak{S}_2\mathfrak{S}_7(\varpi_2\varpi_7 - \varpi_2^2\varpi_5 - \varpi_2\varpi_3\varpi_4 + \varpi_2^3\varpi_3)\\ &\quad + \mathfrak{S}_3\mathfrak{S}_6\biggl(\varpi_3\varpi_6 - \varpi_2\varpi_3\varpi_4 - \frac{1}{2}\varpi_3^3 + \frac{1}{3}\varpi_2^3\varpi_3\biggr)\\ &\quad + \mathfrak{S}_4\mathfrak{S}_5\biggl(\varpi_4\varpi_5 - \frac{1}{2}\varpi_2^2\varpi_5 - \varpi_2\varpi_3\varpi_4 + \frac{1}{2}\varpi_2^3\varpi_3\biggr)\\ &\quad + \mathfrak{S}_2^2\mathfrak{S}_5\biggl(\frac{1}{2}\varpi_2^2\varpi_5 - \frac{1}{2}\varpi_2^3\varpi_3\biggr) + \mathfrak{S}_2\mathfrak{S}_3\mathfrak{S}_4\biggl(\varpi_2\varpi_3\varpi_4 - \frac{1}{2}\varpi_2^3\varpi_3\biggr) + \frac{1}{6}\varpi_3^3\mathfrak{S}_3^3 + \frac{1}{6}\varpi_2^3\varpi_3\mathfrak{S}_2^3\mathfrak{S}_3\\ &= \frac{19\cdot 31\cdot 1607}{2^{10}\cdot 3^8\cdot 5\cdot 7}\mathfrak{S}_9 - \frac{229}{2^6\cdot 3^3\cdot 7^2}\mathfrak{S}_2\mathfrak{S}_7 - \frac{11\cdot 43}{2^6\cdot 3^6\cdot 5}\mathfrak{S}_3\mathfrak{S}_6 - \frac{11\cdot 19}{2^9\cdot 3^2\cdot 5^2}\mathfrak{S}_4\mathfrak{S}_5\\ &\quad + \frac{19}{2^8\cdot 3\cdot 5^2}\mathfrak{S}_2^2\mathfrak{S}_5 + \frac{11}{2^8\cdot 3^3}\mathfrak{S}_2\mathfrak{S}_3\mathfrak{S}_4 + \frac{1}{2\cdot 3^7}\mathfrak{S}_3^3 - \frac{1}{2^7\cdot 3^3}\mathfrak{S}_2^3\mathfrak{S}_3.\end{aligned}$$

Similarly,

$$\begin{aligned}Q_{2,2,6} &= \mathfrak{S}_2^2\mathfrak{S}_6 - 2\mathfrak{S}_2\mathfrak{S}_8 - \mathfrak{S}_4\mathfrak{S}_6 + 2\mathfrak{S}_{10},\\ Q_{2,3,5} &= \mathfrak{S}_2\mathfrak{S}_3\mathfrak{S}_5 - \mathfrak{S}_2\mathfrak{S}_8 - \mathfrak{S}_3\mathfrak{S}_7 - \mathfrak{S}_5^2 + 2\mathfrak{S}_{10},\\ Q_{2,4,4} &= \mathfrak{S}_2\mathfrak{S}_4^2 - \mathfrak{S}_2\mathfrak{S}_8 - 2\mathfrak{S}_4\mathfrak{S}_6 + 2\mathfrak{S}_{10},\\ Q_{3,3,4} &= \mathfrak{S}_3^2\mathfrak{S}_4 - 2\mathfrak{S}_3\mathfrak{S}_7 - \mathfrak{S}_4\mathfrak{S}_6 + 2\mathfrak{S}_{10},\\ Q_{2,2,2,4} &= \mathfrak{S}_2^3\mathfrak{S}_4 - 3\mathfrak{S}_2^2\mathfrak{S}_6 - 3\mathfrak{S}_2\mathfrak{S}_4^2 + 6\mathfrak{S}_2\mathfrak{S}_8 + 5\mathfrak{S}_4\mathfrak{S}_6 - 6\mathfrak{S}_{10},\\ Q_{2,2,3,3} &= \mathfrak{S}_2^2\mathfrak{S}_3^2 - \mathfrak{S}_2^2\mathfrak{S}_6 - 4\mathfrak{S}_2\mathfrak{S}_3\mathfrak{S}_5 - \mathfrak{S}_3^2\mathfrak{S}_4 + 4\mathfrak{S}_2\mathfrak{S}_8 + 4\mathfrak{S}_3\mathfrak{S}_7 + \mathfrak{S}_4\mathfrak{S}_6 + 2\mathfrak{S}_5^2 - 6\mathfrak{S}_{10},\\ Q_{2,2,2,2,2} &= \mathfrak{S}_2^5 - 10\mathfrak{S}_2^3\mathfrak{S}_4 + 20\mathfrak{S}_2^2\mathfrak{S}_6 + 15\mathfrak{S}_2\mathfrak{S}_4^2 - 30\mathfrak{S}_2\mathfrak{S}_8 - 20\mathfrak{S}_4\mathfrak{S}_6 + 24\mathfrak{S}_{10},\end{aligned}$$

and

$$\begin{aligned}\Phi_{10} &= \varpi_{10}Q_{10} + \varpi_2\varpi_8 Q_{2,8} + \varpi_3\varpi_7 Q_{3,7} + \varpi_4\varpi_6 Q_{4,6} + \frac{1}{2}\varpi_5^2 Q_{5,5} + \frac{1}{2}\varpi_2^2\varpi_6 Q_{2,2,6} + \varpi_2\varpi_3\varpi_5 Q_{2,3,5}\\ &\quad + \frac{1}{2}\varpi_2\varpi_4^2 Q_{2,4,4} + \frac{1}{2}\varpi_3^2\varpi_4 Q_{3,3,4} + \frac{1}{6}\varpi_2^3\varpi_4 Q_{2,2,2,4} + \frac{1}{4}\varpi_2^2\varpi_3^2 Q_{2,2,3,3} + \frac{1}{120}\varpi_2^5 Q_{2,2,2,2,2}\\ &= A_{10}\mathfrak{S}_{10} + A_{2,8}\mathfrak{S}_2\mathfrak{S}_8 + A_{3,7}\mathfrak{S}_3\mathfrak{S}_7 + A_{4,6}\mathfrak{S}_4\mathfrak{S}_6 + A_{5,5}\mathfrak{S}_5^2 + A_{2,2,6}\mathfrak{S}_2^2\mathfrak{S}_6 + A_{2,3,5}\mathfrak{S}_2\mathfrak{S}_3\mathfrak{S}_5\\ &\quad + A_{2,4,4}\mathfrak{S}_2\mathfrak{S}_4^2 + A_{3,3,4}\mathfrak{S}_3^2\mathfrak{S}_4 + A_{2,2,2,4}\mathfrak{S}_2^3\mathfrak{S}_4 + A_{2,2,3,3}\mathfrak{S}_2^2\mathfrak{S}_3^2 + A_{2,2,2,2,2}\mathfrak{S}_2^5,\end{aligned}$$

where the coefficients A are

$$\begin{aligned}A_{10} &= \varpi_{10} - \varpi_2\varpi_8 - \varpi_3\varpi_7 - \varpi_4\varpi_6 - \frac{1}{2}\varpi_5^2 + \varpi_2^2\varpi_6 + 2\varpi_2\varpi_3\varpi_5\\ &\qquad + \varpi_2\varpi_4^2 + \varpi_3^2\varpi_4 - \varpi_2^3\varpi_4 - \frac{3}{2}\varpi_2^2\varpi_3^2 + \frac{1}{5}\varpi_2^5 = -\frac{199\cdot 329\,981}{2^{13}\cdot 3^6\cdot 5^4\cdot 7},\\ A_{2,8} &= \varpi_2\varpi_8 - \varpi_2^2\varpi_6 - \varpi_2\varpi_3\varpi_5 - \frac{1}{2}\varpi_2\varpi_4^2 + \varpi_2^3\varpi_4 + \varpi_2^2\varpi_3^2 - \frac{1}{4}\varpi_2^5 = \frac{167\cdot 607}{2^{16}\cdot 3^3\cdot 5\cdot 7},\\ A_{3,7} &= \varpi_3\varpi_7 - \varpi_2\varpi_3\varpi_5 - \varpi_3^2\varpi_4 + \varpi_2^2\varpi_3^2 = \frac{229}{2^4\cdot 3^5\cdot 7^2},\\ A_{4,6} &= \varpi_4\varpi_6 - \frac{1}{2}\varpi_2^2\varpi_6 - \varpi_2\varpi_4^2 - \frac{1}{2}\varpi_3^2\varpi_4 + \frac{5}{6}\varpi_2^3\varpi_4 + \frac{1}{4}\varpi_2^2\varpi_3^2 - \frac{1}{6}\varpi_2^5 = \frac{11^2\cdot 43}{2^{12}\cdot 3^5\cdot 5},\end{aligned}$$

$$A_{5,5}=\frac{1}{2}\,\varpi_5^2-\varpi_2\varpi_3\varpi_5+\frac{1}{2}\,\varpi_2^2\varpi_3^2=\frac{19^2}{2^7\cdot 3^2\cdot 5^4},$$
$$A_{2,2,6}=\frac{1}{2}\varpi_2^2\varpi_6-\frac{1}{2}\,\varpi_2^3\varpi_4-\frac{1}{4}\,\varpi_2^2\varpi_3^2+\frac{1}{6}\,\varpi_2^5=-\frac{11\cdot 43}{2^{11}\cdot 3^4\cdot 5},$$
$$A_{2,3,5}=\varpi_2\varpi_3\varpi_5-\varpi_2^2\varpi_3^2=-\frac{19}{2^5\cdot 3^3\cdot 5^2},$$
$$A_{2,4,4}=\frac{1}{2}\,\varpi_2\varpi_4^2-\frac{1}{2}\,\varpi_2^3\varpi_4+\frac{1}{8}\,\varpi_2^5=-\frac{11^2}{2^{15}\cdot 3^2},$$
$$A_{3,3,4}=\frac{1}{2}\,\varpi_3^2\varpi_4-\frac{1}{4}\,\varpi_2^2\varpi_3^2=-\frac{11}{2^7\cdot 3^5},$$
$$A_{2,2,2,4}=\frac{1}{6}\,\varpi_2^3\varpi_4-\frac{1}{12}\,\varpi_2^5=\frac{11}{2^{13}\cdot 3^2},$$
$$A_{2,2,3,3}=\frac{1}{4}\,\varpi_2^2\varpi_3^2=\frac{1}{2^6\cdot 3^4},$$
$$A_{2,2,2,2,2}=\frac{1}{120}\,\varpi_2^5=-\frac{1}{2^{13}\cdot 3\cdot 5}.$$

1.3. The order of growth of the coefficients Φ_n as $n\to+\infty$. The main term (1.1) in the expression for a fractional moment V_a is obtained by formally integrating the series expansion of the function $H(-u^2/\mathfrak{S}_1)$ along the ray $0\le u<+\infty$.

To prove that series (1.1) converges and is asymptotic, we need a number of statements on the order of growth of the coefficients Φ_n as n unboundedly increases.

One of such assertions (Lemma 1.11) is based on an inequality for the expansion coefficients of the function

$$\varphi(v)=e^{-v}\sum_{m=0}^{+\infty}\frac{v^m}{m!}=e^{-v}J_0(2i\sqrt{v}\,).$$

Lemma 1.10. *For any $n\ge 0$,*

$$|\varpi_n|<\frac{1}{n!}.$$

Proof. The definition of $\varphi(z)$ implies that $(-1)^n\varpi_n$ coincides with the coefficient of v^{2n} in the series expansion of $\varphi(-v^2)$. Observing that

$$J_0(z)=\sum_{m=0}^{+\infty}\frac{(-1)^m}{(m!)^2}\left(\frac{z}{2}\right)^{2m},$$

we obtain

$$\varphi(-v^2)=e^{v^2}\sum_{m=0}^{+\infty}\frac{(-1)^m}{(m!)^2}\,v^{2m}=e^{v^2}J_0(2v).$$

Therefore,

$$(-1)^n\varpi_n=\frac{1}{2\pi i}\oint\frac{e^{v^2}J_0(2v)}{v^{2n+1}}\,dv,$$

where the integration is along a circle of any radius r centered at the origin. Since the function $J_0(2v)$ is even, it follows that, for $v=re^{i\vartheta}$, we have

$$(-1)^n\varpi_n=\frac{r^{-2n}}{\pi}\int_{-\pi/2}^{\pi/2}e^{r^2e^{2i\vartheta}}J_0(2re^{i\vartheta})e^{-2in\vartheta}\,d\vartheta,$$

whence

$$|\varpi_n| \le \frac{r^{-2n}}{\pi}\int_{-\pi/2}^{\pi/2} e^{r^2\cos 2\vartheta}|J_0(2re^{i\vartheta})|\,d\vartheta.$$

Now, we set $r=\sqrt{n}$ and apply Stirling's formula. We obtain

$$r^{-2n}=n^{-n}=\frac{\sqrt{2\pi n}}{n!}\,e^{-n+\theta/12n}\le\frac{\sqrt{2\pi n}}{n!}\,e^{-n+1/12n}$$

and, therefore,

$$|\varpi_n| \le \frac{2}{\pi}\frac{\sqrt{2\pi n}}{n!}\,e^{1/12n}\int_0^{\pi/2} e^{-n+n\cos 2\vartheta}|J_0(2\sqrt{n}e^{i\vartheta})|\,d\vartheta.$$

Applying the bound of Lemma I.2, we arrive at the inequality

$$n!|\varpi_n| \le \frac{2\sqrt{2}}{\pi}\sqrt[4]{n}\,e^{1/12}\biggl(1+\frac{1}{4\sqrt{2n}}\biggr)j(n),$$

in which

$$j(n)=\int_0^{\pi/2} e^{-n+n\cos 2\vartheta+2\sqrt{n}\sin\vartheta}\,d\vartheta=\int_0^{\pi/2} e^{f(\sin\vartheta)}\,d\vartheta,\qquad f(u)=-2nu^2+2u\sqrt{n}.$$

The function $f(u)$ monotonically decreases at $u\ge 1/2\sqrt{n}$; therefore, for any ϑ satisfying the condition $\pi/6\le\vartheta\le\pi/2$, the quantity $f(\sin\vartheta)$ does not exceed $-n/2+\sqrt{n}$. Thus, the integral along the corresponding interval is bounded above by

$$\frac{\pi}{3}\exp\biggl(-\frac{n}{2}+\sqrt{n}\biggr).$$

Next, since $\cos\vartheta\ge\sqrt{3}/2$ for $0\le\vartheta\le\pi/6$, it follows that, for $u=2\sqrt{n}\sin\vartheta$, we have

$$\int_0^{\pi/6} e^{f(\sin\vartheta)}\,d\vartheta=\frac{1}{2\sqrt{n}}\int_0^{\sqrt{n}} e^{-u^2/2+u}\frac{du}{\cos\vartheta}\le\frac{1}{\sqrt{3n}}\int_0^{+\infty} e^{-u^2/2+u}\,du=\sqrt{\frac{2\pi e}{3n}}\,G(1),$$

where

$$G(1)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{x} e^{-u^2/2}\,du.$$

Thus, we obtain the inequality $n!\,|\varpi_n|\le F(n)$, in which

$$F(n)=\frac{2\sqrt{2}}{\pi}\biggl(1+\frac{1}{4\sqrt{2n}}\biggr)\frac{e^{1/12n}}{\sqrt[4]{n}}\biggl\{\sqrt{\frac{2\pi e}{3}}\,G(1)+\frac{\pi\sqrt{n}}{3}\,e^{-n/2+\sqrt{n}}\biggr\}.$$

Computation performed by using the *Wolfram Mathematica* 7.0.0 software package shows that $0<F(n)<1$ for all $n\ge 16$. For $0\le n\le 15$, the required inequality is verified directly (see Table 3).

This completes the proof of the lemma.

Remark 1.1. The quantity $\Omega_n=(-1)^n n!\,\varpi_n$ equals the value of the Kummer confluent hypergeometric function

$$F(a,c;z)=1+\frac{a}{c}\cdot\frac{z}{1}+\frac{a(a+1)}{c(c+1)}\cdot\frac{z^2}{2!}+\cdots$$

with $a=-n$ and $c=1$ at the point $z=1$.

A special case of a statement proved in [31] is the representation

$$\Omega_n=\frac{c}{\sqrt[4]{n}}\cos\biggl(2\sqrt{n}-\frac{\pi}{4}\biggr)+O(n^{-3/4}),\qquad c=\sqrt{\frac{e}{\pi}}\,.$$

This fact was kindly communicated to the author by Yu. A. Neretin.

Table 3

n	$n!\,\varpi_n$	n	$n!\,\varpi_n$
0	1	8	0.153 993
1	0	9	−0.309 744
2	−0.5	10	0.418 946
3	0.666 667	11	−0.480 134
4	−0.625	12	0.496 212
5	0.466 667	13	−0.472 883
6	−0.256 944	14	0.417 443
7	0.040 476	15	−0.337 870

The first of the inequalities necessary for determining Φ_n is given by the following lemma.

Lemma 1.11. *For any n,*

$$|\Phi_n| \le \frac{\mathfrak{S}_1^n}{n!}.$$

Proof. Let $n \ge 2$. Using expression (1.17) and the bound of Lemma 1.10, we obtain

$$|\Phi_n| \le \sum_{1\le s\le n} \sum_{\substack{\alpha_1+\cdots+\alpha_s=n\\ \alpha_1,\ldots,\alpha_s\ge 0}} |\varpi_{\alpha_1}\cdots\varpi_{\alpha_s}| \sum_{p_1<\cdots<p_s\le x} (p_1^{\alpha_1}\cdots p_s^{\alpha_s})^{-1}$$

$$\le \frac{1}{n!} \sum_{1\le s\le n} \sum_{\substack{\alpha_1+\cdots+\alpha_s=n\\ \alpha_1,\ldots,\alpha_s\ge 0}} \frac{n!}{\alpha_1!\cdots\alpha_s!} \sum_{p_1<\cdots<p_s\le x} (p_1^{\alpha_1}\cdots p_s^{\alpha_s})^{-1} = \frac{1}{n!}\biggl(\sum_{p\le x}\frac{1}{p}\biggr)^n = \frac{\mathfrak{S}_1^n}{n!}.$$

For $n = 0$ and 1, the required inequality is obvious. This completes the proof of the Lemma.

Lemma 1.12. *For any n satisfying the condition $2 \le n \le x$,*

$$|\Phi_n| \le \frac{\sqrt{2\pi}}{n!}(\log\log n + 1)^n.$$

Proof. It follows from the definition of $\varphi(z)$ and $H(z)$ that $(-1)^n\Phi_n$ coincides with the coefficient of v^{2n} in the series expansion of the function $\Psi(v) = \Phi(-v^2)$. Therefore,

$$(-1)^n\Phi_n = \frac{1}{2\pi i}\oint \frac{\Psi(v)}{v^{2n+1}}\,dv;$$

for the integration contour a circle of any radius $r > 0$ centered at the origin can be taken. We set $r = \Delta\sqrt{n}$ and assume that the quantity Δ, whose exact value will be chosen later on, satisfies the condition $0 < \Delta \le 4/5$. Thus, we have

$$|\Phi_n| \le \frac{1}{2\pi}\frac{1}{(\Delta\sqrt{n})^{2n}}\int_{-\pi}^{\pi} |\Psi(\Delta\sqrt{n}e^{i\vartheta})|\,d\vartheta \le \frac{\Psi}{(\Delta\sqrt{n})^{2n}},$$

where

$$\Psi = \max_{0\le\vartheta\le\pi/2} |\Psi(\Delta\sqrt{n}e^{i\vartheta})|.$$

Using Stirling's formula, we write the last inequality in the form

$$|\Phi_n| \le \frac{\sqrt{2\pi n}}{n!}\exp\biggl\{-n+\frac{1}{12n}\biggr\}\Delta^{-2n}\Psi = \frac{\sqrt{2\pi}}{n!}\exp\biggl\{-n+\frac{\log n}{2}+\frac{1}{12n}+2n\log\frac{1}{\Delta}\biggr\}\Psi.$$

To estimate Ψ from above, we represent the function $\Psi(v)$ in the form

$$\Psi(v)=\Phi(-v^2)=\prod_{p\le x}\varphi\biggl(-\frac{v^2}{p}\biggr)=\prod_{p\le x}e^{v^2/p}\biggl\{\sum_{m=0}^{+\infty}\frac{(-1)^m}{(m!)^2}\biggl(\frac{v}{\sqrt{p}}\biggr)^{2m}\biggr\}$$
$$=\prod_{p\le x}e^{v^2/p}J_0\biggl(\frac{2v}{\sqrt{p}}\biggr)=\biggl\{\prod_{p\le n}\cdot\prod_{n<p\le x}\biggr\}e^{v^2/p}J_0\biggl(\frac{2v}{\sqrt{p}}\biggr)=\Psi_1(v)\cdot\Psi_2(v).$$

Setting $R=2\Delta\sqrt{n/p}$ in Lemma I.2, we obtain

$$\biggl|J_0\biggl(\frac{2v}{\sqrt{p}}\biggr)\biggr|\le\frac{1}{\sqrt{\pi\Delta}}\biggl(1+\frac{1}{4\Delta\sqrt{2}}\sqrt{\frac{p}{n}}\biggr)\sqrt[4]{\frac{p}{n}}\exp\biggl\{2\Delta\sqrt{\frac{n}{p}}\sin\vartheta\biggr\},$$

whence

$$|\Psi_1(v)|\le\biggl\{\prod_{p\le n}\frac{1}{\sqrt{\pi\Delta}}\biggl(1+\frac{1}{4\Delta\sqrt{2}}\sqrt{\frac{p}{n}}\biggr)\biggr\}\exp\biggl\{\Delta^2n\biggl(\sum_{p\le n}\frac{1}{p}\biggr)\cos2\vartheta+2\Delta\sqrt{n}\biggl(\sum_{p\le n}\frac{1}{\sqrt{p}}\biggr)\sin\vartheta\biggr\}.\tag{1.28}$$

Applying relations (I.2) and (I.3) of Lemma I.3 and noting that

$$A\cos2\vartheta+2B\sin\vartheta\le A+\frac{B^2}{2A}\tag{1.29}$$

for any $A>0$, we see that, for all ϑ under examination, the exponent in (1.28) does not exceed

$$\Delta^2n\biggl(L_n+\frac{\theta}{(\log n)^2}\biggr)\cos2\vartheta+\frac{2a\Delta n}{\log n}\sin\vartheta\le\Delta^2nL_n+\frac{a^2}{2}\frac{n}{(\log n)^2L_n}+\frac{\Delta^2n}{(\log n)^2},$$

where $L_n=\log\log n+c$ (a and c are the same constants as in Lemma I.3).

For $0<\Delta\le4/5$, each factor in the product over $p\le n$ in (1.28) is at most

$$\frac{\Delta^{-3/2}}{\sqrt{\pi}}\biggl(\frac{4}{5}+\frac{1}{4\sqrt{2}}\sqrt{\frac{p}{n}}\biggr)\sqrt[4]{\frac{p}{n}}\le d\Delta^{-3/2}\sqrt[4]{\frac{p}{n}},$$

where

$$d=\frac{1}{\sqrt{\pi}}\biggl(\frac{4}{5}+\frac{1}{4\sqrt{2}}\biggr);$$

thus, we finally obtain

$$|\Psi_1(v)|\le(d\Delta^{-3/2})^{\pi(n)}\biggl\{\prod_{p\le n}\frac{p}{n}\biggr\}^{1/4}\exp\biggl\{\Delta^2nL_n+\frac{a^2}{2}\frac{n}{(\log n)^2L_n}+\frac{\Delta^2n}{(\log n)^2}\biggr\}.\tag{1.30}$$

Now, let us estimate $|\Psi_2(v)|$. For this purpose, we set

$$r=r_p(v)=\frac{1}{(3!)^2}\frac{v^6}{p^3}-\frac{1}{(4!)^2}\frac{v^8}{p^4}+\frac{1}{(5!)^2}\frac{v^{10}}{p^5}-\cdots$$

and

$$q=q_p(v)=\frac{v^2}{p}-\frac{v^4}{4p^2}+r_p(v),$$

so that

$$J_0\biggl(\frac{2v}{\sqrt{p}}\biggr)=1-q_p(v).$$

For each $p > n$, we obviously have

$$\left|\frac{v}{\sqrt{p}}\right| = \Delta\sqrt{\frac{n}{p}} < \Delta.$$

Therefore,

$$|r| \le \frac{1}{36}\frac{|v|^6}{p^3}\left(1 + \frac{1}{4^2}\frac{|v|^2}{p} + \frac{1}{(4\cdot 5)^2}\frac{|v|^4}{p^2} + \cdots\right) \le \frac{1}{36}\frac{|v|^6}{p^3}\left(1 + \left(\frac{\Delta}{4}\right)^2 + \left(\frac{\Delta}{4}\right)^4 + \cdots\right) = \lambda_1\frac{|v|^6}{p^3},$$

where

$$\lambda_1 = \lambda_1(\Delta) = \frac{4}{9(16-\Delta^2)};$$

at the same time,

$$|q| \le \frac{|v|^2}{p} + \frac{|v|^4}{4p^2} + |r| \le \lambda_2\frac{|v|^2}{p} \le \lambda_3,$$

$$\lambda_2 = \lambda_2(\Delta) = 1 + \frac{\Delta^2}{4} + \lambda_1\Delta^4, \qquad \lambda_3 = \lambda_3(\Delta) = \lambda_2\Delta^2.$$

Next, since

$$\left|-\frac{v^4}{4p^2} + r\right| \le \frac{|v|^4}{4p^2} + \lambda_1\frac{|v|^6}{p^3} \le \lambda_4\frac{|v|^4}{p^2}, \qquad \lambda_4 = \lambda_4(\Delta) = \frac{1}{4} + \lambda_1\Delta^2,$$

it follows that

$$q^2 = \left(\frac{v^2}{p} - \frac{v^4}{4p^2} + r\right)^2 = \left(\frac{v^2}{p} + \lambda_4\theta\frac{|v|^4}{p^2}\right)^2 = \frac{v^4}{p^2} + \lambda_5\theta_1\frac{|v|^6}{p^3},$$

where

$$\lambda_5 = \lambda_5(\Delta) = 2\lambda_4 + (\lambda_4\Delta)^2.$$

Finally, for v under consideration, we have

$$\begin{aligned}\left|\sum_{k=3}^{+\infty}\frac{q^k}{k}\right| &\le \frac{1}{3}|q|^3 + \frac{1}{4}|q|^4 + \frac{1}{5}\sum_{k=5}^{+\infty}|q|^k \\ &\le |q|^3\left(\frac{1}{3} + \frac{\lambda_3}{4} + \frac{\lambda_3^2}{5}\sum_{k=0}^{+\infty}\lambda_3^k\right) = |q|^3\left(\frac{1}{3} + \frac{\lambda_3}{4} + \frac{\lambda_3^2}{5(1-\lambda_3)}\right) \le \lambda_6\frac{|v|^6}{p^3},\end{aligned}$$

where

$$\lambda_6 = \lambda_6(\Delta) = \lambda_2^3\left(\frac{1}{3} + \frac{\lambda_3}{4} + \frac{\lambda_3^2}{5(1-\lambda_3)}\right).$$

Thus,

$$\begin{aligned}\log J_0\left(\frac{2v}{\sqrt{p}}\right) &= \log(1-q) = -q - \frac{1}{2}q^2 - \sum_{k=3}^{+\infty}\frac{q^k}{k} \\ &= -\frac{v^2}{p} + \frac{v^4}{4p^2} - r - \frac{1}{2}\left(\frac{v^4}{p^2} + \lambda_5\theta_1\frac{|v|^6}{p^3}\right) + \lambda_6\theta_2\frac{|v|^6}{p^3} = -\frac{v^2}{p} - \frac{v^4}{4p^2} + \lambda\theta\frac{|v|^6}{p^3},\end{aligned}$$

where

$$\lambda = \lambda(\Delta) = \lambda_1 + \frac{1}{2}\lambda_5 + \lambda_6 = \sum_{k=0}^{2}\frac{P_k}{Q_k}(16-\Delta^2)^{-k}$$

and the P_k and Q_k, $k=0,1,2$, are the polynomials

$$P_0(\Delta)\equiv 2^2,\qquad Q_0(\Delta)\equiv 3^2,$$
$$P_1(\Delta)=2^{11}\cdot 3^4+2^{11}\cdot 3^2\Delta^2+2^3\cdot 3^3\cdot 7\Delta^4+7^2\Delta^6,\qquad Q_1(\Delta)\equiv 2^5\cdot 3^4,$$
$$\begin{aligned}P_2(\Delta)&=(2^6\cdot 3^2+2^2\cdot 3^3\Delta^2+7\Delta^4)^3\\&\quad\times\big(-2^{14}\cdot 3^3\cdot 5+2^{11}\cdot 3^4\cdot 5\Delta^2+2^6\cdot 3^3\cdot 227\Delta^4+2^4\cdot 3\cdot 7^2\cdot 53\Delta^6\\&\qquad+2^2\cdot 3\cdot 1609\Delta^8+2^3\cdot 3^3\cdot 7\Delta^{10}+7^2\Delta^{12}\big),\end{aligned}$$

and

$$Q_2(\Delta)=-2^6\cdot 3^2+2^2\cdot 3^2\cdot 17\Delta^2+2^2\cdot 3^3\Delta^4+7\Delta^6.$$

Thus,

$$e^{v^2/p}J_0\biggl(\frac{2v}{\sqrt{p}}\biggr)=\exp\biggl\{\frac{v^2}{p}-\frac{v^2}{p}-\frac{v^4}{4p^2}+\lambda\theta\frac{|v|^6}{p^3}\biggr\}=\exp\biggl\{-\frac{v^4}{4p^2}+\lambda\theta\frac{|v|^6}{p^3}\biggr\},$$

whence

$$|\Psi_2(v)|\le\exp\biggl\{\frac{|v|^4}{4}\sum_{p>n}\frac{1}{p^2}+\lambda|v|^6\sum_{p>n}\frac{1}{p^3}\biggr\}=\exp\biggl\{\frac{\Delta^4n^2}{4}\sum_{p>n}\frac{1}{p^2}+\lambda\Delta^6n^3\sum_{p>n}\frac{1}{p^3}\biggr\}.\tag{1.31}$$

Multiplying out estimates (1.30) and (1.31), we obtain

$$\begin{aligned}\Psi&\le(d\Delta^{-3/2})^{\pi(n)}\biggl\{\prod_{p\le n}\frac{p}{n}\biggr\}^{1/4}\\&\quad\times\exp\biggl\{\Delta^2nL_n+\frac{\Delta^4n^2}{4}\sum_{p>n}\frac{1}{p^2}+\lambda\Delta^6n^3\sum_{p>n}\frac{1}{p^3}+\frac{a^2}{2}\frac{n}{(\log n)^2L_n}+\frac{\Delta^2n}{(\log n)^2}\biggr\}.\end{aligned}\tag{1.32}$$

Finally, we set $\Delta=1/\sqrt{L_n}$ (it is easy to verify that $\Delta\le 4/5$ for all $n\ge 40$). We have

$$\begin{aligned}\Psi&\le\biggl\{\prod_{p\le n}\frac{p}{n}\biggr\}^{1/4}\\&\quad\times\exp\biggl\{n+\pi(n)\biggl(\frac{3}{4}\log L_n+\log d\biggr)+\frac{n^2}{4L_n^2}\sum_{p>n}\frac{1}{p^2}+\frac{\lambda n^3}{L_n^3}\sum_{p>n}\frac{1}{p^3}+\biggl(\frac{a^2}{2}+1\biggr)\frac{n}{(\log n)^2L_n}\biggr\}\end{aligned}$$

and

$$\begin{aligned}|\Phi_n|\le\frac{\sqrt{2\pi}}{n!}\biggl\{\prod_{p\le n}\frac{p}{n}\biggr\}^{1/4}\exp\biggl\{&n\log L_n+\pi(n)\biggl(\frac{3}{4}\log L_n+\log d\biggr)+\frac{n^2}{4L_n^2}\sum_{p>n}\frac{1}{p^2}+\frac{\lambda n^3}{L_n^3}\sum_{p>n}\frac{1}{p^3}\\&+\biggl(\frac{a^2}{2}+1\biggr)\frac{n}{(\log n)^2L_n}+\frac{1}{2}\log n+\frac{1}{12n}\biggr\}.\end{aligned}\tag{1.33}$$

First, suppose that $n\ge n_1=1143$. It is easy to show that, for such n, the number $(3/4)\log L_n+\log d$ is positive, and we can estimate the quantity $\pi(n)$ in the exponent in (1.33) by using inequality (I.4) in Lemma I.3. Applying the same inequality, we obtain

$$\sum_{p>n}\frac{1}{p^2}=2\int_n^{+\infty}(\pi(u)-\pi(n))\,\frac{du}{u^3}\le 2b\int_n^{+\infty}\frac{du}{u^2\log u}\le\frac{2b}{\log n}\int_n^{+\infty}\frac{du}{u^2}=\frac{2b}{n\log n}$$

and

$$\sum_{p>n}\frac{1}{p^3}\le\frac{3b}{2n^2\log n}.$$

Finally, we estimate the product over $p \le n$ from above by 1. Then inequality (1.33) takes the form

$$|\Phi_n| \le \frac{\sqrt{2\pi}}{n!} \exp\big\{n(\log L_n + Q(n))\big\}, \tag{1.34}$$

where

$$Q(n) = \frac{b}{\log n}\left(\frac{3}{4}\log L_n + \log d\right) + \frac{b}{2}\frac{1}{(\log n)L_n^2} + \frac{3\lambda_0 b}{2}\frac{1}{(\log n)L_n^3} + \left(\frac{a^2}{2}+1\right)\frac{1}{(\log n)^2 L_n} + \frac{\log n}{2n} + \frac{1}{12n^2},$$

and λ_0 denotes the maximum value of $\lambda(\Delta)$ over $n \ge n_1$, i.e.,

$$\lambda_0 = \max_{n\ge n_1} \lambda\left(\frac{1}{\sqrt{L_n}}\right) = \lambda\left(\frac{1}{\sqrt{L_{n_1}}}\right) = 1.09559\ldots .$$

The function $f(n) = L_n(e^{Q(n)} - 1) + c$ is of order $L_n(\log L_n)/\log n$ and monotonically decreases with increasing n in the interval under consideration. Therefore, for all such n, we have

$$f(n) \le f(n_1) = 0.47652\ldots < 1, \qquad L_n e^{Q(n)} < L_n - c + 1 = \log\log n + 1,$$

and, finally,

$$|\Phi_n| \le \frac{\sqrt{2\pi}}{n!}(L_n e^{Q(n)})^n < \frac{\sqrt{2\pi}}{n!}(\log\log n + 1)^n.$$

Now, suppose that $43 \le n \le 1142$. We write (1.33) in the form (1.34), where the role of $Q(n)$ is played by the function

$$\frac{\pi(n)}{n}\left(\frac{3}{4}\log L_n + \log d\right) + \frac{1}{4n}\left(\sum_{p\le n}\log p - \pi(n)\log n\right) + \frac{n}{4L_n^2}\left(\zeta_P(2) - \sum_{p\le n}\frac{1}{p^2}\right) + \frac{\lambda n^3}{L_n^3}\left(\zeta_P(3) - \sum_{p\le n}\frac{1}{p^3}\right) + \left(\frac{a^2}{2}+1\right)\frac{1}{(\log n)^2 L_n} + \frac{\log n}{2n} + \frac{1}{12n^2},$$

$$\zeta_P(s) = \sum_p \frac{1}{p^s}, \qquad \text{and} \qquad \lambda = \lambda\left(\frac{1}{\sqrt{L_n}}\right).$$

The computation of $Q(n)$ by the *Wolfram Mathematica* 7.0.0 package shows that, for all n satisfying the condition $43 \le n \le 1142$, we have

$$f(n) = L_n(e^{Q(n)} - 1) + c \le f(43) = 0.999\,985 < 1,$$

which gives the required estimate of $|\Phi_n|$. In the case $6 \le n \le 42$, inequalities (1.28) and (1.29) imply the estimate

$$|\Psi_1(v)| \le \exp\Bigg\{\Delta^2 n\left(\sum_{p\le n}\frac{1}{p}\right) + \frac{1}{2}\left(\sum_{p\le n}\frac{1}{\sqrt{p}}\right)^2\left(\sum_{p\le n}\frac{1}{p}\right)^{-1} + \frac{\pi(n)}{2}\log\frac{1}{\pi\Delta} + \frac{1}{4}\left(\sum_{p\le n}\log p - \pi(n)\log n\right) + \sum_{p\le n}\log\left(1 + \frac{1}{4\Delta\sqrt{2}}\sqrt{\frac{p}{n}}\right)\Bigg\}$$

for any $0 < \Delta \le 4/5$; combining this estimate and (1.31), we obtain inequality (1.34) with

$$Q(n) = -\log L_n - 1 + 2\log\frac{1}{\Delta} + \Delta^2\left(\sum_{p\le n}\frac{1}{p}\right) + \frac{1}{2n}\left(\sum_{p\le n}\frac{1}{\sqrt{p}}\right)^2\left(\sum_{p\le n}\frac{1}{p}\right)^{-1}$$

$$+\frac{\pi(n)}{2n}\log\frac{1}{\pi\Delta}+\frac{1}{4n}\biggl(\sum_{p\le n}\log p-\pi(n)\log n\biggr)+\frac{1}{n}\sum_{p\le n}\log\biggl(1+\frac{1}{4\Delta\sqrt{2}}\sqrt{\frac{p}{n}}\biggr)$$
$$+\frac{n}{4L_n^2}\biggl(\zeta_P(2)-\sum_{p\le n}\frac{1}{p^2}\biggr)+\frac{\lambda n^2}{L_n^3}\biggl(\zeta_P(3)-\sum_{p\le n}\frac{1}{p^3}\biggr)+\frac{\log n}{2n}+\frac{1}{12n^2}\,,\qquad \lambda=\lambda(\Delta).$$

Setting $\Delta=4/5$, we obtain

$$L_n(e^{Q(n)}-1)+c\le 0.923\,714<1$$

for all n in the interval $6\le n\le 42$, which proves the required inequality.

Finally, the explicit expressions for Φ_n, $2\le n\le 5$, in Lemma 1.7 imply the inequalities

$$|\Phi_2|\le\frac{1}{4}\zeta_P(2),\qquad |\Phi_3|\le\frac{1}{9}\zeta_P(3),$$
$$-\frac{11}{192}\zeta_P(4)<\Phi_4<\frac{1}{32}\zeta_P^2(2),\qquad\text{and}\qquad -\frac{1}{36}\zeta_P(2)\zeta_P(3)<\Phi_5<\frac{19}{600}\zeta_P(5),$$

which, in turn, imply estimates of the form $|\Phi_n|<\phi_n$, where

$$\phi_2=\frac{1}{4}\zeta_P(2)=0.113\,061\,855\dots\,,\qquad \phi_3=\frac{1}{9}\zeta_P(3)=0.019\,418\,071\dots\,,$$
$$\phi_4=\frac{1}{32}\zeta_P^2(2)=0.006\,391\,491\dots\,,\qquad \phi_5=\frac{1}{36}\zeta_P(2)\zeta_P(3)=0.001\,132\,242\dots\,.$$

Setting

$$\psi_n=\biggl(\frac{n!\,\phi_n}{\sqrt{2\pi}}\biggr)^{1/n}-\log\log n,$$

we obtain

$$\psi_2=0.666\,863\,230\,3\dots\,,\qquad \psi_3=0.265\,499\,281\,0\dots\,,$$
$$\psi_4=0.170\,737\,224\,5\dots\,,\qquad \psi_5=0.161\,383\,869\,7\dots\,.$$

Thus, $\psi_n<1$ for n under consideration, whence

$$|\Phi_n|<\phi_n=\frac{\sqrt{2\pi}}{n!}(\log\log n+\psi_n)^n<\frac{\sqrt{2\pi}}{n!}(\log\log n+1)^n.$$

This completes the proof of the lemma.

1.4. Fractional moments of $V(t_n)$. In some cases, a fractional power $|z|^a$ of a quantity z can be approximated fairly well by a linear combination of its even powers z^{2k}, $k=0,1,2,\dots$. This fact, which, apparently was first mentioned by Ghosh in [32], [33], makes it possible to obtain asymptotic expressions for fractional moments of $V(t_n)$ and, eventually, for Δ_n. The construction of these approximations is based on the following three lemmas. (The identity in Lemma 1.13 in the general form, i.e., with any admissible parameters a and s, was first applied to evaluate fractional moments by Boyarinov; see [34].)

Lemma 1.13. *If $a>0$ and s is an integer such that $2s>a$, then, for any z, the identity*

$$|z|^a=\kappa^{-1}\int_0^{+\infty}\frac{(\sin u|z|)^{2s}}{u^{1+a}}\,du$$

holds, where

$$\kappa=\kappa(a;s)=\int_0^{+\infty}\frac{(\sin u)^{2s}}{u^{1+a}}\,du.$$

Lemma 1.14. *For any integer $s\ge 1$,*

$$(\sin y)^{2s}=2^{1-2s}\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}\cos(2my)+2^{-2s}\binom{2s}{s}.$$

Lemma 1.15. *Given integer $s \geq 1$ and nonnegative a, let $B(a;s)$ be defined by*

$$B(a;s) = \begin{cases} \dfrac{1}{2}\binom{2s}{s} + \displaystyle\sum_{m=1}^{s} (-1)^m \binom{2s}{s-m} & \text{if } a = 0, \\ \displaystyle\sum_{m=1}^{s} (-1)^m \binom{2s}{s-m} m^a & \text{if } a > 0. \end{cases}$$

Then

(1) $|B(a;s)| \leq 2^{2s-1} s^a$;

(2) $B(2k;s) = 0$ *for any integer* k, $0 \leq k < s$;

(3) *for any integer* s *and* K,

$$(\sin y)^{2s} = 2^{1-2s} \sum_{k=s}^{K} \frac{(-1)^k B(2k;s)(2y)^{2k}}{(2k)!} + \frac{\theta (2ys)^{2K+2}}{(2K+2)!}.$$

Lemma 1.13 is proved by a change of variable in the integral. Lemma 1.14 is a consequence of Euler's formulas

$$\sin u = \frac{e^{iu} - e^{-iu}}{2i} \qquad \text{and} \qquad \cos u = \frac{e^{iu} + e^{-iu}}{2}.$$

Assertion (1) of Lemma 1.15 follows from

$$\sum_{m=1}^{s} \binom{2s}{s-m} = 2^{2s-1};$$

assertions (2) and (3) of Lemma 1.15 are obtained from Lemma 1.14 and Taylor's formula for the function $\cos(2my)$ with remainder in the Lagrange form by evaluating the coefficients of the powers y^{2k}, $k = 0, 1, \ldots, s-1$, in the expansion of $(\sin y)^{2s}$.

Remark 1.2. The following relations hold:

$$\begin{aligned} B(a;1) &= -1, \\ B(a;2) &= 2^a - 4, \\ B(a;3) &= -15 + 6 \cdot 2^a - 3^a, \\ B(a;4) &= -56 + 28 \cdot 2^a - 8 \cdot 3^a + 4^a, \\ B(a;5) &= -210 + 120 \cdot 2^a - 45 \cdot 3^a + 10 \cdot 4^a - 5^a. \end{aligned}$$

Lemma 1.16. *If $s \geq 1$ is an integer and a is any number satisfying the conditions $0 < a < 2s$ and $a \neq 2n$, $n = 1, 2, 3, \ldots$, then the integral $\kappa(a;s)$ can be represented as*

$$\kappa(a;s) = -\frac{2^{a-2s} \pi B(a;s)}{\Gamma(a+1) \sin(\pi a/2)}.$$

Moreover,

$$\kappa(a;s) \geq \frac{1}{2s-a} \left(\frac{2}{\pi}\right)^a.$$

Proof. First, suppose that $0 < a < 2$. Then, using the relation of Lemma 1.14 and integrating by parts, we obtain

$$\kappa(a;s) = -2^{1-2s}\int_0^{+\infty}\biggl\{\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}\cos(2mu) + \frac{1}{2}\binom{2s}{s}\biggr\}d\frac{u^{-a}}{a}$$
$$= -\frac{2^{1-2s}}{a}\int_0^{+\infty}\frac{1}{u^a}\biggl\{\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}2m\sin(2mu)\biggr\}du. \tag{1.35}$$

The assumption $0 < a < 2$ allows us to integrate the sum over m termwise:

$$\kappa(a;s) = -\frac{2^{1-2s}}{a}\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}\int_0^{+\infty}\frac{\sin(2mu)}{u^a}\,du$$
$$= \frac{2^{1-2s}}{a}j(a)\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}(2m)^a = -\frac{2^{a+1-2s}}{a}j(a)B(a;s),$$

where

$$j(a) = \int_0^{+\infty}\frac{\sin u}{u^a}\,du.$$

If $2 < a < 4$, then, twice integrating by parts, we derive the following representation from (1.35):

$$\kappa(a;s) = -\frac{2^{1-2s}}{a(a-1)}\int_0^{+\infty}\biggl\{\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}(2m)^2\cos(2mu)\biggr\}\frac{du}{u^{a-1}}$$
$$= \frac{2^{1-2s}}{a(a-1)(a-2)}\int_0^{+\infty}\biggl\{\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}(2m)^3\sin(2mu)\biggr\}\frac{du}{u^{a-2}}. \tag{1.36}$$

Again integrating termwise (which is legitimate, because $0 < a-2 < 2$), we obtain

$$\kappa(a;s) = \frac{2^{1-2s}}{a(a-1)(a-2)}\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}(2m)^3\int_0^{+\infty}\frac{\sin(2mu)}{u^{a-2}}\,du$$
$$= \frac{2^{1-2s}}{a(a-1)(a-2)}j(a-2)\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}(2m)^3\cdot(2m)^{a-3} = \frac{2^{a+1-2s}j(a-2)B(a;s)}{a(a-1)(a-2)}.$$

In general, if $2k < a < 2(k+1)$, where $k \geq 0$ is an integer, then it follows from (1.35) by $2(k+1)$ times integrating by parts that

$$\kappa(a;s) = \frac{(-1)^{k-1}2^{a+1-2s}j(a-2k)B(a;s)}{a(a-1)(a-2)\cdots(a-2k+1)(a-2k)}.$$

Finally, noting that

$$\int_0^{+\infty}\frac{(\sin u)^2}{u^{1+a}}\,du = \frac{2^a j(a)}{2a}$$

and using a relation in Section 3.823 of book [35], we arrive at

$$j(a) = \frac{\pi}{4\Gamma(a+1)\sin(\pi a/2)}.$$

Therefore,

$$\kappa(a;s) = \frac{(-1)^{k-1}\,2^{a+1-2s}(-1)^k\pi B(a;s)}{2a(a-1)\cdots(a-2k)\Gamma(a-2k)\sin(\pi a/2)} = -\frac{2^{a-2s}\pi B(a;s)}{\Gamma(a+1)\sin(\pi a/2)}.$$

The lower bound for $\kappa(a;s)$ is obtained from the inequalities

$$\kappa(a;s) > \int_0^{\pi/2} \frac{(\sin u)^{2s}}{u^{1+a}}\,du \ge \int_0^{\pi/2} \left(\frac{2u}{\pi}\right)^{2s} \frac{du}{u^{1+a}} = \frac{1}{2s-a}\left(\frac{2}{\pi}\right)^a.$$

This completes the proof of the lemma.

To prove the main assertion of this section, we need the following lemma, which establishes the simplest properties of the function $H(z)$ defined in Section 1.1.

Lemma 1.17. *The following relations hold:*

(1) $H(0) = 1$;

(2) $0 < H^{(k)}(0) \le \mathfrak{S}_1^k$;

(3) $|H(-u)| \le 1$ *for any* $u \ge 0$.

Proof. Relation (1) follows from the equality $H(0) = \Phi(0)$ and the definition of the function $\Phi(z)$. To prove relation (2), we use the expression

$$H^{(k)}(0) = \frac{1}{k!} \sum_{\substack{p_1\cdots p_k = q_1\cdots q_k \\ p_1,\dots,p_k \le x}} (p_1\cdots p_k)^{-1}$$

given by Lemma 1.5. For fixed $p_1,\dots,p_k$, the number of solutions of the equation $p_1\cdots p_k = q_1\cdots q_k$ in primes $q_1,\dots,q_k$ is at most $k!$. Therefore,

$$0 < H^{(k)}(0) \le \frac{1}{k!}k! \sum_{p_1,\dots,p_k \le x} (p_1\cdots p_k)^{-1} = \left(\sum_{p\le x}\frac{1}{p}\right)^k = \mathfrak{S}_1^k.$$

To prove inequality (3) in the lemma, we set $u = v^2$. We have

$$H(-u) = H(-v^2) = \prod_{p\le x} J_0\left(2i\sqrt{-\frac{v^2}{p}}\right) = \prod_{p\le x} J_0\left(\frac{2v}{\sqrt{p}}\right).$$

Applying the representation

$$J_0(z) = \frac{1}{\pi}\int_0^\pi \cos(z\sin t)\,dt$$

(see, e.g., [36, II, Section 17.23]), we obtain $|J_0(z)| \le 1$ for real z. Thus,

$$|H(-u)| = \prod_{p\le x}\left|J_0\left(\frac{2v}{\sqrt{p}}\right)\right| \le 1.$$

This proves the lemma.

Lemma 1.18. *Suppose that* $x_0 < x \le N^\delta$, $400\delta = 1$, $V(t) = V_x(t)$, *a is any real number not being an even integer and satisfying the conditions*

$$0 < a < \frac{1}{3e}\sqrt{\frac{\log N}{\log x}},$$

and, finally, $s = [(a+1)/2] + 1$. *Then*

$$\sum_n |V(t_n)|^a = \frac{M\mathfrak{S}_1^{a/2}}{\kappa(a;s)}(I(a) + r(a)),$$

where

$$I(a) = 2^{1-2s}\int_0^{+\infty}\frac{\psi(u)}{u^{1+a}}\,du,$$
$$\psi(u) = \psi(u;x,s) = \sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}H\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr) + \frac{1}{2}\binom{2s}{s},$$

and

$$|r(a)| \le 2.006a^{-1}\biggl(5es\sqrt{\frac{\log x}{\log N}}\,\biggr)^a.$$

Proof. We set $v_n = V(t_n)/\sqrt{\mathfrak{S}_1}$ and define quantities λ and K by

$$\lambda = \frac{1}{5es}\sqrt{\frac{\log N}{\log x}} \qquad \text{and} \qquad K = \biggl[\frac{1}{22}\frac{\log N}{\log x}\biggr].$$

It is easy to show that

$$K \ge s, \qquad \lambda > 1, \qquad \frac{e(\lambda s)^2}{K} < \frac{1}{e}, \qquad \text{and} \qquad 1 < 2s - a \le 3 \tag{1.37}$$

for $x \le N^{\delta}$. By virtue of Lemma 1.13, we have

$$|v_n|^a = \kappa^{-1}\int_0^{+\infty}\frac{(\sin(uv_n))^{2s}}{u^{1+a}}\,du, \qquad \kappa = \kappa(a;s).$$

Summing each side of this relation over n, we obtain

$$\sum_n |v_n|^a = \kappa^{-1}\sum_n\biggl(\int_0^{\lambda} + \int_{\lambda}^{+\infty}\biggr)\frac{(\sin(uv_n))^{2s}}{u^{1+a}}\,du = W_1 + R_1,$$

where

$$|R_1| \le \kappa^{-1}\sum_n\int_{\lambda}^{+\infty}\frac{du}{u^{1+a}} = \frac{M\kappa^{-1}}{a\lambda^a} \qquad \text{and} \qquad W_1 = \kappa^{-1}\sum_n\int_0^{\lambda}\frac{(\sin(uv_n))^{2s}}{u^{1+a}}\,du.$$

The transformation of the sum W_1 by means of the relation of Lemma 1.15 yields

$$W_1 = \kappa^{-1}\int_0^{\lambda}\frac{1}{u^{1+a}}\biggl\{2^{1-2s}\sum_{k=s}^{K}\frac{(-1)^kB(2k;s)}{(2k)!}(2u)^{2k}\sum_n v_n^{2k} + \theta_1\frac{(2us)^{2K+2}}{(2K+2)!}\sum_n v_n^{2K+2}\biggr\}du.$$

The parameters x, M, N, and K are such that $x_0 < x \le M^{1/(2k)}$ if $1 \le k \le K+1$. Therefore, applying the relations of Lemmas 1.4 and 1.5 to the sums over n, we obtain

$$\sum_n v_n^{2k} = \frac{(2k)!}{k!}\frac{M}{2^{2k}}\frac{H^{(k)}(0)}{\mathfrak{S}_1^k} + \theta_k\frac{x^{2k}}{\mathfrak{S}_1^k}\log N, \qquad k = s,\dots,K+1,$$

and

$$\begin{aligned}W_1 = \kappa^{-1}\int_0^{\lambda}\frac{1}{u^{1+a}}\biggl\{&2^{1-2s}\sum_{k=s}^{K}\frac{(-1)^kB(2k;s)}{(2k)!}(2u)^{2k}\biggl(\frac{(2k)!}{k!}\frac{M}{2^{2k}}\frac{H^{(k)}(0)}{\mathfrak{S}_1^k} + \theta_k\frac{x^{2k}}{\mathfrak{S}_1^k}\biggr)\\ &+ \theta_{K+1}\frac{(2us)^{2K+2}}{(2K+2)!}\biggl(\frac{(2K+2)!}{(K+1)!}\frac{M}{2^{2K+2)}}\frac{H^{(K+1)}(0)}{\mathfrak{S}_1^{K+1}} + \frac{x^{2K+2}}{\mathfrak{S}_1^{K+1}}\biggr)\biggr\}\,du\\ &= W_2 + R_2,\end{aligned}$$

where R_2 is the contribution of the terms of the form $x^{2k}\mathfrak{S}_1^{-k}\log N$, $s\le k\le K+1$. The first inequality in Lemma 1.15 gives

$$
\begin{aligned}
|R_2| &\le \kappa^{-1}\int_0^\lambda \frac{1}{u^{1+a}}\biggl\{\sum_{k=s}^{K}\frac{2^{1-2s}|B(2k;s)|}{(2k)!}(2u)^{2k}\frac{x^{2k}}{\mathfrak{S}_1^k}\log N+\frac{(2us)^{2K+2}}{(2K+2)!}\frac{x^{2K+2}}{\mathfrak{S}_1^{K+1}}\log N\biggr\}\,du\\
&\le \kappa^{-1}x^{2K+2}(\log N)\int_0^\lambda\frac{1}{u^{1+a}}\sum_{k=s}^{K+1}\frac{(2us)^{2k}}{(2k)!}\,\mathfrak{S}_1^{-k}\,du\\
&= \kappa^{-1}x^{2K+2}(\log N)\sum_{k=s}^{K+1}\frac{1}{(2k)!}\biggl(\frac{2s}{\sqrt{\mathfrak{S}_1}}\biggr)^{2k}\frac{\lambda^{2k-a}}{2k-a}\le\frac{\kappa^{-1}x^{2K+2}(\log N)}{\lambda^a(2s-a)}\sum_{k=s}^{K+1}\frac{1}{(2k)!}\biggl(\frac{2s\lambda}{\sqrt{\mathfrak{S}_1}}\biggr)^{2k}\\
&< \kappa^{-1}x^{2K+2}(\log N)\lambda^{-a}\exp\biggl\{\frac{2s\lambda}{\sqrt{\mathfrak{S}_1}}\biggr\}.
\end{aligned}
$$

The definition of the quantities λ, K, and s and the inequality $x\le N^\delta$ imply

$$
\begin{aligned}
|R_2| &\le \kappa^{-1}\lambda^{-a}\exp\biggl\{3\biggl(\frac{1}{22}\frac{\log N}{\log x}+1\biggr)\log x+\frac{2}{5e}\sqrt{\frac{\log N}{\mathfrak{S}_1\log x}}+\log\log N\biggr\}\\
&< \kappa^{-1}\lambda^{-a}\exp\biggl\{\frac{5}{33}\log N\biggr\}<\kappa^{-1}\lambda^{-a}MN^{-1/6-\varepsilon}\le\kappa^{-1}\frac{MN^{-1/6}}{a\lambda^a}.
\end{aligned}
$$

Now, let us transform the quantity

$$
W_2=M\kappa^{-1}\int_0^\lambda\biggl\{2^{1-2s}\sum_{k=s}^{K}\frac{B(2k;s)}{k!}\biggl(-\frac{u^2}{\mathfrak{S}_1}\biggr)^kH^{(k)}(0)+\theta\frac{(us)^{2K+2}}{(K+1)!}\frac{H^{(K+1)}(0)}{\mathfrak{S}_1^{K+1}}\biggr\}\frac{du}{u^{1+a}}.
$$

We replace the sum over k, $s\le k\le K$, by an infinite sum, using the explicit expression and estimate of the numbers $B(2k;s)$ given in Lemma 1.15. We obtain

$$
\begin{aligned}
W_2 &= M\kappa^{-1}\int_0^\lambda\biggl\{2^{1-2s}\sum_{k=s}^{+\infty}\frac{B(2k;s)}{k!}\biggl(-\frac{u^2}{\mathfrak{S}_1}\biggr)^kH^{(k)}(0)\\
&\qquad -2^{1-2s}\sum_{k>K}\frac{B(2k;s)}{k!}\biggl(-\frac{u^2}{\mathfrak{S}_1}\biggr)^kH^{(k)}(0)+\theta\frac{(us)^{2K+2}}{(K+1)!}\frac{H^{(K+1)}(0)}{\mathfrak{S}_1^{K+1}}\biggr\}\frac{du}{u^{1+a}}\\
&= M\kappa^{-1}\int_0^\lambda\biggl\{2^{1-2s}\sum_{k=s}^{+\infty}\frac{B(2k;s)}{k!}\biggl(-\frac{u^2}{\mathfrak{S}_1}\biggr)^kH^{(k)}(0)+2\theta\sum_{k>K}\frac{(us)^k}{k!}\frac{H^{(k)}(0)}{\mathfrak{S}_1^k}\biggr\}\frac{du}{u^{1+a}}\\
&= W_3+R_3;
\end{aligned}
$$

moreover,

$$
|R_3|\le 2M\kappa^{-1}\int_0^\lambda\sum_{k>K}\frac{(us)^{2k}}{k!}\frac{du}{u^{1+a}}=\frac{2M\kappa^{-1}}{\lambda^a}\sum_{k>K}\frac{(\lambda s)^{2k}}{k!}\frac{1}{2k-a}\le\frac{2M\kappa^{-1}}{\lambda^a(2K+2-a)}\sum_{k>K}\frac{(\lambda s)^{2k}}{k!}.
$$

By virtue of Stirling's formula, for $k>K$, we have

$$
\frac{(\lambda s)^{2k}}{k!}=\frac{(\lambda s)^{2k}}{\sqrt{2\pi k}}\biggl(\frac{e}{2k}\biggr)^ke^{-\theta/12k}\le\frac{1}{\sqrt{2\pi k}}\biggl(\frac{e(\lambda s)^2}{k}\biggr)^k\le\frac{1}{\sqrt{2\pi K}}\biggl(\frac{e(\lambda s)^2}{K}\biggr)^k.
$$

The third inequality in (1.37) implies

$$
|R_3|\le\frac{2M\kappa^{-1}}{\lambda^aK}\sum_{k>K}\frac{e^{-k}}{\sqrt{2\pi K}}<\frac{3}{2}\frac{M\kappa^{-1}}{\lambda^aK\sqrt{K}}e^{-(K+1)}.
$$

Note that

$$\log N/\log x > (3ea)^2 \qquad \text{and} \qquad K+1 \geq \frac{1}{22}\frac{\log N}{\log x} > \beta a^2,$$

where $\beta = (3e)^2/22$. Therefore, applying the obvious inequality $e^{-u} < 1/(u+1)$, $u > 0$, we arrive at

$$e^{-(K+1)} < e^{-\beta a^2} < \frac{1}{1+\beta a^2} = \frac{1}{a}\frac{a}{1+\beta a^2} \leq \frac{1}{a}\max_{z\geq 0}\frac{z}{1+\beta z^2} = \frac{1}{a}\frac{1}{2\sqrt{\beta}} = \frac{1}{a}\frac{\sqrt{22}}{6e}.$$

Finally, taking into account the inequality $K \geq 18$, we obtain

$$|R_3| \leq \frac{3}{2}\frac{M\kappa^{-1}}{a\lambda^a}\frac{\sqrt{22}}{6e}\frac{1}{K\sqrt{K}} \leq \frac{\sqrt{11}}{216e}\frac{M\kappa^{-1}}{a\lambda^a} < \frac{1}{177}\frac{M\kappa^{-1}}{a\lambda^a}.$$

Next, according to Lemma 1.15, we have $B(2k;s) = 0$ for $0 \leq k \leq s-1$. Therefore, the change of the range of variation k in the sum W_3 by the interval $k \geq 1$ does not affect the value of this sum:

$$\begin{aligned}
W_3 &= M\kappa^{-1}\int_0^\lambda \biggl\{2^{1-2s}\sum_{k=1}^{+\infty}\frac{B(2k;s)}{k!}\biggl(-\frac{u^2}{\mathfrak{S}_1}\biggr)^k H^{(k)}(0)\biggr\}\frac{du}{u^{1+a}}\\
&= 2^{1-2s}M\kappa^{-1}\int_0^\lambda \biggl\{\sum_{k=1}^{+\infty}\frac{H^{(k)}(0)}{k!}\biggl(-\frac{u^2}{\mathfrak{S}_1}\biggr)^k \sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}m^{2k}\biggr\}\frac{du}{u^{1+a}}\\
&= 2^{1-2s}M\kappa^{-1}\int_0^\lambda \biggl\{\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}\sum_{k=1}^{+\infty}\biggl(-\frac{u^2}{\mathfrak{S}_1}\biggr)^k\frac{H^{(k)}(0)}{k!}\biggr\}\frac{du}{u^{1+a}}\\
&= 2^{1-2s}M\kappa^{-1}\int_0^\lambda \sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}\biggl\{H\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr)-H(0)\biggr\}\frac{du}{u^{1+a}}.
\end{aligned}$$

Applying relation (1) of Lemma 1.17 and the identity

$$B(0;s) = \frac{1}{2}\binom{2s}{s} + \sum_{m=1}^{s}(-1)^m\binom{2s}{s-m} = 0$$

of Lemma 1.15, we obtain

$$W_3 = M\kappa^{-1}\cdot 2^{1-2s}\int_0^\lambda \frac{\psi(u)}{u^{1+a}}\,du,$$

where

$$\psi(u) = \sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}H\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr) + \frac{1}{2}\binom{2s}{s}.$$

By virtue of the last relation of Lemma 1.17, we have

$$|\psi(u)| \leq \sum_{m=1}^{s}\binom{2s}{s-m} + \frac{1}{2}\binom{2s}{s} = 2^{2s-1}.$$

Therefore, replacing the integral with respect to u in the expression for W_3 by improper integrals, we obtain

$$W_3 = M\kappa^{-1}\,2^{1-2s}\biggl(\int_0^{+\infty} - \int_\lambda^{+\infty}\biggr)\frac{\psi(u)}{u^{1+a}}\,du = M\kappa^{-1}\biggl(I(a) + \frac{\theta}{a\lambda^a}\biggr).$$

Summing the found estimates of R_k, $k = 1,2,3$, we arrive at the relation

$$\sum_n |v_n|^a = M\kappa^{-1}(I(a) + r(a)),$$

in which

$$|r(a)| \le \frac{2}{a\lambda^a} + \frac{N^{-1/6}}{a\lambda^a} + \frac{1}{177}\frac{1}{a\lambda^a} < \frac{2.006}{a\lambda^a} = 2.006a^{-1}\left(5es\sqrt{\frac{\log x}{\log N}}\right)^a.$$

This completes the proof of the lemma.

The difficulty of handling $I(a)$ is explained as follows. The functions $H(-m^2u^2/\mathfrak{S}_1)$, $m = 1, 2, \dots, s$, whose linear combination equals the integrand $\psi(u)$ in $I(a)$, expand into infinite series in powers of $\mathfrak{S}_1^{-1}$. These series converge uniformly in any finite interval $0 \le u \le U$ but not on the entire half-line $u \ge 0$. Can we integrate termwise the expressions obtained by substituting these series into $I(a)$?

Examples show that, in some cases, such a trick yields a correct result. For example, Hardy proved (see [37], [38, Section 1.79]) that a power series $\sum_n a_n u^n$ convergent at $u > 0$ can be multiplied by e^{-u} and integrated termwise in the interval $(0, +\infty)$, provided that the resulting series converges.

Moreover, the replacement of e^u in the expression

$$0 = \int_0^{+\infty} \frac{1-1}{u^{1+a}}\,du = \int_0^{+\infty} \frac{1-e^{-u}e^u}{u^{1+a}}\,du, \qquad 0 < a < 1,$$

by a series and termwise integration lead to the true identity

$$0 = \int_0^{+\infty} \frac{1-e^{-u}}{u^{1+a}}\,du - \sum_{n=1}^{+\infty}\int_0^{+\infty} u^{n-1-a}e^{-u}\,du = \frac{1}{a}\Gamma(1-a) - \sum_{n=1}^{+\infty}\frac{\Gamma(n-a)}{n!} = -\sum_{n=0}^{+\infty}\frac{\Gamma(n-a)}{n!}.$$

Lemmas 1.19 and 1.20 given below prove the legitimacy of termwise integration in $I(a)$.

Lemma 1.19. *If*

$$r_N(u) = \sum_{n>N}\frac{u^n}{n!} \qquad \text{and} \qquad R_N(a) = \int_0^{+\infty}\frac{e^{-u}r_N(u)}{u^{1+a}}\,du$$

for $0 < a < N$, *then*

$$0 \le R_N(a) \le \frac{\Gamma(N+1-a)}{a\,N!}.$$

Proof. We apply the following version of Taylor's formula (see [36, Vol. I, Section 5.41]): if a function $f(u)$ is ν-times continuously differentiable in the interval $a \le u \le a+h$, then

$$f(a+h) = \sum_{m=0}^{\nu-1}\frac{h^m}{m!}f^{(m)}(a) + \frac{h^\nu}{(\nu-1)!}\int_0^1 (1-t)^{\nu-1}f^{(\nu)}(a+th)\,dt.$$

Setting $f(u) = e^u$, $a = 0$, $h = u$, and $\nu = N+1$, we obtain

$$r_N(u) = e^u - \sum_{m=0}^{N}\frac{u^m}{m!} = \frac{u^{N+1}}{N!}\int_0^1 (1-t)^N e^{tu}\,dt.$$

For $N > a$, we have

$$R_N(a) = \lim_{Y\to+\infty} R_N(a;Y),$$

where

$$R_N(a;Y) = \int_0^Y \frac{e^{-u}r_N(u)}{u^{1+a}}\,du = \frac{1}{N!}\int_0^Y u^{N-a}e^{-u}\int_0^1 (1-t)^N e^{tu}\,dt\,du. \tag{1.38}$$

The function $u^{N-a}e^{-u}\cdot(1-t)^N e^{tu}$ of two variables is Riemann integrable in the rectangle $0\le u\le Y$, $0\le t\le 1$; therefore, by Fubini's theorem, we can change the order of integration in (1.38). As a result, we obtain

$$R_N(a;Y)=\frac{1}{N!}\int_0^1(1-t)^N\int_0^Y e^{-(1-t)u}u^{N-a}\,du\,dt=\frac{1}{N!}\int_0^1(1-t)^{-1+a}\biggl(\int_0^{(1-t)Y}e^{-y}y^{N-a}\,dy\biggr)\,dt.$$

The inner integral is at most $\Gamma(N+1-a)$ for any t and Y; hence

$$0\le R_N(a;Y)\le\frac{\Gamma(N+1-a)}{N!}\int_0^1(1-t)^{-1+a}\,dt=\frac{\Gamma(N+1-a)}{a\,N!}.$$

Thus,

$$0\le R_N(a)\le\frac{\Gamma(N+1-a)}{a\,N!},$$

which proves the lemma.

Lemma 1.20. *If $a>0$, $a\ne 2,4,6,\dots$, and $s=[(a+1)/2]+1$, then the integral $I(a)$ satisfies the relation*

$$I(a)=2^{-2s}B(a;s)\sum_{n=0}^{+\infty}\frac{(-1)^n\Phi_n}{\mathfrak{S}_1^n}\,\Gamma\biggl(n-\frac{a}{2}\biggr).$$

Proof. Let N be a sufficiently large integer ($N\ge 2s$). Consider the representation

$$\Phi(z)=\biggl\{\sum_{n=0}^{N}+\sum_{n>N}\biggr\}\Phi_n z^n=\Phi_N(z)+\varphi_N(z).$$

For any m such that $1\le m\le s$, we have

$$H\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr)=e^{-m^2u^2}\biggl\{\Phi_N\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr)+\varphi_N\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr)\biggr\}.$$

Substituting these expressions into the expression for $\psi(u)$, we obtain

$$\psi(u)=\frac{1}{2}\binom{2s}{s}+\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}e^{-m^2u^2}\biggl\{\Phi_N\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr)+\varphi_N\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr)\biggr\}=\Psi_N(u)+\psi_N(u);$$

the meaning of the notations $\Psi_N(u)$ and $\psi_N(u)$ is obvious. Thus, the integral $I(a)$ can be represented as

$$2^{1-2s}\int_0^{+\infty}\frac{\Psi_N(u)+\psi_N(u)}{u^{1+a}}\,du=J_N(a)+j_N(a).$$

Let us estimate $|j_N(a)|$ from above. First,

$$j_N(a)=2^{1-2s}\int_0^{+\infty}\frac{\psi_N(u)}{u^{1+a}}\,du=2^{1-2s}\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}\int_0^{+\infty}\frac{e^{-m^2u^2}}{u^{1+a}}\,\varphi_N\biggl(-\frac{m^2u^2}{\mathfrak{S}_1}\biggr)\,du.$$

Setting $v=(mu)^2$ in the integral, we reduce it to the form

$$\frac{m^a}{2}\int_0^{+\infty}\frac{e^{-v}}{v^{1+a/2}}\,\varphi_N\biggl(-\frac{v}{\mathfrak{S}_1}\biggr)\,dv.$$

Therefore,

$$\begin{aligned}j_N(a)&=2^{2s}\biggl\{\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}m^a\biggr\}\int_0^{+\infty}\frac{e^{-v}}{v^{1+a/2}}\,\varphi_N\biggl(-\frac{v}{\mathfrak{S}_1}\biggr)\,dv\\&=2^{-2s}B(a;s)\int_0^{+\infty}\frac{e^{-v}}{v^{1+a/2}}\,\varphi_N\biggl(-\frac{v}{\mathfrak{S}_1}\biggr)\,dv.\end{aligned}$$

Lemma 1.11 implies the inequalities

$$\left|\int_0^{+\infty}\frac{e^{-v}}{v^{1+a/2}}\,\varphi_N\!\left(-\frac{v}{\mathfrak{S}_1}\right)dv\right|\le\int_0^{+\infty}\frac{e^{-v}r_N(v)}{v^{1+a/2}}\,dv=R_N\!\left(\frac{a}{2}\right)\le\frac{2\Gamma(N+1-a/2)}{a\,N!}.$$

Now, let us evaluate the integral $J_N(a)$. For this purpose, we represent the integrand $\Psi_N(u)$ in the form

$$\begin{aligned}&\frac{1}{2}\binom{2s}{s}+\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}e^{-m^2u^2}\Phi_N\!\left(-\frac{m^2u^2}{\mathfrak{S}_1}\right)\\&\quad=\frac{1}{2}\binom{2s}{s}+\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}e^{-m^2u^2}\sum_{n=0}^{N}(-1)^n\Phi_n\!\left(-\frac{m^2u^2}{\mathfrak{S}_1}\right)^n=\sum_{n=0}^{N}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}A_n(u)u^{2n},\end{aligned}$$

where

$$A_n(u)=\begin{cases}\dfrac{1}{2}\binom{2s}{s}+\displaystyle\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}e^{-m^2u^2} & \text{if } n=0,\\[2ex]\displaystyle\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}m^{2n}e^{-m^2u^2} & \text{if } n\ge1.\end{cases}$$

Then $J_N(a)$ is represented as

$$2^{1-2s}\int_0^{+\infty}\left\{\sum_{n=0}^{N}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}A_n(u)u^{2n}\right\}\frac{du}{u^{1+a}}=2^{1-2s}\sum_{n=0}^{N}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}I_n,$$

where I_n denotes the integral

$$\int_0^{+\infty}A_n(u)u^{2n}\frac{du}{u^{1+a}}.$$

All integrals I_n converge. Indeed, their convergence in any infinite interval of the form $(\delta;+\infty)$, $\delta>0$, follows from the estimate

$$|A_n(u)|\le e^{-u^2}s^{2n}\sum_{m=1}^{s}\binom{2s}{s-m}=2^{2s-1}s^{2n}e^{-u^2},$$

and their convergence in $(0;\delta)$ follows from the fact that the integrand $A_n(u)u^{2n-1-a}$ has no singularity at the point $u=0$. If $n\ge s$, then this it is easy to conclude from the inequalities

$$2n-1-a\ge2s-1-a=2\left(1-\left\{\frac{a+1}{2}\right\}\right)=\gamma>0.$$

If $0\le n\le s-1$, then, denoting the coefficient of u^{2k} in the expansion of $A_n(u)$ in powers of u by $a_k^{(n)}$, we obtain

$$\begin{aligned}a_0^{(0)}&=\frac{1}{2}\binom{2s}{s}+\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}=B(0;s),\\a_n^{(k)}&=\sum_{m=1}^{s}(-1)^m\binom{2s}{s-m}m^{2n}\frac{(-m^2)^k}{k!}=\frac{(-1)^k}{k!}\,B(2(n+k);s)\qquad\text{if}\quad(k,n)\ne(0,0).\end{aligned}$$

According to Lemma 1.15, the expansion of $A_n(u)$ begins with a term containing $u^{2(s-n)}$. Therefore, near the point $u=0$, the integrand is of order $u^{2(s-n)}\cdot u^{2n}u^{-1-a}=u^{\gamma}$, where, as above,

$$\gamma=2\left(1-\left\{\frac{a+1}{2}\right\}\right).$$

This implies the convergence of I_n.

Let us evaluate the integrals I_n. First, suppose that $n \geq s$. We have already shown that $2n-1-a \geq \gamma > 0$. Therefore, I_n can be evaluated termwise:

$$I_n = \int_0^{+\infty} A_n(u) u^{2n} \frac{du}{u^{1+a}} = \int_0^{+\infty} \sum_{m=1}^{s} (-1)^m \binom{2s}{s-m} m^{2n} e^{-m^2u^2} u^{2n-1-a}\, du$$
$$= \sum_{m=1}^{s} (-1)^m \binom{2s}{s-m} m^{2n} \int_0^{+\infty} e^{-m^2u^2} u^{2n-1-a}\, du$$
$$= \sum_{m=1}^{s} (-1)^m \binom{2s}{s-m} m^{2n} \frac{m^a}{2m^{2n}} \Gamma\biggl(n - \frac{a}{2}\biggr) = \frac{1}{2} B(a;s) \Gamma\biggl(n - \frac{a}{2}\biggr). \tag{1.39}$$

Now, suppose that $0 \leq n \leq s-1$. Since $a/2$ is not integer, we evaluate I_n by integration by parts; we obtain

$$I_n = \int_0^{+\infty} A_n(u)\, \frac{du^{2n-a}}{2n-a} = \frac{1}{2n-a} \biggl(A_n(u) u^{2n-a}\big|_0^{+\infty} - \int_0^{+\infty} A_n'(u) u^{2n-a}\, du \biggr).$$

It was shown above that $A_n(u) = O(u^{2(s-n)})$ for n under consideration. Hence,

$$A_n(u) u^{2n-a}\big|_{u=0} = 0$$

and, therefore,

$$I_n = -\frac{1}{2n-a} \int_0^{+\infty} A_n'(u) u^{2n-a}\, du.$$

Observing that

$$A_n'(u) = \frac{d}{du} \sum_{m=1}^{s} (-1)^m \binom{2s}{s-m} m^{2n} e^{-m^2u^2} = \sum_{m=1}^{s} (-1)^m \binom{2s}{s-m} m^{2n} (-2m^2u) e^{-m^2u^2}$$
$$= -2u \sum_{m=1}^{s} (-1)^m \binom{2s}{s-m} m^{2(n+1)} e^{-m^2u^2} = -2u A_{n+1}(u),$$

we arrive at the recursive formula

$$I_n = \frac{2}{2n-a} \int_0^{+\infty} A_{n+1}(u) u^{2(n+1)} \frac{du}{u^{1+a}} = \frac{I_{n+1}}{n - a/2}. \tag{1.40}$$

If $n+1 \geq s$, then, by virtue of (1.39), we have

$$I_{n+1} = \frac{1}{2} B(a;s) \Gamma\biggl(n+1-\frac{a}{2}\biggr) = \frac{1}{2} B(a;s) \biggl(n-\frac{a}{2}\biggr) \Gamma\biggl(n-\frac{a}{2}\biggr),$$
$$I_n = \frac{1}{2} B(a;s) \Gamma\biggl(n-\frac{a}{2}\biggr).$$

If $n+1 < s$, then to obtain the required relation, we must apply relation (1.40) several times. Namely, let $s-k \leq n < s+1-k$, where $2 \leq k \leq s$ is an integer. Then, using (1.39) and (1.40), we obtain $n+k \geq s$, so that

$$I_n = \frac{1}{n-a/2} \cdot \frac{1}{n+1-a/2} \cdots \frac{1}{n+k-1-a/2} I_{n+k}$$
$$= \frac{1}{2} B(a;s) \frac{\Gamma(n+k-a/2)}{\prod_{\nu=0}^{k-1} (n+\nu-a/2)} = \frac{1}{2} B(a;s) \Gamma\biggl(n-\frac{a}{2}\biggr),$$

as required.

We proceed to calculate $J_N(a)$. Collecting the results obtained above, we see that

$$J_N(a) = \sum_{n=0}^{N} 2^{1-2s} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \frac{1}{2} B(a;s) \Gamma\biggl(n-\frac{a}{2}\biggr) = \frac{1}{2} B(a;s) 2^{1-2s} \sum_{n=0}^{N} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \Gamma\biggl(n-\frac{a}{2}\biggr),$$

and

$$I(a)=2^{-2s}B(a;s)\biggl\{\sum_{n=0}^{N}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}\Gamma\biggl(n-\frac{a}{2}\biggr)+\frac{2\theta}{a}\frac{\Gamma(N+1-a/2)}{\Gamma(N+1)}\biggr\}.$$

Letting N tend to infinity, we obtain the required relation.

Remark 1.3. Below we mention some special cases of Lemma 1.20. Using the relation

$$\Gamma\biggl(\nu+\frac{1}{2}\biggr)=\frac{(2\nu)!}{\nu!}\frac{\sqrt{\pi}}{2^{2\nu}},$$

we can easily obtain the expansions

$$\begin{aligned}I(1)&=\frac{\sqrt{\pi}}{4}\biggl\{1+\frac{1}{2}\sum_{n=1}^{+\infty}\frac{(-1)^n}{2^{2n}}\frac{(2n)!}{n!}\frac{\Phi_{n+1}}{\mathfrak{S}_1^{n+1}}\biggr\}\\&=\frac{\sqrt{\pi}}{4}\biggl\{1-\frac{1}{4}\frac{\Phi_2}{\mathfrak{S}_1^2}+\frac{3}{8}\frac{\Phi_3}{\mathfrak{S}_1^3}-\frac{15}{16}\frac{\Phi_4}{\mathfrak{S}_1^4}+\frac{105}{32}\frac{\Phi_5}{\mathfrak{S}_1^5}-\frac{945}{64}\frac{\Phi_6}{\mathfrak{S}_1^6}+\cdots\biggr\}\end{aligned}\tag{1.41}$$

and

$$\begin{aligned}I(3)&=\frac{\sqrt{\pi}}{8}\biggl\{1+\frac{3}{4}\sum_{n=0}^{+\infty}\frac{(-1)^n}{2^{2n}}\frac{(2n)!}{n!}\frac{\Phi_{n+2}}{\mathfrak{S}_1^{n+2}}\biggr\}\\&=\frac{\sqrt{\pi}}{8}\biggl\{1+\frac{3}{4}\frac{\Phi_2}{\mathfrak{S}_1^2}-\frac{3}{8}\frac{\Phi_3}{\mathfrak{S}_1^3}+\frac{9}{16}\frac{\Phi_4}{\mathfrak{S}_1^4}-\frac{45}{32}\frac{\Phi_5}{\mathfrak{S}_1^5}+\frac{315}{64}\frac{\Phi_6}{\mathfrak{S}_1^6}-\frac{2\,835}{128}\frac{\Phi_7}{\mathfrak{S}_1^7}+\frac{31\,185}{256}\frac{\Phi_8}{\mathfrak{S}_1^8}-\cdots\biggr\}.\end{aligned}$$

Remark 1.4. Using the reflection formula in the form

$$\Gamma\biggl(n-\frac{a}{2}\biggr)=\frac{(-1)^{n+1}\pi}{\sin(\pi a/2)}\frac{1}{\Gamma(1-n+a/2)},$$

we can write the expression for $I(a)$ in the form

$$I(a)=2^{2s-a-1}\pi\kappa(a;s)\Gamma(a+1)\sum_{n=0}^{+\infty}\frac{\Phi_n\mathfrak{S}_1^{-n}}{\Gamma(1-n+a/2)}=-\frac{\pi}{2}B(a;s)\biggl(\sin\frac{\pi a}{2}\biggr)^{-1}\sum_{n=0}^{+\infty}\frac{\Phi_n\mathfrak{S}_1^{-n}}{\Gamma(1-n+a/2)}.$$

Theorem 1.2. *If $x_0<x\le N^{0.0001}$ and a is any real number not being an even integer and satisfying the conditions*

$$0<a<\frac{1}{3e}\sqrt{\frac{\log N}{\log x}},$$

then

$$\sum_n|V(t_n)|^a=\frac{M}{\sqrt{\pi}}\Gamma\biggl(\frac{a+1}{2}\biggr)\biggl\{\sum_{n=0}^{+\infty}\frac{(-1)^n\Phi_n}{\mathfrak{S}_1^{n-a/2}}\frac{\Gamma(n-a/2)}{\Gamma(-a/2)}+\theta b\biggl(5ec\sqrt{\frac{\log x}{\log N}\cdot\frac{\max(a,1)}{\mathfrak{S}_1}}\,\biggr)^a\biggr\},$$

where

$$b=\begin{cases}4.01 & \text{if } 0<a<1,\\ 10.67 & \text{if } a\ge 1\end{cases}\qquad\text{and}\qquad c=\begin{cases}1 & \text{if } 0<a<1,\\ \pi & \text{if } a\ge 1.\end{cases}$$

Proof. According to Lemma 1.18, we have

$$\sum_n|V(t_n)|^a=M\mathfrak{S}_1^{a/2}\biggl\{\frac{I(a)}{\kappa(a;s)}+\theta\frac{r(a)}{\kappa(a;s)}\biggr\},$$

where

$$s=\left[\frac{a+1}{2}\right]+1,\qquad \kappa(a;s)=\int_0^{+\infty}\frac{(\sin u)^{2s}}{u^{1+a}}\,du,\qquad r(a)=2.006a^{-1}\biggl(5es\sqrt{\frac{\log x}{\log N}}\,\biggr)^a,$$

and $I(a)$ is the integral represented in Lemma 1.20 as

$$I(a)=2^{-2s}B(a;s)\sum_{n=0}^{+\infty}\frac{(-1)^n\Phi_n}{\mathfrak{S}_1^n}\,\Gamma\biggl(n-\frac{a}{2}\biggr).$$

By virtue of Lemma 1.16, we have

$$\frac{B(a;s)}{\kappa(a;s)}=-\frac{2^{2s-a}}{\pi}\,\Gamma(a+1)\sin\frac{\pi a}{2}.$$

Applying the duplication and reflection formulas for the gamma-function, we obtain

$$\Gamma(a+1)=\frac{2^a}{\sqrt{\pi}}\,\Gamma\biggl(\frac{a+1}{2}\biggr)\Gamma\biggl(\frac{a}{2}+1\biggr)\qquad\text{and}\qquad \Gamma\biggl(\frac{a}{2}+1\biggr)=-\frac{\pi}{\Gamma(-a/2)}\biggl(\sin\frac{\pi a}{2}\biggr)^{-1},$$

whence

$$\frac{1}{\pi}\,\Gamma(a+1)\sin\frac{\pi a}{2}=-\frac{2^a}{\sqrt{\pi}}\,\frac{\Gamma((a+1)/2)}{\Gamma(-a/2)}.$$

Thus,

$$2^{-2s}\frac{B(a;s)}{\kappa(a;s)}=2^{-2s}\cdot(-2^{2s-a})\cdot\frac{(-2^a)}{\sqrt{\pi}}\,\frac{\Gamma((a+1)/2)}{\Gamma(-a/2)}=\frac{1}{\sqrt{\pi}}\,\frac{\Gamma((a+1)/2)}{\Gamma(-a/2)}$$

and, therefore,

$$\sum_n|V(t_n)|^a=\frac{M}{\sqrt{\pi}}\,\Gamma\biggl(\frac{a+1}{2}\biggr)\biggl\{\sum_{n=0}^{+\infty}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^{n-a/2}}\,\frac{\Gamma(n-a/2)}{\Gamma(-a/2)}+\theta R(a)\biggr\},$$

where

$$R(a)=\frac{\sqrt{\pi}}{\Gamma((a+1)/2)}\,\frac{\mathfrak{S}_1^{-a/2}r(a)}{\kappa(a;s)}.$$

First, suppose that $0<a<1$. Then

$$s=1,\qquad B(a;s)=-1,\qquad\text{and}\qquad \kappa(a;s)=\frac{2^a\pi}{4\Gamma(a+1)\sin(\pi a/2)},$$

so that

$$\begin{aligned}R(a)&=\frac{\sqrt{\pi}}{\Gamma((a+1/2)}\cdot\frac{4\Gamma(a+1)\sin(\pi a/2)}{2^a\pi}\cdot\mathfrak{S}_1^{-a/2}\cdot\frac{2.006}{a}\biggl(\frac{5e}{2}\sqrt{\frac{\log x}{\log N}}\,\biggr)^a\\&=1.003\cdot\frac{\sin(\pi a/2)}{\pi a/2}\cdot\frac{4\sqrt{\pi}\,\Gamma(a+1)}{\Gamma((a+1)/2)}\biggl(\frac{5e}{2}\sqrt{\frac{\log x}{\mathfrak{S}_1\log N}}\,\biggr)^a.\end{aligned}$$

Since

$$\frac{\Gamma(a+1)}{\Gamma((a+1)/2)}=-\frac{2^a\sqrt{\pi}}{\Gamma(-a/2)}\biggl(\sin\frac{\pi a}{2}\biggr)^{-1}=\frac{a}{2}\,\frac{2^a\sqrt{\pi}}{\Gamma(1-a/2)}\biggl(\sin\frac{\pi a}{2}\biggr)^{-1},$$

we finally arrive at

$$\begin{aligned}R(a)&=1.003\cdot\frac{\sin(\pi a/2)}{\pi a/2}\cdot4\sqrt{\pi}\cdot\frac{a}{2}\,\frac{2^a\sqrt{\pi}}{\Gamma(1-a/2)}\biggl(\sin\frac{\pi a}{2}\biggr)^{-1}\biggl(\frac{5e}{2}\sqrt{\frac{\log x}{\mathfrak{S}_1\log N}}\,\biggr)^a\\&=\frac{4.012}{\Gamma(1-a/2)}\biggl(\frac{5e}{2}\sqrt{\frac{\log x}{\mathfrak{S}_1\log N}}\,\biggr)^a<4.012\biggl(5e\sqrt{\frac{\log x}{\mathfrak{S}_1\log N}}\,\biggr)^a.\end{aligned}$$

Now, consider the case $a \geq 1$. Using the lower bound of Lemma 1.16 for $\kappa(a;s)$, we obtain

$$|R(a)| \leq \frac{\sqrt{\pi}}{\Gamma((a+1)/2)} \cdot 3\left(\frac{\pi}{2}\right)^a \cdot \frac{2.006}{a}\left(5es\sqrt{\frac{\log x}{\mathfrak{S}_1 \log N}}\right)^a$$
$$= \frac{6.018\sqrt{\pi}}{\Gamma((a+1)/2)}\left\{\frac{5\pi e}{2}\, s\sqrt{\frac{\log x}{\mathfrak{S}_1 \log N}}\right\}^a \leq \frac{10.67}{a}\left\{\frac{5\pi e}{2}\,\varrho(a)\sqrt{\frac{a\log x}{\mathfrak{S}_1 \log N}}\right\}^a,$$

where

$$\varrho(a) = \frac{1 + [(a+1)/2]}{\sqrt{a}\{\Gamma((a+1)/2)\}^{1/a}}.$$

It is easy to verify that $\varrho(a) \leq \varrho(1) = 2$ for $a \geq 1$. Therefore, we finally arrive at

$$|R(a)| \leq \frac{10.67}{a}\left\{5\pi e\sqrt{\frac{a\log x}{\mathfrak{S}_1 \log N}}\right\}^a.$$

This completes the proof of the theorem.

To conclude this section, we show that the infinite series mentioned in the relation of Theorem 1.2 is asymptotic. We precede the proof of this fact by the following auxiliary assertion.

Lemma 1.21. *If x is so large that $\mathfrak{S}_1 = \sum_{p\leq x} p^{-1} \geq 5$, then the sequence of numbers*

$$g(n) = \left(\frac{\log\log n + 1}{\mathfrak{S}_1}\right)^n, \qquad n = 2, 3, 4, \ldots,$$

decreases in the interval $2 \leq n \leq x^{2/5}$.

Proof. The inequalities

$$g(2) > \frac{0.40}{\mathfrak{S}_1^2}, \qquad \frac{1.30}{\mathfrak{S}_1^3} < g(3) < \frac{1.31}{\mathfrak{S}_1^3}, \qquad \frac{3.09}{\mathfrak{S}_1^4} < g(4) < \frac{3.10}{\mathfrak{S}_1^4},$$
$$\frac{7.00}{\mathfrak{S}_1^5} < g(5) < \frac{7.01}{\mathfrak{S}_1^5}, \qquad \text{and} \qquad \frac{15.74}{\mathfrak{S}_1^6} < g(6) < \frac{15.75}{\mathfrak{S}_1^6}$$

imply that each of the inequalities $g(n) \geq g(n+1)$, $n = 2, 3, 4, 5$, holds already for $\mathfrak{S}_1 > 1.31/0.40 = 3.275$. Therefore, to prove the lemma, it suffices to show that the function $f(u) = \log g(u)$ monotonically decreases in the interval $6 \leq u \leq x^{2/5}$. In turn, to show this, it suffices to verify that the derivative

$$f'(u) = \log(\log\log u + 1) + \frac{1}{(\log u)(\log\log u + 1)} - \mathfrak{S}_1$$

is negative for u specified above. Let us write it in the form $h(\log\log u + 1)$, where

$$h(v) = \log v + \frac{e}{ve^v} - \log \mathfrak{S}_1$$

and show that $h(v) < 0$ for $w_1 < v < w_2$, where

$$w_1 = \log\log 6 + 1 \qquad \text{and} \qquad w_2 = \log\log(x^{2/5}) + 1.$$

The function $e^{v-1} - 1 - v^{-1}$ monotonically increases at positive v. Let v_0 denote the only zero of this function, which approximately equals $1.508\,55\ldots$. Then, for $v > v_0$, the derivative

$$h'(v) = \frac{e}{ve^v}\left(e^{v-1} - 1 - \frac{1}{v}\right)$$

is positive, and the function $h(v)$ itself monotonically increases.

Now, we set $v_1 = \mathfrak{S}_1 - 3e^{-\mathfrak{S}_1}$ and $v_2 = \mathfrak{S}_1$. We have

$$h(v_2) = \frac{3}{\mathfrak{S}_1 e^{\mathfrak{S}_1}} > 0.$$

At the same time,

$$h(v_1) = \log\frac{v_1}{\mathfrak{S}_1} + \frac{e}{v_1 e^{v_1}} = \log\left(1 - \frac{3}{\mathfrak{S}_1 e^{\mathfrak{S}_1}}\right) + \frac{e}{v_1 e^{v_1}} < -\frac{3}{\mathfrak{S}_1 e^{\mathfrak{S}_1}} + \frac{e}{v_1 e^{v_1}}.$$

Using the inequality $e^{-u} > 1 - u$, we obtain

$$\begin{aligned} v_1 e^{v_1} &= \mathfrak{S}_1 e^{\mathfrak{S}_1}\left(1 - \frac{3}{\mathfrak{S}_1 e^{\mathfrak{S}_1}}\right)\exp\left\{-\frac{3}{e^{\mathfrak{S}_1}}\right\} \\ &> \mathfrak{S}_1 e^{\mathfrak{S}_1}\left(1 - \frac{3}{\mathfrak{S}_1 e^{\mathfrak{S}_1}}\right)^2 > \mathfrak{S}_1 e^{\mathfrak{S}_1}\left(1 - \frac{3}{e^5}\right)^2 > 0.95\mathfrak{S}_1 e^{\mathfrak{S}_1}, \end{aligned}$$

whence

$$h(v_1) < \left(-3 + \frac{e}{0.95}\right)\frac{1}{\mathfrak{S}_1 e^{\mathfrak{S}_1}} < -\frac{0.1}{\mathfrak{S}_1 e^{\mathfrak{S}_1}} < 0.$$

By virtue of the above considerations, the function $h(v)$ has a unique zero between v_1 and v_2. Therefore, $h(v) < 0$ at any v, $v_0 < v < v_1$.

Now, note that the condition $\mathfrak{S}_1 \geq 5$ implies $\log\log x \geq 4.5$ and $\log x \geq e^{4.5}$. Hence, by virtue of relation (I.2) of Lemma I.3, we have

$$\begin{aligned} v_1 = \mathfrak{S}_1 - 3e^{-\mathfrak{S}_1} &\geq \mathfrak{S}_1 - 3e^{-5} > \log\log x + 0.2614 - \frac{0.5}{\log^2 x} - \frac{3}{e^5} \geq \log\log x + 0.2614 - \frac{0.5}{4.5^2} - \frac{3}{e^5} \\ &> \log\log x + 0.2 > \log\log x + 1 - \log\frac{5}{2} = \log\log\left(x^{2/5}\right) + 1 = w_2. \end{aligned}$$

Moreover, obviously, $w_1 = \log\log 6 + 1 = 1.583\,198\ldots > v_0$. Thus, the interval (w_1, w_2) is contained entirely in the interval (v_0, v_1), in which $h(v)$ is negative. Therefore, $f'(u) < 0$ for $6 \leq u \leq x^{2/5}$. This completes the proof of the lemma.

Lemma 1.22. *Let a be a positive number not being an even integer, and let $m \geq 0$ be an integer not exceeding the minimum of the numbers $\sqrt[3]{x}$ and $(a/30)(\log x)/\log\mathfrak{S}_1 - 1$. Then*

$$\left|\sum_{n>m}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}\frac{\Gamma(n - a/2)}{\Gamma(-a/2)}\right| < c\Gamma\left(1 + \frac{a}{2}\right)g(m + d + 1),$$

where, as above,

$$g(u) = \left(\frac{\log\log u + 1}{\mathfrak{S}_1}\right)^u$$

and the quantities c and d are defined by

$$c = \begin{cases} 0.32 & \text{if } m = 0 \text{ or } m = 1, \\ 4.56 & \text{if } m \geq 2 \end{cases} \qquad \text{and} \qquad d = \begin{cases} 1 & \text{if } m = 0, \\ 0 & \text{if } m \geq 1. \end{cases}$$

Proof. First, we prove the required inequality for all $m \geq 2$. We set

$$S_m = \sum_{n>m}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}\Gamma\left(n - \frac{a}{2}\right)$$

and consider two cases.

Case 1: $m > 1 + a/2$. Stirling's formula implies that, for any $n > m$, we have

$$\Gamma\biggl(n - \frac{a}{2}\biggr) = \frac{\Gamma(n - a/2)}{n - a/2} = \frac{\sqrt{2\pi(n - (1/2)a)}}{n - (1/2)a} \cdot \biggl(\frac{n - (1/2)a}{e}\biggr)^{n-a/2} \exp\biggl\{\frac{\theta}{12(n - (1/2)a)}\biggr\}$$
$$\le \sqrt{\frac{2\pi}{n - (1/2)a}}\biggl(\frac{e}{n - (1/2)a}\biggr)^{a/2}\biggl(\frac{n - (1/2)a}{e}\biggr)^{n} e^{1/12}$$
$$= \frac{\sqrt{2\pi}\, e^{1/12}}{(n - (1/2)a)^{(1+a)/2}}\biggl(\frac{n}{e}\biggr)^{n} \cdot e^{a/2}\biggl(1 - \frac{a}{2n}\biggr)^{n} < \frac{\sqrt{2\pi}\, e^{1/12}}{(n - (1/2)a)^{(1+a)/2}}\biggl(\frac{n}{e}\biggr)^{n}.$$

Observing that if $n \ge m + 1 > 2 + a/2$, then

$$n - \frac{a}{2} = n\biggl(1 - \frac{a}{2n}\biggr) > n\biggl(1 - \frac{a}{a+4}\biggr) = \frac{4n}{a+4},$$

we finally obtain

$$\Gamma\biggl(1 - \frac{a}{2}\biggr) < \frac{\sqrt{2\pi}\, e^{1/12}}{((n/(a+4))a)^{(1+a)/2}}\biggl(\frac{n}{e}\biggr)^{n} = \frac{\sqrt{2\pi}\, e^{1/12}}{n^{(1+a)/2}}\biggl(\frac{n}{e}\biggr)^{n}\biggl(1 + \frac{a}{4}\biggr)^{(1+a)/2}$$

and

$$\frac{1}{n!}\Gamma\biggl(1 - \frac{a}{2}\biggr) < \frac{e^{1/12}}{n^{1+a/2}}\biggl(1 + \frac{a}{4}\biggr)^{(1+a)/2}. \tag{1.42}$$

Let us represent S_m as

$$\biggl\{\sum_{m<n\le x^{2/5}} + \sum_{n>x^{2/5}}\biggr\}(-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n}\Gamma\biggl(1 - \frac{a}{2}\biggr) = S_m^{(1)} + S_m^{(2)}.$$

It follows from the bound of Lemma 1.12 that

$$|S_m^{(1)}| \le \sqrt{2\pi} \sum_{m<n\le x^{2/5}} \frac{g(n)}{n!}\Gamma\biggl(1 - \frac{a}{2}\biggr), \qquad g(n) = \biggl(\frac{\log\log n + 1}{\mathfrak{S}_1}\biggr)^{n}.$$

According to Lemma 1.21, the fraction $g(n)$ attains its maximum value in the interval under consideration at $n = m + 1$. Therefore, applying (1.42), we obtain

$$|S_m^{(1)}| \le \sqrt{2\pi}\, g(m+1) e^{1/12}\biggl(1 + \frac{a}{4}\biggr)^{(1+a)/2} \sum_{n>m} n^{-1-a/2} < 2\sqrt{2\pi}\, e^{1/12}\, \frac{g(m+1)}{am^{a/2}}\biggl(1 + \frac{a}{4}\biggr)^{(1+a)/2}.$$

Since $m > 1 + a/2$, it follows that the quantity $m^{-a/2}(1 + a/4)^{(1+a)/2}$ does not exceed the maximum of the function

$$\frac{(1 + a/4)^{(1+a)/2}}{(1 + a/2)^{a/2}} = \sqrt{1 + \frac{a}{4}}\biggl(\frac{1 + (1/4)a}{1 + (1/2)a}\biggr)^{a/2}$$

over $a \ge 0$, which equals $b_0 = 1.033\,68\ldots$. Therefore,

$$|S_m^{(2)}| < \frac{b_1}{a}\, g(m+1), \qquad b_1 = 2b_0\sqrt{2\pi}\, e^{1/12} = 5.63245\ldots .$$

Next, using Lemma 1.11 and inequality (1.42), we obtain

$$|S_m^{(2)}| < \sum_{n>x^{2/5}} \frac{1}{n!}\Gamma\biggl(n - \frac{a}{2}\biggr) < e^{1/12}\biggl(1 + \frac{a}{4}\biggr)^{(1+a)/2} \sum_{n>x^{2/5}} n^{-1-a/2} < \frac{2e^{1/12}}{a}\biggl(1 + \frac{a}{4}\biggr)^{(1+a)/2} x^{-a/2}.$$

The inequality $m+1<(a/30)(\log x)/\log\mathfrak{S}_1$ implies

$$\frac{1}{g(m+1)}=\left(\frac{\mathfrak{S}_1}{\log\log(m+1)+1}\right)^{m+1}<\mathfrak{S}_1^{m+1}<x^{a/30}.$$

Since $m\le\sqrt[3]{x}$, it follows that $x^{a/30}=x^{a/5}\cdot x^{-a/6}\le x^{a/5}\cdot m^{-a/2}$, so that we finally obtain

$$\frac{1}{g(m+1)}<\frac{x^{a/5}}{m^{a/2}},\qquad\text{i.e.,}\qquad x^{-a/5}<\frac{g(m+1)}{m^{a/2}}.$$

Thus,

$$|S_m^{(2)}|<\frac{2e^{1/12}}{a}\left(1+\frac{a}{4}\right)^{(1+a)/2}\frac{g(m+1)}{m^{a/2}}<\frac{b_2}{a}\,g(m+1),\qquad b_2=2e^{1/12}b_0=2.247\,02\ldots;$$

therefore, in the case $m>1+a/2$, we have

$$|S_m|<|S_m^{(1)}|+|S_m^{(2)}|<\frac{(b_1+b_2)}{a}\,g(m+1).\tag{1.43}$$

Case 2: $m<1+a/2$ (obviously, the condition $m\ge2$ can hold in this case only at $a>2$). Setting $\nu=[a/2]+2$, we represent S_m in the form

$$\left\{\sum_{m<n\le\nu}+\sum_{n>\nu}\right\}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}\,\Gamma\left(n-\frac{a}{2}\right)=S_1^{(1)}+S_1^{(2)}.$$

By virtue of (1.43), we have

$$|S_m^{(2)}|<\frac{(b_1+b_2)}{a}\,g(\nu+1)\le\frac{(b_1+b_2)}{a}\,g(m+1).\tag{1.44}$$

Again applying Lemmas 1.12 and 1.21, we obtain

$$\begin{aligned}|S_m^{(1)}|&\le\sqrt{2\pi}\sum_{m<n\le\nu}\frac{g(n)}{n!}\left|\Gamma\left(n-\frac{a}{2}\right)\right|\le\sqrt{2\pi}\,g(m+1)\max_{m<n\le\nu}\left|\Gamma\left(n-\frac{a}{2}\right)\right|\sum_{n=3}^{+\infty}\frac{1}{n!}\\&\le\sqrt{2\pi}\left(e-\frac{5}{2}\right)g(m+1)\max_{2\le n\le a/2+2}\left|\Gamma\left(n-\frac{a}{2}\right)\right|.\end{aligned}$$

The reflection formula implies

$$\left|\Gamma\left(n-\frac{a}{2}\right)\right|=\frac{\pi}{|\sin(\pi a/2)|}\left|\Gamma\left(1+\frac{a}{2}-n\right)\right|^{-1}.$$

Since the maximum value of the fraction $|\Gamma((1+a/2-n)|^{-1}$ in the interval $1\le n\le a/2+2$ does not exceed Γ_0^{-1}, where

$$\Gamma_0=\min_{u>0}\Gamma(u)=0.885\,603\ldots,$$

it follows that

$$|S_m^{(1)}|\le\frac{\sqrt{2\pi}}{\Gamma_0}\left(e-\frac{5}{2}\right)g(m+1)\pi\left|\sin\frac{\pi a}{2}\right|^{-1}.\tag{1.45}$$

Summing (1.44) and (1.45), we obtain

$$|S_m|<\left\{\frac{b_1+b_2}{a}+\frac{\sqrt{2\pi}}{\Gamma_0}\left(e-\frac{5}{2}\right)\pi\left|\sin\frac{\pi a}{2}\right|^{-1}\right\}g(m+1).\tag{1.46}$$

Obviously, estimate (1.46) remains valid in the case $m \geq 1 + a/2$. Roughing this estimate, we see that, for all $m \geq 2$,

$$|S_m| < \frac{\pi g(m+1)}{|\sin(\pi a/2)|}\left\{\frac{\sqrt{2\pi}}{\Gamma_0}\left(e - \frac{5}{2}\right) + \frac{1}{2}(b_1 + b_2)\frac{|\sin(\pi a/2)|}{\pi a/2}\right\} \leq \frac{\pi b g(m+1)}{|\sin(\pi a/2)|},$$

where

$$b = \frac{\sqrt{2\pi}}{\Gamma_0}\left(e - \frac{5}{2}\right) + \frac{1}{2}(b_1 + b_2) < 4.56.$$

Thus, the absolute value of the sum in the statement of the lemma does not exceed

$$\frac{|S_m|}{|\Gamma(-a/2)|} < \frac{\pi b g(m+1)}{|\Gamma(-a/2)\sin(\pi a/2)|} = b g(m+1)\Gamma\left(1 + \frac{a}{2}\right),$$

as required.

Now, let us prove the lemma for $m = 0, 1$. If $m = 1$, then it follows from what was proved above that

$$\begin{aligned}\left|\sum_{n>1}(-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n}\frac{\Gamma(n - a/2)}{\Gamma(-a/2)}\right| &\leq \frac{|\Phi_2|}{\mathfrak{S}_1^2}\frac{|\Gamma(2 - a/2)|}{|\Gamma(-a/2)|} + b g(3)\Gamma\left(1 + \frac{a}{2}\right)\\ &\leq \frac{\zeta_P(2)}{4\mathfrak{S}_1^2}\frac{a}{2}\left|1 - \frac{a}{2}\right| + b g(3)\Gamma\left(1 + \frac{a}{2}\right)\\ &= g(2)\Gamma\left(1 + \frac{a}{2}\right)\left\{b\frac{g(3)}{g(2)} + \frac{\zeta_P(2)}{4(\log\log 2 + 1)^2}\frac{(a/2)|1 - a/2|}{\Gamma(1 + a/2)}\right\}. \quad (1.47)\end{aligned}$$

Observing that the maximum value of the function

$$\frac{(a/2)|1 - a/2|}{\Gamma(1 + a/2)} = \frac{|1 - a/2|}{\Gamma(a/2)}$$

over $a \geq 0$ is at most $1.129\,18\ldots$, we conclude that the right-hand side of (1.47) is bounded by

$$g(2)\Gamma\left(1 + \frac{a}{2}\right)\left(\frac{15}{\mathfrak{S}_1} + 0.319\right) < 0.32 g(2)\Gamma\left(1 + \frac{a}{2}\right). \quad (1.48)$$

Since $\Phi_1 = 0$, estimate (1.48) remains valid in the case $m = 0$. This completes the proof of the lemma.

Corollary 1.1. *The series*

$$\sum_{n=0}^{+\infty}(-1)^n \Phi_n \frac{\Gamma(n - a/2)}{\Gamma(-a/2)}\,\mathfrak{S}_1^{a/2-n}$$

is asymptotic. In other words, for any fixed integer $m \geq 0$,

$$\sum_{n=0}^{+\infty}(-1)^n \Phi_n \frac{\Gamma(n - a/2)}{\Gamma(-a/2)}\,\mathfrak{S}_1^{a/2-n} = \sum_{n=0}^{m}(-1)^n \Phi_n \frac{\Gamma(n - a/2)}{\Gamma(-a/2)}\,\mathfrak{S}_1^{a/2-n} + O_{m,a}(\mathfrak{S}_1^{a/2-(m+1)}).$$

Example 1.9. As mentioned above, most of the results of this chapter can be carried over, virtually without change, to the "continuous" case, i.e., to the case where summation over the discrete set of Gram points is replaced by integration with respect to a continuous parameter t. In particular, for the integral first moment of the polynomial $V(t)$, we have

$$I(T) = \int_0^T \left|\sum_{p \leq x}\frac{\sin(t\log p)}{\sqrt{p}}\right| dt = I^*(T) + O\left(T\sqrt{\frac{\log x}{\log T}}\right)$$

and

$$I^*(T) = \frac{T}{\sqrt{\pi}}\sum_{n=0}^{+\infty}(-1)^n \Phi_n \frac{\Gamma(n - 1/2)}{\Gamma(-1/2)}\,\mathfrak{S}_1^{1/2-n} = \frac{T}{\sqrt{\pi}}\left\{\mathfrak{S}_1^{1/2} + \sum_{n=1}^{+\infty}\frac{(-1)^{n+1}}{2^{2n-1}}\frac{(2n-2)!}{(n-1)!}\,\Phi_{n+1}\mathfrak{S}_1^{-1/2-n}\right\}$$

$$= \frac{T}{\sqrt{\pi}}\biggl\{\mathfrak{S}_1^{1/2} - \frac{1}{4}\,\Phi_2\mathfrak{S}_1^{-3/2} + \frac{3}{8}\,\Phi_3\mathfrak{S}_1^{-5/2} - \frac{15}{16}\,\Phi_4\mathfrak{S}_1^{-7/2} + \frac{105}{32}\,\Phi_5\mathfrak{S}_1^{-9/2} - \frac{945}{64}\,\Phi_6\mathfrak{S}_1^{-11/2} + \cdots\biggr\}.$$

For $T = 10^4$ and $x = 10^2$, using the *Wolfram Mathematica* 7.0.0 package, we obtain

$$I = 7645.924\,933\,515$$

accurate to one part in 10^9. The contribution of the first seven terms of the asymptotic series I^* equals

$$\begin{aligned}
&7575.318\,726\,470\\
&+65.615\,082\,789\\
&+9.413\,648\,855\\
&-1.296\,960\,483\\
&-1.376\,016\,060\\
&-0.442\,231\,162\\
&-0.004\,219\,519\\
\hline
&7647.228\,030\,890
\end{aligned}$$

and differs from the true one by no more than 1.456, i.e., by less than 0.02%. We also mention that the further increase of the number of summands taken into account does not improve the accuracy of approximation to I. The summands decrease fairly rapidly, and the sum of the series differs from

$$7647.368\,143\,7$$

by no more than 10^{-6}.

The value of the same integral corresponding to the parameters $T = 10^5$ and $x = 10^2$ equals

$$76\,463.353\,421\,946\ldots,$$

and the contribution of the first nine summands in the series amounts to

$$\begin{aligned}
&75\,753.187\,264\,698\\
&+656.150\,827\,892\\
&+94.136\,485\,470\\
&-12.969\,604\,826\\
&-13.760\,160\,599\\
&-4.422\,311\,6184\\
&-0.042\,195\,191\\
&+0.822\,280\,202\\
&+0.167\,703\,383\\
&-0.015\,333\,537\\
\hline
&76\,473.786\,823\,421
\end{aligned}$$

The difference between the two obtained values does not exceed 10.434, i.e., is less than 0.014%.

1.5. Improving the estimate of the remainder in the expression for the first moment. The identity

$$|z| = \frac{2}{\pi}\int_0^{+\infty}\left(\frac{\sin|z|u}{u}\right)^2 du \tag{1.49}$$

makes it possible to approximate the first power of the quantity $|z|$ by a linear combination of its even powers $|z|^2, |z|^4, |z|^6, \dots$. It is this fact which, together with the explicit expressions of Theorem 1.1, allows us to eventually obtain an asymptotic expression for the first moment of the quantities $V(t_n)$, i.e., the sum

$$\sum_n |V(t_n)|. \tag{1.50}$$

At the same time, identity (1.49) has the following drawback. The error resulting from the change of the ray of integration in (1.49) by a finite interval $0 \le u \le \lambda$ equals

$$\frac{2}{\pi}\int_\lambda^{+\infty}\left(\frac{\sin|z|u}{u}\right)^2 du. \tag{1.51}$$

Since the role of z is played by $V(t_n)/\sqrt{\mathfrak{S}_1}$, generally, the function $\sin^2|z|u$ may take values close to 1 for arbitrarily large u. Therefore, we are not able to obtain an estimate of (1.51) which is better than the trivial one, namely,

$$\frac{2}{\pi}\int_\lambda^{+\infty}\left(\frac{\sin|z|u}{u}\right)^2 du \le \frac{2}{\pi}\int_\lambda^{+\infty}\frac{du}{u^2} = \frac{2}{\pi\lambda}.$$

This explains the fact that the remainder term in the expression for the first moment of $V(t_n)$ has order

$$\lambda^{-1} \asymp \sqrt{\frac{\log x}{\log N}}. \tag{1.52}$$

In [39] (see also [40, Chapter 2]), Boyarinov obtained new relations, which, as well as (1.49), reduce calculating the first (and, in general, any positive) moment of a given quantity to calculating the even moments of this quantity; he also proposed a way to overcome the difficulties mentioned above.

Combining Boyarinov's method with the technique of Sections 1.1–1.4 of this paper, we can substantially improve the remainder term in the asymptotic expression for a fractional moment given in Theorem 1.2. In particular, in the case of the first moment, we estimate the remainder by a quantity of order

$$\frac{\log x}{\log N}\log\left(\frac{\log N}{\log x}\right);$$

this estimate is more accurate than (1.52) for $(\log x)/\log N \to 0$.

We mention at once that results related to Gram's law, in particular, to calculating fractional moments of $\Delta(n)$ and Δ_n, that are obtained by using expressions of Theorem 1.2 cannot be substantially improved by applying the asymptotic expressions for (1.50) sharpened as above.

Nevertheless, asymptotic expansions of fractional moments of trigonometric polynomials may be of independent interest. For this reason, we deemed it appropriate to devote an additional section to this question.

Since Boyarinov's method requires a large amount of computation, we consider only a special case and evaluate only the first moment. In the general case, the calculation is performed in a similar way, without substantial change.

Like Theorems 1.1 and 1.2, all results of the preceding sections can be carried over to the "continuous" case of the moments

$$\int_T^{T+H}|V(t)|\,dt, \qquad H = T^{\alpha+\varepsilon}$$

virtually without change. As above, we precede the proof of the main result (Theorem 1.3) by a technical lemma.

Lemma 1.23. *If $n \ge 0$ is an integer, $\Lambda, V > 0$, and $0 < \beta \le 1$, then the integrals*

$$j_n(\Lambda) = \int_0^V e^{-(\Lambda t)^2} t^{2n}\, dt \tag{1.53}$$

satisfy the inequalities

$$j_0(\Lambda) \le \frac{\sqrt{\pi}}{2\Lambda} \qquad \textit{and} \qquad j_n(\Lambda) \le \frac{\sqrt{\pi}}{2\Lambda}\frac{1}{\Lambda^{2n}}\frac{n!}{\sqrt{n}} \min\left(\frac{1}{2}, \left(\frac{\Lambda V}{\sqrt{n}}\right)^{2\beta}\right), \quad n \ge 1.$$

Proof. First, for any $n \ge 0$, we have

$$j_n(\Lambda) \le \int_0^{+\infty} e^{-(\Lambda t)^2} t^{2n}\, dt = \frac{\Gamma(n+1/2)}{2\Lambda^{2n+1}}.$$

Next, if $n \ge 1$, then the multiplier t^{2n} in (1.53) does not exceed $V^{2\beta} t^{2(n-\beta)}$ on the whole interval of integration. Therefore,

$$j_n(\Lambda) \le V^{2\beta} \int_0^{+\infty} e^{-(\Lambda t)^2} t^{2(n-\beta)}\, dt = \frac{(\Lambda V)^{2\beta}}{2\Lambda^{2n+1}} \Gamma\biggl(n + \frac{1}{2} - \beta\biggr).$$

The required inequalities follows from the bounds

$$\frac{n^{1/2+\beta}}{n!}\Gamma\biggl(n + \frac{1}{2} - \beta\biggr) \le \sqrt{\pi} \qquad \text{and} \qquad \frac{\sqrt{n}}{n!}\Gamma\biggl(n + \frac{1}{2}\biggr) \le \frac{\sqrt{\pi}}{2},$$

which are valid for any $n \ge 1$ and $0 < \beta \le 1$. This completes the proof of the lemma.

Theorem 1.3. *If $x_0 < x \le N^\delta$ for a sufficiently small absolute constant δ, $0 < \delta < 0.01$, then*

$$\sum_n |V(t_n)| = \frac{M}{\sqrt{\pi}} \sqrt{\mathfrak{S}_1} \biggl\{ \sum_{n=0}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \frac{\Gamma(n-1/2)}{\Gamma(-1/2)} + \theta R \biggr\}, \tag{1.54}$$

where

$$R = c(\log r)\biggl(\frac{1}{r} + \frac{x^{-0.1}}{20\sqrt[4]{r}}\biggr), \qquad r = \frac{\log N}{\log x}, \qquad c = 5.5 \cdot 10^3.$$

In particular, for $(\log N)^{7.5} \le x \le N^\delta$, the quantity R satisfies the inequality

$$R \le c_1 \frac{\log r}{r}, \qquad c_1 = 5765.$$

The series in negative powers of $\mathfrak{S}_1$ on the right-hand side of (1.54) *is asymptotic.*

Proof. Relation (1.54) is derived in three steps.

Step 1: determination of the principal term of the asymptotics. As above, we set $v_n = V(t_n)/\sqrt{\mathfrak{S}_1}$ and define numbers K, λ, and V by

$$K = \biggl[\frac{r}{22}\biggr], \qquad \lambda = \frac{1}{10e}\sqrt{\frac{r}{\log r}}, \qquad \text{and} \qquad V = \sqrt{8\log r},$$

where $r = (\log N)/\log x$. In the identity

$$|z| = \kappa^{-1} \int_0^{+\infty} e^{-(zu)^2}(1 - e^{-(zu)^2})\, \frac{du}{u^2}, \qquad \kappa = \int_0^{+\infty} e^{-u^2}(1 - e^{-u^2})\, \frac{du}{u^2} = \frac{\sqrt{\pi}}{\sqrt{2}+1}$$

given in [39], we set $z = v_n$. This yields

$$S = \sum_n |v_n| = \kappa^{-1} \sum_n \biggl(\int_0^\lambda + \int_\lambda^{+\infty}\biggr)(e^{-(uv_n)^2} - e^{-2(uv_n)^2})\, \frac{du}{u^2} = S_1 + R_1.$$

Again following Boyarinov's paper [39] and applying the identity

$$e^{-v^2} = \frac{1}{\sqrt{\pi}} \int_0^{+\infty} e^{-t^2/4} \cos(vt)\, dt, \tag{1.55}$$

we obtain

$$\begin{aligned} S_1 &= \kappa^{-1} \sum_n \int_0^\lambda \frac{1}{u^2} \left\{ \frac{1}{\sqrt{\pi}} \int_0^{+\infty} e^{-t^2/4} \cos(utv_n)\, dt - \frac{1}{\sqrt{\pi}} \int_0^{+\infty} e^{-t^2/4} \cos(utv_n\sqrt{2})\, dt \right\} du \\ &= \frac{\kappa^{-1}}{\sqrt{\pi}} \sum_n \int_0^\lambda \frac{1}{u^2} \int_0^{+\infty} e^{-t^2/4} \{\cos(utv_n) - \cos(utv_n\sqrt{2})\}\, dt\, du \\ &= \frac{\kappa^{-1}}{\sqrt{\pi}} \sum_n \int_0^\lambda \frac{1}{u^2} \left(\int_0^V + \int_V^{+\infty} \right) e^{-t^2/4} \{\cos(utv_n) - \cos(utv_n\sqrt{2})\}\, dt\, du = S_2 + R_2. \end{aligned}$$

The Taylor expansion of the cosines with Lagrange remainder gives

$$\begin{aligned} S_2 = \frac{\kappa^{-1}}{\sqrt{\pi}} \sum_n \int_0^\lambda \frac{1}{u^2} \int_0^V e^{-t^2/4} \Bigg\{ &\sum_{k=1}^K \frac{(-1)^k}{(2k)!} (uv_nt)^{2k} + \theta_1 \frac{(uv_nt)^{2(K+1)}}{(2K+2)!} \\ &- \sum_{k=1}^K \frac{(-1)^k}{(2k)!} (uv_nt\sqrt{2})^{2k} + \theta_1 \frac{(uv_nt\sqrt{2})^{2(K+1)}}{(2K+2)!} \Bigg\} dt\, du \\ = \frac{\kappa^{-1}}{\sqrt{\pi}} \int_0^\lambda \frac{1}{u^2} \int_0^V e^{-t^2/4} \Bigg\{ &\sum_{k=1}^K \frac{(-1)^{k-1}}{(2k)!} (2^k - 1)\, (ut)^k \sum_n v_n^{2k} \\ &+ \theta_3 \frac{(2^{K+1}+1)}{(2K+2)!} (ut)^{2K+2} \sum_n v_n^{2(K+1)} \Bigg\} dt\, du. \end{aligned}$$

Replacing the sums over n by their expressions given in Lemmas 1.4 and 1.5, we obtain

$$\begin{aligned} S_2 = \frac{\kappa^{-1}}{\sqrt{\pi}} \int_0^\lambda \frac{1}{u^2} \int_0^V e^{-t^2/4} \Bigg\{ &\sum_{k=1}^K \frac{(-1)^{k-1}}{(2k)!} (2^k - 1) \frac{(ut)^{2k}}{\mathfrak{S}_1^k} \left(M \frac{(2k)!}{k!} \frac{H^{(k)}(0)}{2^{2k}} + \theta x^{2k} \log N \right) \\ &+ \theta_4 \frac{(2^{K+1}+1)}{(2K+2)!} \frac{(ut)^{2K+2}}{\mathfrak{S}_1^{K+1}} \left(M \frac{(2K+2)!}{(K+1)!} \frac{H^{(K+1)}(0)}{2^{2(K+1)}} + x^{2(K+1)} \log N \right) \Bigg\} dt\, du. \end{aligned}$$

Let R_3 denote the contribution to S_2 of the terms containing the factor x^{2k}. A further transformation of S_2 yields

$$\begin{aligned} S_2 = \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_0^\lambda \frac{1}{u^2} \int_0^V e^{-t^2/4} \Bigg\{ &\sum_{k=1}^K \frac{(-1)^{k-1}}{(2k)!} \frac{(2^k-1)}{2^{2k}} (ut)^{2k} \frac{(2k)!}{k!} \frac{H^{(k)}(0)}{\mathfrak{S}_1^k} \\ &+ \theta_5 \frac{(ut)^{2K+2}}{(2K+2)!} \frac{(2^{K+1}+1)}{2^{2(K+1)}} \frac{(2K+2)!}{(K+1)!} \frac{H^{(K+1)}(0)}{\mathfrak{S}_1^{K+1}} \Bigg\} dt\, du + R_3 \\ = S_3 + R_3. \end{aligned}$$

Next, we replace the sum over k in the expression for S_3 by an infinite sum:

$$\begin{aligned} S_3 = \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_0^\lambda \frac{1}{u^2} \int_0^V e^{-t^2/4} \Bigg\{ &\sum_{k=0}^{+\infty} \frac{(-1)^{k-1}}{k!} \frac{(2^k-1)}{2^{2k}} \left(\frac{ut}{\sqrt{\mathfrak{S}_1}} \right)^{2k} H^{(k)}(0) \\ &+ \frac{\theta_6}{(K+1)!} \frac{(2^{K+1}+1)}{2^{2(K+1)}} \left(\frac{ut}{\sqrt{\mathfrak{S}_1}} \right)^{2(K+1)} H^{(K+1)}(0) \end{aligned}$$

$$- \sum_{k>K} \frac{(-1)^{k-1}}{k!} \frac{(2^k-1)}{2^{2k}} \biggl(\frac{ut}{\sqrt{\mathfrak{S}_1}} \biggr)^{2k} H^{(k)}(0) \biggr\} \, dt \, du$$

$$= S_4 + R_4,$$

where S_4 denotes the contribution to S_3 of the sum over $k \geq 0$. Note that this sum over k is the difference of two Taylor series; hence,

$$S_4 = \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_0^\lambda \frac{1}{u^2} \int_0^V e^{-t^2/4} \biggl\{ \sum_{k=0}^{+\infty} \frac{(-1)^k}{k!} \biggl(\frac{(ut)^2}{\mathfrak{S}_1} \biggr)^k H^{(k)}(0) - \sum_{k=0}^{+\infty} \frac{(-1)^k}{k!} \biggl(\frac{(ut)^2}{2\mathfrak{S}_1} \biggr)^k H^{(k)}(0) \biggr\} \, dt \, du$$

$$= \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_0^\lambda \frac{1}{u^2} \int_0^V e^{-t^2/4} \biggl\{ H\biggl(-\frac{(ut)^2}{4\mathfrak{S}_1} \biggr) - H\biggl(-\frac{(ut)^2}{2\mathfrak{S}_1} \biggr) \biggr\} \, dt \, du.$$

Replacing the integral with respect to u by improper ones, we obtain

$$S_4 = \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_0^V e^{-t^2/4} \biggl(\int_0^{+\infty} - \int_\lambda^{+\infty} \biggr) \frac{1}{u^2} \biggl\{ H\biggl(-\frac{(ut)^2}{4\mathfrak{S}_1} \biggr) - H\biggl(-\frac{(ut)^2}{2\mathfrak{S}_1} \biggr) \biggr\} \, dt \, du = S_5 - R_5.$$

Finally, we set $u = 2v/t$ in the integral with respect to u and arrive at

$$S_5 = \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_0^V e^{-t^2/4} \frac{t}{2} \int_0^{+\infty} \biggl\{ H\biggl(-\frac{v^2}{\mathfrak{S}_1} \biggr) - H\biggl(-\frac{2v^2}{\mathfrak{S}_1} \biggr) \biggr\} \frac{du}{u^2} \, dt$$

$$= \frac{Mj\kappa^{-1}}{\sqrt{\pi}} \int_0^V \frac{t}{2} e^{-t^2/4} \, dt = \frac{Mj\kappa^{-1}}{\sqrt{\pi}} (1 - e^{-V^2/4}),$$

where

$$j = \int_0^{+\infty} \biggl\{ H\biggl(-\frac{v^2}{\mathfrak{S}_1} \biggr) - H\biggl(-\frac{2v^2}{\mathfrak{S}_1} \biggr) \biggr\} \frac{dv}{v^2}.$$

Step 2: evaluation of the integral j. We represent this integral as the difference

$$\int_0^{+\infty} \biggl\{ 1 - H\biggl(-\frac{2v^2}{\mathfrak{S}_1} \biggr) \biggr\} \frac{dv}{v^2} - \int_0^{+\infty} \biggl\{ 1 - H\biggl(-\frac{v^2}{\mathfrak{S}_1} \biggr) \biggr\} \frac{dv}{v^2} = j_1 - j_2.$$

Note that

$$j_1 = \sqrt{2} \int_0^{+\infty} \biggl\{ 1 - H\biggl(-\frac{v^2}{\mathfrak{S}_1} \biggr) \biggr\} \frac{dv}{v^2} = \sqrt{2} j_2;$$

hence, $j = (\sqrt{2} - 1) j_2$. Integrating j_2 termwise (the legitimacy of this trick was substantiated in Lemmas 1.19 and 1.20) and observing that $\Phi_1 = 0$, we arrive at the relations

$$j_2 = \int_0^{+\infty} \biggl(1 - e^{-v^2} \sum_{n=0}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} v^{2n} \biggr) \frac{dv}{v^2} = \int_0^{+\infty} (1 - e^{-v^2}) \frac{dv}{v^2} - \sum_{n=2}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \int_0^{+\infty} e^{-v^2} v^{2n} \, dv$$

$$= \sqrt{\pi} - \sum_{n=2}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \frac{1}{2} \Gamma\biggl(n - \frac{1}{2} \biggr) = \sqrt{\pi} \biggl(1 - \sum_{n=2}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \frac{\Gamma(n - 1/2)}{2\sqrt{\pi}} \biggr).$$

Since $\Gamma(-1/2) = -2\sqrt{\pi}$, we finally obtain

$$j_2 = \sqrt{\pi} \sum_{n=0}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \frac{\Gamma(n - 1/2)}{\Gamma(-1/2)},$$

$$j = (\sqrt{2} - 1) \sqrt{\pi} \sum_{n=0}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \frac{\Gamma(n - 1/2)}{\Gamma(-1/2)} = \kappa \sum_{n=0}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \frac{\Gamma(n - 1/2)}{\Gamma(-1/2)}.$$

As shown in Lemma 1.22, all these series are asymptotic.

Step 3: evaluating R_k, $1 \le k \le 5$. First, we prove an auxiliary relation for the integral

$$j(u) = \int_0^V e^{-t^2/4} H\left(-\frac{(ut)^2}{4\mathfrak{S}_1}\right) dt, \qquad u \ge \lambda.$$

Termwise integrating the series resulting from replacing H by the expression

$$e^{-u^2t^2/4} \sum_{n=0}^{+\infty} (-1)^n \Phi_n \left(\frac{u^2t^2}{4\mathfrak{S}_1}\right)^n,$$

we find

$$j(u) = \sum_{n=0}^{+\infty} (-1)^n \Phi_n \left(\frac{u^2}{4\mathfrak{S}_1}\right)^n j_n(\Lambda),$$

where

$$j_n(\Lambda) = \int_0^V e^{-(\Lambda t)^2} t^{2n}\, dt, \qquad \Lambda = \frac{1}{2}\sqrt{u^2+1}\,.$$

We split the sum thus obtained into the three parts corresponding to $n = 0$, $2 \le n \le x^{2/5}$, and $n > x^{2/5}$. Applying the bounds of Lemmas 1.12 (for $2 \le n \le x^{2/5}$) and 1.11 (for $n > x^{2/5}$), we obtain

$$|j(u)| \le j_0(\Lambda) + \sqrt{2\pi} \sum_{2 \le n \le x^{2/5}} g(n) \left(\frac{u}{2}\right)^{2n} \frac{|j_n(\Lambda)|}{n!} + \sum_{n > x^{2/5}} \left(\frac{u}{2}\right)^{2n} \frac{|j_n(\Lambda)|}{n!}\,,$$

where, as above,

$$g(n) = \left(\frac{\log\log n + 1}{\mathfrak{S}_1}\right)^n.$$

To estimate the integrals $j_n(\Lambda)$, we apply Lemma 1.23 with $\beta = 3/4$, which gives

$$|j(u)| \le \frac{\sqrt{\pi}}{2\Lambda} + \sqrt{2\pi} \sum_{2 \le n \le x^{2/5}} g(n) \left(\frac{u}{2\Lambda}\right)^{2n} \frac{\sqrt{\pi}}{2\Lambda} \frac{1}{2\sqrt{n}} + \sum_{n > x^{2/5}} \frac{\sqrt{\pi}}{2} \frac{\Lambda^{1/2} V^{3/2}}{n^{5/4}} \left(\frac{u}{2\Lambda}\right)^{2n}.$$

Taking into account the inequality $2\Lambda > u$, we obtain

$$|j(u)| \le \frac{\sqrt{\pi}}{2\Lambda} \left\{ 1 + \sqrt{\frac{\pi}{2}} \sum_{2 \le n \le x^{2/5}} \frac{g(n)}{\sqrt{n}} \right\} + \frac{5\sqrt{\pi}}{2} \frac{\Lambda^{1/2} V^{3/2}}{x^{1/10}}\,.$$

We split the sum over n in the last expression into the three parts corresponding to $2 \le n \le \mathfrak{S}_1$, $\mathfrak{S}_1 < n \le e^{\mathfrak{S}_1}$, and $e^{\mathfrak{S}_1} < n \le x^{2/5}$. By Lemma 1.21, the function $g(n)$ monotonically decreases on each of these intervals, so that

$$\begin{aligned}
\sum_{2 \le n \le x^{2/5}} \frac{g(n)}{\sqrt{n}} &\le g(2) \sum_{2 \le n \le \mathfrak{S}_1} \frac{1}{\sqrt{n}} + g(\mathfrak{S}_1) \sum_{\mathfrak{S}_1 \le n \le e^{\mathfrak{S}_1}} \frac{1}{\sqrt{n}} + g(e^{\mathfrak{S}_1}) \sum_{e^{\mathfrak{S}_1} \le n \le x^{2/5}} \frac{1}{\sqrt{n}} \\
&\le \left(\frac{\log\log 2 + 1}{\mathfrak{S}_1}\right)^2 \cdot 2\sqrt{\mathfrak{S}_1} + \left(\frac{\log\log \mathfrak{S}_1 + 1}{\mathfrak{S}_1}\right)^{\mathfrak{S}_1} \cdot 2e^{\mathfrak{S}_1/2} + \left(\frac{\log \mathfrak{S}_1 + 1}{\mathfrak{S}_1}\right)^{e^{\mathfrak{S}_1}} \cdot x^{1/5} \\
&< \frac{0.9}{\mathfrak{S}_1\sqrt{\mathfrak{S}_1}} + \frac{2}{\mathfrak{S}_1^2} + 2\left(\frac{1}{\sqrt{\mathfrak{S}_1}}\right)^{\mathfrak{S}_1} + 2\left(\frac{1}{\sqrt{\mathfrak{S}_1}}\right)^{e^{\mathfrak{S}_1}} x^{1/5} \\
&< \frac{0.9}{\mathfrak{S}_1\sqrt{\mathfrak{S}_1}} + \frac{2}{\mathfrak{S}_1^2} + 2\exp\left\{\frac{1}{5}\log x - \frac{1}{2}(\log x)\log\log x\right\} < \frac{1}{\mathfrak{S}_1\sqrt{\mathfrak{S}_1}}\,.
\end{aligned}$$

Thus,

$$|j(u)| \le \frac{\sqrt{\pi}}{2\Lambda}\left(1+\sqrt{\frac{\pi}{2}}\frac{1}{\mathfrak{S}_1\sqrt{\mathfrak{S}_1}}\right)+\frac{5\sqrt{\pi}}{2}\frac{\Lambda^{1/2}V^{3/2}}{x^{1/10}}. \tag{1.56}$$

Estimating R_1. First, we have

$$\begin{aligned}0 \le R_1 &= \kappa^{-1}\sum_n \int_\lambda^{+\infty} e^{-(uv_n)^2}(1-e^{-(uv_n)^2})\,\frac{du}{u^2}\\ &\le \kappa^{-1}\sum_n \int_\lambda^{+\infty} e^{-(uv_n)^2}\frac{du}{u^2} \le \kappa^{-1}\sum_n e^{-(\lambda v_n)^2}\int_\lambda^{+\infty}\frac{du}{u^2} = \frac{\kappa^{-1}}{\lambda}\sum_n e^{-(\lambda v_n)^2}.\end{aligned}$$

Let us transform the last sum, again applying identity (1.55). We obtain

$$\begin{aligned}0 \le R_1 &\le \frac{\kappa^{-1}}{\lambda\sqrt{\pi}}\sum_n \int_0^{+\infty} e^{-t^2/4}\cos(\lambda v_n t)\,dt\\ &= \frac{\kappa^{-1}}{\lambda\sqrt{\pi}}\sum_n\left(\int_0^V+\int_V^{+\infty}\right)e^{-t^2/4}\cos(\lambda v_n t)\,dt = R_1^{(2)}+R_1^{(1)}.\end{aligned}$$

The last term is estimated trivially:

$$|R_1^{(1)}| \le \frac{\kappa^{-1}}{\lambda\sqrt{\pi}}\sum_n\int_V^{+\infty} e^{-t^2/4}\,dt = \frac{M\kappa^{-1}}{\lambda\sqrt{\pi}}\int_{V^2/4}^{+\infty}\frac{e^{-v}}{\sqrt{v}}\,dv \le \frac{2M\kappa^{-1}}{\lambda V\sqrt{\pi}}\exp\left(-\frac{V^2}{4}\right).$$

Using the notation $r=(\log N)/\log x$, we finally obtain

$$|R_1^{(1)}| \le 20e\frac{M\kappa^{-1}}{\sqrt{\pi}}\frac{\sqrt{\log r}}{r^2\sqrt{r}}.$$

To estimate $R_1^{(2)}$, we replace the cosine in the integrand by a finite Taylor series:

$$\begin{aligned}R_1^{(2)} &= \frac{\kappa^{-1}}{\lambda\sqrt{\pi}}\sum_n\int_0^V e^{-t^2/4}\left\{\sum_{k=0}^K\frac{(-1)^k}{(2k)!}(\lambda t)^{2k}v_n^{2k}+\theta_1\frac{(\lambda t)^{2K+2}}{(2K+2)!}v_n^{2K+2}\right\}dt\\ &= \frac{\kappa^{-1}}{\lambda\sqrt{\pi}}\int_0^V e^{-t^2/4}\left\{\sum_{k=0}^K\frac{(-1)^k}{(2k)!}(\lambda t)^{2k}\sum_n v_n^{2k}+\theta_2\frac{(\lambda t)^{2K+2}}{(2K+2)!}\sum_n v_n^{2K+2}\right\}dt.\end{aligned}$$

Using the relations of Lemmas 1.4 and 1.5, we reduce the last expression to the form

$$\begin{aligned}R_1^{(2)} &= \frac{\kappa^{-1}}{\lambda\sqrt{\pi}}\int_0^V e^{-t^2/4}\left\{\sum_{k=0}^K\frac{(-1)^k}{(2k)!}(\lambda t)^{2k}\left(\frac{(2k)!}{k!}\frac{M}{2^{2k}}\frac{H^{(k)}(0)}{\mathfrak{S}_1^k}+\theta_3\frac{x^{2k}}{\mathfrak{S}_1^k}\log N\right)\right.\\ &\qquad\left.+\theta_4\frac{(\lambda t)^{2K+2}}{(2K+2)!}\left(\frac{(2K+2)!}{(K+1)!}\frac{M}{2^{2K+2}}\frac{H^{(K+1)}(0)}{\mathfrak{S}_1^{K+1}}+\frac{x^{2K+2}}{\mathfrak{S}_1^{K+1}}\log N\right)\right\}dt\\ &= \frac{M\kappa^{-1}}{\lambda\sqrt{\pi}}\int_0^V e^{-t^2/4}\left\{\sum_{k=0}^K\frac{(-1)^k}{k!}\left(\frac{\lambda t}{2}\right)^{2k}\frac{H^{(k)}(0)}{\mathfrak{S}_1^k}+\frac{\theta_5}{(K+1)!}\left(\frac{\lambda t}{2}\right)^{2K+2}\frac{H^{(K+1)}(0)}{\mathfrak{S}_1^{K+1}}\right\}dt\\ &\quad+\frac{\theta_6\kappa^{-1}}{\lambda\sqrt{\pi}}\int_0^V e^{-t^2/4}\left\{\sum_{k=0}^{K+1}\frac{(\lambda t)^{2k}}{(2k)!}\frac{x^{2k}}{\mathfrak{S}_1^k}\log N\right\}.\end{aligned}$$

Let us replace the sum over k in the first integral by the infinite sum and denote the term thus obtained by $R_1^{(3)}$; we also denote the error involved in this replacement by $R_1^{(4)}$ and the contribution of the terms containing the factor x^{2k} by $R_1^{(5)}$. We obtain

$$R_1^{(2)} = R_1^{(3)}+R_1^{(4)}+R_1^{(5)}.$$

First, we estimate $R_1^{(5)}$. We have

$$\begin{aligned}|R_1^{(5)}| &\le \frac{x^{2(K+1)}\kappa^{-1}}{\lambda\sqrt{\pi}}(\log N)\int_0^V e^{-t^2/4}\sum_{k=0}^{+\infty}\frac{1}{(2k)!}\biggl(\frac{\lambda t}{\sqrt{\mathfrak{S}_1}}\biggr)^{2k}dt\\ &\le \frac{x^{2(K+1)}\kappa^{-1}}{\lambda\sqrt{\pi}}(\log N)\int_0^V \exp\biggl\{-\frac{t^2}{4}+\frac{\lambda t}{\sqrt{\mathfrak{S}_1}}\biggr\}dt\\ &\le \frac{x^{2(K+1)}\kappa^{-1}}{\lambda\sqrt{\pi}}(\log N)\exp\biggl\{\frac{\lambda t}{\sqrt{\mathfrak{S}_1}}\biggr\}\int_0^V e^{-t^2/4}\,dt \le \frac{x^{2(K+1)}\kappa^{-1}}{\lambda}(\log N)\exp\biggl\{\frac{\lambda t}{\sqrt{\mathfrak{S}_1}}\biggr\}.\end{aligned}$$

The definition of K, λ, and V implies

$$|R_1^{(5)}| \le 10e\kappa^{-1}\sqrt{\frac{\log r}{r}}\exp\biggl\{3\biggl(\frac{1}{22}\frac{\log N}{\log x}+1\biggr)\log x+\log\log N+\frac{1}{5e}\sqrt{\frac{2\log N}{\mathfrak{S}_1\log x}}\biggr\}.$$

Since

$$\frac{3}{22}=\biggl(\frac{27}{82}+\varepsilon\biggr)-\frac{147}{451}-\varepsilon<\biggl(\frac{27}{82}+\varepsilon\biggr)-\frac{5}{13}-\varepsilon,$$

it follows that

$$\begin{aligned}|R_1^{(5)}| &\le \sqrt{\frac{\log r}{r}}\exp\biggl\{\biggl(\frac{27}{82}+\varepsilon\biggr)\log N-\biggl(\frac{5}{13}+\varepsilon\biggr)\log N+3\log x+\log\log N+\sqrt{\log N}\biggr\}\\ &< M\sqrt{\frac{\log r}{r}}\exp\biggl\{-\frac{5}{13}\log N+3\log x\biggr\}.\end{aligned}$$

Finally, observing that $x\le N^{1/60}$, we obtain

$$|R_1^{(5)}| \le M\sqrt{\frac{\log r}{r}}\exp\biggl\{-\frac{1}{3}\log N\biggr\}=M\sqrt{\frac{\log r}{r}}\,N^{-1/3}.$$

We proceed to estimate $R_1^{(4)}$. Note that, for $0\le t\le V$ and $k>K$, the second relation in Lemma 1.17 gives

$$\frac{1}{k!}\biggl(\frac{(\lambda t)^2}{4\mathfrak{S}_1}\biggr)^k H^{(k)}(0)\le\frac{1}{k!}\biggl(\frac{\lambda t}{2}\biggr)^{2k}\le\frac{1}{k!}\biggl(\frac{\lambda V}{2}\biggr)^{2k}\le\frac{1}{\sqrt{2\pi K}}\biggl(\frac{e}{k}\frac{(\lambda V)^2}{4}\biggr)^k\le\frac{1}{\sqrt{2\pi K}}\biggl(\frac{e(\lambda V)^2}{4K}\biggr)^k.$$

But it is easy to see that

$$\frac{e(\lambda V)^2}{4K}=\frac{r}{50eK}\le\frac{r}{50e}\frac{25}{r}=\frac{1}{2e}.$$

Therefore,

$$\begin{aligned}|R_1^{(4)}| &\le \frac{2M\kappa^{-1}}{\lambda\sqrt{\pi}}\int_0^V e^{-t^2/4}\sum_{k>K}\frac{H^{(k)}(0)}{k!}\biggl(\frac{(\lambda t)^2}{4\mathfrak{S}_1}\biggr)^k dt\le\frac{2M\kappa^{-1}}{\lambda\sqrt{\pi}}\frac{1}{\sqrt{2\pi K}}\biggl(\sum_{k>K}(2e)^{-k}\biggr)\int_0^V e^{-t^2/4}\,dt\\ &< \frac{2M\kappa^{-1}}{\lambda\sqrt{\pi}}\frac{e^{-(K+1)}}{\sqrt{2K}}<\frac{2M\kappa^{-1}}{\lambda\sqrt{\pi}}20e\sqrt{3}\frac{\sqrt{\log r}}{r}\,e^{-r/22}<\frac{M\kappa^{-1}}{\lambda\sqrt{\pi}}\frac{190\sqrt{\log r}}{r}\,e^{-r/22}.\end{aligned}$$

Finally, let us estimate $R_1^{(3)}$:

$$R_1^{(3)}=\frac{M\kappa^{-1}}{\lambda\sqrt{\pi}}\int_0^V e^{-t^2/4}\biggl\{\sum_{k=0}^{+\infty}\frac{H^{(k)}(0)}{k!}\biggl(-\frac{(\lambda t)^2}{4\mathfrak{S}_1}\biggr)^k\biggr\}dt=\frac{M\kappa^{-1}}{\lambda\sqrt{\pi}}\int_0^V e^{-t^2/4}H\biggl(-\frac{(\lambda t)^2}{4\mathfrak{S}_1}\biggr)dt.$$

Setting $u=\lambda$ in inequality (1.56) and noting that

$$2\Lambda=\sqrt{\lambda^2+1}>\lambda \qquad \text{and} \qquad \sqrt{\Lambda}=\frac{\sqrt[4]{\lambda^2+1}}{\sqrt{2}}\le\frac{\sqrt{\lambda}}{\sqrt[4]{2}},$$

we obtain

$$\begin{aligned}|R_1^{(3)}|&\le\frac{M\kappa^{-1}}{\lambda\sqrt{\pi}}\biggl\{\frac{\sqrt{\pi}}{\lambda}\biggl(1+\sqrt{\frac{\pi}{2}}\,\mathfrak{S}_1^{-3/2}\biggr)+\frac{5\sqrt{\pi}}{2\sqrt[4]{2}}\frac{\lambda^{1/2}V^{3/2}}{x^{0.1}}\biggr\}\\&\le\frac{M\kappa^{-1}}{\lambda^2}\biggl(1+\sqrt{\frac{\pi}{2}}\,\mathfrak{S}_1^{-3/2}\biggr)+\frac{5M\kappa^{-1}}{2\sqrt[4]{2}}\frac{V^{3/2}}{\lambda^{1/2}x^{0.1}}\\&\le(10e)^2M\kappa^{-1}\frac{\log r}{r}\biggl(1+\sqrt{\frac{\pi}{2}}\,\mathfrak{S}_1^{-3/2}\biggr)+5\sqrt{10e}\frac{M\kappa^{-1}}{x^{0.1}}\frac{\log r}{\sqrt[4]{r}}.\end{aligned}$$

Now, the sought estimate of $R_1^{(2)}$ takes the form

$$\begin{aligned}|R_1^{(2)}|&\le M\kappa^{-1}\frac{\log r}{r}\biggl\{(10e)^2\biggl(1+\sqrt{\frac{\pi}{2}}\,\mathfrak{S}_1^{-3/2}\biggr)+\frac{190}{\sqrt{\pi}}\frac{e^{-q/22}}{\sqrt{\log q}}+\kappa N^{-1/3}\biggr\}\\&\quad+M\kappa^{-1}\frac{5\sqrt{10e}}{x^{0.1}}\frac{\log q}{\sqrt[4]{q}}<M\kappa^{-1}\biggl(739.9\frac{\log r}{r}+\frac{26.1}{x^{0.1}}\frac{\log r}{\sqrt[4]{r}}\biggr).\end{aligned}$$

Thus,

$$\begin{aligned}|R_1|&\le|R_1^{(1)}|+|R_1^{(2)}|\le M\kappa^{-1}\biggl(\frac{20e}{\sqrt{\pi}}\frac{\sqrt{\log r}}{r^2\sqrt{r}}+739.9\frac{\log r}{r}+\frac{26.1}{x^{0.1}}\frac{\log r}{\sqrt[4]{r}}\biggr)\\&<M\kappa^{-1}\biggl(740\frac{\log r}{r}+\frac{26.1}{x^{0.1}}\frac{\log r}{\sqrt[4]{r}}\biggr).\end{aligned}$$

Estimating R_2. For fixed t, we set

$$f_n(u)=\cos(utv_n)-\cos(utv_n\sqrt{2})$$

and represent R_2 in the form

$$\frac{\kappa^{-1}}{\sqrt{\pi}}\int_V^{+\infty}e^{-t^2/4}\biggl\{\sum_n\int_0^{\lambda}\frac{f_n(u)}{u^2}\,du\biggr\}dt. \tag{1.57}$$

Take any number $t\ge V$. We split the sum over n in (1.57) into two sums, one of which consists of the terms satisfying the inequality $|v_n|>(\lambda t)^{-1}$ and the other, of all remaining terms. We denote the contributions of these sums to R_2 by $R_2^{(1)}$ and $R_2^{(2)}$, respectively.

Take any number n for which $|v_n|>(\lambda t)^{-1}$. For this n, we have $\lambda>(t|v_n|)^{-1}$, so that the interval of integration with respect to u can be split into two intervals:

$$\int_0^{\lambda}\frac{f_n(u)}{u^2}\,du=\biggl\{\int_0^{(t|v_n|)^{-1}}+\int_{(t|v_n|)^{-1}}^{\lambda}\biggr\}\frac{f_n(u)}{u^2}\,du. \tag{1.58}$$

Obviously, $|f_n(u)|\le2$ for all u; hence, the absolute value of the second integral in (1.58) does not exceed

$$2\int_{(t|v_n|)^{-1}}^{+\infty}\frac{du}{u^2}=2t|v_n|.$$

On the other hand, using the inequality $|\sin w|\le|w|$, we obtain

$$|f_n(u)|\le2\biggl|\sin\biggl(uv_nt\cdot\frac{\sqrt{2}-1}{2}\biggr)\sin\biggl(uv_nt\cdot\frac{\sqrt{2}+1}{2}\biggr)\biggr|\le\frac{1}{2}(uv_nt)^2.$$

Therefore, the absolute value of the first integral on the right-hand side of (1.58) is at most

$$\int_0^{(t|v_n|)^{-1}} \frac{1}{2}(uv_nt)^2\,du = \frac{1}{2}t|v_n|.$$

Thus,

$$\left|\int_0^{\lambda} \frac{f_n(u)}{u^2}\,du\right| \le \frac{5}{2}t|v_n|$$

and

$$|R_2^{(1)}| \le \frac{\kappa^{-1}}{\sqrt{\pi}}\int_V^{+\infty} e^{-t^2/4}\frac{5t}{2}\biggl(\sum_n |v_n|\biggr)\,dt = \frac{5\kappa^{-1}}{\sqrt{\pi}}\exp\left\{-\frac{V^2}{4}\right\}\biggl(\sum_n |v_n|\biggr).$$

Applying Cauchy's inequality to the last sum and using the relation of Theorem 1.1, we obtain

$$\sum_n |v_n| \le \biggl(M\sum_n v_n^2\biggr)^{1/2} < \biggl(M\biggl\{\frac{M}{2\mathfrak{S}_1} + \frac{x^2}{\mathfrak{S}_1}\log N\biggr\}\biggr)^{1/2} < M$$

and

$$|R_2^{(1)}| \le \frac{5M\kappa^{-1}}{\sqrt{\pi}}\exp\left\{-\frac{V^2}{4}\right\} = \frac{5M\kappa^{-1}}{r^2\sqrt{\pi}}\,.$$

Now, consider those n for which $0 < |v_n| \le (\lambda t)^{-1}$. We have $\lambda \le (t|v_n|)^{-1}$ for these n, so that

$$\left|\int_0^{\lambda} \frac{f_n(u)}{u^2}\,du\right| \le \int_0^{\lambda} \frac{1}{2}(uv_nt)^2\,\frac{du}{u^2} = \frac{\lambda}{2}(v_nt)^2 \le \frac{1}{2\lambda}.$$

Thus,

$$\begin{aligned}|R_2^{(2)}| &\le \frac{\kappa^{-1}}{\sqrt{\pi}}\int_V^{+\infty} e^{-t^2/4}\sum_n (2\lambda)^{-1}\,dt \le \frac{M\kappa^{-1}}{2\lambda\sqrt{\pi}}\int_V^{+\infty} e^{-t^2/4}\,dt\\ &\le \frac{M\kappa^{-1}}{2\lambda\sqrt{\pi}}\frac{2}{V}\exp\left\{-\frac{V^2}{4}\right\} \le \frac{5eM\kappa^{-1}}{\sqrt{2\pi}}\frac{1}{r^2\sqrt{r}}.\end{aligned}$$

Therefore,

$$|R_2| \le |R_2^{(1)}| + |R_2^{(2)}| \le \frac{5M\kappa^{-1}}{\sqrt{\pi}}\frac{1}{r^2} + \frac{5eM\kappa^{-1}}{\sqrt{2\pi}}\frac{1}{r^2\sqrt{r}} < \frac{10M\kappa^{-1}}{\sqrt{\pi}}\frac{1}{r^2}.$$

Estimating R_3 and R_4. Note that

$$|R_3| \le \frac{M\kappa^{-1}}{\sqrt{\pi}}\int_0^{\lambda}\frac{1}{u^2}\int_0^V e^{-t^2/4}\frac{(2^{K+1}+1)}{(K+1)!\,2^{2K+2}}\frac{(ut)^{2K+2}}{\mathfrak{S}_1^{K+1}}\,H^{(K+1)}(0)\,dt\,du$$

and

$$R_4 \le \frac{M\kappa^{-1}}{\sqrt{\pi}}\int_0^{\lambda}\frac{1}{u^2}\int_0^V e^{-t^2/4}\sum_{k>K}\frac{H^{(k)}(0)}{k!}\frac{(2^k-1)}{2^{2k}}\frac{(ut)^{2k}}{\mathfrak{S}_1^k}\,dt\,du;$$

hence,

$$\begin{aligned}|R_3| + |R_4| &\le \frac{2M\kappa^{-1}}{\sqrt{\pi}}\int_0^{\lambda}\frac{1}{u^2}\int_0^V e^{-t^2/4}\sum_{k>K}\frac{\mathfrak{S}_1^k}{k!}\frac{(2^k+1)}{2^{2k}}\frac{(ut)^{2k}}{\mathfrak{S}_1^k}\,dt\,du\\ &\le \frac{3M\kappa^{-1}}{\sqrt{\pi}}\int_0^{\lambda}\frac{1}{u^2}\int_0^V e^{-t^2/4}\sum_{k>K}\frac{1}{k!}\biggl(\frac{ut}{\sqrt{2}}\biggr)^{2k}\,dt\,du\end{aligned}$$

$$= \frac{3M\kappa^{-1}}{\sqrt{\pi}} \int_0^V e^{-t^2/4} \sum_{k>K} \frac{1}{k!} \biggl(\frac{t}{\sqrt{2}}\biggr)^{2k} \frac{\lambda^{2k-1}}{2k-1}\, dt < \frac{3M\kappa^{-1}}{\lambda K\sqrt{\pi}} \int_0^V e^{-t^2/4} \sum_{k>K} \frac{1}{k!} \biggl(\frac{\lambda t}{\sqrt{2}}\biggr)^{2k} dt.$$

For any t such that $0 \le t \le V$, the sum under the integral sign is at most

$$\frac{1}{\sqrt{2\pi K}} \sum_{k>K} \biggl(\frac{e}{k}\biggr)^k \biggl(\frac{\lambda V}{\sqrt{2}}\biggr)^{2k} < \frac{1}{\sqrt{2\pi K}} \sum_{k>K} \biggl(\frac{e(\lambda V)^2}{2K}\biggr)^k.$$

Since $e(\lambda V)^2/(2K) \le 1/e$, it follows that

$$\begin{aligned} |R_3| + |R_4| &\le \frac{3M\kappa^{-1}}{\lambda K\sqrt{\pi}} \frac{1}{\sqrt{2\pi K}} \biggl(\sum_{k>K} e^{-k}\biggr) \int_0^V e^{-t^2/4}\, dt \\ &\le \frac{3M\kappa^{-1}}{\sqrt{2\pi}} \frac{1}{\lambda K\sqrt{K}} \frac{e^{-K}}{(e-1)} \le \frac{6(25e)^2}{e-1} \frac{M\kappa^{-1}}{\sqrt{2\pi}} \frac{\sqrt{\log r}}{r^2}\, e^{-r/22} < 6434 M\kappa^{-1} \frac{\sqrt{\log r}}{r^2}\, e^{-r/22}. \end{aligned}$$

Estimating R_5. In the expression

$$R_5 = \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_0^V e^{-t^2/4} \int_\lambda^{+\infty} \frac{1}{u^2} \biggl\{ H\biggl(-\frac{(ut)^2}{4\mathfrak{S}_1}\biggr) - H\biggl(-\frac{(ut)^2}{2\mathfrak{S}_1}\biggr) \biggr\}\, du\, dt\,,$$

we split the integral with respect to u into two integrals over $\lambda \le u \le \lambda^2$ and $u > \lambda^2$. Accordingly, R_5 is represented by the sum $R_5^{(1)} + R_5^{(2)}$; the meaning of the notation is obvious. Using Lemma 1.17, we obtain the inequalities

$$|R_5^{(2)}| \le \frac{2M\kappa^{-1}}{\sqrt{\pi}} \int_0^V e^{-t^2/4} \int_{\lambda^2}^{+\infty} \frac{du\, dt}{u^2} \le \frac{2M\kappa^{-1}}{\lambda^2\sqrt{\pi}} \int_0^{+\infty} e^{-t^2/4}\, dt = \frac{2M\kappa^{-1}}{\lambda^2}\,.$$

To estimate $R_5^{(1)}$, we again apply the inequality for the integral $j(u)$ obtained above. We have

$$\begin{aligned} R_5^{(1)} &= \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_\lambda^{\lambda^2} \frac{1}{u^2} \int_0^V e^{-t^2/4} \biggl\{ H\biggl(-\frac{(ut)^2}{4\mathfrak{S}_1}\biggr) - H\biggl(-\frac{(ut)^2}{2\mathfrak{S}_1}\biggr) \biggr\}\, dt\, du \\ &= \frac{M\kappa^{-1}}{\sqrt{\pi}} \int_\lambda^{\lambda^2} \frac{1}{u^2} (j(u) - j(u\sqrt{2}))\, \frac{du}{u^2} = \frac{M\kappa^{-1}}{\sqrt{\pi}} \biggl\{ \int_\lambda^{\lambda^2} \frac{j(u)}{u^2}\, du - \sqrt{2} \int_{\lambda\sqrt{2}}^{\lambda^2\sqrt{2}} \frac{j(u)}{u^2}\, du \biggr\}. \end{aligned}$$

Applying estimate (1.56) and the inequalities $2\Lambda > u$ and $\sqrt{\Lambda} \le \sqrt{u}/\sqrt[4]{2}$, we obtain

$$\begin{aligned} \biggl| \int_\lambda^{\lambda^2} \frac{j(u)}{u^2}\, du \biggr| &\le \int_\lambda^{\lambda^2} \biggl\{ \frac{\sqrt{\pi}}{u} \biggl(1 + \sqrt{\frac{\pi}{2}}\, \mathfrak{S}_1^{-3/2}\biggr) + \frac{5\sqrt{\pi}}{2} \frac{\sqrt{u}}{\sqrt[4]{2}} \frac{V^{3/2}}{x^{0.1}} \biggr\} \frac{du}{u^2} \\ &\le \frac{\sqrt{\pi}}{2\lambda^2} \biggl(1 + \sqrt{\frac{\pi}{2}}\, \mathfrak{S}_1^{-3/2}\biggr) + \frac{5\sqrt{\pi}}{\sqrt[4]{2}} \frac{V^{3/2}}{x^{0.1}\sqrt{\lambda}} \end{aligned}$$

and, similarly,

$$\biggl| \int_{\lambda\sqrt{2}}^{\lambda^2\sqrt{2}} \frac{j(u)}{u^2}\, du \biggr| \le \frac{\sqrt{\pi}}{4\lambda^2} \biggl(1 + \sqrt{\frac{\pi}{2}}\, \mathfrak{S}_1^{-3/2}\biggr) + \frac{5\sqrt{\pi}}{\sqrt{2}} \frac{V^{3/2}}{x^{0.1}\sqrt{\lambda}}.$$

Adding these estimates yields

$$|R_5^{(1)}| \le M\kappa^{-1} \biggl(\frac{3}{4\lambda^2} \biggl(1 + \sqrt{\frac{\pi}{2}}\, \mathfrak{S}_1^{-3/2}\biggr) + \frac{5}{\sqrt[4]{2}} \biggl(1 + \frac{1}{\sqrt[4]{2}}\biggr) \frac{V^{3/2}}{x^{0.1}\sqrt{\lambda}} \biggr)$$

and

$$\begin{aligned} |R_5| \le |R_5^{(1)}| + |R_5^{(2)}| &\le M\kappa^{-1} \biggl(\frac{11}{4\lambda^2} \biggl(1 + \sqrt{\frac{\pi}{2}}\, \mathfrak{S}_1^{-3/2}\biggr) + \frac{5}{\sqrt[4]{2}} \biggl(1 + \frac{1}{\sqrt[4]{2}}\biggr) \frac{V^{3/2}}{x^{0.1}\sqrt{\lambda}} \biggr) \\ &< M\kappa^{-1} \biggl(2032 \frac{\log r}{r} + 84.1 \frac{\log r}{x^{0.1}\sqrt[4]{r}} \biggr). \end{aligned}$$

Summing the estimates of R_j, $1 \le j \le 5$, obtained above, we conclude that

$$\sum_{j=1}^{5} |R_5| \le M\kappa^{-1}\left\{740\frac{\log r}{r} + \frac{26.1}{x^{0.1}}\frac{\log r}{\sqrt[4]{r}} + \frac{10}{\sqrt{\pi}}\frac{1}{r^2} + 6434\frac{\sqrt{\log r}}{r^2}\,e^{-r/22} + 2032\frac{\log r}{r} + \frac{84.1}{x^{0.1}}\frac{\log r}{\sqrt[4]{r}}\right\}$$
$$< M\kappa^{-1}\left\{2273\frac{\log r}{r} + \frac{110.2}{x^{0.1}}\frac{\log r}{\sqrt[4]{r}}\right\}.$$

Therefore,

$$\sum_n |v_n| = \frac{M\kappa^{-1}}{\sqrt{\pi}}\,j(1-e^{-V^2/4}) + \theta\sum_{j=1}^{5}|R_j| = \frac{M\kappa^{-1}}{\sqrt{\pi}}\,j + M\theta\kappa^{-1}\left(\frac{a\log r}{r} + \frac{b\log r}{x^{0.1}\sqrt[4]{r}}\right)$$
$$= \frac{M}{\sqrt{\pi}}\left\{\sum_{n=0}^{+\infty}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}\frac{\Gamma(n-1/2)}{\Gamma(-1/2)} + \theta c(\log r)\left(\frac{1}{r} + \frac{x^{-0.1}}{20\sqrt[4]{r}}\right)\right\},$$

where $a = 2274$, $b = 110.2$, and $c = 2274(\sqrt{2}+1) < 5.49\cdot 10^3$. Now it is easy to see that, for $x \ge (\log N)^{7.5}$, we have

$$x \ge r^{7.5} \qquad \text{and} \qquad \frac{x^{-0.1}}{\sqrt[4]{r}} \le \frac{1}{r}\,.$$

Therefore, for such x, the absolute value of the remainder term in the above expression is at most

$$\frac{21}{20}\,c\frac{\log r}{r} < 5765\frac{\log r}{r}\,.$$

This completes the proof of the theorem.

2. THE BEHAVIOR OF THE ARGUMENT OF THE RIEMANN ZETA-FUNCTION AT THE GRAM POINTS

As mentioned above, the main problems related to the numbers Δ_n, that is, evaluating the moments of Δ_n and constructing the distribution function for them, reduce to similar problems for the sequence of numbers $\Delta(n) = S(t_n + 0)$. For this reason, we precede the proof of Selberg's formulas by a study of properties of the quantities $\Delta(n)$, which is performed in this chapter.

Quite satisfactory, although not the most accurate, results concerning the problems mentioned above are obtained by directly applying methods of [32, 41]: the passage from a "continuous" parameter t, $T < t \le T + H$, to a "discrete" parameter t_n, $N < n \le N + M$, involves no difficulties. We list these results below in order to compare them with those obtained in what follows.

First, in the case of integer $k \ge 1$, the asymptotic relation

$$\sum_n \Delta^{2k}(n) = \frac{(2k)!}{k!}\frac{M}{(2\pi)^{2k}}(\log\log N)^k(1+r_{2k}) \tag{2.1}$$

holds, where

$$|r_{2k}| \le \frac{4^k\sqrt{k}}{\sqrt{\log\log N}}\,. \tag{2.2}$$

It is easy to see that relation (2.1) is instructive only if

$$1 \le k \le (b_1 - \delta)\log\log\log N,$$

where $4b_1\log 2 = 1$.

Next, for noninteger a satisfying the condition $0 < a < b_2\log\log\log N$, a method for dealing with fractional moments going back to Ghosh [32] gives the relation

$$\sum_n |\Delta(n)|^{2a} = \frac{\Gamma(a+1/2)}{\sqrt{\pi}}\frac{M}{\pi^{2a}}(\log\log N)^a(1+r_{2a}), \tag{2.3}$$

in which

$$|r_{2a}| \le (\log\log\log N)^{-b_3}, \qquad b_3 = b_3(a) > 0 \tag{2.4}$$

(cf. [42, Chapter II, Section 3, Theorem 2]). A fundamental improvement of this method due to Boyarinov [39, 40] makes it possible, in some cases, to replace this estimate of r_{2a} by the more accurate estimate

$$|r_{2a}| \le \exp\{-b_4\sqrt{\log\log\log N}\}, \qquad b_4 = b_4(a) > 0. \tag{2.5}$$

The standard scheme for obtaining an asymptotic expression for a distribution function is based on a good approximation of the characteristic function of the random variable under examination and the application of the Berry–Esseen inequality. This scheme was first applied to the problem of determining the distribution of $S(t)$ and related functions by Ghosh.

Applying Ghosh's method to the function

$$\Phi(u) = \frac{1}{M}\#\left\{n \;\Big|\; N < n \le N+M,\ \Delta(n) \le \frac{u}{\pi}\sqrt{\frac{L}{2}}\right\}$$

yields

$$\Phi(u) = M(G(u) + r(u)), \tag{2.6}$$

where

$$G(u) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{u} e^{-v^2/2}\,dv \qquad \text{and} \qquad |r(u)| \le \frac{b_5}{\sqrt{\log\log\log N}}, \quad b_5 > 0; \tag{2.7}$$

by virtue of the relations

$$G(-u) = 1 - G(u) \asymp (ue^{u^2/2})^{-1}, \qquad u \to +\infty,$$

(2.6) is instructive only for

$$|u| \le \left(\frac{1}{2} - \delta\right)\log\log\log\log N.$$

A more perfect method for finding distribution functions, which was developed by Selberg and implemented (in the continuous case) by Tsang in [43] (see also Theorem 5 in [44]) allows us to replace the estimate of $r(u)$ in (2.7) by the inequality

$$|r(u)| \le \frac{b_6\log\log\log N}{\sqrt{\log\log N}};$$

accordingly, the range of applicability of such an expression for $\Phi(u)$ extends to

$$|u| \le \left(\frac{1}{2} - \delta\right)\log\log\log\log N.$$

In this chapter, we combine the technique of [32, 33, 41] with certain additional considerations (including theorems of Chapter 1) and obtain a series of new results improving those listed above.

First, estimate (2.2) is replaced by the inequality

$$|r_{2k}| \le \frac{b_7 k\sqrt{k}}{\sqrt{\log\log N}}. \tag{2.8}$$

Accordingly, the range of applicability of asymptotics (2.1) extends to

$$1 \le k \le b_8\sqrt[3]{\log\log N}.$$

Next, the estimates (2.4) and (2.5) of the remainder term r_{2a} are replaced by the general inequality

$$|r_{2a}| \le \frac{b_9(a\sqrt{a} + a\log\log\log N)}{\sqrt{\log\log N}},$$

which is valid for $1 \le a \le b_{10}\sqrt[3]{\log\log N}$. Thus, the gap between the accuracies of the estimates of r_{2a} for integer and noninteger a is virtually bridged.

Finally, under a certain condition on u (namely, $u \ge 3/2$), relations (2.7) can be replaced by

$$\frac{\Phi(-u)}{G(-u)} = 1 + O(r(u)) \qquad \text{and} \qquad \frac{1-\Phi(u)}{1-G(u)} = 1 + O(r(u)),$$

where the quantity

$$r(u) = \frac{u(u^2 + \log\log\log N)}{\sqrt{\log\log N}}$$

remains infinitesimal at $u = o(\sqrt[6]{\log\log N})$.

The main considerations underlying the improved results mentioned above are as follows.

The appearance of the multiplier 4^k in (2.2) is explained by the application of Hölder's inequality. Namely, if u and v are any real numbers, then this inequality, together with Lagrange's finite increment formula, leads to the estimates

$$|(u+v)^{2k} - u^{2k}| = 2k|v| \cdot |u+\theta v|^{2k-1} \le 2k|v| \cdot 2^{2k-2}(|u|^{2k-1} + |v|^{2k-1}) \le 4^k k(|u|^{2k-1}|v| + v^{2k}).$$

If the number $|v|$ is very small in comparison with $|u|$, then these estimates are too rough, and it is expedient to replace them by the sharper inequality

$$|(u+v)^{2k} - u^{2k}| \le \sum_{\nu=1}^{2k} \binom{2k}{\nu} |u|^{2k-\nu}|v|^{\nu}.$$

It is this consideration on which the proof of inequality (2.8) is based.

Next, in Ghosh's method, a fractional moment of the function under examination is approximated by a linear combination of a very large number of even moments of this function. Accordingly, the remainder term r_{2a} turns out to be a linear combination of a very large number of the remainders r_{2k}, $k = 1, 2, \dots$. Such an accumulation of error is the main reason why the estimation accuracy of r_{2a} is lower than that of r_{2k}.

In this paper, we proceed in a different order. Thus, Ghosh's method first approximates the function $S(t)$ by a trigonometric polynomial of the form

$$W_x(t) = -\frac{1}{\pi}\sum_{p \le x} \frac{\sin(t\log p)}{\sqrt{p}}$$

and after this evaluates the even moments of $S(t)$, which are then used to determine a fractional moment of $S(t)$. In this paper, we first calculate a fractional moment of $W_x(t)$ (as we did in Chapter 1) and then estimate the difference between this moment and the fractional moment of $S(t)$. In this way, we avoid the accumulation of error mentioned above.

Let us explain this on the example of the first moment. Representing $S(t)$ in the form

$$W_y(t) + (W_x(t) - W_y(t)) + R(t),$$

where $R(t) = S(t) - W_x(t)$, $y < x$, we readily conclude that the sums

$$\sum_n |S(t_n + 0)| = \sum_n |\Delta(n)| \qquad \text{and} \qquad \sum_n |W_y(t_n)|$$

differ from each other by no more than

$$\sum_n |W_x(t_n) - W_y(t_n)| + \sum_n |R(t_n + 0)| = \Sigma_1 + \Sigma_2.$$

The estimate $\Sigma_1 \ll M$, which goes back to Selberg, and the chain of obvious inequalities

$$\Sigma_2^2 \le M\sum_n \left|\sum_{y<p\le x} \frac{p^{it_n}}{\sqrt{p}}\right|^2 \ll M\sum_{y<p\le x}\frac{1}{p} \ll M\log\left(\frac{\log x}{\log y}\right)$$

with $y=\exp\left(c(\log x)/\log\log N\right)$ give

$$\sum_n|\Delta(n)|=\sum_n|W_y(t_n)|+O(M\sqrt{\log\log\log N}).$$

Theorems of Chapter 1 make it possible to calculate the sum over n on the right-hand side, roughly speaking, with any accuracy. Considering only the first two terms of the asymptotic expansion in Theorem 1.2, we obtain

$$\sum_n|W_y(t_n)|=\frac{M}{\pi\sqrt{\pi}}\sqrt{\log\log N}+O\biggl(\frac{M}{(\log\log N)^{3/2}}\biggr)$$

and

$$\sum_n|\Delta(n)|=\frac{M}{\pi\sqrt{\pi}}\sqrt{\log\log N}\biggl(1+O\biggl(\sqrt{\frac{\log\log\log N}{\log\log N}}\biggr)\biggr).$$

Finally, obtaining an asymptotic expression for the distribution function of $\Phi(u)$ for sufficiently large u has become possible thanks to Radziwiłł's recent paper [30], in which probability methods making it possible to study so-called large deviations for distributions close to normal were applied to $S(t)$.

Let us explain this. For simplicity, consider the classical Bernoulli scheme. Let $\xi_1,\dots,\xi_N$ be independent random variables taking the values 0 and 1 with the same probability (equal to $1/2$). The central limit theorem asserts that, for any fixed x, the function

$$F(x)=\mathsf{P}\biggl\{\frac{\xi_1+\dots+\xi_N-N/2}{\sqrt{N}/2}\le x\biggr\}$$

tends to $G(x)$ as N unboundedly increases, or, equivalently,

$$\lim_{N\to+\infty}\frac{F(x)}{G(x)}=1. \tag{2.9}$$

In the 1930s, there appeared a series of papers studying the question of for how large x (as a function N) satisfying (2.9) can be (see [47] and the references therein). In the 1938 paper [47], Cramer developed a general method and, applying it to the special case of the Bernoulli scheme mentioned above, proved that (2.9) holds up to $x=O(\sqrt{N}/\log N)$. This result of Cramer was subsequently repeatedly generalized and improved by many authors. One of such generalizations due to Hwang [48], which was applied to study the function $S(t)$ by Radziwiłł [30], is also used in this paper.

Applying Hwang's theorem, we have been able to prove that the large deviations of $\Delta(n)$ from the mean value obey the normal law and establish the existence of a very large number of n for which $\Delta(n)$ is of order $(\log\log n)^{1-\varepsilon}$ (this corresponds to an overshoot in the mean value of $\Delta(n)$ of order $\sqrt{\log\log n}$, i.e., by a factor of $(\log\log n)^{0.5-\varepsilon}$).

Section 2.1 is devoted to estimating the mean difference between the function $S(t)$ and its polynomial approximation $W_x(t)$. All statements of this section go back to Selberg's paper [41]. We included their proofs for two purposes: first, to take into account the specifics of the discrete case and short summation intervals and, secondly, to evaluate constants in each of the inequalities which we use. The main statement of Section 2.1 is Lemma 2.7.

In Section 2.2, we establish a relationship between the numbers $\Delta(n)$ and the function $S(t)$, derive an asymptotic expression for the odd moments of these quantities (Theorem 2.1), and estimate the even moments (Theorem 2.2).

In Section 2.3, we prove an asymptotic expression for a fractional moment of $\Delta(n)$ (Theorem 2.3), which is based on an estimate of the accuracy of the approximation of the sum $|x+y|^a$ by $|x|^a$ for noninteger a (Lemma 2.11).

Section 2.4 contains two theorems on the distribution function of $\Phi(u)$. One of them is based on Ghosh's method and applies to small $|u|$ (Theorem 2.4), and the other uses Hwang's result on large deviations from the normal law mentioned above (Theorem 2.5).

2.6. Approximation of the function $S(t)$ by special trigonometric polynomials. The purpose of this section is to obtain an upper bound for the mean value of the difference between the quantity $\Delta(n)$ and its polynomial approximation

$$W_x(t_n) = -\frac{1}{\pi}\sum_{p\le x}\frac{\sin(t_n\log p)}{\sqrt{p}}.$$

All statements of this section are discrete analogues of Lemmas 12 and 13 and Theorems 2 and 4 in Selberg's paper [41], and we use Selberg's methods of proof.

Lemma 2.1. *Let p range over the primes in the interval $(1,x]$, $2\le x\le (N/\log N)^{1/(4k)}$, and let a_p be any sequence of real numbers. Suppose that*

$$U_0(t)=\sum_{p\le x}a_p p^{it},\qquad U_1(t)=\operatorname{Re}U_0(t),\qquad \textit{and}\qquad U_2(t)=\operatorname{Im}U_0(t).$$

Let

$$S_j=S_{j,k}=\sum_n |U_j(t_n)|^{2k}.$$

Then, for any integer $k\ge 1$,

$$S_0\le k!M\mathfrak{s}_2^k+x^k\mathfrak{s}_1^{2k}\log N\qquad \textit{and}\qquad S_1,S_2\le\frac{(2k)!}{k!}\frac{M}{2^{2k}}\mathfrak{s}_2^k+x^k\mathfrak{s}_1^{2k}\log N,$$

where

$$\mathfrak{s}_r=\sum_{p\le x}|a_p|^r.$$

Proof. Estimating S_0, we obtain

$$S_0=\sum_n\sum_{p_1,\dots,p_k,q_1,\dots,q_k\le x}a_{p_1}\cdots a_{p_k}a_{q_1}\cdots a_{q_k}\left(\frac{p_1\cdots p_k}{q_1\cdots q_k}\right)^{it_n}=M\Sigma_1+\Sigma_2,$$

where

$$\Sigma_1=\sum_{\substack{p_1\cdots p_k=q_1\cdots q_k\\ p_1,\dots,p_k,q_1,\dots,q_k\le x}}a_{p_1}\cdots a_{p_k}a_{q_1}\cdots a_{q_k}=\sum_{\substack{p_1\cdots p_k=q_1\cdots q_k\\ p_1,\dots,p_k,q_1,\dots,q_k\le x}}a_{p_1}^2\cdots a_{p_k}^2,$$

$$\Sigma_2=\sum_{\substack{p_1\cdots p_k\ne q_1\cdots q_k\\ p_1,\dots,p_k,q_1,\dots,q_k\le x}}a_{p_1}\cdots a_{p_k}a_{q_1}\cdots a_{q_k}\sum_n\left(\frac{p_1\cdots p_k}{q_1\cdots q_k}\right)^{it_n}.$$

For fixed primes $p_1,\dots,p_k$, the number of solutions of the equation $p_1\cdots p_k=q_1\cdots q_k$ is at most $k!$. Therefore,

$$\Sigma_1\le k!\sum_{p_1,\dots,p_k\le x}a_{p_1}^2\cdots a_{p_k}^2=k!\mathfrak{s}_2^k.$$

Applying Lemma 1.2 to the estimate of Σ_2, we obtain

$$|\Sigma_2|\le\sum_{p_1,\dots,p_k,q_1,\dots,q_k\le x}|a_{p_1}\cdots a_{p_k}a_{q_1}\cdots a_{q_k}|\sqrt{p_1\cdots p_kq_1\cdots q_k}\log N\le x^k\mathfrak{s}_1^{2k}\log N,$$

as required.

The relation $U_1=(U_0+\overline{U}_0)/2$ gives

$$S_1=2^{-2k}\sum_n(U_0(t_n)+\overline{U}_0(t_n))^{2k}=2^{-2k}\sum_{\nu=0}^{2k}\binom{2k}{\nu}u_\nu,$$

where

$$u_\nu=\sum_n U_0^\nu\overline{U}_0^{2k-\nu}(t_n)=\sum_n\sum_{\substack{p_1,\dots,p_\nu\le x\\ q_1,\dots,q_{2k-\nu}\le x}} a_{p_1}\cdots a_{p_\nu}a_{q_1}\cdots a_{q_{2k-\nu}}\biggl(\frac{p_1\cdots p_\nu}{q_1\cdots q_{2k-\nu}}\biggr)^{it_n}.$$

Lemma 1.2 implies the following inequalities for $\nu\ne k$:

$$|u_\nu|\le\log N\sum_{p_1,\dots,p_\nu,q_1,\dots,q_{2k-\nu}\le x}|a_{p_1}\cdots a_{p_\nu}a_{q_1}\cdots a_{q_{2k-\nu}}|\sqrt{p_1\cdots p_\nu q_1\cdots q_{2k-\nu}}\le x^k\mathfrak{s}_1^{2k}\log N.$$

The contribution to u_k of the terms for which $p_1\cdots p_k\ne q_1\cdots q_k$ is estimated in a similar way. Thus,

$$S_1\le 2^{-2k}\binom{2k}{k}M\Sigma_1+2^{-2k}\sum_{\nu=0}^{2k}\binom{2k}{\nu}x^k\mathfrak{s}_1^k\log N\le\frac{(2k)!}{k!}\frac{M}{2^{2k}}\mathfrak{s}_2^k+x^k\mathfrak{s}_1^{2k}\log N.$$

The proof of the estimate of S_2 is similar. This proves the lemma.

Remark 2.1. In the special case $a_p=p^{-0.5}$, the estimates of the sums S_0, S_1, and S_2 for $k\ge 2$ can be somewhat improved by omitting the terms $x^k\mathfrak{s}_1^{2k}\log N$ from the inequalities of the lemma.

Indeed, the sum Σ_1 has the form

$$\sum_{p_1,\dots,p_k\le x}\frac{k!}{\alpha_1!\cdots\alpha_s!}(p_1\cdots p_k)^{-1},$$

where s is the number of different primes among $p_1,\dots,p_k$ and $\alpha_1,\dots,\alpha_s$ are the numbers of occurrences of these primes.

Let $q\le x$ be a prime. The coefficient multiplying the term corresponding to $p_1=\cdots=p_k=q$ in Σ_1 is $k!/k!=1$. At the same time, the coefficient of any such term in the expression $k!\,s_2^k$ is $k!$. Therefore,

$$\Sigma_1\le k!\,s_2^k-(k!-1)\sum_{2\le q\le x}\frac{1}{q^k}\le k!\,s_2^k-\frac{k!-1}{2^k}.$$

To complete the proof, it remains to check that

$$M\frac{k!-1}{2^k}>x^k\mathfrak{s}_1^{2k}\log N\qquad\text{and}\qquad 2^{-2k}\binom{2k}{k}M\frac{k!-1}{2^k}>x^k\mathfrak{s}_1^{2k}\log N.$$

But these inequalities follow from the estimates

$$\binom{2k}{k}^{-1}8^kx^k\mathfrak{s}_1^{2k}<(8x\mathfrak{s}_1^2)^k\le\biggl(8x\frac{a^2x}{(\log x)^2}\biggr)^k=\biggl(\frac{2ax\sqrt{2}}{\log x}\biggr)^{2k}<\frac{M}{\log N}\le(k!-1)\frac{M}{\log N},$$

in which $a=2.784$ is the constant from Lemma I.3.

Remark 2.2. The lemma remains valid under the replacement of $U_0(t)$ by a sum of the form

$$\sum_{p\le x}a_pp^{2it},$$

the sums $U_1(t)$ and $U_2(t)$ by the real and imaginary parts of this sum, and x by x^2 in its statement.

Lemma 2.2. *Suppose that $c>0$, k is an integer, $1\le k\le 0.1\log M$, $18\le y\le(0.2M/\log N)^{1/(3k)}$, and $a_1(p)$ and $a_2(p)$ are real numbers satisfying the conditions*

$$|a_1(p)|\le c\frac{\log p}{\log y}\qquad\textit{and}\qquad|a_2(p)|\le c$$

for p ranging over the primes in the interval $(2,y]$. Then the sums S_1 and S_2 defined by

$$S_1=\sum_n\Big|\sum_{p\le y}\frac{a_1(p)}{\sqrt{p}}\,p^{it_n}\Big|^{2k}\qquad\text{and}\qquad S_2=\sum_n\Big|\sum_{p\le y}\frac{a_2(p)}{p}\,p^{2it_n}\Big|^{2k},$$

are bounded as

$$S_1\le 1.1(kc^2)^kM\qquad\text{and}\qquad S_2\le 1.1(0.5kc^2)^kM.$$

Proof. Observing that $y\le(M/\log N)^{1/(3k)}$ and setting $a_p=a(p)/\sqrt{p}$ in Lemma 2.1, we obtain

$$S_1\le k!\,M\mathfrak{s}_2^k+y^k\mathfrak{s}_1^{2k}\log N.$$

Inequalities (I.5) and (I.3) of Lemma I.3 imply

$$\mathfrak{s}_2=\sum_{p\le y}\frac{a_1^2(p)}{p}\le\frac{c^2}{(\log y)^2}\sum_{p\le y}\frac{(\log p)^2}{p}\le\frac{c^2}{\log y}\sum_{p\le y}\frac{\log p}{p}<c^2,$$

$$\mathfrak{s}_1=\sum_{p\le y}\frac{|a_1(p)|}{\sqrt{p}}\le\frac{c}{\log y}\sum_{p\le y}\frac{\log p}{\sqrt{p}}\le c\sum_{p\le y}\frac{1}{\sqrt{p}}\le\frac{ac\sqrt{y}}{\log y}\le c\sqrt{y}$$

and, finally,

$$y^k\mathfrak{s}_1^{2k}\log N<(cy)^{2k}\log N<0.1c^{2k}M.$$

Thus,

$$S_1\le k!\,Mc^{2k}+0.1c^{2k}M\le 1.1(kc^2)^kM.$$

Next, applying the same Lemma I.3 to the second sum, we obtain

$$S_2\le k!\,M\mathfrak{s}_2^k+y^k\mathfrak{s}_1^{2k}\log N,$$

where

$$\mathfrak{s}_2=\sum_{p\le y}\frac{a_2^2(p)}{p^2}\le c^2\sum_{p\le y}\frac{1}{p^2}<0.5c^2$$

and

$$\mathfrak{s}_1=\sum_{p\le y}\frac{|a_2(p)|}{p}\le c\sum_{p\le y}\frac{1}{p}\le c(\log\log y+1)<c\sqrt{\frac{y}{2}}\,.$$

Therefore,

$$S_2\le k!(0.5c^2)^kM+\left(c\sqrt{\frac{y}{2}}\right)^{2k}\log N\le k!(0.5c^2)^kM+0.1(0.5c^2)^kM\le 1.1(0.5c^2k)^kM.$$

This completes the proof of the lemma.

Suppose that $2\le x\le t^2$. Following Selberg [41], we set

$$\sigma_{x,t}=\frac{1}{2}+2\max\left(\left|\beta-\frac{1}{2}\right|,\frac{1}{\log x}\right),$$

where the maximum is over all zeros $\varrho=\beta+i\gamma$ of the Riemann zeta-function that satisfy the condition

$$|t-\gamma|\le\frac{x^{3|\beta-1/2|}}{\log x}\,.$$

Finally, we set $X=N^{0.1\varepsilon}$.

Lemma 2.3. *If $\nu \ge 0$ is an integer, $x \ge 2$, $y \ge 1$, and $x^3y^2 \le X$, then*

$$\sum_n \biggl(\sigma_{x,t_n} - \frac{1}{2}\biggr)^\nu y^{\sigma_{x,t_n}-1/2} < \biggl(\frac{2}{\log x}\biggr)^\nu M\biggl(e^{(2\log y)/\log x} + 52.1\nu!\,\frac{\log N}{\log x}\biggl(\frac{\log x}{\log X}\biggr)^\nu\biggr).$$

Proof. We divide all numbers n, $N < n \le N + M$, into two classes. The first class consists of all n for which any zero $\beta + i\gamma$ satisfying the condition

$$|t_n - \gamma| \le \frac{x^{3|\beta-1/2|}}{\log x}$$

satisfies also the inequality $|\beta - 1/2| \le 1/\log x$. The remaining numbers n form the second class. Note that the sum V_1 over all n in the first class is at most

$$\biggl(\frac{2}{\log x}\biggr)^\nu y^{2/\log x} M = \biggl(\frac{2}{\log x}\biggr)^\nu M e^{(2\log y)/\log x}.$$

Next, for any point t_n with n in the second class, there exists a zero $\beta_0 + i\gamma_0$ satisfying the condition

$$\gamma_0 - \delta_0 \le t_n \le \gamma_0 + \delta_0, \qquad \delta_0 = \frac{x^{3|\beta_0-1/2|}}{\log x} < \sqrt{X},$$

and such that $|\beta_0 - 1/2| > 1/\log x$. Therefore, all such Gram points t_n are covered by the neighborhoods $(\gamma - \delta, \gamma + \delta)$, $\delta = x^{3|\beta-1/2|}/\log x$, of all ordinates γ in the interval

$$(T - \sqrt{X}, T + H + \sqrt{X}], \qquad T = t_N, \quad H = t_{N+M} - t_N.$$

Thus, the sum V_2 over n in the second class satisfies the inequality

$$V_2 \le 2 \sum_{\substack{T-\sqrt{X}<\gamma\le T+H+\sqrt{X}\\ \beta>1/2}} \sum_{\gamma-\delta\le t_n\le\gamma+\delta} 2^\nu \biggl(\beta - \frac{1}{2}\biggr)^\nu y^{2(\beta-1/2)}$$

(the multiplier 2 takes into account the symmetry of the zeros of $\zeta(s)$ with respect to the critical line).

Let $t_{k+1}, \dots, t_{k+r}$ be all Gram points in the interval $[\gamma - \delta, \gamma + \delta]$. Lagrange's finite increment formula implies that, for some η such that $\gamma - \delta < \eta < \gamma + \delta$, we have

$$r = \pi^{-1}(\vartheta(t_{k+r}) - \vartheta(t_{k+1})) + 1 = \pi^{-1}(t_{k+r} - t_{k+1})\vartheta'(\eta) + 1 < \frac{\delta}{\pi}\log N = \frac{\log N}{\pi \log x}\, x^{3(\beta-1/2)}.$$

Therefore,

$$V_2 < \frac{2^{\nu+1}}{\pi}\frac{\log N}{\log x} \sum_{\substack{T-\sqrt{X}<\gamma\le T+\sqrt{X}+H\\ \beta>1/2}} \biggl(\beta - \frac{1}{2}\biggr)^\nu (x^3y^2)^{\beta-1/2}.$$

Using the identity

$$\biggl(\beta - \frac{1}{2}\biggr)^\nu z^{\beta-1/2} = \int_0^{\beta-1/2} (u^\nu \log z + \nu u^{\nu-1}) z^u\, du$$

and the inequality $x^3y^2 \le X$, we obtain

$$\begin{aligned} V_2 &< \frac{2^{\nu+1}}{\pi}\frac{\log N}{\log x} \sum_{\substack{T-\sqrt{X}<\gamma\le T+H+\sqrt{X}\\ \beta>1/2}} \int_0^{\beta-1/2} (u^\nu \log X + \nu u^{\nu-1}) X^u\, du \\ &= \frac{2^{\nu+1}}{\pi}\frac{\log N}{\log x} \int_0^{1/2} (u^\nu \log X + \nu u^{\nu-1}) X^u \sum_{\substack{T-\sqrt{X}<\gamma\le T+H+\sqrt{X}\\ \beta>1/2}} g(\beta + i\gamma; u)\, du, \end{aligned}$$

where

$$g(\beta+i\gamma;u)=\begin{cases}1 & \text{if } 0\le u\le \beta-\dfrac{1}{2},\\ 0 & \text{otherwise.}\end{cases}$$

But the sum over γ under the integral sign coincides with the quantity

$$N\biggl(\frac{1}{2}+u,T+H+\sqrt{X}\biggr)-N\biggl(\frac{1}{2}+u,T-\sqrt{X}\biggr);$$

by virtue of Lemma I.4 and the Riemann–Mangoldt formula, this quantity does not exceed the sum

$$\begin{aligned}&\biggl\{N\biggl(\frac{1}{2}+u,T+H\biggr)-N\biggl(\frac{1}{2}+u,T\biggr)\biggr\}\\ &\qquad+\{N(T+H+\sqrt{X})-N(T+H)\}+\{N(T)-N(T-\sqrt{X})\}\\ &\le\frac{13}{2}HX^{-2u}\log(T+0.5H)+\frac{1}{3}\sqrt{X}\log N\\ &<13\pi MX^{-2u}(1+o(1))+\frac{1}{3}\sqrt{X}\log N<13\pi MX^{-2u}(1+o(1)).\end{aligned}$$

Therefore,

$$\begin{aligned}V_2&<\frac{2^{\nu+1}}{\pi}\frac{\log N}{\log x}13\pi M(1+o(1))\int_0^{+\infty}(u^\nu\log X+\nu u^{\nu-1})X^u\cdot X^{-2u}\,du\\ &=26(1+o(1))2\nu!\cdot M\frac{\log N}{\log x}\biggl(\frac{2}{\log X}\biggr)^\nu<52.01M\nu!\biggl(\frac{2}{\log X}\biggr)^\nu\frac{\log N}{\log x}.\end{aligned}$$

This implies the assertion of the lemma.

Given an integer $n\ge 2$ and a number $x\ge 2$, we set

$$\Lambda_x(n)=\begin{cases}\Lambda(n) & \text{if } 1\le n\le x,\\ \Lambda(n)\dfrac{\log^2(x^3/n)-2\log^2(x^2/n)}{2\log^2 x} & \text{if } x<n\le x^2,\\ \Lambda(n)\dfrac{\log^2(x^3/n)}{2\log^2 x} & \text{if } x^2<n\le x^3.\end{cases}$$

Lemma 2.4. *If $e^{16}\le x\le t^\delta$ for $\delta=1/250$, then*

$$S(t)=-\frac{1}{\pi}\sum_{m\le x^3}\frac{\Lambda_x(m)}{m^{\sigma_{x,t}}}\frac{\sin(t\log m)}{\log m}+\theta\biggl(\sigma_{x,t}-\frac{1}{2}\biggr)\biggl(a_1\log t+a_2\biggl|\sum_{p\le x^3}\frac{\Lambda_x(p)}{p^{\sigma_{x,t}+it}}\biggr|\biggr),\tag{2.10}$$

where $a_1=4.5$ and $a_2=8.54$.

Proof. Let $R(t)$ be the difference between $S(t)$ and the sum over m on the right-hand side of (2.10). According to Lemma I.6, we have

$$\pi|R(t)|\le\biggl(\sigma_{x,t}-\frac{1}{2}\biggr)\biggl(a_1^{(1)}\log t+a_2^{(1)}|r(x,t)|+a_3^{(1)}+\sum_\varrho\frac{\kappa(\varrho)}{(\sigma_{x,t}-1/2)^2+(t-\gamma)^2}\biggr),$$

where

$$a_1^{(1)}=\frac{13}{6e}+1.8,\qquad a_2^{(1)}=\frac{13}{3e}+2.6,\qquad a_3^{(1)}=\frac{52}{3e}+10.82,$$

$$r(x,t)=\sum_{m\le x^3}\frac{\Lambda_x(m)}{m^{\sigma_{x,t}+it}},\qquad\text{and}\qquad\kappa(\varrho)=\int_{1/2}^{\sigma_{x,t}}\frac{|t-\gamma|\cdot|\sigma_{x,t}+\sigma-2\beta|}{(\sigma-\beta)^2+(t-\gamma)^2}\,d\sigma.$$

By virtue of the same Lemma I.6, in the case where $|\beta - 1/2| > (1/2)(\sigma_{x,t} - 1/2)$, the integral $\kappa(\varrho)$ satisfies the inequality

$$\kappa(\varrho) < 2\biggl(\sigma_{x,t} - \frac{1}{2}\biggr).$$

Let $|\beta - 1/2| \le (1/2)(\sigma_{x,t} - 1/2)$. Then, noting that $|\sigma_{x,t} - \beta| \le (3/2)(\sigma_{x,t} - 1/2)$, we obtain

$$\begin{aligned}\kappa(\varrho) &\le \int_{1/2}^{\sigma_{x,t}} \frac{|t-\gamma|(|\sigma_{x,t}-\beta| + |\sigma-\beta|)}{(\sigma-\beta)^2 + (t-\gamma)^2}\,d\sigma \\ &\le \frac{3}{2}\biggl(\sigma_{x,t} - \frac{1}{2}\biggr)\int_{-\infty}^{+\infty} \frac{|t-\gamma|\,d\sigma}{(\sigma-\beta)^2 + (t-\gamma)^2} + \int_{1/2}^{\sigma_{x,t}} \frac{|t-\gamma|\cdot|\sigma-\beta|}{(\sigma-\beta)^2 + (t-\gamma)^2}\,d\sigma \\ &\le \frac{3\pi}{2}\biggl(\sigma_{x,t} - \frac{1}{2}\biggr) + \frac{1}{2}\biggl(\sigma_{x,t} - \frac{1}{2}\biggr).\end{aligned}$$

Lemma I.5 gives

$$\pi|R(t)| \le \biggl(\sigma_{x,t} - \frac{1}{2}\biggr)(a_1^{(2)}\log t + a_2^{(2)}|r(x,t)| + a_3^{(2)}),$$

where

$$a_1^{(2)} = a_1^{(1)} + \frac{13}{12}(3\pi+1), \qquad a_2^{(2)} = a_2^{(1)} + \frac{13}{6}(3\pi+1), \qquad \text{and} \qquad a_3^{(2)} = a_3^{(1)} + \frac{13}{2}(3\pi+1).$$

The definition of $\Lambda_x(m)$ and the asymptotic law of the distribution of primes imply

$$|r(x,t)| \le \biggl|\sum_{p\le x^3} \frac{\Lambda_x(p)}{p^{\sigma_{x,t}+it}}\biggr| + \sum_{p^2\le x^3} \frac{\Lambda_x(p^2)}{p} + \sum_{r\ge 3}\sum_{p^r\le x^3} \frac{\log p}{p^{r/2}} \le \biggl|\sum_{p\le x^3} \frac{\Lambda_x(p)}{p^{\sigma_{x,t}+it}}\biggr| + 2\log x.$$

Finally, noting that

$$a_1^{(2)}\log t + 2a_2^{(2)}\log x + a_3^{(2)} < 4.5\pi\log t \qquad \text{for} \quad x \le t^{\delta}, \quad t \ge e^{16/\delta},$$

we arrive at the assertion of the lemma.

Lemma 2.5. *Let k be an integer such that $1 \le k \le c\log X$, where c is a sufficiently small absolute constant, and let $x = X^{1/(8k+3)}$. Then*

$$\sum_n \biggl|S(t_n+0) + \frac{1}{\pi}\sum_{m\le x^3} \frac{\Lambda_x(m)}{m^{\sigma_{x,t_n}}}\frac{\sin(t_n\log m)}{\log m}\biggr|^{2k} < \frac{100.4}{\varepsilon}\biggl(\frac{1440k}{\varepsilon}\biggr)^{2k} M.$$

Proof. Using Lemma 2.4 and the inequality $\log t_N < \log N$, we conclude that the sum under consideration does not exceed

$$2^{2k-1}((a_1\log N)^{2k}\Sigma_1 + a_2^{2k}\Sigma_2),$$

where

$$\Sigma_2 = \sum_n \biggl(\sigma_{x,t_n} - \frac{1}{2}\biggr)^{2k} \qquad \text{and} \qquad \Sigma_1 = \sum_n \biggl(\sigma_{x,t_n} - \frac{1}{2}\biggr)^{2k}\biggl|\sum_{p\le x^3} \frac{\Lambda_x(p)}{p^{\sigma_{x,t_n}+it_n}}\biggr|^{2k}.$$

According to Lemma 2.3, we have

$$\begin{aligned}\Sigma_1 &= \biggl(\frac{2}{\log x}\biggr)^{2k} M\biggl(1 + \frac{521\varepsilon^{-1}(2k)!}{(8k+3)^{2k-1}}\biggr) < \frac{94.8}{\varepsilon}\biggl(\frac{2}{\log x}\biggr)^{2k} M \\ &= \frac{94.8}{\varepsilon}\biggl(\frac{160k}{\varepsilon}\biggr)^{2k}\frac{(1+3/8k)^{2k}}{(\log N)^{2k}} M < \frac{94.8e^{3/4}}{\varepsilon}\biggl(\frac{160k}{\varepsilon\log N}\biggr)^{2k} M < \frac{200.7}{\varepsilon}\biggl(\frac{160k}{\varepsilon\log N}\biggr)^{2k} M.\end{aligned}$$

Transforming the sum over $p \le x^3$ and following Selberg's argument in [41], we obtain

$$\begin{aligned}\left|\sum_{p\le x^3}\frac{\Lambda_x(p)}{p^{\sigma_{x,t}+it}}\right| &\le \left|\sum_{p\le x^3}\frac{\Lambda_x(p)}{\sqrt{p}}p^{it}\right| + \left|\sum_{p\le x^3}\frac{\Lambda_x(p)}{\sqrt{p}}(1-p^{1/2-\sigma_{x,t}})p^{it}\right| \\ &\le \left|\sum_{p\le x^3}\frac{\Lambda_x(p)}{\sqrt{p}}p^{it}\right| + \frac{x^{\sigma_{x,t}-1/2}}{\log x}\int_0^{+\infty} x^{-v}\left|\sum_{p\le x^3}\frac{\Lambda_x(p)(\log p)(\log xp)}{p^{1/2+v+it}}\right| dv.\end{aligned}$$

Twice applying Hölder's inequality, we arrive at

$$\begin{aligned}&\left|\sum_{p\le x^3}\frac{\Lambda_x(p)}{p^{\sigma_{x,t}+it}}\right|^{2k} \\ &\quad\le 2^{2k-1}\Bigg(\left|\sum_{p\le x^3}\frac{\Lambda_x(p)}{\sqrt{p}}p^{it}\right|^{2k} \\ &\qquad + \frac{x^{2k(\sigma_{x,t}-1/2)}}{(\log x)^{2k}}\left(\int_0^{+\infty} x^{-v}\,dv\right)^{2k-1}\int_0^{+\infty} x^{-v}\left|\sum_{p\le x^3}\frac{\Lambda_x(p)(\log p)(\log xp)}{p^{1/2+v+it}}\right|^{2k} dv\Bigg).\end{aligned}$$

We proceed to estimate the sum Σ_2:

$$\begin{aligned}\Sigma_2 &< 2^{2k-1}\Bigg(\sum_n\left(\sigma_{x,t_n}-\frac{1}{2}\right)^{2k}\left|\sum_{p\le x^3}\frac{\Lambda_x(p)}{\sqrt{p}}p^{it_n}\right|^{2k} \\ &\quad + \frac{1}{(\log x)^{4k-1}}\int_0^{+\infty} x^{-v}\sum_n\left(\sigma_{x,t_n}-\frac{1}{2}\right)^{2k} x^{2k(\sigma_{x,t_n}-1/2)}\left|\sum_{p\le x^3}\frac{\Lambda_x(p)(\log p)(\log xp)}{p^{1/2+v+it_n}}\right|^{2k} dv\Bigg).\end{aligned}$$

Now, we set

$$a_1(p)=\frac{\Lambda_x(p)}{\log x}\qquad\text{and}\qquad a_2(p)=\frac{\Lambda_x(p)(\log p)(\log xp)}{p^{v_0}(\log x)^3},$$

where v_0 is the value of $v\ge 0$ at which the sum over n in the last inequality under the integral sign attains its maximum value. We have

$$\Sigma_2 < \frac{1}{2}(2\log x)^{2k}(\Sigma_3+\Sigma_4),$$

where

$$\Sigma_3=\sum_n\left(\sigma_{x,t_n}-\frac{1}{2}\right)^{2k}\left|\sum_{p\le x^3}\frac{a_1(p)}{\sqrt{p}}p^{it_n}\right|^{2k}$$

and

$$\Sigma_4=\sum_n\left(\sigma_{x,t_n}-\frac{1}{2}\right)^{2k} x^{2k(\sigma_{x,t_n}-1/2)}\left|\sum_{p\le x^3}\frac{a_2(p)}{\sqrt{p}}p^{it_n}\right|^{2k}.$$

Applying Cauchy's inequality to each of the sums Σ_3 and Σ_4, we obtain

$$\Sigma_3\le(\Sigma_5\Sigma_7)^{1/2}\qquad\text{and}\qquad\Sigma_4\le(\Sigma_6\Sigma_8)^{1/2},$$

where

$$\Sigma_5=\sum_n\left(\sigma_{x,t_n}-\frac{1}{2}\right)^{4k},\qquad\Sigma_6=\sum_n\left(\sigma_{x,t_n}-\frac{1}{2}\right)^{4k} x^{2k(\sigma_{x,t_n}-1/2)},$$

and

$$\Sigma_7,\,\Sigma_8=\sum_n\biggl|\sum_{p\le x^3}\frac{a_j(p)}{\sqrt{p}}\,p^{it_n}\biggr|^{4k},\qquad j=1,2.$$

Estimating the sums Σ_5 and Σ_6 by using Lemma 2.3, we find

$$\Sigma_5<\biggl(\frac{2}{\log x}\biggr)^{4k}M\biggl(1+\frac{521\varepsilon^{-1}\,(4k)!}{(8k+3)^{4k-1}}\biggr)<\frac{9.4M}{\varepsilon}\biggl(\frac{2}{\log x}\biggr)^{4k}$$

and

$$\Sigma_6<\biggl(\frac{2}{\log x}\biggr)^{4k}M\biggl(e^{8k}+\frac{521\varepsilon^{-1}\,(4k)!}{(8k+3)^{4k-1}}\biggr)<\frac{M}{240\varepsilon}\biggl(\frac{2e^2}{\log x}\biggr)^{4k}.$$

For $p=x^\lambda$, we have

$$a_1(p)=\varphi_1(\lambda)\frac{\log p}{\log(x^3)}\qquad\text{and}\qquad a_2(p)=\varphi_2(\lambda)\frac{\log p}{\log(x^3)},$$

where

$$\varphi_1(\lambda)=\begin{cases}3 & \text{if } 0\le\lambda\le 1,\\ \dfrac{3}{2}(1+2\lambda-\lambda^2) & \text{if } 1\le\lambda\le 2,\\ \dfrac{3}{2}(\lambda-3)^2 & \text{if } 2\le\lambda\le 3\end{cases}\qquad\text{and}\qquad \varphi_2(\lambda)=\lambda(\lambda+1)\varphi_1(\lambda).$$

It is easy to show that

$$\max_{0\le\lambda\le 3}\varphi_1(\lambda)=\varphi_1(1)=3\qquad\text{and}\qquad\max_{0\le\lambda\le 3}\varphi_2(\lambda)=\varphi_2(\lambda_0)=3a,$$

where $a=3.446\,088\,218\ldots$ and $\lambda_0=1.711\,668\,606\ldots$. Therefore, applying Lemma 2.2 to the sums Σ_7 and Σ_8, we obtain

$$\Sigma_7\le 1.1(3^2\cdot 2k)^{2k}M\qquad\text{and}\qquad\Sigma_8\le 1.1(9a^2\cdot 2k)^{2k}M.$$

Thus,

$$\Sigma_3\le\frac{3.3}{\sqrt{\varepsilon}}\frac{M}{(\log x)^{2k}}(72k)^k,\qquad\Sigma_4\le\frac{0.07}{\sqrt{\varepsilon}}\frac{M}{(\log x)^{2k}}(72e^4a^2k)^k,\qquad\text{and}\qquad\Sigma_2\le\frac{M}{25\sqrt{\varepsilon}}(288e^4a^2k)^k.$$

Therefore, the sum under consideration does not exceed

$$\begin{aligned}&\frac{M}{2}\biggl(\frac{200.7}{\varepsilon}\biggl(\frac{2a_1\cdot 160k}{\varepsilon}\biggr)^{2k}+\frac{1}{25\sqrt{\varepsilon}}(a_2^2\cdot 288e^4a^2k)^k\biggr)\\ &\qquad<\frac{100.4}{\varepsilon}\biggl(\frac{320a_1k}{\varepsilon}\biggr)^{2k}M=\frac{100.4}{\varepsilon}\biggl(\frac{1440k}{\varepsilon}\biggr)^{2k}M.\end{aligned}$$

This completes the proof of the lemma.

Lemma 2.6. *If the conditions of Lemma* 2.5 *hold, then*

$$\sum_n\biggl|\sum_{m\le x^3}\frac{\Lambda_x(m)}{m^{\sigma_{x,t_n}}}\frac{\sin\,(t_n\log m)}{\log m}-\sum_{p\le x}\frac{\sin\,(t_n\log p)}{\sqrt{p}}\biggr|^{2k}<\frac{M}{8\varepsilon}(1152e^4k)^k.\tag{2.11}$$

Proof. Let $R(t)$ denote the quantity obtained from the difference in (2.11) under the modulus sign under the replacement of t_n by t. We have

$$R(t)=\sum_{x<p\le x^3}\frac{\Lambda_x(p)}{\log p}\frac{\sin\,(t\log p)}{\sqrt{p}}+\frac{1}{2}\sum_{p^2\le x^3}\frac{\Lambda_x(p^2)}{\log p}\frac{\sin\,(2t\log p)}{p}$$

$$-\sum_{p\le x^3}\frac{\Lambda_x(p)}{\log p}(1-p^{1/2-\sigma_{x,t}})\frac{\sin(t\log p)}{\sqrt{p}}-\frac{1}{2}\sum_{p^2\le x^3}\frac{\Lambda_x(p^2)}{\log p}(1-p^{1-2\sigma_{x,t}})\frac{\sin(2t\log p)}{p}$$
$$+\sum_{r\ge 3}\sum_{p^r\le x^3}\frac{\Lambda_x(p^r)}{p^{r\sigma_{x,t}}}\frac{\sin(rt\log p)}{r\log p}.$$

The inequalities $0\le 1-p^{1-2\sigma_{x,t}}\le 2(\sigma_{x,t}-1/2)\log p$ imply that the contribution to $R(t)$ of the two last sums does not exceed

$$\frac{1}{2}\sum_{p^2\le x^3}\frac{\Lambda_x(p^2)}{\log p}2\biggl(\sigma_{x,t}-\frac{1}{2}\biggr)\frac{\log p}{p}+\sum_p\sum_{r\ge 3}\frac{1}{rp^{r/2}}\le\biggl(\sigma_{x,t}-\frac{1}{2}\biggr)\sum_{p^2\le x^3}\frac{\log p}{p}+\frac{3}{5}<2\biggl(\sigma_{x,t}-\frac{1}{2}\biggr)\log x$$

in absolute value. Thus,

$$|R(t)|\le\sum_{j=1}^{4}R_j(t),$$

where

$$R_1(t)=\biggl|\sum_{p\le x^3}\frac{a_1(p)}{\sqrt{p}}p^{it}\biggr|,\qquad R_2(t)=\frac{1}{2}\biggl|\sum_{p\le x^{3/2}}\frac{a_2(p)}{p}p^{2it}\biggr|,$$
$$R_3(t)=\biggl|\sum_{p\le x^3}\frac{\Lambda_x(p)}{\log p}(1-p^{1/2-\sigma_{x,t}})\frac{p^{it}}{\sqrt{p}}\biggr|,\qquad R_4(t)=2\biggl(\sigma_{x,t}-\frac{1}{2}\biggr)\log x,$$

$$a_1(p)=\begin{cases}0 & \text{if}\quad p\le x,\\ \dfrac{\Lambda_x(p)}{\log p} & \text{if } x<p\le x^3,\end{cases}\qquad\text{and}\qquad a_2(p)=\frac{\Lambda_x(p^2)}{\log p}.$$

According to Hölder's inequality, the sum in the statement of the lemma is at most

$$4^{2k-1}(\Sigma_1+\Sigma_2+\Sigma_3+\Sigma_4),\qquad\Sigma_j=\sum_n R_j^{2k}(t_n).$$

Let us estimate each of the sums Σ_j. First, noting that

$$|a_1(p)|\le\frac{3\log p}{\log(x^3)}\qquad\text{and}\qquad|a_2(p)|\le 1$$

and applying Lemma 2.2, we obtain

$$\Sigma_1<1.1M(3^2k)^k\qquad\text{and}\qquad\Sigma_2<1.1M(0.5k)^k.$$

Next, the estimate

$$|R_3(t)|\le\frac{x^{\sigma_{x,t}-1/2}}{\log x}\int_0^\infty x^{-v}\biggl|\sum_{p\le x^3}\frac{\Lambda_x(p)(\log xp)}{p^{1/2+v}}p^{it}\biggr|\,dv$$

in Selberg's paper [41] and Hölder's inequality imply

$$\Sigma_3\le\frac{1}{(\log x)^{4k-1}}\int_0^{+\infty}x^{-v}\sum_n x^{2k(\sigma_{x,t_n}-1/2)}\biggl|\sum_{p\le x^3}\frac{\Lambda_x(p)(\log xp)}{p^{1/2+v}}p^{it_n}\biggr|^{2k}dv$$
$$\le\sum_n x^{2k(\sigma_{x,t_n}-1/2)}\biggl|\sum_{p\le x^3}\frac{a_3(p)}{\sqrt{p}}p^{it_n}\biggr|^{2k},$$

where

$$a_3(p)=\frac{\Lambda_x(p)(\log xp)}{p^{v_0}(\log x)^2}$$

and v_0 is the value $v\ge 0$ at which the sum over n under the integral sign attains its maximum. Applying Cauchy's inequality, we obtain

$$\Sigma_3\le(\Sigma_5\,\Sigma_6)^{1/2},\qquad \Sigma_5=\sum_n x^{4k(\sigma_{x,t_n}-1/2)},\qquad \text{and}\qquad \Sigma_6=\sum_n\biggl|\sum_{p\le x^3}\frac{a_3(p)}{\sqrt{p}}\,p^{it_n}\biggr|^{4k}.$$

By virtue of Lemma 2.3, we have

$$\Sigma_5<M\biggl(e^{8k}+52.1\frac{\log N}{\log x}\biggr)<\frac{2M}{\varepsilon}\,e^{8k}.$$

Now, noting that

$$|a_3(p)|\le\frac{\Lambda_x(p)\log xp}{(\log x)^2}\le\frac{6\log p}{\log(x^3)}$$

and applying Lemma 2.2, we obtain the estimate $\Sigma_6<1.1M(6^2\cdot 2k)^{2k}$. Therefore,

$$\Sigma_3<\frac{1.5}{\sqrt{\varepsilon}}(72e^4k)^kM.$$

Finally, applying Lemma 2.3 to the estimate Σ_4, we find

$$\begin{aligned}\Sigma_4&<(2\log x)^{2k}\biggl(\frac{2}{\log x}\biggr)^{2k}M\biggl(1+52.1\,(2k)!\,\frac{\log N}{\log x}\biggl(\frac{\log x}{\log X}\biggr)^{2k}\biggr)\\&<4^{2k}M\biggl(1+\frac{521\varepsilon^{-1}\,(2k)!}{(8k+3)^{2k-1}}\biggr)<\frac{94.8M}{\varepsilon}\,16^k.\end{aligned}$$

Thus, for the initial sum, we have

$$\begin{aligned}&4^{2k-1}(\Sigma_1+\cdots+\Sigma_4)\\&\quad<4^{2k-1}M\biggl(1.1(9k)^k+1.1(0.5k)^k+\frac{1.5}{\sqrt{\varepsilon}}(72e^4k)^k+16^k\cdot\frac{94.8}{\varepsilon}\biggr)<\frac{M}{8\varepsilon}(1152e^4k)^k.\end{aligned}$$

This completes the proof of the lemma.

Lemma 2.7. *If the conditions of Lemma* 2.5 *hold, then*

$$\sum_n\biggl|S(t_n+0)+\frac{1}{\pi}\sum_{p\le x}\frac{\sin(t_n\log p)}{\sqrt{p}}\biggr|^{2k}<\frac{c_0}{57}(c_0k)^{2k}M,$$

where $c_0=2880\varepsilon^{-1}$.

Proof. Using the estimates of Lemmas 2.4 and 2.5 and Hölder's inequality, we conclude that the sum under consideration is at most

$$\begin{aligned}&\sum_n\biggl|S(t_n+0)+\frac{1}{\pi}\sum_{m\le x^3}\frac{\Lambda_x(m)}{m^{\sigma_{x,t_n}}}\frac{\sin(t_n\log m)}{\log m}\\&\qquad-\frac{1}{\pi}\biggl(\sum_{p\le x}\frac{\sin(t_n\log p)}{\sqrt{p}}-\sum_{m\le x^3}\frac{\Lambda_x(m)}{m^{\sigma_{x,t_n}}}\frac{\sin(t_n\log m)}{\log m}\biggr)\biggr|^{2k}\\&\quad\le 2^{2k-1}\biggl(\sum_n\biggl|S(t_n+0)+\frac{1}{\pi}\sum_{m\le x^3}\frac{\Lambda_x(m)}{m^{\sigma_{x,t_n}}}\frac{\sin(t_n\log m)}{\log m}\biggr|^{2k}\end{aligned}$$

$$+\pi^{-2k}\sum_n\left|\sum_{p\le x}\frac{\sin(t_n\log p)}{\sqrt{p}}-\sum_{m\le x^3}\frac{\Lambda_x(m)}{m^{\sigma_{x,t_n}}}\frac{\sin(t_n\log m)}{\log m}\right|^{2k}\Bigg)$$

$$\le 2^{2k-1}M\left(\frac{100.4}{\varepsilon}\left(\frac{1440k}{\varepsilon}\right)^{2k}+\frac{1}{8\varepsilon}\left(\frac{1152e^4k}{\pi^2}\right)^k\right)<\frac{50.3}{\varepsilon}\left(\frac{2880k}{\varepsilon}\right)^{2k}M.$$

This proves the lemma.

2.7. The integer moments of $\Delta(n)$. In this section, we calculate the even moments of $\Delta(n)$ and estimate the absolute values of the odd moments from above. The argument used below has the special feature that rejecting the application of Hölder's inequality when dealing with the sum

$$\sum_n|R(t_n+0)|(|\Delta(n)|+|R(t_n+0)|)^{2k-1}$$

leads to a more accurate (with respect to the parameter k) estimate of this sum.

Lemma 2.8. *For any n, $\Delta(n)=S(t_n+0)$.*

Proof. According to the Riemann–Mangoldt formula, we have

$$N(t)=\frac{1}{\pi}\vartheta(t)+1+S(t).$$

Taking $t=t_n+h$ and assuming that $\gamma_m\le t_n<\gamma_{m+1}$ as $h\to+0$, we obtain

$$m=N(t_n+0)=\frac{1}{\pi}\vartheta(t_n)+1+S(t_n+0)=n+S(t_n+0),$$

whence $S(t_n+0)=m-n=\Delta(n)$. This proves the lemma.

Lemma 2.9. *For any $k\ge1$,*

$$\mathfrak{C}_k=\sum_{\substack{p_1\cdots p_k=q_1\cdots q_k\\ p_1,\dots,p_k,q_1,\dots,q_k\le x}}(p_1\cdots p_kq_1\cdots q_k)^{-1}=k!\left(\mathfrak{S}_1^k-\frac{\theta}{4}k^2\mathfrak{S}_1^{k-2}\right),$$

where $\theta=0$ for $k=1$ and $0\le\theta\le1$ for $k\ge2$.

Proof. For $k=1$, the assertion of the lemma is obvious. For $k\ge2$, the required relation follows from the inequalities

$$k!\left(\mathfrak{S}_1^k-\frac{k^2}{4}\mathfrak{S}_1^{k-2}\right)\le\mathfrak{C}_k\le k!\,\mathfrak{S}_1^k.$$

The upper bound for $\mathfrak{C}_k$ is obtained from assertion (2) of Lemma 1.17 and the equality in Lemma 1.15. To prove the lower bound, note that

$$\mathfrak{C}_k\ge\sum_{\substack{p_1\cdots p_k=q_1\cdots q_k\\ \text{all } p_1,\dots,p_k,q_1,\dots,q_k\le x \text{ are different}}}(p_1\cdots p_kq_1\cdots q_k)^{-1}=k!\sum_{\text{all } p_1,\dots,p_k\le x \text{ are different}}(p_1\cdots p_k)^{-1}$$

$$=k!\left\{\sum_{p_1,\dots,p_k\le x}(p_1\cdots p_k)^{-1}-\sum_{p_1,\dots,p_k\le x}{}'(p_1\cdots p_k)^{-1}\right\},$$

where the prime denotes summation over all k-tuples $(p_1,\dots,p_k)$ containing at least two equal numbers. Let us fix any $(k-2)$-tuple $(p_1,\dots,p_{k-2})$ and a prime $p\le x$. The k-tuples obtained from $(p_1,\dots,p_{k-2})$ by inserting two p's equals $\binom{k}{2}<k^2/2$. Therefore,

$$\sum_{p_1,\dots,p_k\le x}{}'(p_1\cdots p_k)^{-1}<\frac{k^2}{2}\sum_{p_1,\dots,p_{k-2}\le x}(p_1\cdots p_{k-2})^{-1}\sum_{p\le x}p^{-2}<\frac{k^2}{4}\left(\sum_{p\le x}\frac{1}{p}\right)^{k-2}=\frac{k^2}{4}\,\mathfrak{S}_1^{k-2}.$$

Thus,

$$\mathfrak{C}_k \geq k!\biggl(\mathfrak{S}_1^k - \frac{k^2}{4}\,\mathfrak{S}_1^{k-2}\biggr).$$

This completes the proof of the lemma.

To simplify notation, we set $L = \log\log N$.

Together with the expression in Theorem 2.1, we need its simplified version for an even moment of $W(t_n)$, which is more convenient for obtaining an upper bound.

Lemma 2.10. *Let k be an integer such that $1 \leq k \leq L/\log L$, and let $x = X^{1/(8k+3)}$. Then*

$$\sum_n W^{2k}(t_n) = \frac{(2k)!}{k!}\frac{ML^k}{2^{2k}}\biggl(1 - 1.9\theta\frac{k}{L}\log\frac{k}{\varepsilon}\biggr),$$

where $0 \leq \theta \leq 1$.

Proof. By virtue of Lemma 1.4, the sum under consideration equals

$$\binom{2k}{k}\frac{M}{2^{2k}}\,\mathfrak{C}_k + x^{2k}\theta\log N.$$

According to Lemma 2.9, we have

$$\mathfrak{C}_k = k!\biggl(\mathfrak{S}_1^k - \frac{k^2}{4}\,\theta\mathfrak{S}_1^{k-2}\biggr).$$

Applying relation (I.2) in Lemma I.3, we obtain

$$\mathfrak{S}_1 = \sum_{p\leq x}\frac{1}{p} = L + c - \log\frac{10(8k+3)}{\varepsilon} + \frac{\theta_1}{(\log x)^2}\,,$$

where $c = 0.261497\ldots$ and $-0.5 \leq \theta_1 \leq 1$. Since $0 < \varepsilon < 10^{-3}$, it easily follows that

$$11.28 < \log\frac{80k}{\varepsilon} < \log\frac{10(8k+3)}{\varepsilon} < \log\frac{110k}{\varepsilon} < 1.7\log\frac{k}{\varepsilon}.$$

Therefore,

$$\mathfrak{S}_1 < L + 0.27 - 11.28 + \frac{1}{(\log x)^2} < L$$

and

$$\mathfrak{S}_1 > L + 0.26 - \log\frac{110k}{\varepsilon} - \frac{0.5}{(\log x)^2} > L - 1.7\log\frac{k}{\varepsilon},$$

whence

$$\mathfrak{S}_1 = L - 1.7\theta_2\log\frac{k}{\varepsilon}$$

(here and in what follows, $0 \leq \theta_j \leq 1$). Next, the finite increment formula implies

$$\mathfrak{S}_1^k = L^k - 1.7\theta_3 k\biggl(\log\frac{k}{\varepsilon}\biggr)L^{k-1},$$

so that

$$\mathfrak{S}_1^k - \theta_4\frac{k^2}{4}\mathfrak{S}_1^{k-2} = L^k\biggl\{1 - \theta_5\biggl(\frac{1.7k}{L}\log\frac{k}{\varepsilon} + \frac{k^2}{4L^2}\biggr)\biggr\} = L^k\biggl\{1 - \frac{1.7\theta_6 k}{L}\log\frac{k}{\varepsilon}\biggr\}.$$

This proves the required assertion.

Corollary 2.1. *For any integer k such that $1 \le k \le L/\log L$,*

$$\sum_n W^{2k}(t_n) \le \frac{(2k)!}{k!}\frac{ML^k}{2^{2k}}.$$

Theorem 2.1. *Let k be an integer such that $1 \le k \le c_1\sqrt[3]{\log\log N}$, where $c_1 = (1/c_0)(7\pi^2/4)^{1/3}$. Then*

$$\sum_n \Delta^{2k}(n) = \frac{(2k)!}{k!}\frac{M}{(2\pi)^{2k}}(\log\log N)^k\left(1 + \frac{2.4\theta(c_0k)^{3/2}}{\sqrt{\log\log N}}\right). \tag{2.12}$$

Proof. We set

$$x = X^{1/(8k+3)}, \qquad W(t) = -\frac{1}{\pi}\sum_{p\le x}\frac{\sin(t\log p)}{\sqrt{p}}, \qquad \text{and} \qquad U(t) = -\sum_{p\le x}\frac{p^{it}}{\sqrt{p}},$$

so that

$$W(t) = \frac{1}{2\pi i}(U(t) - \overline{U}(t)) \qquad \text{and} \qquad R(t) = S(t) + R(t).$$

By virtue of Lemma 2.8, the sum in the statement of the theorem can be written in the form

$$\sum_n (W(t_n) - R(t_n+0))^{2k} = \sum_{\nu=0}^{2k}(-1)^\nu\binom{2k}{\nu}v_\nu, \qquad v_\nu = \sum_n W^{2k-\nu}(t_n)R^\nu(t_n+0).$$

Lemma 2.10 implies

$$v_0 = \sum_n W^{2k}(t_n) = \mathcal{L}_k\left(1 - 1.9\theta_1\frac{k}{L}\log\frac{k}{\varepsilon}\right),$$

where we use the abbreviated notation

$$\mathcal{L}_k = \frac{(2k)!}{k!}\frac{ML^k}{(2\pi)^{2k}}.$$

It follows from the estimate $v_0 < \mathcal{L}_k$ of the corollary to Lemma 2.10 and Hölder's inequality that, for any ν such that $1 \le \nu \le 2k$, we have

$$|v_\nu| \le \left(\sum_n W^{2k}(t_n)\right)^{1-\nu/(2k)}\left(\sum_n R^{2k}(t_n+0)\right)^{\nu/(2k)} < \mathcal{L}_k^{1-\nu/(2k)}\left(\frac{c_0}{56}(c_0k)^{2k}M\right)^{\nu/(2k)} = \mathcal{L}_k\delta_\nu,$$

where

$$\begin{aligned}\delta_\nu &= \left\{\frac{(2\pi)^{2k}k!}{(2k)!}\frac{c_0}{18}(c_0k)^{2k}\right\}^{\nu/(2k)}L^{-\nu/2} < \left\{(2\pi)^{2k}\frac{2}{e}\left(\frac{e}{4k}\right)^k\frac{c_0}{56}(c_0k)^{2k}\right\}^{\nu/(2k)}L^{-\nu/2}\\ &\le \left\{\left(\frac{c_0}{28e}\right)^{1/k}\frac{\pi^2ec_0^2k}{L}\right\}^{\nu/2} \le \left\{\frac{\pi^2}{28}\frac{c_0^3k}{L}\right\}^{\nu/2}.\end{aligned}$$

Therefore,

$$\binom{2k}{\nu}\delta_\nu \le (2k)^\nu\delta_\nu \le \left\{\frac{\pi^2}{7}\frac{(c_0k)^3}{L}\right\}^{\nu/2}.$$

Note that the quantity in braces does not exceed $1/4$; hence,

$$\left|\sum_{\nu=1}^{2k}(-1)^\nu\binom{2k}{\nu}v_\nu\right| \le \sum_{\nu=1}^{2k}\mathcal{L}_k\binom{2k}{\nu}\delta_\nu < 2\mathcal{L}_k\left\{\frac{4\pi^2}{7}\frac{(c_0k)^3}{L}\right\}^{1/2} = \mathcal{L}_k\frac{2\pi}{\sqrt{7}}\frac{(c_0k)^{3/2}}{\sqrt{L}},$$

$$\sum_n \Delta^{2k}(n) = v_0 + \theta\mathcal{L}_k\frac{2\pi}{\sqrt{7}}\frac{(c_0k)^{3/2}}{\sqrt{L}}.$$

Replacing v_0 by expression (2.12), we obtain

$$\sum_n \Delta^{2k}(n) = \mathcal{L}_k(1+\theta\delta),$$

where

$$\delta = \frac{2\pi}{\sqrt{7}}\frac{(c_0k)^{3/2}}{\sqrt{L}} + \frac{1.9k}{L}\log\frac{k}{\varepsilon} < \frac{(c_0k)^{3/2}}{\sqrt{L}}\left(\frac{2\pi}{\sqrt{7}} + \frac{1.9\cdot 2880^{-1.5}\varepsilon}{\sqrt{kL}}\frac{\log(k/\varepsilon)}{\sqrt{k/\varepsilon}}\right)$$
$$< \frac{(c_0k)^{3/2}}{\sqrt{L}}\left(\frac{2\pi}{\sqrt{7}} + \frac{10^{-8}}{\sqrt{L}}\right) < 2.4\frac{(c_0k)^{3/2}}{\sqrt{L}}.$$

This completes the proof of the theorem.

Theorem 2.2. *Let k be an integer such that $1 \le k \le c_1\sqrt[3]{\log\log N}$. Then*

$$\left|\sum_n \Delta^{2k-1}(n)\right| < \frac{c_0^{3/2}}{3}\frac{(2k)!}{(k-1)!}\frac{M}{(2\pi)^{2k-2}}(\log\log N)^{k-1}.$$

Proof. In the notation of Theorem 2.1, we have

$$\left|\sum_n \Delta^{2k-1}(n)\right| \le \sum_{\nu=0}^{2k-1}\binom{2k-1}{\nu}|w_\nu|, \qquad w_\nu = \sum_n W^{2k-\nu-1}(t_n)R^\nu(t_n+0).$$

By virtue of Lemma 1.3,

$$w_0 = \sum_n W^{2k-1}(t_n) = \frac{1}{(2\pi i)^{2k-1}}\sum_n(\overline{U}(t_n) - U(t_n))^{2k-1}$$
$$= \frac{1}{(2\pi i)^{2k-1}}\sum_{\nu=0}^{2k-1}(-1)^{\nu-1}\binom{2k-1}{\nu}\sum_n \overline{U}^{\nu}(t_n)U^{2k-\nu-1}(t_n)$$
$$= \frac{\theta}{(2\pi)^{2k-1}}\sum_{\nu=0}^{2k-1}\binom{2k-1}{\nu}\sqrt[4]{X} = \frac{\theta_1\sqrt[4]{X}}{\pi^{2k-1}}.$$

In the case $1 \le \nu \le 2k-1$, twice applying Hölder's inequality, the estimate in Corollary 2.1, and Lemma 2.7, we obtain

$$|w_\nu| \le \sum_n |W(t_n)|^{2k-\nu-1}|R(t_n+0)|^\nu \le v_0^{1-\nu+1/(2k)}\left(\sum_n |R(t_n+0)|^{2k\nu\nu+1}\right)^{\nu+1/(2k)}$$
$$\le v_0^{1-\nu+1/(2k)}\left(v_{2k}^{\nu/\nu+1}M^{1/\nu+1}\right)^{\nu+1/(2k)} \le \mathcal{L}_k^{1-\nu+1/(2k)}\left(\frac{c_0}{18}(c_0k)^{2k}M\right)^{\nu/(2k)}M^{1/(2k)} = \mathcal{L}_k\delta_\nu,$$

where

$$\delta_\nu = \mathcal{L}_k^{-(\nu+1)/(2k)}\left(\frac{c_0}{56}(c_0k)^{2k}M\right)^{\nu/(2k)}M^{1/(2k)} = \left\{\frac{(2\pi)^{2k}\,k!}{(2k)!}\right\}^{(\nu+1)/(2k)}\left(\frac{c_0}{56}(c_0k)^{2k}\right)^{\nu/(2k)}L^{-\nu+1/2}$$
$$\le \left\{\frac{c_0}{28e}\left(\frac{e}{4k}\cdot 4\pi^2c_0^2k^2\right)^k\right\}^{\nu/(2k)}\left\{\frac{2}{e}\left(\frac{e}{4k}\cdot 4\pi^2\right)^k\right\}^{1/(2k)}L^{-\nu+1/2}$$
$$= \left\{\left(\frac{c_0}{28e}\right)^{1/k}\cdot\frac{\pi^2ec_0^2k}{L}\right\}^{\nu/2}\left\{\left(\frac{2}{e}\right)^{1/k}\frac{\pi^2e}{kL}\right\}^{1/2}$$
$$\le \left\{\frac{c_0}{28e}\frac{\pi^2ec_0^2k}{L}\right\}^{\nu/2}\left\{\frac{\pi^2e}{kL}\right\}^{1/2} = \frac{\pi\sqrt{e}}{\sqrt{kL}}\left\{\frac{\pi^2}{28}\frac{c_0^3k}{L}\right\}^{\nu/2}.$$

Thus,

$$\binom{2k-1}{\nu}\delta_\nu < (2k)^\nu\delta_\nu \le \frac{\pi\sqrt{e}}{\sqrt{kL}}\left\{\frac{\pi^2}{7}\frac{(c_0k)^3}{L}\right\}^{\nu/2},$$

$$\sum_{\nu=1}^{2k-1}\binom{2k-1}{\nu}\delta_\nu < \frac{2\pi\sqrt{e}}{\sqrt{kL}}\left\{\frac{\pi^2}{7}\frac{(c_0k)^3}{L}\right\}^{1/2} = 2\pi^2\sqrt{\frac{e}{7}}\frac{c_0^{3/2}k}{L},$$

whence

$$\sum_{\nu=0}^{2k-1}\binom{2k-1}{\nu}|w_\nu| < \frac{\sqrt[4]{X}}{\pi^{2k-1}} + 2\mathcal{L}_k\pi^2\sqrt{\frac{e}{7}}\frac{c_0^{3/2}k}{L}$$

$$= \frac{\sqrt{X}}{\pi^{2k-1}} + c_0^{3/2}\sqrt{\frac{e}{28}}\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}} < \frac{c_0^{3/2}}{3}\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}.$$

This completes the proof of the theorem.

2.8. Fractional moments of $\Delta(n)$. In this section, we calculate a fractional moment of $|\Delta(n)|$, preliminarily approximating it by a fractional moment of $|W_x(t_n)|$. To avoid applying Hölder's inequality to estimate the accuracy of such an approximation, we precede Theorem 2.3 by Lemma 2.11, which gives an estimate of the difference $|x+y|^a - |x|^a$ for noninteger a.

Lemma 2.11. *Let $a > 0$. Then, for any real x and y, the absolute value of the difference $|x+y|^a - |x|^a$ is at most $|y|^a$ if $0 < a \le 1$; if $a > 1$, then it does not exceed $t_1 + 3t_2 + 9t_3$, where*

$$t_1 = \begin{cases} 0 & \text{if } k = 0, \\ \displaystyle\sum_{\nu=1}^{k}\binom{k}{\nu}|x|^{a-\nu}|y|^\nu & \text{if } k \ge 1, \end{cases}$$

$$t_2 = \sum_{\nu=0}^{k}\binom{k}{\nu}|x|^{a-\nu-1}|y|^{\nu+1}, \qquad t_3 = \sum_{\nu=1}^{k}\binom{k}{\nu}|x|^{a-\nu-b}|y|^{\nu+b},$$

and k and b are numbers such that $a = k + b$, $k \ge 0$ is integer, and $1 \le b \le 2$.

Proof. In the case $0 < a \le 1$, the required assertion is obvious. Suppose that $a = k + b > 1$. Note that, for any λ satisfying the condition $|\lambda| \le 1/2$, we have

$$(1+\lambda)^b = 1 + 3\theta_1|\lambda|.$$

Therefore, if $1 \le |y| \le |x|/2$, $x \ne 0$, then

$$|x+y|^b = |x|^b\left|1+\frac{y}{x}\right|^b = |x|^b\left(1+3\theta_2\left|\frac{y}{x}\right|\right) = |x|^b + 3\theta_2|x|^{b-1}|y|, \tag{2.13}$$

and if $0 \le |x|/2 < y$, then

$$|x+y|^b = |x|^b + (|x+y|^b - |x|^b) = |x|^b + \theta_3(3|y|)^b = |x|^b + 9\theta_4|y|^b. \tag{2.14}$$

Combining (2.13) and (2.14), we obtain

$$|x+y|^b = |x|^b + \theta_5(3|x|^{b-1}|y| + 9|y|^b). \tag{2.15}$$

Next,

$$|x+y|^a = |(x+y)^k|\cdot|x+y|^b = \left|\sum_{\nu=0}^{k}\binom{k}{\nu}x^{k-\nu}y^\nu\right|\cdot|x+y|^b$$

$$= \left(|x|^k + \theta_6\sum_{\nu=1}^{k}\binom{k}{\nu}|x|^{a-b-\nu}|y|^\nu\right)\cdot|x+y|^b.$$

Using (2.15), we finally obtain

$$|x+y|^a = \left(|x|^{a-b} + \theta_6 \sum_{\nu=1}^{k} \binom{k}{\nu} |x|^{a-b-\nu}|y|^\nu\right) \cdot (|x|^b + \theta_5(3|x|^{b-1}|y| + 9|y|^b))$$
$$= |x|^a + \theta_6 \sum_{\nu=1}^{k} \binom{k}{\nu} |x|^{a-\nu}|y|^\nu + \theta_7(|x|^b + 3|x|^{b-1}|y| + 9|y|^b) \sum_{\nu=0}^{k} \binom{k}{\nu} |x|^{a-b-\nu}|y|^\nu,$$

which implies the assertion of the lemma.

Theorem 2.3. *Let a be any real number not being an even integer and such that $0 < a \le c_2 \sqrt[3]{\log\log N}$, where $c_2 = (11^{2/3}c_0)^{-1}$. Then*

$$\sum_n |\Delta(n)|^a = \frac{M}{\pi^a\sqrt{\pi}} \Gamma\left(\frac{a+1}{2}\right) (\log\log N)^{a/2}(1 + \theta r(a)),$$

where

$$r(a) = \begin{cases} 14\left(\dfrac{\log\log\log N}{\log\log N}\right)^{a/2} & \text{if } 0 < a \le 1, \\ \dfrac{781a}{\sqrt{\log\log N}}(\sqrt{\log\log\log N} + 2.2c_0^{3/2}\sqrt{a}) & \text{if } a > 1. \end{cases}$$

Remark 2.3. In 2011, Boyarinov obtained a more accurate estimate of the remainder term in the case $a = 1$ by the method of Selberg and Tsang (see Theorem 6 in [44]). The estimates of $r(a)$ with $0 < a < 1$ can be improved further by using results of [39, 40].

Proof of Theorem 2.3. First, consider the case $0 < a \le 1$. Setting

$$y = X^{1/11}, \qquad W(t) = -\frac{1}{\pi} \sum_{p \le y} \frac{\sin(t\log p)}{\sqrt{p}}, \qquad \text{and} \qquad R(t) = S(t) - W(t) \tag{2.16}$$

and applying Lemmas 2.8 and 2.11, we obtain

$$|\Delta(n)|^a = |W(t_n) - R(t_n + 0)|^a = |W(t_n)|^a + \theta_1|R(t_n + 0)|^a,$$

whence

$$\sum_n |\Delta(n)|^a = W_a + \theta_2 R_a,$$

where

$$W_a = \sum_n |W(t_n)|^a \qquad \text{and} \qquad R_a = \sum_n |R(t_n + 0)|^a.$$

By virtue of Hölder's inequality and Lemma 2.7, we have

$$R_a \le M^{1-a/2}R_2^{a/2} \le M^{1-a/2}\left\{\frac{c_0^3}{57}M\right\}^{a/2} < c_0^{3a/2}M.$$

Suppose that

$$x = \exp\left(\frac{\log N}{L^3}\right), \qquad U(t) = \frac{1}{\pi} \sum_{x < p \le y} \frac{\sin(t\log p)}{\sqrt{p}}, \qquad \text{and} \qquad V(t) = \frac{1}{\pi} \sum_{p \le x} \frac{\sin(t\log p)}{\sqrt{p}}. \tag{2.17}$$

Then $W_a = V_a + \theta_3 U_a$, where

$$V_a = \sum_n |V(t_n)|^a \qquad \text{and} \qquad U_a = \sum_n |U(t_n)|^a.$$

Again applying Hölder's inequality, we obtain

$$U_a \le M^{1-a/2}U_2^{a/2}.$$

Lemma I.3 and the estimate of Lemma 1.2 lead to the chain of inequalities

$$U_2 = \frac{1}{\pi^2}\sum_n\biggl|\operatorname{Im}\sum_{x<p\le y}\frac{p^{it_n}}{\sqrt{p}}\biggr|^2 \le \frac{1}{\pi^2}\sum_n\biggl|\sum_{x<p\le y}\frac{p^{it_n}}{\sqrt{p}}\biggr|^2 = \frac{1}{\pi^2}\biggl(M\sum_{x<p\le y}\frac{1}{p}+\sum_{\substack{p,q\le x\\ p\ne q}}\frac{1}{\sqrt{pq}}\sum_n\biggl(\frac{p}{q}\biggr)^{it_n}\biggr)$$

$$\le \frac{1}{\pi^2}\biggl(M\biggl(\log\log y-\log\log x+\frac{1.5}{(\log x)^2}\biggr)+(\log N)\sum_{p,q\le x}\frac{1}{\sqrt{pq}}\sqrt{pq}\biggr) < \frac{3M}{\pi^2}\log L.$$

Therefore,

$$U_a < \frac{M}{\pi^a}(3\log L)^{a/2}.$$

Theorem 1.2 gives the following representation for the sum V_a:

$$V_a = \frac{M}{\pi^a\sqrt{\pi}}\Gamma\biggl(\frac{a+1}{2}\biggr)\mathfrak{S}_1^{a/2}(1+r_1(a)),$$

$$r_1(a) = \sum_{n=2}^{+\infty}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}\frac{\Gamma(n-a/2)}{\Gamma(a/2)}+10.67\theta\biggl(5\pi e\sqrt{\frac{\log x}{\log N}}\,\biggr)^a.$$

Setting $m=1$ in Lemma 1.22, we conclude that the absolute value of the sum over n does not exceed

$$0.32\Gamma\biggl(\frac{3}{2}\biggr)g(2) = 0.16\sqrt{\pi}\biggl(\frac{\log\log 2+1}{\mathfrak{S}_1}\biggr)^2 < \frac{1}{8\mathfrak{S}_1^2},$$

whence

$$|r_1(a)| < \frac{1}{8\mathfrak{S}_1^2}+10.67\biggl(\frac{5\pi e}{L^{3/2}}\biggr)^a < \frac{10.8}{L^a}.$$

Passing to W_a, we obtain

$$W_a = \frac{M}{\pi^a\sqrt{\pi}}\Gamma\biggl(\frac{a+1}{2}\biggr)\mathfrak{S}_1^{a/2}(1+r_2(a)),$$

where

$$|r_2(a)| \le \frac{10.8}{L^a}+\frac{\sqrt{\pi}}{\Gamma((a+1)/2)}\biggl(\frac{\log L}{\mathfrak{S}_1}\biggr)^{a/2} < 12\biggl(\frac{\log L}{\mathfrak{S}_1}\biggr)^{a/2}.$$

Finally, for the initial sum, we have

$$\sum_n|\Delta(n)|^a = \frac{M}{\pi^a\sqrt{\pi}}\Gamma\biggl(\frac{a+1}{2}\biggr)L^{a/2}(1+r(a)),$$

where

$$|r(a)| < \biggl(\frac{\mathfrak{S}_1}{L}\biggr)^{a/2}\biggl\{12\biggl(\frac{\log L}{\mathfrak{S}_1}\biggr)^{a/2}+\frac{\sqrt{\pi}}{\Gamma((a+1)/2)}\biggl(\frac{\pi c_0^{3/2}}{\sqrt{\mathfrak{S}_1}}\biggr)^a\biggr\}$$

$$\le \frac{12+\sqrt{\pi}}{L^a}((\log L)^{a/2}+(\pi c_0^{3/2})^a) < 14\biggl(\frac{\log L}{L}\biggr)^{a/2}.$$

Now, suppose that $a>1$. Let us represent a as the sum $k+b$ in which $k\ge 0$ is integer, $1\le b<2$ if $k=2m$, and $1<b<2$ if $k=2m-1$. We set

$$y = X^{1/\mu} \qquad \text{and} \qquad \mu = 8(m+1)^2+3$$

and define $W(t)$ and $R(t)$ by (2.17). Applying Lemma 2.11 to the initial sum, we obtain

$$\sum_n |\Delta(n)|^a = \sum_n |W(t_n) + R(t_n + 0)|^a = W_a + \theta(u + 3v + 9w),$$

where

$$u = \sum_{\nu=1}^{k} \binom{k}{\nu} u_\nu, \qquad v = \sum_{\nu=0}^{k} \binom{k}{\nu} v_\nu, \qquad w = \sum_{\nu=0}^{k} \binom{k}{\nu} w_\nu, \tag{2.18}$$

and each of the sums u_ν, v_ν, and w_ν has the form

$$t_{\nu,\delta} = \sum_n |W(t_n)|^{a-\nu-\delta} |R(t_n + 0)|^{\nu+\delta},$$

where δ takes the values 0, 1, and b, respectively.

It follows from the definition of the numbers m and y that

$$\frac{\log N}{\log y} = \frac{10\mu}{\varepsilon} > \frac{10}{\varepsilon} 8 \left(\frac{a}{2}\right)^2 = \frac{20a^2}{\varepsilon},$$

whence

$$a < \frac{1}{2}\sqrt{\frac{\varepsilon}{5}}\sqrt{\frac{\log N}{\log y}} < \frac{1}{3e}\sqrt{\frac{\log N}{\log y}}.$$

Therefore, for the sum W_a, the relation of Theorem 1.2 holds. Applying Lemma 1.22 with $m = 1$, we obtain

$$W_a = \frac{M}{\pi^a \sqrt{\pi}} \Gamma\left(\frac{a+1}{2}\right) \mathfrak{S}_1^{a/2} (1 + r_1(a)),$$

where

$$\begin{aligned} |r_1(a)| &\le \left| \sum_{n=2}^{+\infty} (-1)^n \frac{\Phi_n}{\mathfrak{S}_1^n} \frac{\Gamma(n - a/2)}{\Gamma(a/2)} \right| + 10.67\theta \left(5\pi e \sqrt{\frac{a \log x}{\log N}}\right)^a \\ &< \frac{1}{8\mathfrak{S}_1^2} + 10.67 \left\{ \frac{\pi e}{2} \sqrt{\frac{5\varepsilon}{a}} \right\}^a < \frac{1}{8\mathfrak{S}_1^2} + 10.67 \frac{\pi e}{2} \sqrt{5\varepsilon} < \frac{13}{4}. \end{aligned}$$

Thus,

$$W_a < I_a, \qquad I_a = \frac{17}{4} \frac{M}{\pi^a \sqrt{\pi}} \Gamma\left(\frac{a+1}{2}\right) \mathfrak{S}_1^{a/2}. \tag{2.19}$$

Applying Hölder's inequality, Lemma 2.7, and the inequality $m + 1 < a/2 + 1$, we obtain

$$R_a \le M^{1-a/(2(m+1))} R_{2(m+1)}^{a/(2(m+1))} \le M^{1-a/2(m+1)} \{c_0 (c_0 (m+1))^{2(m+1)}\}^{a/2(m+1)} < J_a, \tag{2.20}$$

$$J_a = \left\{ c_0^{3/2} \left(\frac{a}{2} + 1\right) \right\}^a M.$$

Again applying Hölder's inequality together with estimates (2.19) and (2.20), we arrive at

$$\begin{aligned} t_{\nu,\delta} &\le \left\{ \sum_n |W(t_n)|^{(a-\nu-\delta)a/(a-\nu-\delta)} \right\}^{1-(\nu+\delta)/a} \left\{ \sum_n |R(t_n + 0)|^{(\nu+\delta)a/(\nu+\delta)} \right\}^{(\nu+\delta)/a} \\ &\le I_a^{1-(\nu+\delta)/a} J_a^{(\nu+\delta)/a} = I_a q^{\nu+\delta}, \end{aligned}$$

where $q = (J_a/I_a)^{1/a}$. Since $\Gamma((a+1)/2) > 2(a/2e)^{a/2}$ for all $a \ge 1$, it follows that

$$q^a = \frac{4\sqrt{\pi}}{17} \left\{ \frac{\pi c_0^{3/2}}{\sqrt{\mathfrak{S}_1}} \left(\frac{a}{2} + 1\right) \right\}^a \Gamma^{-1}\left(\frac{a+1}{2}\right) < \frac{2\sqrt{\pi}}{17} \left\{ \frac{\pi \sqrt{e} c_0^{3/2}}{\sqrt{\mathfrak{S}_1}} \left(\sqrt{\frac{a}{2}} + \sqrt{\frac{2}{a}}\right) \right\}^a < \left(11 c_0^{3/2} \sqrt{\frac{a}{L}}\right)^a,$$

so that

$$q < 11c_0^{3/2}\sqrt{\frac{a}{L}}\,.$$

The inequality $a < (11^{2/3}c_0)^{-1}\sqrt[3]{L}$ easily implies $q < a^{-1}$. Therefore,

$$u \le \sum_{\nu=1}^{k}\binom{k}{\nu}I_a q^\nu = I_a\{(1+q)^k - 1\} < I_a kq(1+q)^{k-1} < I_a aq\left(1+\frac{1}{a}\right)^a < I_a aeq,$$

$$v, w \le \sum_{\nu=0}^{k}\binom{k}{\nu}I_a q^{\nu+1} = I_a q(1+q)^k < I_a eq,$$

whence

$$u + 3v + 3w < I_a(a+12)eq < 13I_a aeq < \frac{1652.1M}{\pi^a\sqrt{\pi}}\,\Gamma\left(\frac{a+1}{2}\right)\mathfrak{S}_1^{a/2}\frac{(ac_0)^{3/2}}{\sqrt{L}}\,. \tag{2.21}$$

Defining x, $U(t)$, and $V(t)$ by (2.17) and again applying Lemma 2.11, we arrive at the relation

$$W_a = \sum_n |V(t_n) + U(t_n)|^a = V_a + \theta(U + 3V + 9W),$$

in which U, V and W are defined as u, v, and w in (2.18), but the role of u_ν, v_ν, and w_ν is played by the sums

$$T_{\nu,\delta} = \sum_n |V(t_n)|^{a-\nu-\delta}|W(t_n)|^{\nu+\delta}, \qquad \delta = 0, 1, b.$$

Repeating the proof of the estimate U_2 obtained above almost word for word, we see that, for any integer $r \ge 1$ satisfying the condition $x^{2r}\log N < \sqrt{M}$, we have

$$\begin{aligned} U_{2r} = \sum_n U^{2r}(t_n) &\le \frac{1}{\pi^{2r}}\sum_n\left|\sum_{x<p\le y}\frac{p^{it_n}}{\sqrt{p}}\right|^{2r} \\ &\le \frac{1}{\pi^{2r}}\left\{r!\,M\left(\sum_{x<p\le y}\frac{1}{p}\right)^r + x^{2r}\log N\right\} \le \frac{1}{\pi^{2r}}(r!\,M(3\log L)^r + \sqrt{M}) < 3\sqrt{r}\left(\frac{3r}{\pi^2 e}\log L\right)^r M, \end{aligned}$$

which, in turn, implies the estimate

$$U_{2r}^{1/(2r)} \le \frac{3}{\pi\sqrt{e}}\,M^{1/(2r)}\sqrt{r\log L}\,. \tag{2.22}$$

In the case $r = m + 1 \le a/2 + 1$, inequality (2.22) gives

$$U_a \le M^{1-a/(2(m+1))}U_{2(m+1)}^{a/(2(m+1))} \le M\left\{\frac{3}{\pi\sqrt{e}}\sqrt{(m+1)\log L}\right\}^a \le \frac{M}{\pi^a}\left\{\frac{9}{e}\left(\frac{a}{2}+1\right)\log L\right\}^{a/2} = K_a.$$

Estimating $T_{\nu,\delta}$ in the same way as above, we obtain

$$U < I_a aeQ \qquad \text{and} \qquad V, W < I_a eQ,$$

where

$$Q = \left(\frac{K_a}{I_a}\right)^{1/a}.$$

Next, it follows from the definition of I_a and K_a that

$$Q^a = \frac{K_a}{I_a} = \frac{4\sqrt{\pi}}{17}\left\{\frac{9}{e}\left(\frac{a}{2}+1\right)\frac{\log L}{L}\right\}^{a/2}\frac{1}{\Gamma((a+1)/2)}$$

$$< \frac{2\sqrt{\pi}}{17}\left\{\frac{9}{e}\left(\frac{a}{2}+1\right)\frac{2e}{a}\frac{\log L}{L}\right\}^{a/2} < \left(\frac{27\log L}{L}\right)^{a/2},$$

whence

$$Q < 3\sqrt{\frac{3\log L}{L}}.$$

Therefore,

$$U+3V+9W < I_a(a+12)eQ < \frac{M}{\pi^a\sqrt{\pi}}\Gamma\left(\frac{a+1}{2}\right)\mathfrak{S}_1^{a/2}r_3(a),$$

where

$$r_3(a) = \frac{221}{4}aeQ < \frac{663e}{4}a\sqrt{\frac{3\log L}{L}} < 780.4a\sqrt{\frac{\log L}{L}}. \tag{2.23}$$

Again using Theorem 1.2 and applying inequalities (2.21) and (2.23), we obtain

$$\sum_n |\Delta(n)|^a = \frac{M}{\pi^a\sqrt{\pi}}\Gamma\left(\frac{a+1}{2}\right)L^{a/2}(1+r(a)),$$

where

$$\begin{aligned}|r(a)| &\le \left|\sum_{n=2}^{+\infty}(-1)^n\frac{\Phi_n}{\mathfrak{S}_1^n}\frac{\Gamma(n-a/2)}{\Gamma(a/2)}\right| + 10.67\theta\left(5\pi e\sqrt{\frac{a\log x}{\log N}}\right)^a\\ &\quad + \left(\frac{L}{\mathfrak{S}_1}\right)^{a/2}\left\{780.4a\sqrt{\frac{\log L}{L}} + 1652.1\frac{(ac_0)^{3/2}}{\sqrt{L}}\right\}\\ &< \left(\frac{L}{\mathfrak{S}_1}\right)^{a/2}\left\{\frac{1}{8\mathfrak{S}_1^2} + 10.67\left(5\pi e\sqrt{\frac{c_1\sqrt[3]{L}}{L^3}}\right)^a + 780.4a\sqrt{\frac{\log L}{L}} + 1652.1\frac{(ac_0)^{3/2}}{\sqrt{L}}\right\}\\ &< \frac{781a}{\sqrt{L}}(\sqrt{\log L} + 2.2\sqrt{a}\,c_0^{3/2}).\end{aligned}$$

This completes the proof of the theorem.

2.9. Distribution of $\Delta(n)$. Consider the probability space (Ω, P), where $\Omega = \{n \mid N < n \le N+M\}$ is the set of all integers in the interval $(N, N+M]$ and the measure P is defined by $\mathsf{P}(\mathcal{A}) = |\mathcal{A}|/M$ for any $\mathcal{A} \subset \Omega$. Let $\delta\colon \Omega \mapsto \mathbb{R}$ be a random variable taking the values $\delta(n) = \pi\Delta(n)\sqrt{2/L}$, $N < n \le N+M$, and let

$$\Phi(u) = P(\{n : n \in \Omega,\ \delta(n) \le u\})$$

be its distribution function. Then the number $\nu(u)$ of those n for which $\delta(n) \le u$ equals $M\Phi(u)$.

In this section, we prove two theorems on the distribution function of δ, in which $\Phi(u)$ is approximated by the Gaussian distribution

$$G(u) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{u} e^{-v^2/2}\,dv.$$

Theorem 2.4. *For any u, the following relation holds:*

$$\nu(u) = M\left(G(u) + \frac{1.78\theta}{\sqrt{\log\log\log N}}\right).$$

Remark 2.4. After the author published Theorem 2.4, there appeared Boyarinov's paper [44] (see also [40, Chapter 3]) containing a much more accurate estimate of the remainder term in the expression for $\nu(u)$, namely,

$$O\left(\frac{\log\log\log N}{\sqrt{\log\log N}}\right) \qquad \text{instead of} \qquad O\left(\frac{1}{\sqrt{\log\log\log N}}\right).$$

Proof of Theorem 2.4. The characteristic function $f(t)$ of the distribution of $\Phi(u)$ has the form

$$f(t)=\mathsf{E}(e^{it\delta})=\frac{1}{M}\sum_n e^{it\delta(n)}. \tag{2.24}$$

Choose an integer $P\le c_1\sqrt[3]{L}$. Transforming expression (2.24) and applying Theorems 2.1 and 2.2, we successively obtain

$$\begin{aligned}f(t)&=\frac{1}{M}\sum_n\biggl\{\sum_{k=0}^P(-1)^k\frac{(t\delta(n))^{2k}}{(2k)!}+i\sum_{k=0}^P(-1)^k\frac{(t\delta(n))^{2k+1}}{(2k+1)!}+2\theta_1\frac{(t\delta(n))^{2P+2}}{(2P+2)!}\biggr\}\\&=\sum_{k=0}^P\frac{(-1)^k}{k!}\biggl(\frac{t^2}{2}\biggr)^k\biggl(1+2.4\theta_2\frac{(c_0k)^{3/2}}{\sqrt{L}}\biggr)+\frac{\pi\sqrt{2}}{3}\frac{\theta_3c_0}{\sqrt{L}}\sum_{k=0}^P\frac{2(k+1)}{k!}\frac{|t|^{2k+1}}{2^k}\\&\quad+\frac{2\theta_4}{(P+1)!}\biggl(\frac{t^2}{2}\biggr)^{P+1}\biggl(1+2.4\frac{(c_0(P+1))^{3/2}}{\sqrt{L}}\biggr).\end{aligned}$$

Let $f_1(t)$ denote the contribution of the odd moments, and let $f_2(t)$ be that of the remainder terms in the corresponding asymptotic relations; then

$$\begin{aligned}f(t)&=\sum_{k=0}^P\frac{1}{k!}\biggl(-\frac{t^2}{2}\biggr)^k+\frac{2\theta_5}{(P+1)!}\biggl(\frac{t^2}{2}\biggr)^{P+1}+f_1(t)+f_2(t)\\&=g(t)+\frac{3\theta_6}{(P+1)!}\biggl(\frac{t^2}{2}\biggr)^{P+1}+f_1(t)+f_2(t),\end{aligned}$$

where $g(t)=e^{-t^2/2}$ is the characteristic function of the Gaussian distribution. We have

$$|f_1(t)|\le\frac{2\pi\sqrt{2}}{3\sqrt{L}}c_0^{3/2}\sum_{k=0}^P\frac{(k+1)}{k!\,2^k}|t|^{2k+1}\qquad\text{and}\qquad|f_2(t)|\le\frac{4.8}{\sqrt{L}}c_0^{3/2}\sum_{k=1}^{P+1}\frac{k^{3/2}}{k!}\biggl(\frac{t^2}{2}\biggr)^k.$$

Now, setting $\delta=10^{-3}$, $\lambda=(1-\delta)\sqrt{\log L}$, and $P=[e\lambda^2]+1$, we obtain

$$\begin{aligned}I(\lambda)&=\int_0^\lambda\biggl|\frac{f(t)-g(t)}{t}\biggr|\,dt\\&\le\frac{3}{(P+1)!}\frac{\lambda^{2P+2}}{2^{P+1}(2P+2)}+\frac{2\pi\sqrt{2}}{3\sqrt{L}}c_0^{3/2}\sum_{k=0}^{P+1}\frac{k+1}{2k+1}\frac{\lambda^{2k+1}}{2^k\,k!}+\frac{4.8}{\sqrt{L}}c_0^{3/2}\sum_{k=1}^P\frac{k^{3/2}}{k!}\frac{\lambda^{2k}}{2k\cdot2^k}\\&<\frac{1.5}{P^2}\frac{1}{P!}\biggl(\frac{\lambda^2}{2}\biggr)^{P+1}+\frac{2\pi\sqrt{2}}{3\sqrt{L}}c_0^{3/2}\lambda\sum_{k=0}^P\frac{1}{k!}\biggl(\frac{\lambda^2}{2}\biggr)^k+\frac{2.4}{\sqrt{L}}c_0^{3/2}\sum_{k=1}^{P+1}\frac{\sqrt{k}}{k!}\biggl(\frac{\lambda^2}{2}\biggr)^k\\&<\frac{3\lambda^2}{4\sqrt{2\pi}\,P^2\sqrt{P}}\biggl(\frac{e\lambda^2}{2P}\biggr)^P+\frac{3c_0^{3/2}}{\sqrt{L}}(\lambda+\sqrt{P+1})\sum_{k=0}^{P+1}\frac{1}{k!}\biggl(\frac{\lambda^2}{2}\biggr)^k\\&<\frac{2^{-P}}{40\lambda^2}+8c_0^{3/2}\sqrt{\frac{\log L}{L}}\cdot L^{0.5(1-\delta)^2}<L^{-0.5\delta}.\end{aligned}$$

Lemma I.7 with

$$G(u)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^u e^{-v^2/2}\,dv,\qquad B=\sup_u|G'(u)|=\frac{1}{\sqrt{2\pi}},\qquad\text{and}\qquad g(t)=e^{-t^2/2}$$

implies

$$\sup_u|\Phi(u)-G(u)|\le1.771\lambda^{-1}+0.298I(\lambda)<\frac{1.78}{\sqrt{\log L}}\qquad\text{and}\qquad\Phi(u)=G(u)+\frac{1.78\theta}{\sqrt{\log L}}.$$

This completes the proof of the theorem.

We set

$$y = \exp\left(\frac{\log N}{220L}\right) \qquad \text{and} \qquad \mathfrak{S} = \sum_{p \le y} \frac{1}{p} = L - \log L + O(1).$$

To approximate $\Phi(u)$ at large $|u|$ by a normal distribution, we proceed as follows. Let $\xi\colon \Omega \mapsto \mathbb{R}$ be a discrete random variable taking the values $W_y(t_n)$, $N < n \le N + M$, and let

$$F(u) = \frac{1}{M} \#\left\{ n \;\middle|\; N < n \le N + M,\, W_y(t_n) \le \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}} \right\}$$

be its distribution function. It turns out that, for appropriate y, the function $F(u)$ is close to $\Phi(u)$. We replace the distribution $F(u)$ by a close distribution function $F_{\mathcal{A}}(u)$ of a random variable $\eta\colon \Omega \mapsto \mathbb{R}$ taking the values $W_y(t_n)$, $n \in \mathcal{A}$. The set $\mathcal{A}$ is obtained from Ω by removing all numbers n for which $|W_y(t_n)|$ is abnormally large. The constraint $n \in \mathcal{A}$ allows us to construct the moment generating function

$$K(\eta; s) = \mathsf{E}(e^{s\eta})$$

for η and apply Lemma I.8. The subsequent transitions from $F_{\mathcal{A}}(u)$ to $F(u)$ and from $F(u)$ to $\Phi(u)$ involve no difficulties.

Lemma 2.12. *Let $\delta_1 = 4e^{-\pi^2\mathfrak{S}}$. Then, for all n such that $N < n \le N + M$, except at most $M\delta_1$ of them, the following inequality holds*

$$|W_y(t_n)| \le \mathfrak{S}.$$

Proof. Let $\mathcal{A}_1$ denote the set of those n for which $|W_y(t_n)| > \mathfrak{S}$. Setting $k_1 = [\pi^2\mathfrak{S}]$ and taking into account the first remark to Lemma 2.1, we obtain

$$|\mathcal{A}_1|\mathfrak{S}^{2k_1} \le \sum_{n \in \mathcal{A}_1} W^{2k_1}(t_n) \le \sum_n W^{2k_1}(t_n) \le \frac{(2k_1)!}{k_1!}\frac{M}{(2\pi)^{2k_1}}\left(\sum_{p \le y}\frac{1}{p}\right)^{k_1} \le M\sqrt{2}\left(\frac{4k_1}{e}\frac{\mathfrak{S}}{4\pi^2}\right)^{k_1},$$

whence

$$|\mathcal{A}_1| \le M\sqrt{2}\left(\frac{k_1}{\pi^2\mathfrak{S}e}\right)^{k_1} < M\sqrt{2}e^{-k_1} < 4Me^{-\pi^2\mathfrak{S}},$$

as required.

Lemma 2.13. *For any $u \ge 1$,*

$$F(-u) = G(-u)\left(1 + O\left(\frac{u}{\sqrt{\mathfrak{S}}}\right)\right) + 2\theta_1\delta_1$$

and

$$1 - F(u) = (1 - G(u))\left(1 + O\left(\frac{u}{\sqrt{\mathfrak{S}}}\right)\right) + 2\theta_2\delta_1,$$

where the constants in the big-O notations are absolute.

Proof. Let $\mathcal{A}$ denote the complement of the set $\mathcal{A}_1$ specified in Lemma 2.12 to the set of all integers in the interval $N < n \le N + M$. Let also $K(\eta; s)$ be the moment generating function of the random variable $\eta = \eta_N$ taking the values $W_y(t_n)$, $n \in \mathcal{A}$, i.e.,

$$K(\eta; s) = \mathsf{E}(e^{s\eta}) = |\mathcal{A}|^{-1}\sum_{n \in \mathcal{A}} \exp\{sW_y(t_n)\} = |\mathcal{A}|^{-1}K_0(s).$$

Then, for all s with $|s| \le 1$ and $K = [e^2\mathfrak{S}] + 1$, we have

$$K_0(s) = \sum_{n \in \mathcal{A}}\sum_{k=0}^{+\infty}\frac{s^k}{k!}W_y^k(t_n) = \left(\sum_{k=0}^{K} + \sum_{k>K}\right)\frac{s^k}{k!}\sum_{n \in \mathcal{A}} W_y^k(t_n) = K_1(s) + R_1(s)$$

and

$$|R_1(s)| \le \sum_{k>K} \frac{|s|^k}{k!}\, M\mathfrak{S}^k < \frac{M}{\sqrt{2\pi K}} \sum_{k>K} \left(\frac{e\mathfrak{S}}{k}\right)^k < \frac{M}{\sqrt{2\pi K}} \sum_{k>K} \left(\frac{e\mathfrak{S}}{K}\right)^k$$
$$\le \frac{M}{e\sqrt{2\pi\mathfrak{S}}} \sum_{k>K} e^{-k} < \frac{M}{e\sqrt{\mathfrak{S}}}\, e^{-e^2\mathfrak{S}} < Me^{-9\mathfrak{S}}.$$

Next, representing $K_1(s)$ by the difference

$$\sum_{k=0}^{K} \frac{s^k}{k!} \left(\sum_n W_y^k(t_n) - \sum_{n\in\mathcal{A}} W_y^k(t_n) \right) = K_2(s) - R_2(s),$$

we obtain

$$|R_2(s)| \le |\mathcal{A}_1| + \sum_{k=1}^{K} \frac{1}{k!} \sum_{n\in\mathcal{A}_1} |W_y(t_n)|^k.$$

Cauchy's inequality and the first remark to Lemma 2.1 imply

$$|R_2(s)| \le M\delta_1 + \sum_{k=1}^{K} \frac{1}{k!} \left(M\delta_1 \sum_n W^{2k}(t_n) \right)^{1/2} \le M\delta_1 + \sqrt{M\delta_1} \sum_{k=1}^{K} \frac{1}{k!} \left(\frac{(2k)!}{k!} \frac{M\mathfrak{S}^k}{2^{2k}} \right)^{1/2}$$
$$\le M\delta_1 + \sqrt{M\delta_1} \sum_{k=1}^{K} \frac{1}{k!} \left(M\sqrt{2} \left(\frac{k\mathfrak{S}}{e} \right)^k \right)^{1/2} \le M\delta_1 + \frac{M\sqrt{\delta_1}}{\sqrt[4]{2}\sqrt{\pi}} \sum_{k=1}^{K} \frac{1}{\sqrt{k}} \left(\frac{e\mathfrak{S}}{k} \right)^{k/2}.$$

Since the maximum value of the fraction $(e\mathfrak{S}/k)^{k/2}$ in the interval $1 \le k \le K$ is at most $e^{0.5\mathfrak{S}}$, it follows that

$$|R_2(s)| \le M\delta_1 + M\sqrt{\delta_1} \frac{2\sqrt{K}}{\sqrt[4]{2}\sqrt{\pi}} < 1.5M\sqrt{\mathfrak{S}} e^{-4\mathfrak{S}}.$$

To calculate $K_2(s)$, we set

$$w_k = \sum_n W_y^k(t_n) \qquad \text{and} \qquad U(t) = \frac{1}{\pi} \sum_{p\le y} \frac{p^{it}}{\sqrt{p}}.$$

Lemma 1.3 gives the estimate $|w_k| \le y^k \log N < M^{1/3}$ for odd k. In the case of even $k = 2m$, Lemma 1.4 leads to the relation

$$w_k = (2\pi)^{-2m} \binom{2m}{m} M\mathfrak{C}_m + \theta M^{1/3}, \qquad \mathfrak{C}_m = \sum_{\substack{p_1\cdots p_m = q_1\cdots q_m \\ p_1,\dots,p_m,q_1,\dots,q_m \le y}} (p_1\cdots p_m)^{-1}.$$

According to Lemma 1.5, we have $\mathfrak{C}_m = m!\, H_y^{(m)}(0)$, where

$$H_y(z) = \prod_{p\le y} J_0\left(2i\sqrt{\frac{z}{p}} \right).$$

Thus,

$$K_2(s) = M \sum_{0\le m\le 0.5K} \frac{s^{2m}}{(2m)!} (2\pi)^{-2m} \binom{2m}{m} m!\, H_y^{(m)}(0) + \theta_1 M^{1/3} \sum_{k=1}^{+\infty} \frac{1}{k!}$$
$$= M \sum_{0\le m\le 0.5K} \frac{1}{m!} \left(\frac{s}{2\pi} \right)^{2m} H_y^{(m)}(0) + 2\theta_2 M^{1/3}.$$

Replacing the sum over m by the infinite sum, we obtain

$$K_2(s) = MH_y\left(\frac{s^2}{4\pi^2}\right) - R_3(s) + 2\theta_2 M^{1/3},$$

where, by virtue of Lemma 1.17,

$$|R_3(s)| \le M \sum_{m>0.5K} \frac{1}{m!}\left(\frac{\mathfrak{S}}{4\pi^2}\right)^m < \frac{M}{\sqrt{\pi K}} \sum_{m>0.5K} \left(\frac{e\mathfrak{S}}{4\pi^2 m}\right)^m$$
$$< \frac{M}{\sqrt{\pi K}} \sum_{m>0.5K} \left(\frac{e\mathfrak{S}}{2\mathfrak{S}e^2\pi^2}\right)^m < \frac{M}{\sqrt{K}}(2\pi^2 e)^{-0.5K} < Me^{-14\mathfrak{S}}.$$

Adding these estimates yields

$$K_0(s) = MH_y\left(\frac{s^2}{4\pi^2}\right) + \theta M(4e^{-\pi^2\mathfrak{S}} + 1.5\sqrt{\mathfrak{S}}e^{-4\mathfrak{S}} + 2M^{-2/3} + e^{-14\mathfrak{S}})$$
$$= M\left\{H_y\left(\frac{s^2}{4\pi^2}\right) + 2\theta\sqrt{\mathfrak{S}}e^{-4\mathfrak{S}}\right\}.$$

Now, let us estimate the quantity $H_y(s^2/4\pi^2)$ from below. For this purpose, we represent this quantity in the form

$$\prod_{p\le y} \exp\left(\frac{s^2}{4\pi^2 p}\right) \cdot \exp\left(-\frac{s^2}{4\pi^2 p}\right) J_0\left(\frac{is}{\pi\sqrt{p}}\right) = \exp\left\{\frac{s^2\mathfrak{S}}{4\pi^2}\right\}\Phi_y(s).$$

By virtue of Lemmas 1.11 and 1.12, for any complex s, the function $\Phi_y(s)$ in this representation expands in a series of the form

$$\Phi_y(s) = 1 + \sum_{n=2}^{+\infty} \Phi_n\left(\frac{s}{2\pi}\right)^{2n};$$

moreover,

$$|\Phi_n| \le \begin{cases} \dfrac{\sqrt{2\pi}}{n!}(\log\log n + 1)^n & \text{if } 2 \le n \le y, \\ \dfrac{\mathfrak{S}^n}{n!} & \text{if } n > y. \end{cases}$$

Therefore, for any s in the disk $|s| \le 1$, we have

$$|\Phi_y(s) - 1| \le \sum_{n=2}^{+\infty} \frac{|\Phi_n|}{(2\pi)^{2n}}\sqrt{2\pi} \sum_{2\le n\le y} \frac{1}{n!}\left(\frac{\log\log n + 1}{4\pi^2}\right)^n + \frac{1}{\sqrt{2\pi y}} \sum_{n>y}\left(\frac{e\mathfrak{S}}{4\pi^2 n}\right)^n$$
$$< \sqrt{2\pi}\sum_{n=2}^{+\infty} \frac{1}{n!}\left(\frac{\log\log n + 1}{4\pi^2}\right)^n + \frac{1}{\sqrt{2\pi y}} \sum_{n>y} e^{-n} < 0.01.$$

Thus,

$$\min_{|s|\le 1}\left|H_y\left(\frac{s^2}{4\pi^2}\right)\right| \ge (1 - 0.01)\min_{|s|\le 1}\left|\exp\left(\frac{s^2}{4\pi^2}\right)\right| > 0.99\exp\left(-\frac{\mathfrak{S}}{4\pi^2}\right).$$

Returning to the expression for $K_0(s)$, we obtain

$$K_0(s) = MH_y\left(\frac{s^2}{4\pi^2}\right)\left\{1 + 2\theta\sqrt{\mathfrak{S}}H_y^{-1}\left(\frac{s^2}{4\pi^2}\right)e^{-4\mathfrak{S}}\right\} = MH_y\left(\frac{s^2}{4\pi^2}\right)\{1 + 2.2\theta\sqrt{\mathfrak{S}}e^{-3.97\mathfrak{S}}\}.$$

Now, let us approximate $\Phi_y(s)$ by a function $\Psi(s)$ not depending on y. For this purpose, we write $\Phi_y(s)$ in the form $\Psi(s)\Psi_y^{-1}(s)$, where

$$\Psi(s)=\prod_p \varphi\biggl(\frac{s^2}{4\pi^2 p}\biggr) \qquad \text{and} \qquad \Psi_y(s)=\prod_{p>y} \varphi\biggl(\frac{s^2}{4\pi^2 p}\biggr);$$

here, $\varphi(v)$ is the function defined in Lemma 1.6. Using the inequality in Lemma 1.10, we obtain

$$\biggl|\varphi\biggl(\frac{s^2}{4\pi^2 p}\biggr)-1\biggr| \le \sum_{m=2}^{+\infty} \frac{|\varpi_m|}{(4\pi^2 p)^m} < \sum_{m=2}^{+\infty} \frac{1}{m!}\frac{1}{(4\pi^2 p)^m} < \frac{1}{194p^2}$$

for $|s|\le 1$, whence

$$\log\varphi\biggl(\frac{s^2}{4\pi^2 p}\biggr)=\log\biggl(1+\frac{\theta_1}{194p^2}\biggr)=\frac{\theta_2}{192p^2},$$
$$\Psi_y(s)=\exp\biggl\{\frac{\theta_3}{192}\sum_{p>y}\frac{1}{p^2}\biggr\}=\exp\biggl\{\frac{\theta_4}{191y}\biggr\}=1+\frac{\theta_5}{190y},$$

and

$$K_0(s)=M\exp\biggl\{\frac{s^2\mathfrak{S}}{4\pi^2}\biggr\}\Psi(s)\biggl(1+\frac{\theta_5}{190y}\biggr)(1+2.2\theta\sqrt{\mathfrak{S}}e^{-3.97\mathfrak{S}}).$$

Lemma 2.12 and the definition of $K_0(s)$ imply

$$K(\eta\,;s)=\exp\biggl\{\frac{s^2\mathfrak{S}}{4\pi^2}\biggr\}\Psi(s)(1+\theta(\log N)^{-3}).$$

Now, let $F_{\mathcal{A}}(u)$ be the distribution function of the random variable $\pi\eta\sqrt{2/\mathfrak{S}}$, so that

$$|\mathcal{A}|F_{\mathcal{A}}(u)=\#\biggl\{n\;\Big|\; n\in A,\, W_y(t_n)\le\frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}\,\biggr\}.$$

Then, setting $\phi(N)=\mathfrak{S}/\pi^2$ and $\kappa(N)=(\log N)^3$ in Lemma I.8, we see that, for any $u\ge 1$,

$$F_{\mathcal{A}}(-u)=G(-u)\biggl(1+O\biggl(\frac{u}{\sqrt{\mathfrak{S}}}\biggr)\biggr), \quad \text{i.e.,} \quad 1-F_{\mathcal{A}}(u)=(1-G(u))\biggl(1+O\biggl(\frac{u}{\sqrt{\mathfrak{S}}}\biggr)\biggr),$$

where the constants in the big-O notations are absolute. The definition of $F_{\mathcal{A}}(u)$ and $F(u)$ and Lemma 2.12 imply that the quantities

$$MF(u)=\#\biggl\{n\;\Big|\; N<n\le N+M,\, W_y(t_n)\le\frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}\,\biggr\}$$

and $|\mathcal{A}|F_{\mathcal{A}}(u)$ differ from each other by no more than $M\delta_1$:

$$|\mathcal{A}|F_{\mathcal{A}}(u)-MF(u)=\theta_1 M\delta_1.$$

Since $|\mathcal{A}|=M-\theta_2 M\delta_1$, it follows that

$$F(u)=F_{\mathcal{A}}(u)+2\theta_3\delta_1. \tag{2.25}$$

Now, the assertion of the lemma follows from (2.24) and (2.25).

Lemma 2.14. *Let $1\le|u|\le\sqrt{L}$, and let*

$$\delta_2=(c_0+2)e^2e^{-u^2/2}L^{-1}.$$

Then, for all n such that $N<n\le N+M$, except at most $M\delta_2$ of them, the inequalities

$$W_y(t_n)-2\tau\le\Delta(n)\le W_y(t_n)+2\tau \tag{2.26}$$

hold, where $\tau=(ec_0/4)(u^2+2\log L)$.

Proof. We set $k_1 = [\tau/(ec_0)]$ and $x = X^{1/(8k_1+3)}$ and let $\mathcal{A}_1$ denote the set of those n, $N < n \le N + M$, for which the quantity

$$R(t) = S(t) - W_x(t) = S(t) + \frac{1}{\pi}\sum_{p\le x}\frac{\sin(t\log p)}{\sqrt{p}}$$

satisfies the inequality $|R(t_n + 0)| > \tau$. Then, by virtue of Lemma 2.7, we have

$$\tau^{2k_1}|\mathcal{A}_1| \le \sum_{n\in\mathcal{A}_1} R^{2k_1}(t_n + 0) \le \sum_n R^{2k_1}(t_n + 0) < c_0(c_0k_1)^{2k_1}M,$$

whence

$$|\mathcal{A}_1| \le c_0\biggl(\frac{c_0k_1}{\tau}\biggr)^{2k_1} M \le c_0e^{-2k_1}M.$$

Next, let $\mathcal{A}_2$ be the set of those n for which the difference

$$W_x(t) - W_y(t) = -\frac{1}{\pi}\sum_{y<p\le x}\frac{\sin(t\log p)}{\sqrt{p}}$$

satisfies the condition $|W_x(t_n) - W_y(t_n)| > \tau$. Then

$$\begin{aligned}\tau^{2k_1}|\mathcal{A}_2| &\le \sum_{n\in\mathcal{A}_2}(W_x(t_n) - W_y(t_n))^{2k_1} \le \sum_n (W_x(t_n) - W_y(t_n))^{2k_1} \le \sum_n\biggl|\sum_{y<p\le x}\frac{p^{it_n}}{\sqrt{p}}\biggr|^{2k_1}\\ &= k_1!\,M\biggl(\sum_{y<p\le x}\frac{1}{p}\biggr)^{k_1} + x^{2k_1}\log N < k_1!\,M(\log L)^{k_1} + X^{1/4} < 5M\sqrt{k_1}\biggl(\frac{k_1\log L}{e}\biggr)^{k_1},\end{aligned}$$

whence

$$|\mathcal{A}_2| < 5M\sqrt{k_1}\biggl(\frac{k_1\log L}{e\tau^2}\biggr)^{k_1} < 5M\sqrt{k_1}e^{-3k_1} < 2Me^{-2k_1}.$$

Therefore, we can throw away $(c_0 + 2)Me^{-2k_1} \le M\delta_2$ or fewer numbers n belonging to the union $\mathcal{A}_1 \cup \mathcal{A}_2$ so that, for all remaining n, we have

$$|R(t_n + 0)| + |W_x(t_n) - W_y(t_n)| \le 2\tau$$

and

$$\begin{aligned}\Delta(n) + S(t_n + 0) &= W_x(t_n) + R(t_n + 0)\\ &= W_y(t_n) + R(t_n + 0) + (W_x(t_n) - W_y(t_n)) = W_y(t_n) + 2\theta\tau,\end{aligned}$$

where $-1 \le \theta \le 1$. This completes the proof of the lemma.

Lemma 2.15. *Let*

$$\Phi_0(u) = \frac{1}{M}\#\biggl\{n \;\Big|\; N < n \le N + M,\ \Delta(n) \le \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}\biggr\},$$

so that

$$\Phi(u) = \Phi_0\biggl(u\sqrt{\frac{L}{\mathfrak{S}}}\biggr).$$

Then, for any u, $1 \le |u| \le \sqrt{L}$, the inequalities

$$F(u_1) - 3\delta_2 \le \Phi_0(u) \le F(u_2) + 3\delta_2$$

hold, where $u_1 = u - 2\pi\tau\sqrt{2/\mathfrak{S}}$ and $u_2 = u + 2\pi\tau\sqrt{2/\mathfrak{S}}$ for $\tau = (ec_0/4)(u^2 + 2\log L)$.

Proof. Let $\mathcal{A}$ denote the set of those n, $N < n \le N + M$, for which inequalities (2.26) hold, and let $\mathcal{B}$ denote the complement of $\mathcal{A}$. We set

$$\Phi_{\mathcal{A}}(u) = |\mathcal{A}|^{-1}\#\left\{n \;\middle|\; n \in A,\, \Delta(n) \le \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}\right\}$$

and

$$F_{\mathcal{A}}(u) = |\mathcal{A}|^{-1}\#\left\{n \;\middle|\; n \in A,\, W_y(t_n) \le \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}\right\}.$$

Consider a number $n \in \mathcal{A}$ for which

$$W_y(t_n) \le \frac{u_1}{\pi}\sqrt{\frac{\mathfrak{S}}{2}} = \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}} - 2\tau. \tag{2.27}$$

According to (2.26), we have

$$\Delta(n) \le W_y(t_n) + 2\tau = \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}\,.$$

Therefore, the number of those $n \in \mathcal{A}$ for which $\Delta(n) \le (u/\pi)\sqrt{\mathfrak{S}/2}$ does not exceed the number of those $n \in \mathcal{A}$ for which (2.27) holds, or, equivalently, $|\mathcal{A}|\Phi_{\mathcal{A}}(u) \ge |\mathcal{A}|F_{\mathcal{A}}(u_1)$. But since the left-hand side of the inequality is at most $M\Phi(u)$, it follows that $M\Phi(u) \ge |\mathcal{A}|F_{\mathcal{A}}(u_1)$. Noting that $|\mathcal{A}| = M - |\mathcal{B}|$ and using the estimate of Lemma 2.14, we obtain

$$\Phi_0(u) \ge (1 - \delta_2)F_{\mathcal{A}}(u_1). \tag{2.28}$$

Now, let $n \in \mathcal{A}$ be any number satisfying the condition

$$W_y(t_n) > \frac{u_2}{\pi}\sqrt{\frac{\mathfrak{S}}{2}} = \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}} + 2\tau. \tag{2.29}$$

From (2.26) we conclude that

$$\Delta(n) \ge W_y(t_n) - 2\tau > \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}\,.$$

Thus, the number of those $n \in \mathcal{A}$ for which $\Delta(n) > (u/\pi)\sqrt{\mathfrak{S}/2}$ is no less than the number of those $n \in \mathcal{A}$ for which (2.29) holds, or, equivalently,

$$|\mathcal{A}|(1 - \Phi_{\mathcal{A}}(u)) \ge |\mathcal{A}|(1 - F_{\mathcal{A}}(u_2)).$$

Since the left-hand side of this inequality does not exceed $M(1 - \Phi_0(u))$, it follows that

$$M(1 - \Phi(u)) \ge M(1 - \delta_2)(1 - F_{\mathcal{A}}(u_2))$$

or, in other terms,

$$\Phi_0(u) \le (1 - \delta_2)F_{\mathcal{A}}(u_2) + \delta_2. \tag{2.30}$$

Combining (2.28) and (2.30), we obtain

$$(1 - \delta_2)F_{\mathcal{A}}(u_1) \le \Phi_0(u) \le (1 - \delta_2)F_{\mathcal{A}}(u_2) + \delta_2.$$

Now, note that, for any v, the quantity $|\mathcal{A}|F_{\mathcal{A}}(v)$ cannot differ from $MF(v)$ by more than the number of elements in the set $\mathcal{B}$. Therefore,

$$|\mathcal{A}|F_{\mathcal{A}}(v) = MF(v) + \theta_1 M\delta_2,$$

whence

$$(1 - \theta_2\delta_2)F_{\mathcal{A}}(v) = F(v) + \theta_1\delta_2, \qquad \text{i.e.,} \qquad F_{\mathcal{A}}(v) = F(v) + 2\theta_3\delta_2.$$

The estimates

$$(1 - \delta_2)F_{\mathcal{A}}(u_1) \ge (1 - \delta_2)(F(u_1) - 2\delta_2) > F(u_1) - 3\delta_2,$$
$$(1 - \delta_2)F_{\mathcal{A}}(u_2) + \delta_2 \le (1 - \delta_2)(F(u_2) + 2\delta_2) + \delta_2 < F(u_2) + 3\delta_2$$

complete the proof of the lemma.

Theorem 2.5. *Let* $1.5 \le u \le \sqrt[6]{\log\log N}$. *Then the function*

$$\Phi(v) = \frac{1}{M}\#\left\{n \;\middle|\; N < n \le N+M,\ \Delta(n) \le \frac{v}{\pi\sqrt{2}}\sqrt{\log\log N}\right\}$$

satisfies the relations

$$\frac{\Phi(-u)}{G(-u)} = 1 + O(R) \qquad \textit{and} \qquad \frac{1-\Phi(u)}{1-G(u)} = 1 + O(R),$$

where

$$R = \frac{u(u^2 + \log\log\log N)}{\sqrt{\log\log N}}$$

and the constants in the big-O notations depend on ε.

Proof. Let us prove the first relation. By virtue of Lemma 2.15, we have

$$F(-v_1) - 3\delta_2 \le \Phi_0(-u) \le F(-v_2) + 3\delta_2, \tag{2.31}$$

where

$$v_1 = u + h, \qquad v_2 = u - h, \qquad h = 2\pi\tau\sqrt{\frac{2}{\mathfrak{S}}},$$
$$\tau = \frac{ec_0}{4}(u^2 + 2\log L), \qquad \text{and} \qquad \delta_2 = (c_0+2)e^2 \cdot e^{-u^2/2}L^{-1}.$$

Since

$$h = \frac{\pi e c_0}{\sqrt{2\mathfrak{S}}}(u^2 + 2\log L) \le \frac{\pi e c_0}{\sqrt{L}}(\sqrt[3]{L} + 2\log L) < 0.5,$$

it follows that $v_1 > v_2 > u - 0.5 \ge 1.5 - 0.5 = 1$. Therefore, applying Lemma 2.13 and the obvious inequalities $2\delta_1 < \delta_2$ and $v_j/\sqrt{\mathfrak{S}} < R$, we obtain

$$\Phi_0(-u) \ge G(-v_1)\left(1 + O\left(\frac{v_1}{\sqrt{\mathfrak{S}}}\right)\right) - 2\delta_1 - 3\delta_2 > G(-v_1)(1 + O(R)) - 4\delta_2$$

and

$$\Phi_0(-u) < G(-v_2)(1 + O(R)) + 4\delta_2.$$

Lagrange's mean value theorem gives

$$G(-v_j) = G(-u) - hG'(-u - \theta_1 h), \qquad j = 1, 2, \quad |\theta| \le 1.$$

Since

$$G'(-u-\theta_1 h) = \frac{1}{\sqrt{2\pi}}\exp\left\{-\frac{1}{2}(u+\theta_1 h)^2\right\} < \frac{1}{\sqrt{2\pi}}e^{-u^2/2 + u|h|}$$

and

$$G(-u) = \frac{1}{\sqrt{2\pi}}\left(\frac{e^{-u^2/2}}{u} - \frac{1}{2\sqrt{2}}\int_{u^2/2}^{+\infty} e^{-w}\frac{dw}{w^{3/2}}\right) \ge \frac{1}{\sqrt{2\pi}}\left(\frac{e^{-u^2/2}}{u} - \frac{e^{-u^2/2}}{u^3}\right) > \frac{e^{-u^2/2}}{5u}$$

for $u \ge 1.5$, it follows that

$$|h|\frac{G'(-u-\theta_1 h)}{G(-u)} \le \frac{5r}{\sqrt{2\pi}}e^r, \qquad \text{where} \quad r = \frac{\pi e c_0 u}{\sqrt{L}}(u^2 + 2\log L) \asymp R.$$

By virtue of the estimates

$$r \le \frac{\pi e c_0}{\sqrt{L}}(\sqrt{L} + 2\sqrt[6]{L}\log L) < 9c_0 \qquad \text{and} \qquad \frac{\delta_2}{G(-u)} < 5(c_0+2)e^2\frac{u}{L} < r,$$

we have

$$\Phi_0(-u) = G(-u)(1+O(R))^2 + 4\theta\delta_2 = G(-u)(1+O(R)).$$

Finally, noting that

$$u\sqrt{\frac{L}{\mathfrak{S}}} = u + h_1, \qquad \text{where} \quad |h_1| \le \frac{u\log L}{L},$$

we obtain

$$G(-u-h_1) = G(-u)\left(1 - h_1\frac{G'(-u-\theta h_1)}{G(-u)}\right) = G(-u)(1+O(R))$$

and

$$\Phi(-u) = \Phi_0\left(-u\sqrt{\frac{L}{\mathfrak{S}}}\right) = \Phi_0(-u-h_1) = G(-u-h_1)(1+O(R)) = G(-u)(1+O(R)).$$

The second relation is proved in a similarly way. This completes the proof of the theorem.

Theorem 2.6. *Let $f(x)$ be any monotonically increasing function unbounded as $x \to +\infty$ and satisfying the conditions $a \le f(x) \le b\sqrt{\log\log x}$ with absolute constants a and b for $N \le x \le N+M$, and let*

$$\mathcal{F} = f(N) \qquad \textit{and} \qquad \delta = \exp\left\{-(ec_0)^2\frac{\log\log N}{\mathcal{F}^2}\right\}.$$

Then, in the interval $N \le n \le N+M$, there exist at least $M\delta$ numbers n for which $\Delta(n) \le -(\pi\mathcal{F})^{-1}\log\log N$ and at least $M\delta$ numbers n for which $\Delta(n) > (\pi\mathcal{F})^{-1}\log\log N$.

Proof. We set $u_0 = \sqrt{L/\mathcal{F}}$ and define $\mathcal{A}_1$ as the set of those n for which

$$\Delta(n) \le W_y(t_n) + 2\tau,$$

where

$$y = \exp\left\{\frac{\log N}{220L}\right\} \qquad \text{and} \qquad \tau = \frac{ec_0}{4}(u^2 + 2\log L).$$

By virtue of Lemma 2.14, the complement $\mathcal{B}_1$ of $\mathcal{A}_1$ consists of no more than $M\delta_2$ numbers, where

$$\delta_2 = (c_0+2)e^2e^{-u_0^2/2}L^{-1} < \frac{9c_0}{L}\exp\left\{-\frac{L}{2\mathcal{F}}\right\}.$$

Next, we set

$$u = \frac{L}{\mathcal{F}}\sqrt{\frac{2}{\mathfrak{S}}} \qquad \text{and} \qquad u_1 = u - 2\pi\tau\sqrt{\frac{2}{\mathfrak{S}}}$$

and consider the set $\mathcal{A}_2$ of those n for which $W_y(t_n) \le -(u_1/\pi)\sqrt{\mathfrak{S}/2}$. By virtue of Lemma 2.13, the number of such n satisfying the condition $N < n \le N+M$ equals

$$MF(u_1) = MG(-u_1)\left(1 + O\left(\frac{u_1}{\sqrt{\mathfrak{S}}}\right)\right) + 2\theta M\delta_1,$$

where $\delta_1 = 4e^{-\pi^2\mathfrak{S}} < 0.5(\log N)^{-9}$. This easily implies that any number n belonging to both $\mathcal{A}_1$ and $\mathcal{A}_2$ must satisfy the condition

$$\Delta(n) \le -\frac{u_1}{\pi}\sqrt{\frac{\mathfrak{S}}{2}} + 2\tau = -\frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}} - 2\tau + 2\tau = -\frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}.$$

Thus, the required number ν of those n for which $\Delta(n) \le -(u_1/\pi)\sqrt{\mathfrak{S}/2}$ is no less than the number of those n which belong to the intersection $\mathcal{A}_1 \cap \mathcal{A}_2$. In turn, this intersection consists of at least $|\mathcal{A}_2| - |\mathcal{B}_1|$ integers. Therefore,

$$\nu \ge MF(-u_1) - 3M\delta_2 \ge M\left\{G(-u_1)\left(1 + O\left(\frac{u_1}{\sqrt{\mathfrak{S}}}\right)\right) - 2\delta_1\right\} - 3M\delta_2.$$

Assuming that the constant a in the condition of the theorem is so large that

$$1 + O\left(\frac{u_1}{\sqrt{\mathfrak{S}}}\right) = 1 + O\left(\frac{1}{\mathcal{F}}\right) \ge \frac{1}{\sqrt{2}}$$

and using the estimate

$$G(-u_1) = \frac{1}{\sqrt{2\pi}}\int_{u_1}^{+\infty} e^{-v^2/2}\,dv > \frac{1}{\pi\sqrt{2}}\frac{e^{-u_1^2/2}}{u_1},$$

we obtain

$$\nu \ge M\left(\frac{1}{2\pi}\frac{e^{-u_1^2/2}}{u_1} - (\log N)^{-9} - \frac{9c_0}{L}\exp\left\{-\frac{L}{2\mathcal{F}}\right\}\right) > \frac{M}{2\pi}\frac{e^{-u_1^2/2}}{u_1}\left(1 - \frac{20\pi c_0 u_1}{L}\exp\left\{\frac{u_1^2}{2} - \frac{L}{2\mathcal{F}}\right\}\right).$$

Since

$$\begin{aligned} u_1^2 - \frac{L}{\mathcal{F}} &< 2\left(u^2 + \frac{8\tau^2}{\mathfrak{S}}\right) - \frac{L}{\mathcal{F}} < 2\left(u^2 + \left(\frac{ec_0}{2}\right)^2\frac{u_0^4}{\mathfrak{S}}\right) - \frac{L}{\mathcal{F}} \\ &< 2\left\{\frac{2L^2}{\mathcal{F}^2\mathfrak{S}} + \left(\frac{ec_0}{2}\right)^2\frac{L^2}{\mathcal{F}^2\mathfrak{S}}\right\} - \frac{L}{\mathcal{F}} < (ec_0)^2\frac{L}{\mathcal{F}^2} - \frac{L}{\mathcal{F}} < -\frac{L}{2\mathcal{F}}, \end{aligned}$$

it follows that

$$\nu > \frac{M}{2\pi}\frac{e^{-u_1^2/2}}{u_1}\left(1 - \frac{20\pi c_0 u_1}{L}\exp\left\{-\frac{L}{4\mathcal{F}}\right\}\right) > M\exp\left\{-(ec_0)^2\frac{L}{\mathcal{F}^2}\right\} = M\delta,$$

as required.

The inequality for the number of those n for which

$$\Delta(n) \ge \frac{L}{\pi\mathcal{F}} = \frac{u}{\pi}\sqrt{\frac{\mathfrak{S}}{2}}$$

is proved in a similar way. This completes the proof of the theorem.

Remark 2.5. In the case where the parameter t varies continuously, a series of conditional and unconditional upper and lower bounds for the measure of the set of t values in the interval $T \le t \le T + H$ at which the function $S(t)$ is large in absolute value were obtained in [45, 46]. In particular, it was proved in [46] that, for any $\lambda > \log\log T$, the measure of the set $E_\lambda(T, H)$ of those t in $(T, T + H]$ for which $|S(t)| \ge \lambda$ is at most $e^2 He^{-\kappa\lambda}$, where $\kappa = \pi\varepsilon^{1.5}e^{-19.5}$. If the Riemann hypothesis is valid, then the measure of the set E_j, $j = 0, 1$, of those t in $(T, 2T]$ for which

$$(-1)^j S(t) > \frac{1}{300}\sqrt{\frac{\log T}{(\log\log T)\log\log\log T}}$$

is estimated as

$$\operatorname{mes}(E_j) \ge \frac{2}{5}\frac{T}{\sqrt{\log T}(\log\log T)}\exp\left(-\frac{\log T}{\log\log T}\right), \qquad j = 0, 1.$$

3. SELBERG'S FORMULAS AND THEIR CONSEQUENCES

As far as we know, the proof of Selberg's formulas (7) and (8) has never been published. However, in his 1976 paper [49], Fujii derived similar formulas for the quantities $\Delta_n(\chi_1,\chi_2)$, where χ_1 and χ_2 are different primitive characters modulo $q\ge 5$ and

$$0\le\gamma_1(\chi_r)\le\gamma_2(\chi_r)\le\cdots\le\gamma_n(\chi_r)\le\gamma_{n+1}(\chi_r)\le\cdots$$

are the ordinates of zeros of the functions $L(s,\chi_r)$, $r=1,2$, in the critical strip numbered in increasing order. The quantity $\Delta_n(\chi_1,\chi_2)$ was defined for given n as the difference $m-n$, where the integer $m=m(n)$ is uniquely determined by the inequalities

$$\gamma_m(\chi_1)<\gamma_n(\chi_2)\le\gamma_{m+1}(\chi_1).$$

The proof of Fujii's formulas

$$\sum_{n\le N}\Delta_n^{2k}(\chi_1,\chi_2)=\frac{(2k)!}{k!}\frac{N}{(2\pi)^{2k}}(\log\log N)^k+O((Ak)^{4k}N(\log\log N)^{k-1/2})+O\biggl(\frac{(\log N)^{2k+1}}{2k+1}\biggr)$$

and

$$\sum_{n\le N}\Delta_n^{2k-1}(\chi_1,\chi_2)=O((Ak)^{3k}N(\log\log N)^{k-1})+O\biggl(\frac{(\log N)^{2k}}{2k}\biggr)$$

used the estimate

$$\mathcal{J}=\int_T^{T+H}(S(t,\chi_1)-S(t,\chi_2))^l\,dS(t,\chi_2)=O\biggl(\frac{(\log N)^{l+1}}{l+1}\biggr).\tag{3.1}$$

Here, $S(t,\chi)=\pi^{-1}\arg(L(1/2+it,\chi))$ is the argument of the Dirichlet function $L(s,\chi)$, which is defined by analogy with $S(t)$. The derivation of estimate (3.1) was not given in [49].

This proof cannot be considered satisfactory, because $\mathcal{J}$ is not defined as a Stieltjes integral, since the points of discontinuity of the piecewise smooth function $S(t,\chi_2)$, with respect to which the integration is performed, coincide with the ordinates $\gamma_n(\chi_2)$. At these points, the integrable function $(S(t,\chi_1)-S(t,\chi_2))^{2k}$ is discontinuous. In this situation, it is easy to construct two sequences of integral sums for $\mathcal{J}$ which converge to different limits (see, e.g., [50, Section 584]). Therefore, manipulating the integral $\mathcal{J}$ (at least without any explanations) is not legitimate.

The formal application of Fujii's argument to the problem of evaluating the even moments of Δ_n yields integrals of the form

$$\int_T^{T+H}S^{2k}(t)\,dS(t),\qquad k=1,2,3,\ldots,$$

to which all of the above considerations of $\mathcal{J}$ apply in full measure.

This chapter is devoted to the proof of Selberg's formulas (7) and (8), their generalizations (to the cases of a short range of variation of n, noninteger k, and k growing with increasing N), and consequences (moments of $S(\gamma_n\pm 0)$ and $t_n-\gamma_n$). The key point, which makes it possible to obtain Selberg's formulas, is that the sequence of differences Δ_n considered by Selberg is closely related to the sequence of quantities $\Delta(n)=S(t_n+0)$ studied in Chapter 2. This relationship is reflected in Lemmas 3.3, 3.5, and 3.6. Its presence makes it possible to easily pass from asymptotic expressions and estimates of Theorems 2.1–2.3 to asymptotic expressions for the corresponding moments of Δ_n.

Note that such a passage can be performed in different ways, three of which are described in this paper (in Sections 3.1–3.3). We found it appropriate to describe all of these ways in the paper because each of them leads to new results, which are of independent interest.

Let γ be the ordinate of r different zeros of the Riemann zeta-function, and let $k_1,\ldots,k_r$ be the multiplicities of these zeros. We define the *multiplicity $\kappa=\kappa(\gamma)$ of the ordinate γ* as the sum of the multiplicities of the corresponding zeros:

$$\kappa=k_1+\cdots+k_r.$$

It follows from properties of the function $S(t)$ (see, e.g., [6, Theorem 2]) that

$$\kappa(\gamma) = S(\gamma + 0) - S(\gamma - 0). \tag{3.2}$$

In the case $\gamma = \gamma_l$, we write κ_l instead of $\kappa(\gamma)$. Thus, in the case

$$\gamma_{l-1} < \gamma_l = \gamma_{l+1} = \cdots = \gamma_{l+p-1} < \gamma_{l+p}, \tag{3.3}$$

we have

$$\kappa_l = \kappa_{l+1} = \cdots = \kappa_{l+p-1} = p.$$

Lemma 3.1. *For all sufficiently large n,*

$$|\Delta(n)| < 0.14 \log n \qquad \textit{and} \qquad |\Delta_n| < 0.42 \log n. \tag{3.4}$$

Proof. The first inequality follows at once from Lemmas 1.19 and I.9. Suppose that $t_{m-1} < \gamma_n \le t_m$. Then the definition of the function $N(t)$ implies

$$N(t_{m-1} + 0) < N(\gamma_n + 0) \le N(t_m + 0). \tag{3.5}$$

Note that

$$N(\gamma_n + 0) = n + \theta_n(\kappa_n - 1),$$

where $0 \le \theta_n \le 1$. Indeed, in the case $\kappa_n = 1$, this equality is obvious. Suppose that, for some $p \ge 2$, inequalities (3.5) hold. If $n = l + s$ and $0 \le s \le p - 1$, then we have $\kappa_n = p$, so that

$$N(\gamma_n + 0) = l + p - 1 = n + (p - s - 1) = n + \kappa_n - s - 1 = n + \theta_n(\kappa_n - 1), \tag{3.6}$$

where

$$0 \le \theta_n = \frac{\kappa_n - s - 1}{\kappa_n - 1} \le 1.$$

Transforming the left- and right-hand sides of (3.5) by means of the Riemann–Mangoldt formula, we obtain

$$m - 1 + S(t_{m-1} + 0) < n + \theta_n(\kappa_n - 1) \le m + S(t_m + 0),$$

whence

$$\theta_n(\kappa_n - 1) - S(t_m + 0) \le m - n < \theta_n(\kappa_n - 1) + 1 - S(t_{m-1} + 0).$$

Applying relation (3.2) and the estimate of Lemma I.9, we arrive at

$$|\Delta_n| \le \kappa_n + 0.14 \log t_m < 0.28 \log \gamma_n + 0.14 \log t_m < 0.42 \log n.$$

This completes the proof of the lemma.

Remark 3.1. The constant on the right-hand side of the second inequality in (3.4) is not the best one. A somewhat more accurate estimate is given in Chapter 4, where the main statements about the order of growth of the quantities Δ_n are collected.

3.10. The first proof of Selberg's formulas. Let a and b be any real numbers such that $a < b$. We denote the number of solutions of the inequality $a < \Delta_n \le b$ satisfying the condition $N < n \le N + M$ by $e(a, b)$ and the number of solutions of the inequality $a < \Delta(n) \le b$ satisfying the same condition by $f(a, b)$.

The following lemma is key in the first proof of Selberg's formulas.

Lemma 3.2. *For integers a and b such that $a < b$,*

$$e(a, b) = f(-(b + 1), -(a + 1)) + \theta(|a| + |b|).$$

Proof. First, we have

$$e(a,b) = M - e_1 - e_2,$$

where e_1 and e_2 denote, respectively, the numbers of solutions of the inequalities

$$\Delta_n \le a \qquad \text{and} \qquad \Delta_n > b \tag{3.7}$$

satisfying the condition $N < n \le N + M$. The first inequality in (3.7) is equivalent to

$$\gamma_n \le t_{n+a}. \tag{3.8}$$

Indeed, if $t_{m-1} < \gamma_n \le t_m$ and $\Delta_n = m - n \le a$, then $m \le n + a$ and $\gamma_n \le t_m \le t_{n+a}$. Conversely, if condition (3.8) holds, then we have $\gamma_n \le t_m \le t_{n+a}$ for m specified above, whence $m \le n + a$ and $\Delta_n \le a$.

Setting $\nu = n + a$ in (3.8), we see that e_1 coincides with the number of solutions of the inequality

$$\gamma_{\nu-a} \le t_\nu \tag{3.9}$$

satisfying the condition $N + a < \nu \le N + M + a$ and, hence, differs from the number f_1 of solutions of (3.9) satisfying the condition $N < \nu \le N + M$ by no more than $|a|$:

$$e_1 = f_1 + \theta_1|a|.$$

A similar argument proves that the inequality $\Delta_n > b$ is equivalent to $\gamma_n > t_{n+b}$ and the number e_2 coincides with the number of those ν, $N + b < \nu \le N + M + b$, for which

$$\gamma_{\nu-b} > t_\nu. \tag{3.10}$$

Therefore,

$$e_2 = f_2 + \theta_2|b|,$$

where f_2 is the number of solutions of (3.10) satisfying the condition $N < \nu \le N + M$.

But if the Gram point t_ν satisfies none of inequalities (3.9) and (3.10), then

$$\gamma_{\nu-b} \le t_\nu < \gamma_{\nu-a}. \tag{3.11}$$

Let us show that (3.11) is equivalent to the double inequality

$$-(b+1) < \Delta(\nu) \le -(a+1). \tag{3.12}$$

Indeed, defining m by the inequalities $\gamma_m \le t_\nu < \gamma_{m+1}$, we conclude from (3.11) that $\nu - b \le m \le \nu - a - 1$, which implies (3.12). Conversely, (3.12) implies that, for m defined above, we have $\nu - b \le m \le \nu - a - 1$; therefore,

$$\gamma_{\nu-b} \le \gamma_m \le t_\nu < \gamma_{m+1} \le \gamma_{\nu-a}.$$

Thus, the number of solutions of (3.11) in the interval $N < \nu \le N + M$ coincides with $f(-(b+1), -(a+1))$, whence

$$e(a,b) = M - e_1 - e_2 = M - f_1 - f_2 - \theta_1|a| - \theta_2|b| = f(-(b+1), -(a+1)) + \theta(|a| + |b|).$$

This completes the proof of the lemma.

We proceed to prove Selberg's formulas.

Theorem 3.1. *Let k be an integer such that $1 \le k \le c_1 \sqrt[3]{\log\log N}$, where $c_1 = c_0^{-1}(1.75\pi^2)^{1/3}$. Then*

$$\sum_n \Delta_n^{2k} = \frac{(2k)!}{k!}\frac{M}{(2\pi)^{2k}}(\log\log N)^k\left(1 + \frac{2.5\theta(c_0 k)^{3/2}}{\sqrt{\log\log N}}\right)$$

and

$$\left|\sum_n \Delta_n^{2k-1}\right| < 0.5c_0^{3/2}\frac{(2k)!}{(k-1)!}\frac{M}{(2\pi)^{2k-2}}(\log\log N)^{k-1}.$$

Theorem 3.2. *Let a be any number not being an even integer and satisfying the condition $0 < a \le c_2 \sqrt[3]{\log\log N}$, where $c_2 = (11^{2/3}c_0)^{-1}$. Then*

$$\sum_n |\Delta_n|^a = \frac{M}{\pi^a\sqrt{\pi}}\,\Gamma\left(\frac{a+1}{2}\right)(\log\log N)^{a/2}(1+\theta r(a)),$$

where[14]

$$r(a) = \begin{cases} 14.1\left(\dfrac{\log\log\log N}{\log\log N}\right)^{a/2} & \text{if } 0 < a \le 1, \\ \dfrac{782a}{\sqrt{\log\log N}}(\sqrt{\log\log\log N} + 2.2c_0^{3/2}\sqrt{a}) & \text{if } a > 1. \end{cases}$$

Proof of Theorems 3.1 and 3.2. Theorems 3.1 and 3.2 have similar proofs, and we prove them simultaneously. Since the numbers Δ_n are integer, we have

$$\sum_n |\Delta_n|^a = \sum_{\nu=1}^{l} \nu^a(e(\nu-1,\nu) + e(-(\nu+1),-\nu))$$

for any $a > 0$ and $l = [0.45\log N]$; this number l is chosen so that Δ_n and $\Delta(n)$ do not exceed l in absolute value for $N < n \le N + M$. Using Lemma 3.2, we obtain

$$e(\nu-1,\nu) = f(-(\nu+1),-\nu) + \theta_1(2\nu-1), \qquad e(-(\nu+1),-\nu)) = f(\nu-1,\nu) + \theta_2(2\nu+1)$$

for $\nu \ge 2$. Therefore,

$$\sum_n |\Delta_n|^a = \sum_{\nu=1}^{l} \nu^a(f(\nu-1,\nu) + f(-(\nu+1),-\nu) + 4\theta_3\nu) = \sum_n |\Delta(n)|^a + 4\theta_4 \sum_{\nu=1}^{l} \nu^{a+1}.$$

Now, the first assertion of Theorem 3.1 and whole Theorem 3.2 follow from Theorems 2.1 and 2.3 and the estimate

$$4\sum_{\nu=1}^{l} \nu^{a+1} < \frac{4(l+1)^{a+2}}{a+2} < (\log N)^{a+2}.$$

Next, for any integer $k \ge 1$, we have

$$\begin{aligned} \sum_n \Delta_n^{2k-1} &= \sum_{\nu=1}^{l}(\nu^{2k-1}e(\nu-1,\nu) + (-\nu)^{2k-1}e(-(\nu+1),-\nu)) \\ &= \sum_{\nu=1}^{l} \nu^{2k-1}(e(\nu-1,\nu) - e(-(\nu+1),-\nu)) \\ &= \sum_{\nu=1}^{l} \nu^{2k-1}(f(-(\nu+1),-\nu) - f(\nu-1,\nu) + 4\theta_1\nu) \\ &= -\sum_{\nu=1}^{l}(\nu^{2k-1}f(\nu-1,\nu) + (-\nu)^{2k-1}f(-(\nu+1),-\nu)) + \theta_2(\log N)^{a+2} \\ &= -\sum_n \Delta^{2k-1}(n) + \theta_2(\log N)^{a+2}. \end{aligned}$$

Thus, the second assertion of Theorem 3.1 is a consequence of the estimate of Theorem 2.2.

[14] See Remark 2.3 on the improvement of the estimate of $r(a)$ for $0 < a \le 1$.

Theorem 3.3. *Let $\Psi(v)$ be the distribution function of the quantities Δ_n, that is,*

$$\Psi(v)=\frac{1}{M}\#\left\{n\;\Big|\;N<n\le N+M,\;\Delta_n\le\frac{v}{\pi\sqrt{2}}\sqrt{\log\log N}\right\}.$$

Then, for any u,

$$\Psi(u)=G(u)+\frac{1.79\theta}{\sqrt{\log\log\log N}}.$$

Moreover, if $1.5\le u\le\sqrt[6]{\log\log N}$, then $\Psi(v)$ satisfies the relations

$$\frac{\Psi(-u)}{G(-u)}=1+O(R)\qquad\text{and}\qquad\frac{1-\Psi(u)}{1-G(u)}=1+O(R),$$

in which

$$R=\frac{u(u^2+\log\log\log N)}{\sqrt{\log\log N}},$$

and the constants in the big-O notations depend[15] *on ε.*

Proof. We begin by proving the first assertion of the theorem. According to Lemma 3.1, if

$$|u|\ge\frac{\pi\log N}{\sqrt{2\log\log N}},$$

then this assertion is trivial. Otherwise, using the notation and assertions of Lemma 3.2 and Theorem 2.4, we obtain

$$\begin{aligned}M\Psi(u)&=e\left(-\infty,\frac{u}{\pi}\sqrt{\frac{L}{2}}\right)=e\left(-\frac{1}{2}\log N,\frac{u}{\pi}\sqrt{\frac{L}{2}}\right)=f\left(-\frac{u}{\pi}\sqrt{\frac{L}{2}}-1,\frac{1}{2}\log N+1\right)+\theta_1\log N\\&=f\left(-\frac{u}{\pi}\sqrt{\frac{L}{2}}-1,+\infty\right)+\theta_1\log N=M\left(\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{u}e^{-v^2/2}\,dv+\theta\Delta\right),\end{aligned}$$

where $\Delta=1.79(\log\log\log N)^{-0.5}$.

Let us prove the second assertion of the theorem. Lemmas 3.1 and 3.2 imply the chain of relations

$$\begin{aligned}M\Psi(-u)&=e\left(-\infty,-\frac{u}{\pi}\sqrt{\frac{L}{2}}\right)=e\left(-\frac{1}{2}\log N,-\frac{u}{\pi}\sqrt{\frac{L}{2}}\right)\\&=f\left(\frac{u}{\pi}\sqrt{\frac{L}{2}}-1,\frac{1}{2}\log N+1\right)+\theta\log N=f\left(\frac{u_1}{\pi}\sqrt{\frac{L}{2}},+\infty\right)+\theta\log N\\&=M(1-\Phi(u_1))+\theta\log N,\end{aligned}$$

where $u_1=u-h$ and $h=\pi\sqrt{2/L}$. By virtue of Theorem 2.4, we have

$$1-\Phi(u_1)=(1-G(u_1))(1+O(R))=G(-u_1)(1+O(R))=G(-u)(1+O(R)),$$

which implies the required relation for $\Psi(-u)/G(-u)$. The relation for $(1-\Psi(u))/(1-G(u))$ is proved in a similar way. This completes the proof of the theorem.

Corollary 3.1. *Let $f(x)$ be a monotone positive function unboundedly increasing as $x\to+\infty$. Then, for almost all n in the interval $N<n\le N+M$,*

$$\frac{1}{f(n)}\sqrt{\log\log n}<|\Delta_n|\le f(n)\sqrt{\log\log n}.$$

In other words, Selberg's conjecture remains true in the case of a short range of variation of n.

[15] The remainder term in the first expression for $\Psi(u)$ can be improved; see Remark 2.4.

Theorem 3.4. *Let $f(x)$ be any monotonically increasing function unbounded as $x \to +\infty$ and satisfying the conditions $a \le f(x) \le b\sqrt{\log\log x}$ for $N \le x \le N + M$, where a and b are absolute constants, and let*

$$\mathcal{F} = f(N) \qquad \text{and} \qquad \delta = 0.5 \exp\left\{-(ec_0)^2 \frac{\log\log N}{\mathcal{F}^2}\right\}.$$

Then the interval $N \le n \le N + M$ contains at least $M\delta$ numbers n satisfying the condition $\Delta_n \le -(\pi\mathcal{F})^{-1}\log\log N$ and at least $M\delta$ numbers n satisfying the condition $\Delta_n > (\pi\mathcal{F})^{-1}\log\log N$.

Proof. The required assertion follows at once from Theorem 2.5 and Lemma 3.2.

3.11. The second proof of Selberg's formulas. In this section, we derive Selberg's formulas from Theorems 2.1 and 2.2 in a different way. This way leads to somewhat less accurate results (in comparison with Theorem 3.1) and does not allow us to derive an asymptotic expression for fractional moments directly from Theorem 2.3 (at least, without all those tricks which were used in its proof). Nevertheless, it seems to be more natural, because it uses no information about the distribution of the numbers $\Delta(n)$ and Δ_n (such as properties of the quantities $e(a, b)$ and $f(a, b)$) and is based only on the calculation of sums containing the values $S(t_n + 0)$. Moreover, the sums

$$T_k = \sum_n S^k(t_n + 0)(S(t_n + 0) - S(t_{n-1} + 0)),$$

which arise in this approach, turn out to be useful in solving some other problems related to the distribution of zeros of $\zeta(s)$ (see Sections 3.6 and 3.7).

As above, we set $X = N^{0.1\varepsilon}$. We also set

$$r(n) = S(t_n + 0) - S(t_{n-1} + 0) = \Delta(n) - \Delta(n - 1).$$

The second proof of Selberg's formulas uses several auxiliary assertions.

Theorem 3.5. *Let k be an integer satisfying the condition $1 \le k \le c\log X$, where c is a sufficiently small absolute constant. Then*

$$\sum_n r^{2k}(n) < c_0(2c_0k)^{2k}M.$$

Proof. As above, we set

$$x = X^{1/(8k+3)}, \qquad W(t) = -\frac{1}{\pi}\sum_{p\le x}\frac{\sin(t\log p)}{\sqrt{p}}, \qquad R(t) = S(t) + W(t)$$

and

$$w_n = W(t_{n+1}) - W(t_n), \qquad u_n = R(t_{n+1} + 0) - R(t_n + 0).$$

By virtue of Lemma 1.1, if $N < n \le N + M$, then $t_{n+1} - t_n = 2h + \varepsilon_n$, where

$$h = \frac{\pi}{2\vartheta'(t_N)} \qquad \text{and} \qquad |\varepsilon_n| < MN^{-1}.$$

Note that

$$|W'(t)| \le \frac{1}{\pi}\sum_{p\le x}\frac{\log p}{\sqrt{p}} < \sqrt{x};$$

hence,

$$\begin{aligned} w_n &= W(t_n + 2h + \varepsilon_n) - W(t_n) = W(t_n + 2h) - W(t_n) + \theta_1 MN^{-1}\sqrt{x} \\ &= -\frac{1}{\pi}\sum_{p\le x}\frac{1}{\sqrt{p}}(\sin((t_n + 2h)\log p) - \sin(t_n\log p)) + \theta_1 MN^{-1}\sqrt{x} \\ &= -\frac{1}{\pi}\sum_{p\le x} a_p\, p^{it_n} + \theta_1 MN^{-1}\sqrt{x}, \end{aligned}$$

where $a_p = p^{-0.5+ih}\sin(h\log p)$. According to Lemma 2.1, we have the estimate

$$\sum_n w_n^{2k} \le \frac{(2k)!}{k!}\frac{M}{2^{2k}}\mathfrak{s}^k + \sqrt{X}, \qquad \text{where} \quad \mathfrak{s} = \sum_{p\le x}|a_p|^2 = \sum_{p\le x}\frac{\sin^2(h\log p)}{p}.$$

By virtue of Lemma I.2, the sum $\mathfrak{s}$ does not exceed

$$\sum_{p\le x}\frac{(h\log p)^2}{p} < (h\log x)^2 < \left(\frac{\varepsilon}{80k}\right)^2.$$

Therefore,

$$\sum_n w_n^{2k} \le \frac{(2k)!}{k!}\frac{M}{2^{2k}}\left(\frac{\varepsilon}{80k}\right)^{2k} + \sqrt{X} < \frac{3}{2}\left(\frac{\varepsilon}{80}\right)^{2k}M. \tag{3.13}$$

Returning to the sum in the statement of the theorem and applying the inequality in Lemma I.9, we obtain

$$\sum_{N\le n<N+M}(S(t_n+0)-S(t_{n-1}+0))^{2k} = \sum_n (u_n-w_n)^{2k} + 2\theta_1(0.5\log N)^{2k}$$
$$= \sum_{\nu=0}^{2k}(-1)^\nu\binom{2k}{\nu}\sum_n u_n^\nu w_n^{2k-\nu} + 2\theta_1(0.5\log N)^{2k} \le \sum_{\nu=0}^{2k}\binom{2k}{\nu}2^{\nu-1}(W'_\nu + W''_\nu),$$

where

$$W_\nu^{(j)} = \sum_n |R(t_{n+j-1}+0)|^\nu v_\nu^{2k-\nu}, \qquad j=1,2.$$

Hölder's inequality (3.13) and the estimate of Lemma 2.7 imply

$$W'_\nu \le \left(\sum_n v_n^{2k}\right)^{1-\nu/(2k)}\left(\sum_n R^{2k}(t_n+0)\right)^{\nu/(2k)}$$
$$\le M\left\{\frac{3}{2}\left(\frac{\varepsilon}{80}\right)^{2k}\right\}^{1-\nu/(2k)}\left\{\frac{c_0}{18}(c_0k)^{2k}\right\}^{\nu/(2k)} < \frac{3}{2}M\left(\frac{\varepsilon}{80}\right)^{2k}\left\{\frac{80c_0k}{\varepsilon}\left(\frac{c_0}{27}\right)^{1/(2k)}\right\}^\nu$$

(this estimate is also valid for W''_ν). Hence,

$$\sum_n r^{2k}(n) < \frac{3}{2}M\left(\frac{\varepsilon}{80}\right)^{2k}\sum_{\nu=0}^{2k}\binom{2k}{\nu}\left\{\frac{80c_0k}{\varepsilon}\left(\frac{c_0}{27}\right)^{1/(2k)}\right\}^\nu + 2(0.5\log N)^{2k}$$
$$= \frac{3}{2}M\left\{\frac{\varepsilon}{80}\cdot\frac{160c_0k}{\varepsilon}\left(\frac{c_0}{27}\right)^{1/(2k)}\right\}^{2k}\left(1+\frac{\varepsilon}{160c_0k}\right)^{2k} + 2(0.5\log N)^{2k} < c_0(2c_0k)^{2k}M.$$

This completes the proof of the theorem.

Theorem 3.6. *Let k be an integer satisfying the condition $1\le k\le c_1\sqrt[3]{\log\log N}$. Then*

$$|T_{2k-1}| < 1.5c_0^3k4^k\frac{(2k)!}{(k-1)!}\frac{M}{(2\pi)^{2(k-1)}}(\log\log N)^{k-1}$$

and

$$|T_{2k}| < 42(c_0k)^{5/2}4^k\frac{(2k)!}{k!}\frac{M}{(2\pi)^{2k}}(\log\log N)^{k-1/2}.$$

Proof. First, we estimate the sum T_{2k-1}. Setting $a=\Delta(n)$ and $b=r(n)$ in the easy-to-verify identity

$$(a-b)^{2k} = a^{2k} - 2ka^{2k-1}b + \theta_1k^22^{2k-2}(a^{2k-2}b^2+b^{2k})$$

and performing simple transformations, we obtain

$$2k\Delta^{2k-1}(n)r(n) = \Delta^{2k}(n) - \Delta^{2k}(n-1) + \theta_2 k^2 2^{2k-2}(r^{2k}(n) + \Delta^{2k-2}(n)r^2(n)).$$

Summing over n, we arrive at the relation

$$2kT_{2k-1} = \Delta^{2k}(N+M) - \Delta^{2k}(N) + \theta_3 k^2 2^{2k-2}(W_1 + W_2),$$

where

$$W_1 = \sum_n r^{2k}(n) \qquad \text{and} \qquad W_2 = \sum_n \Delta^{2k-2}(n)r^2(n).$$

According to Theorem 3.5, we have $W_1 < c_0(2c_0k)^{2k}M$. Applying Hölder's inequality and using the notation and assertion of Theorem 2.1, we obtain

$$\begin{aligned} W_2 &\le \left(\sum_n \Delta^{2k}(n)\right)^{1-1/k} W_1^{1/k} < (2.01\mathcal{L}_k)^{1-1/k}(c_0(2c_0k)^{2k}M)^{1/k} \\ &< 2.01\frac{\mathcal{L}_k}{L}\left\{\frac{k!}{(2k)!}c_0(4\pi c_0 k)^{2k}\right\}^{1/k} < 10.95c_0^3\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}. \end{aligned}$$

Therefore,

$$\begin{aligned} W_1 + W_2 &< c_0(2c_0k)^{2k}M + 10.95c_0^3\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}} \\ &< 10.95c_0^3\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}\left(1 + \frac{L}{40(\pi c_0)^2}\frac{(k-1)!}{(2k)!}\left(\frac{4\pi c_0 k}{\sqrt{L}}\right)^k\right) \\ &< 11c_0^3\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}, \end{aligned}$$

whence

$$|T_{2k-1}| < (0.5\log N)^{2k} + \frac{4^k k}{8}11c_0^3\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}} < 1.5c_0^3k4^k\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}.$$

Now, let us estimate the sum T_{2k}. Setting $a = \Delta(n)$ and $b = r(n)$ in the identity

$$(a-b)^{2k+1} = a^{2k+1} - (2k+1)a^{2k}b + \theta_1 k(2k+1)2^{2k-2}(|b|^{2k+1} + |a|^{2k-1}b^2)$$

and performing transformations, we obtain

$$(2k+1)T_{2k+1} = \Delta^{2k}(N+M) - \Delta^{2k+1}(N) + \theta_2 k(2k+1)2^{2k-2}(W_3 + W_4),$$

where

$$W_3 = \sum_n |r(n)|^{2k+1} \qquad \text{and} \qquad W_4 = \sum_n |\Delta(n)|^{2k-1}r^2(n).$$

By virtue of Hölder's inequality and Theorem 3.5, we have

$$W_3 = \sum_n |r(n)|^{2k-1}r^2(n) \le W_1^{1-1/(2k)}\left(\sum_n r^{4k}(n)\right)^{1/(2k)} < 4c_0(2c_0k)^{2k+1}M.$$

Next,

$$\begin{aligned} W_4 &\le \left(\sum_n \Delta^{2k}(n)\right)^{1-1/(2k)}\left(\sum_n r^{4k}(n)\right)^{1/(2k)} \\ &< 2\frac{\mathcal{L}_k}{\sqrt{L}}\left\{\frac{k!}{(2k)!}c_0(2\pi)^{2k}(4c_0k)^{4k}\right\}^{1/(2k)} < 165.8\frac{c_0^{5/2}k^{3/2}}{\sqrt{L}}\mathcal{L}_k. \end{aligned}$$

Therefore,

$$W_3+W_4<165.8\frac{c_0^{5/2}k^{3/2}}{\sqrt{L}}\,\mathcal{L}_k\biggl(1+\frac{1}{48}\sqrt{\frac{L}{c_0k}}\biggl(\frac{(2\pi c_0)^2ek}{L}\biggr)^k\biggr)<166\frac{c_0^{5/2}k^{3/2}}{\sqrt{L}}\,\mathcal{L}_k,$$

and

$$|T_{2k}|<41.5k\cdot 4^k\frac{c_0^{5/2}k^{3/2}}{\sqrt{L}}\,\mathcal{L}_k<42(c_0k)^{5/2}4^k\frac{\mathcal{L}_k}{\sqrt{L}}.$$

This proves the theorem.

The following lemma reveals the geometric meaning of the quantity $r(n)$.

Lemma 3.3. *The quantity $r(n)+1$ equals the number of the ordinates of $\zeta(s)$ zeros in the Gram interval $G_n=(t_{n-1},t_n]$, or, equivalently,*

$$r(n)=N(t_n+0)-N(t_{n-1}+0)-1.$$

In particular, $r(n)\ge -1$ for any n.

Proof. Using the Riemann–Mangoldt formula, we obtain

$$\begin{aligned}N(t_n+0)-N(t_{n-1}+0)&=(\pi^{-1}\vartheta(t_n)+1+S(t_n+0))-(\pi^{-1}\vartheta(t_{n-1})+1+S(t_{n-1}+0))\\&=1+\Delta(n)-\Delta(n-1)=1+r(n).\end{aligned}$$

This proves the lemma.

The second proof of Selberg's formulas is based on the following lemma, which establishes a relationship between the sum of the numbers Δ_m^k over all ordinates γ_m in the nth Gram interval G_n and the quantity $\Delta^k(n)$.

Lemma 3.4. *Let k and n be any positive integers and suppose that the Gram interval $G_n=(t_{n-1},t_n]$ contains $r+1$ ordinates of zeros of $\zeta(s)$, where $r=r(n)\ge -1$. Then*

$$\sum_{t_{n-1}<\gamma_m\le t_n}\Delta_m^{2k}=(r+1)\Delta^{2k}(n)+\theta_1k2^{2k}(|\Delta(n)|^{2k-1}r^2+|r|^{2k+1}) \tag{3.14}$$

and

$$\sum_{t_{n-1}<\gamma_m\le t_n}\Delta_m^{2k-1}=-(r+1)\Delta^{2k-1}(n)+\theta_2k2^{2k}(\Delta^{2k-2}(n)r^2+r^{2k}). \tag{3.15}$$

Proof. First, consider the case $r=-1$, in which G_n contains no ordinates. In this case, the sums on the left-hand sides of (3.14) and (3.15) are empty, and the assertion of the lemma holds for $\theta_1=\theta_2=0$.

Now, suppose that $r\ge 0$ and

$$\gamma_{s-1}\le t_{n-1}<\gamma_s\le\gamma_{s+1}\le\cdots\le\gamma_{s+r}\le t_n<\gamma_{s+r+1}$$

for some $s\ge 1$. Then

$$\Delta(n)=S(t_n+0)=N(t_n+0)-\pi^{-1}\vartheta(t_n)-1=s+r-n$$

and, therefore,

$$\begin{aligned}\Delta_s&=n-s=r-\Delta(n),\\\Delta_{s+1}&=n-s-1=r-1-\Delta(n),\\&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\\\Delta_{s+r}&=n-s-r=-\Delta(n).\end{aligned}$$

Thus,

$$\sum_{t_{n-1}<\gamma_m\le t_n}\Delta_m^{2k}=\sum_{m=s}^{s+r}\Delta_m^{2k}=\sum_{j=0}^{r}(j-\Delta(n))^{2k}=\sum_{j=0}^{r}(\Delta^{2k}(n)+\theta_1 2kj2^{2k-2}(|\Delta(n)|^{2k-1}+j^{2k-1}))$$
$$=(r+1)\Delta^{2k}(n)+\theta_2 k2^{2k}(|\Delta(n)|^{2k-1}r^2+r^{2k+1}).$$

Relation (3.15) is derived in a similar way. This completes the proof of the lemma.

The second proof of Selberg's formulas. We define integers μ and ν by the inequalities

$$\gamma_\mu\le t_N<\gamma_{\mu+1}\qquad\text{and}\qquad\gamma_\nu\le t_{N+M}<\gamma_{\nu+1}$$

and consider the sum

$$V_{2k-1}=\sum_{m=\mu+1}^{\nu}\Delta_m^{2k-1}.$$

By virtue of the definition of $\Delta(n)$ and Lemma 3.1, we have

$$|\mu-N|=|N(t_N+0)-N|=|\pi^{-1}\vartheta(t_N)+1+S(t_N+0)-N|=|\Delta(N)|<0.15\log N$$

and, similarly,

$$|\nu-(N+M)|<0.15\log N.$$

Thus, the sum V_{2k-1} differs from $\sum_m\Delta_m^{2k-1}$ by no more than $2\cdot0.15\log N$ summands, each of which does not exceed $(0.15\log N)^{2k-1}$ in absolute value by Lemma 3.1, i.e.,

$$V_{2k-1}=\sum_m\Delta_m^{2k-1}+2\theta_1(0.15\log N)^{2k}.$$

On the other hand, Lemma 3.4 implies

$$V_{2k-1}=\sum_n\sum_{t_{n-1}<\gamma_m\le t_n}\Delta_m^{2k}=-\sum_n((r(n)+1)\Delta^{2k-1}(n)+\theta_2k2^{2k}(\Delta^{2k-2}(n)r^2(n)+r^{2k}(n)))$$
$$=-\sum_n\Delta^{2k-1}(n)-T_{2k-1}+\theta_3k2^{2k}(W_1+W_2),$$

where W_1 and W_2 denote the same sums as in the proof of Theorem 3.6. Using the estimates of these sums and Theorems 1.2 and 3.6, we obtain

$$|V_{2k-1}|<\frac{c_0^{3/2}}{2}\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}+c_0^3k4^k\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}+6c_0^3k4^k\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}$$
$$<7.1c_0^3k4^k\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}$$

and, finally,

$$\left|\sum_m\Delta_m^{2k-1}\right|<7.2c_0^3k4^k\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}.$$

In the case of even moments, a similar argument leads to the relation

$$\sum_{m=\mu+1}^{\nu}\Delta_m^{2k}=\sum_n\Delta^{2k}(n)+T_{2k}+\theta_3\,k2^{2k}(W_3+W_4),$$

where W_3 and W_4 are the same as in Theorem 3.6. Therefore,

$$\sum_m \Delta_m^{2k} = \mathcal{L}_k\left(1+\frac{2.4\theta_4(c_0k)^{3/2}}{\sqrt{L}}\right) + 41.5\theta_5\mathcal{L}_k\cdot\frac{(c_0k)^{5/2}}{\sqrt{L}} + 166\theta_6\mathcal{L}_k\cdot\frac{(c_0k)^{5/2}}{\sqrt{L}} + 2\theta_7(0.15\log N)^{2k+1}$$
$$= \mathcal{L}_k\left(1+\frac{208\theta_8(c_0k)^{5/2}4^k}{\sqrt{L}}\right).$$

This completes the proof of the theorem.

The results obtained above are less precise than Theorems 2.1 and 2.2. For example, it is easy to see that the expressions for even moments are asymptotic only for

$$k \le ((4\log 2)^{-1} - \delta)\log\log\log N.$$

Nevertheless, even these results make it possible to prove the first assertion of Theorem 3.3 on the distribution of the quantities Δ_n and, in particular, Selberg's conjecture on the "normal" order of these quantities.

Let us make the following observation. A trivial estimate of the sum T_k, such as

$$|T_k| = \left|\sum_n \Delta^k(n)r(n)\right| \le \left(\sum_n \Delta^{2k}(n)\sum_n r^2(n)\right)^{1/2} \le c(k)M(\log\log N)^{k/2},$$

is too rough to be used for deriving Selberg's formulas. The reason why accuracy drops is the loss of the information about the signs of summands, in particular, of the integers $r(n)$. According to Lemma 3.3, we have $r(n) \ge -1$ for any n. It turns out that, in a positive fraction of cases, $r(n)$ takes the value (-1), and in a positive fraction of cases, it is strictly positive (this is proved below). This means that the sums T_k are alternating (at least for even k). But the trick used in the proof of Theorem 3.6 makes it possible to detect the mutual cancellation of summands of opposite signs.

3.12. The third proof of Selberg's formulas. Although estimating alternating sums T_k does not cause significant difficulties, the consideration of the explicit form of the remainders in (3.14) and (3.15) makes it possible to avoid the application of such estimates and derive Selberg's formulas with the same remainders as in Theorem 3.1 without relying on any assertions about the distribution of Δ_n. This is the purpose of this section.

As above, we set $r(n) = \Delta(n) - \Delta(n-1)$. For nonnegative integers p and q, we define

$$T_{p,q} = \sum_n r^p(n)\Delta^q(n),$$

so that

$$\sum_n \Delta^k(n) = T_{0,k} \qquad \text{and} \qquad T_k = \sum_n \Delta^k(n)r(n) = T_{1,k}.$$

Finally, by B_ν we denote the Bernoulli numbers.

Lemma 3.5. *For any integers $k \ge 1$ and $n \ge 1$,*

$$\sum_{t_{n-1}<\gamma_m\le t_n} \Delta_m^k = \sum_{q=0}^{k}(-1)^q\binom{k}{q}\Delta^q(n)(r^{k-q} + F_{k+1-q}(r)), \tag{3.16}$$

where $r = r(n)$ and

$$F_p(x) = \frac{1}{p}\sum_{\nu=0}^{p-1}\binom{p}{\nu}B_\nu x^{p-\nu}.$$

Proof. First, consider the case where the Gram interval $G_n = (t_{n-1}, t_n]$ contains at least one ordinate of a zero of $\zeta(s)$. Namely, suppose that

$$\gamma_{s-1} \le t_{n-1} < \gamma_s \le \gamma_{s+1} \le \cdots \le \gamma_{s+r} \le t_n < \gamma_{s+r-1},$$

where $r = r(n) \ge 0$ by Lemma 3.3. Then

$$\Delta(n) = s + r - n$$

and

$$\begin{aligned}
\Delta_s &= n - s = r - \Delta(n),\\
\Delta_{s+1} &= n - s - 1 = r - 1 - \Delta(n),\\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\\
\Delta_{s+r} &= n - s - r = -\Delta(n),
\end{aligned}$$

whence

$$\begin{aligned}
\sum_{t_{n-1}<\gamma_m\le t_n} \Delta_m^k &= \sum_{m=s}^{s+r} \Delta_m^k = \sum_{j=0}^{r} (j - \Delta(n))^k\\
&= \sum_{j=0}^{r}\sum_{q=0}^{k} (-1)^q \binom{k}{q} \Delta^q(n) j^{k-q} = \sum_{q=0}^{k} (-1)^q \binom{k}{q} \Delta^q(n) \sum_{j=0}^{r} j^{k-q}.
\end{aligned}$$

Using a well-known property of Bernoulli numbers (see, e.g., [51, Section 1, Chapter IV]), we find that, for $r \ge 1$,

$$\sum_{j=0}^{r} j^{k-q} = r^{k-q} + \sum_{j=0}^{r-1} j^{k-q} = r^{k-q} + \frac{1}{k-q+1} \sum_{\nu=0}^{k-q} \binom{k-q+1}{\nu} B_\nu r^{k-q+1-\nu} = r^{k-q} + F_{k-q+1}(r).$$

This relation remains valid for $r = 0$, because $F_{k-q+1}(0) = 0$ for any q, $0 \le q \le k$. Thus,

$$\sum_{t_{n-1}<\gamma_m\le t_n} \Delta_m^k = \sum_{q=0}^{k} (-1)^q \binom{k}{q} \Delta^q(n)(r^{k-q} + F_{k-q+1}(r)).$$

Now, consider the case where the interval $(t_{n-1}, t_n]$ contains no ordinates. In this case, according to Lemma 3.3, we have

$$r(n) = N(t_n + 0) - N(t_{n-1} + 0) - 1 = -1.$$

The sum over $t_{n-1} < \gamma_m \le t_n$ in the statement of the lemma turns out to be empty. Thus, it suffices to check that the right-hand side of (3.16) vanishes under the substitution of $r = -1$. But this immediately follows from the relations

$$F_{p+1}(x+1) - F_{p+1}(x) = x^p \qquad \text{and} \qquad F_{p+1}(0) = 0.$$

This completes the proof of the lemma.

Example 3.1. Let $k = 2$. Then, for $r = r(n) \ge 0$, we have

$$\begin{aligned}
\sum_{t_{n-1}<\gamma_m\le t_n} \Delta_m^2 &= \sum_{j=0}^{r} (j - \Delta(n))^2 = \sum_{j=0}^{r} (j^2 - 2j\Delta(n) + \Delta^2(n))\\
&= \frac{1}{6} r(r+1)(2r+1) - r(r+1)\Delta(n) + (r+1)\Delta^2(n).
\end{aligned}$$

At $r = -1$, the right-hand side vanishes and, thereby, coincides with the left-hand side.

Theorem 3.7. *For any $k \ge 1$, the identity*

$$\sum_{t_N<\gamma_m\le t_{N+M}} \Delta_m^k = (-1)^k T_{0,k} + \alpha^k - \beta^k + F_{k+1}(\alpha) - F_{k+1}(\beta) \tag{3.17}$$

holds, where $\alpha = -\Delta(N)$ and $\beta = -\Delta(N+M)$.

Proof. According to Lemma 3.5, we have

$$\sum_{t_{n-1}<\gamma_m\le t_n}\Delta_m^k=\sum_{q=0}^{k}(-1)^q\binom{k}{q}r^{k-q}(n)\Delta^q(n)$$
$$+\sum_{q=0}^{k}\frac{(-1)^q}{k+1-q}\binom{k}{q}\sum_{p=1}^{k-q+1}\binom{k+1-q}{p}B_{k+1-q-p}r^p(n)\Delta^q(n).$$

Summing each side of this equality over n and denoting the left-hand side of (3.17) by V_k, we obtain

$$V_k=\sum_{q=0}^{k}(-1)^q\binom{k}{q}T_{k-q,q}+\sum_{q=0}^{k}\frac{(-1)^q}{k+1-q}\binom{k}{q}\sum_{p=1}^{k-q+1}\binom{k+1-q}{p}B_{k+1-q-p}T_{p,q}$$
$$=\frac{1}{k+1}T_{k+1,0}+\frac{1}{2}T_{k,0}+\frac{1}{k+1}\sum_{p=1}^{k-1}\binom{k+1}{p}B_{k+1-p}T_{p,0}$$
$$+\sum_{q=1}^{k}(-1)^q\binom{k}{q}T_{k-q,q}+\sum_{q=1}^{k}\frac{(-1)^q}{k+1-q}\binom{k}{q}\sum_{p=1}^{k-q+1}\binom{k+1-q}{p}B_{k+1-q-p}T_{p,q}. \tag{3.18}$$

Let us express each of the sums $T_{p,0}$, $1\le p\le k+1$, in terms of the sums $T_{p-q,q}$, $1\le q\le p-1$. For this purpose, we use the identity

$$\Delta^p(n)-\Delta^p(n-1)=\Delta^p(n)-(\Delta(n)-r(n))^p=\sum_{q=0}^{p-1}(-1)^{p-q-1}\binom{p}{q}r^{p-q}(n)\Delta^q(n).$$

Summing each side of this identity over n and setting $D_p=\Delta^p(N+M)-\Delta^p(N)$, we obtain

$$D_p=\sum_{q=0}^{p-1}(-1)^{p-q-1}\binom{p}{q}T_{p-q,q},$$
$$T_{p,0}=(-1)^{p-1}D_p+\sum_{q=1}^{p-1}(-1)^{q-1}\binom{p}{q}T_{p-q,q}. \tag{3.19}$$

Let us replace the quantities $T_{p,0}$ in (3.18) by their expressions (3.19). First, note that the term containing $T_{0,k}$ in not contained in (3.19) for any p; therefore, the coefficient multiplying $T_{0,k}$ in (3.18) equals $(-1)^k\binom{k}{k}=(-1)^k$.

Next, the substitution of the expression for $T_{p,0}$, $p\ne k$, into (3.18) entails the appearance of the summand

$$\frac{(-1)^{p-1}D_p}{k+1}\binom{k+1}{p}B_{k+1-p}=\binom{k+1}{p}\frac{B_{k+1-p}}{k+1}(\alpha^p-\beta^p);$$

the coefficient multiplying the sum $T_{p-q,q}$, $1\le q\le p-1$ in the resulting expression equals

$$\frac{(-1)^{q-1}}{k+1}\binom{k+1}{p}\binom{p}{q}B_{k+1-p}+\frac{(-1)^q}{k+1-q}\binom{k}{q}\binom{k+1-q}{p-q}B_{k+1-p}$$
$$=(-1)^{q-1}B_{k+1-p}\left(\frac{1}{k+1}\binom{k+1}{p}\binom{p}{q}-\frac{1}{k+1-q}\binom{k}{q}\binom{k+1-q}{p-q}\right)=0.$$

Finally, after the substitution into (3.18) of the expression for $T_{k,0}$, there arises the summand

$$\frac{1}{2}(-1)^{k-1}D_k=\frac{1}{2}(\alpha^k-\beta^k);$$

the coefficient multiplying the sum $T_{k-q,q}$, $1 \le q \le k-1$, turns out to equal

$$\frac{1}{2}(-1)^{q-1}\binom{k}{q} + (-1)^q\binom{k}{q} + \frac{(-1)^q B_1}{k+1-q}\binom{k}{q}\binom{k+1-q}{k-q} = 0.$$

Thus,

$$\begin{aligned} V_k &= (-1)^k T_{0,k} + \frac{1}{2}(\alpha^k - \beta^k) + \sum_{p=1,\, p\ne k}^{k+1}\binom{k+1}{p}\frac{B_{k+1-p}}{k+1}(\alpha^p - \beta^p) \\ &= (-1)^k T_{0,k} + \alpha^k - \beta^k + \frac{1}{k+1}\sum_{q=0}^{k}\binom{k+1}{q}B_q(\alpha^{k+1-q} - \beta^{k+1-q}) \\ &= (-1)^k T_{0,k} + \alpha^k - \beta^k + F_{k+1}(\alpha) - F_{k+1}(\beta). \end{aligned}$$

This completes the proof of the theorem.

Example 3.2. Let $k = 2$. Summing each side of the equality

$$\sum_{t_{n-1}<\gamma_m\le t_n}\Delta_m^2 = \frac{1}{3}r^3 + \frac{1}{2}r^2 + \frac{1}{6}r - r^2\Delta(n) - r\Delta(n) + r\Delta^2(n) + \Delta^2(n)$$

from Example 3.1 over n, we obtain

$$V_2 = \frac{1}{3}T_{3,0} + \frac{1}{2}T_{2,0} + \frac{1}{6}T_{1,0} - T_{2,1} - T_{1,1} + T_{1,2} + T_{0,2}. \tag{3.20}$$

Summing the identities

$$\begin{aligned} &\Delta^3(n) - \Delta^3(n-1) = \Delta^3(n) - (\Delta(n) - r)^3 = 3r\Delta^2(n) - 3r^2\Delta(n) + r^3, \\ &\Delta^2(n) - \Delta^2(n-1) = 2r\Delta(n) - r^2, \end{aligned}$$

and

$$\Delta(n) - \Delta(n-1) = r$$

over n yields

$$D_3 = 3T_{1,2} - 3T_{2,1} + T_{3,0}, \qquad D_2 = 2T_{1,1} - T_{2,0}, \qquad \text{and} \qquad D_1 = T_{1,0},$$

whence

$$T_{3,0} = D_3 - 3T_{1,2} + 3T_{2,1}, \qquad T_{2,0} = -D_2 + 2T_{1,1}, \qquad \text{and} \qquad T_{1,0} = D_1.$$

Substituting these expressions into (3.20), we arrive at

$$\begin{aligned} V_2 &= \left(\frac{1}{3}D_3 - T_{1,2} + T_{2,1}\right) + \left(-\frac{1}{2}D_2 + T_{1,1}\right) + \frac{1}{6}D_1 - T_{2,1} - T_{1,1} + T_{1,2} + T_{0,2} \\ &= T_{0,2} + \frac{1}{3}D_3 - \frac{1}{2}D_2 + \frac{1}{6}D_1. \end{aligned}$$

Corollary 3.2. *For any integer $k \ge 1$,*

$$\left|\sum_n \Delta_n^k - (-1)^k\sum_n \Delta^k(n)\right| \le 16k!(0.15\log N)^{k+1}.$$

Proof. First, using the estimate of Lemma I.9, we obtain

$$|D_p| \le 2H^p,$$

where $H = 0.15\log N$. Next, note that

$$|B_{2s}| = \frac{2(2s)!}{(2\pi)^{2s}}\zeta(2s) < \frac{4(2s)!}{(2\pi)^{2s}};$$

hence,

$$|B_q| < 4q!\,(2\pi)^{-q}$$

for any $q \geq 1$, so that

$$|V_k - (-1)^k T_{0,k}| \leq 2H^k + |F_{k+1}(\alpha) - F_{k+1}(\beta)| \leq 2H^k + \frac{1}{k+1}\sum_{q=0}^{k}\binom{k+1}{q}|B_q|2H^{k+1-q}$$

$$\leq 2H^k + \frac{2}{k+1}\sum_{q=0}^{k}\frac{(k+1)!}{(k+1-q)!\,q!}\frac{4q!}{(2\pi H)^q}H^{k+1}$$

$$\leq 2H^k + 8k!\,H^{k+1}\sum_{q=0}^{k}\frac{1}{(k+1-q)!} < 2H^k + 8(e-1)k!\,H^{k+1} < 14k!\,H^{k+1}.$$

The relations

$$N(t_N + 0) = N + \Delta(N) \qquad \text{and} \qquad N(t_{N+M} + 0) = N + M + \Delta(N+M)$$

imply that the sum V_k differs from $\sum_n \Delta_n^k$ by no more than $|\Delta(N)| + |\Delta(N+M)|$ summands, each of which does not exceed H^k in absolute value. Observing that

$$2H \cdot H^k + 14k!\,H^{k+1} \leq 16k!\,H^{k+1},$$

we obtain the required result.

The third proof of Selberg's formulas (in the form given in Theorem 3.1). It suffices to apply the corollary of Theorem 3.7 combined with Theorems 2.1 and 2.2.

Remark 3.2. The proof given above has yet another special feature, which makes it similar to the first method of derivation of Selberg's formulas.

The proofs of Lemma 3.5, Theorem 3.7, and its corollary do not essentially use the specifics of the sequences $\{\gamma_n\}$ and $\{t_n\}$. In fact, these statements have analogues in a more general situation.

Namely, let $\{\alpha_n\}$ and $\{\beta_n\}$ be unbounded nondecreasing sequences such that $\beta_1 < \alpha_1 \leq \beta_2$. Given $n \geq 1$, let us define the number $m = m(n)$ by the inequalities $\beta_{m-1} < \alpha_n \leq \beta_m$ and set $\Delta_n = m - n$. Similarly, given $\nu \geq 2$, we define $\mu = \mu(\nu)$ by the conditions $\alpha_\mu \leq \beta_\nu < \alpha_{\mu+1}$ and set $\Delta(\nu) = \mu - \nu$. Finally, let $D = D(N)$ denote the maximum of the quantities $|\Delta_n|$ and $|\Delta(n)|$ over all $n \leq 2N$. Then, for any integer $k \geq 1$, we have

$$\left|\sum_n \Delta_n^k - (-1)^k \sum_n \Delta^k(n)\right| \leq 16k!\,D^{k+1}.$$

Lemma 3.2, on which the first proof of Selberg's formulas is based, does not use the specifics of the sequences of Gram points and ordinates of zeros of $\zeta(s)$ either. It has an analogue in the situation where the points γ_n and t_n are replaced by the members of sequences $\{\alpha_n\}$ and $\{\beta_n\}$ satisfying the above conditions.

3.13. The behavior of the differences $t_n - \gamma_n$. Selberg's formulas show that, for almost every n, the number of the ordinate γ_n differs from the number m of the Gram interval containing this ordinate by a quantity of order $\sqrt{\log\log n}$.

The statements of this section express the same fact in terms of the differences $\gamma_n - t_n$. It turns out that the distance between an ordinate γ_n and the Gram point t_n with the same number is greater than the length of the Gram interval $G_n = (t_{n-1}, t_n]$ by a factor of $\sqrt{\log\log n}$ "on average."

In this section, we also prove theorems on the fractional moments and distributions of the differences $\gamma_n - t_n$ similar to Theorems 3.2–3.4.

Theorem 3.8. *Let k be an integer such that $1 \le k \le c_1 \sqrt[3]{\log\log N}$. Then*

$$\sum_n (\gamma_n - t_n)^{2k} = \frac{(2k)!}{k!} M \left(\frac{\sqrt{\log\log N}}{2\log N} \right)^{2k} \left(1 + \frac{2.4\theta (c_0 k)^{3/2}}{\sqrt{\log\log N}} \right)$$

and

$$\left| \sum_n (\gamma_n - t_n)^{2k-1} \right| < 4c_0^3 \frac{(2k)!}{(k-1)!} \frac{M}{\sqrt{\log\log N}} \left(\frac{\sqrt{\log\log N}}{\log N} \right)^{2k-1}.$$

Proof. Suppose that $t_{m-1} < \gamma_n \le t_m$ for given n, so that $\Delta_n = m - n$. Then

$$t_{m-1} - t_n < \gamma_n - t_n \le t_m - t_n.$$

Lemma 1.1 implies the relations

$$t_m - t_n = \frac{\pi \Delta_n}{\vartheta'(t_N)} + \frac{2\pi\theta_1 |\Delta_n| M}{N(\log N)^2} = \frac{\pi\Delta_n}{\vartheta'(t_N)} \left(1 + \frac{\pi\theta_2 M}{N\log N} \right)$$

and

$$t_{m-1} - t_n = \frac{\pi(\Delta_n - 1)}{\vartheta'(t_N)} \left(1 + \frac{\pi\theta_3 M}{N\log N} \right),$$

which, in turn, imply

$$\gamma_n - t_n = \frac{\pi(\Delta_n - \theta_4)}{\vartheta'(t_N)} \left(1 + \frac{\pi\theta_5 M}{N\log N} \right),$$

where θ_4 and θ_5 depend on n.

Let us prove the first relation. It follows from the above considerations that the difference of the quantities

$$\sum_n (\gamma_n - t_n)^{2k} \qquad \text{and} \qquad \left(\frac{\pi}{\vartheta'(t_N)} \right)^{2k} \sum_n \Delta_n^{2k}$$

does not exceed the sum

$$\left(\frac{\pi}{\vartheta'(t_N)} \right)^{2k} \left\{ \frac{M}{N} \sum_n \Delta_n^{2k} + \left(1 + \frac{M}{N} \right) \sum_{\nu=1}^{2k} \binom{2k}{\nu} \sum_n |\Delta_n|^{2k-\nu} \right\} \tag{3.21}$$

in absolute value. According to Lemma 3.1, the first summand in braces is at most

$$\frac{M}{N} \cdot M(0.15\log N)^{2k} < N^{-1/3}.$$

We split the sum over ν in (3.21) into the three subsums Σ_1, Σ_2, and Σ_3 corresponding to $\nu = 1$, even ν, and odd ν, respectively. Applying Theorem 3.1, we obtain

$$\Sigma_1 = 2k \sum_n |\Delta_n|^{2k-1} \le 2kM^{1/(2k)} \left(\sum_n \Delta_n^{2k} \right)^{1-1/(2k)} \le 2kM^{1/(2k)} \{2.03\mathcal{L}_k\}^{1-1/(2k)}$$
$$\le 2k \cdot 2.03\mathcal{L}_k \left(\frac{k!}{(2k)!} (2\pi)^{2k} \right)^{1/(2k)} < 21.1 \sqrt{\frac{k}{L}}\, \mathcal{L}_k.$$

Similarly,

$$\Sigma_2 \le \sum_{\mu=1}^{k} \binom{2k}{2\mu} \cdot 2.03 \frac{(2k-2\mu)!}{(k-\mu)!} \frac{ML^{k-\mu}}{(2\pi)^{2(k-\mu)}}$$
$$= 2.03\mathcal{L}_k \sum_{\mu=1}^{k} \frac{\mu!}{(2\mu)!} \binom{k}{\mu} \left(\frac{4\pi^2}{L} \right)^{\mu} \le 1.02 L_k \left(\left\{ 1 + \frac{4\pi^2}{L} \right\}^k - 1 \right) < \frac{9k}{L} \mathcal{L}_k.$$

Finally, since the numbers Δ_n are integer, it follows that

$$\begin{aligned}\Sigma_3 &= \sum_{\mu=1}^{k-1}\binom{2k}{2\mu+1}\sum_n|\Delta_n|^{2k-2\mu-1}\\ &\le \sum_{\mu=1}^{k-1}\binom{2k}{2\mu+1}\sum_n\Delta_n^{2k-2\mu}\le 2.03\sum_{\mu=1}^{k-1}\binom{2k}{2\mu+1}\frac{(2k-2\mu)!}{(k-\mu)!}\frac{ML^{k-\mu}}{(2\pi)^{2(k-\mu)}}\\ &= 4.06\mathcal{L}_k\sum_{\mu=1}^{k-1}\frac{(\mu+1)!}{(2\mu+1)!}\binom{k}{\mu+1}\left(\frac{4\pi^2}{L}\right)^{\mu}<4.06\mathcal{L}_k\frac{L}{6\pi^2}\sum_{\mu=2}^{k}\binom{k}{\mu}\left(\frac{4\pi^2}{L}\right)^{\mu}\\ &< 2.71\mathcal{L}_k\frac{L}{4\pi^2}\left(\left\{1+\frac{4\pi^2}{L}\right\}^k-1-\frac{4\pi^2}{L}\right)<\frac{54k^2}{L}\mathcal{L}_k.\end{aligned}$$

Adding up the obtained estimates, we see that (3.21) does not exceed

$$\left(\frac{\pi}{\vartheta'(t_N)}\right)^{2k}\left\{N^{-1/3}+\left(1+\frac{M}{N}\right)\mathcal{L}_k\left(21\sqrt{\frac{k}{L}}+\frac{9k}{L}+\frac{54k^2}{L}\right)\right\}<22\sqrt{\frac{k}{L}}\left(\frac{\pi}{\vartheta'(t_N)}\right)^{2k}\mathcal{L}_k.$$

Thus, the expression for the initial sum takes the form

$$\begin{aligned}&\left(\frac{\pi}{\vartheta'(t_N)}\right)^{2k}\mathcal{L}_k\left(1+\frac{4.3\theta_1(c_0k)^{3/2}}{\sqrt{L}}\right)+22\theta_2\sqrt{k}L\left(\frac{\pi}{\vartheta'(t_N)}\right)^{2k}\mathcal{L}_k\\ &\quad=\left(\frac{\pi}{\vartheta'(t_N)}\right)^{2k}\mathcal{L}_k\left(1+\frac{4.35\theta_1(c_0k)^{3/2}}{\sqrt{L}}\right)=\left(\frac{2\pi}{\log N}\right)^{2k}\mathcal{L}_k\left(1+\frac{4.4\theta_3(c_0k)^{3/2}}{\sqrt{L}}\right).\end{aligned}$$

Let us prove the second relation. Repeating the above argument almost word for word, we conclude that the absolute value of the $(2k-1)$th moment is at most

$$\left(\frac{\pi}{\vartheta'(t_N)}\right)^{2k-1}\left\{\left|\sum_n\Delta_n^{2k-1}\right|+\frac{M}{N}\sum_n|\Delta_n|^{2k-1}+\left(1+\frac{M}{N}\right)\sum_{\nu=1}^{2k-1}\binom{2k-1}{\nu}|\Delta_n|^{2k-\nu-1}\right\}.$$

Let us denote the contribution to the last sum of odd numbers ν by Σ_1 and the contribution of even numbers ν by Σ_2; we have

$$\begin{aligned}\Sigma_1&\le\sum_{\mu=1}^{k}\binom{2k-1}{2\mu-1}\sum_n\Delta_n^{2k-2\mu}\le 2.03\sum_{\mu=1}^{k}\binom{2k-1}{2\mu-1}\frac{(2k-2\mu)!}{(k-\mu)!}\frac{ML^{k-\mu}}{(2\pi)^{2(k-\mu)}}<\frac{40k}{L}\mathcal{L}_k,\\ \Sigma_2&\le\sum_{\mu=1}^{k-1}\binom{2k-1}{2\mu}\sum_n|\Delta_n|^{2k-2\mu-1}\le\sum_{\mu=1}^{k-1}\binom{2k-1}{2\mu}\sum_n\Delta_n^{2k-2\mu}\\ &\le 2.03\sum_{\mu=1}^{k-1}\binom{2k-1}{2\mu}\frac{(2k-2\mu)!}{(k-\mu)!}\frac{ML^{k-\mu}}{(2\pi)^{2(k-\mu)}}<\frac{40k}{L}\mathcal{L}_k.\end{aligned}$$

Using Theorem 3.1, we finally obtain

$$\begin{aligned}\left|\sum_n(\gamma_n-t_n)^{2k-1}\right|&<\left(\frac{\pi}{\vartheta'(t_N)}\right)^{2k-1}\left\{\frac{c_0^{3/2}}{2}\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}+N^{-1/3}+\frac{80k}{L}\mathcal{L}_k\right\}\\ &<\frac{3}{5}c_0^3\left(\frac{2\pi}{\log N}\right)^{2k-1}\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}\\ &<4c_0^3\frac{(2k)!}{(k-1)!}\frac{M}{\sqrt{\log\log N}}\left(\frac{\sqrt{\log\log N}}{\log N}\right)^{2k-1}.\end{aligned}$$

This completes the proof of the theorem.

Theorem 3.9. *Let a be any number not being an even integer and satisfying the condition $0 < a \le c_2\sqrt[3]{\log\log N}$. Then*

$$\sum_n |\gamma_n - t_n|^a = \frac{M}{\sqrt{\pi}}\Gamma\left(\frac{a+1}{2}\right)\left(\frac{2\sqrt{\log\log N}}{\log N}\right)^a(1+\theta r(a)),$$

where

$$r(a) = \begin{cases} 14.2\left(\dfrac{\log\log\log N}{\log\log N}\right)^{0.5a} & \text{if } 0 < a \le 1, \\ \dfrac{783a}{\sqrt{\log\log N}}(\sqrt{\log\log\log N} + 2.2c_0^{3/2}\sqrt{a}) & \text{if } a > 1. \end{cases}$$

The proof of this theorem repeats that of Theorem 3.2 almost word for word.

Theorem 3.10. *Let $\Xi(v)$ be the distribution function of $\gamma_n - t_n$, i.e.,*

$$\Xi(v) = \frac{1}{M}\#\left\{n \;\middle|\; N < n \le N+M,\ \gamma_n - t_n \le v\frac{\sqrt{2\log\log N}}{\log N}\right\}.$$

Then, for any u, the function $\Xi(u)$ can be expresses as[16]

$$\Xi(u) = G(u) + \frac{1.79\theta}{\sqrt{\log\log\log N}}.$$

Moreover, if $1.5 \le u \le \sqrt[6]{\log\log N}$, then this function satisfies the relations

$$\frac{\Xi(-u)}{G(-u)} = 1 + O(R) \qquad \text{and} \qquad \frac{1-\Xi(u)}{1-G(u)} = 1 + O(R),$$

in which

$$R = \frac{u(u^2 + \log\log\log N)}{\sqrt{\log\log N}}$$

and the constants in the big-O notations depend on ε.

The proof of this theorem is quite similar to those of Theorems 2.5 and 3.3.

Corollary 3.3. *Let $f(x)$ be a positive monotone function unboundedly increasing as $x \to +\infty$. Then the inequality*

$$\frac{1}{f(n)}\frac{\sqrt{\log\log n}}{\log n} < |\gamma_n - t_n| < f(n)\frac{\sqrt{\log\log n}}{\log n}$$

holds for almost all n.

Theorem 3.11. *Let $f(x)$ be any monotonically increasing function unbounded for $x \to +\infty$ and satisfying the conditions $a \le f(x) \le b\sqrt{\log\log x}$, where a and b are absolute constants, for $N \le x \le N+M$, and let*

$$\mathcal{F} = f(N) \qquad \text{and} \qquad \delta = 0.5\exp\left\{-(ec_0)^2\frac{\log\log N}{\mathcal{F}^2}\right\}.$$

Then the interval $N \le n \le N+M$ contains at least $M\delta$ numbers n for which

$$\gamma_n - t_n \le -\frac{\log\log N}{\mathcal{F}\log N},$$

and at least $M\delta$ numbers for which

$$\gamma_n - t_n > \frac{\log\log N}{\mathcal{F}\log N}.$$

The proof of this theorem repeats that of Theorem 2.6 almost word for word.

[16] The remainder term in the expression for $\Xi(u)$ can be improved; see Remark 2.4.

3.14. The behavior of the argument of the Riemann zeta-function at the points of discontinuity. The function $S(t)$ is piecewise smooth, has discontinuities at the points coinciding with the ordinates of zeros of $\zeta(s)$, and monotonically decreases in each interval of continuity (γ_n, γ_{n+1}). Thus, the right and left limits of $S(t)$ at the points of discontinuity, i.e., the quantities $S(\gamma_n \pm 0)$, have an obvious geometric meaning: these are the peaks of the saw-tooth graph of the function $S(t)$ (see Fig. 2).

It turns out that, just as the numbers $\Delta(n)$ are related to the values of the function $S(t)$ at the Gram points (see Lemma 2.8), the numbers Δ_n are related to the values of $S(t)$ at the points $t = \gamma_n \pm 0$, although this relation is of more complex form (see Lemmas 3.6 and 3.7).

We set

$$q_n = \frac{\gamma_n - t_n}{t_{n+1} - t_n}.$$

Lemma 3.6. *The relation*

$$q_n = \Delta_n - \theta_1 + \frac{\theta_2 \log n}{4n},$$

holds, where $0 \le \theta_1 \le 1$ *and* $|\theta_2| \le 1$.

Proof. Let $t_{m-1} < \gamma_n \le t_m$. Then

$$\frac{t_{m-1} - t_n}{t_{n+1} - t_n} < q_n \le \frac{t_m - t_n}{t_{n+1} - t_n}. \tag{3.22}$$

Using Lemma 1.1, we obtain

$$\begin{aligned}\frac{t_m - t_n}{t_{n+1} - t_n} &= \left(\frac{\pi(m-n)}{\vartheta'(t_n)} + \frac{\theta_1|m-n|}{n\log n}\right)\left\{\frac{\pi}{\vartheta'(t_n)}\left(1 + \frac{\theta_2}{n\log n}\right)\right\}^{-1} \\ &= \Delta_n\left(1 + (1+o(1))\frac{\theta_3(|\Delta_n|+1)}{n\log n}\right) = \Delta_n\left(1 + \frac{\theta_4}{2n}\right)\end{aligned}$$

and, similarly,

$$\frac{t_{m-1} - t_n}{t_{n+1} - t_n} = (\Delta_n - 1)\left(1 + \frac{\theta_5}{2n}\right).$$

Due to these relations and Lemma 3.1, it follows from (3.22) that

$$q_n = (\Delta_n - \theta_6)\left(1 + \frac{\theta_7}{2n}\right) = \Delta_n - \theta_6 + \frac{\theta_7 \log n}{4n},$$

where $0 \le \theta_6 \le 1$. This proves the lemma.

Lemma 3.7. *For any* n, $N < n \le N + M$, *the relations*

$$S(\gamma_n + 0) = -\Delta_n + \theta\kappa_n + \frac{\theta_1(\log N)^2}{80N} \qquad \text{and} \qquad S(\gamma_n - 0) = -\Delta_n - (1-\theta)\kappa_n + \frac{\theta_2(\log N)^2}{80N}$$

hold, where $0 \le \theta \le 1$.

Proof. According to the Riemann–Mangoldt formula and relation (3.6), we have

$$S(\gamma_n + 0) = N(\gamma_n + 0) - \frac{1}{\pi}\vartheta(\gamma_n) - 1 = n - 1 - \frac{1}{\pi}\vartheta(\gamma_n) + \theta_1(\kappa_n - 1).$$

From the definition of the fractions q_n we obtain

$$\gamma_n = t_n + h_n, \qquad h_n = q_n(t_{n+1} - t_n)$$

and

$$\vartheta(\gamma_n) = \vartheta(t_n) + h_n\vartheta'(t_n) + \frac{h_n^2}{2}\vartheta''(\xi), \quad \xi = t_n + \theta_2 h_n.$$

Next, by virtue of Lemmas 1.1 and 3.6, we have

$$h_n\vartheta'(t_n) = \pi q_n$$

and

$$\frac{1}{2}h_n^2\vartheta''(\xi) \le \frac{1}{2}h_n^2\vartheta''(t_n) = \frac{\pi q_n^2}{2N}\left(1 + (1+o(1))\frac{\log\log N}{\log N}\right) < \frac{2\pi|q_n|}{25}\frac{\log N}{N},$$

whence

$$\frac{\vartheta(\gamma_n)}{\pi} = n - 1 + q_n + \frac{2\theta_2|q_n|}{25}\frac{\log N}{N}.$$

Thus,

$$S(\gamma_n + 0) = -q_n + \theta_1(\kappa_n - 1) + \frac{2\theta_2|q_n|}{25}\frac{\log N}{N} = -\Delta_n + \theta_3\kappa_n + \frac{\theta_4(\log N)^2}{80N},$$

where $0 \le \theta_3 \le 1$. The second assertion of the lemma follows from the relation $S(\gamma_n + 0) - S(\gamma_n - 0) = \kappa_n$. This completes the proof of the lemma.

The relations of Lemma 3.7 show that, to calculate the moments of $S(\gamma_n \pm 0)$, we must estimate the moments of multiplicities κ_n.

The statements given below are aimed at showing that, on average, these multiplicities behave as constants. The proof of this phenomenon is based on the fact that, near ordinates of large multiplicity, the difference

$$S(t+2h) - S(t), \qquad h \asymp \left(\log\frac{t}{2\pi}\right)^{-1} \tag{3.23}$$

turns out to be sufficiently large and positive (Lemma 3.16). However, an upper bound for the mean value of function (3.23) (Lemma 3.14) shows that the measure of the set of those points t at which difference (3.23) is large in absolute value is very small (Lemma 3.15). The combination of these assertions leads to an upper estimate for the number of ordinates of very high multiplicity and gives the required estimate of the moments of κ_n (Lemmas 3.17 and 3.18).

Lemmas 3.8–3.13 are auxiliary; they are used to estimate the mean value of difference (3.23). These are integral analogues of Lemmas 2.2, 2.3, 1.3, 2.5, 2.6, and 2.7. As above, we assume that

$$T \ge T_0(\varepsilon) > 0 \qquad \text{and} \qquad T^{\alpha+\varepsilon_1} \le H \le T^{\alpha+\varepsilon}, \quad \alpha = \frac{27}{82}, \quad \varepsilon_1 = 0.9\varepsilon.$$

We also set

$$Y = T^{0.1\varepsilon}.$$

Lemma 3.8. *Suppose that $c > 0$, m is an integer,*

$$1 \le m \le 0.1\log M, \qquad 18 \le z \le \left(\frac{M}{5\log N}\right)^{1/(3m)},$$

and sequences of real numbers $a_1(p)$ and $a_2(p)$ satisfy the conditions $|a_1(p)| \le (c\log p)/\log z$ and $|a_2(p)| \le c$ for p ranging over all primes in the interval $(1, z]$. Then the integrals

$$I_1 = \int_T^{T+H}\left|\sum_{p\le z}\frac{a_1(p)}{\sqrt{p}}p^{it}\right|^{2m}dt \qquad \textit{and} \qquad I_2 = \int_T^{T+H}\left|\sum_{p\le z}\frac{a_2(p)}{p}p^{2it}\right|^{2m}dt$$

are estimated as

$$I_1 < 1.1H(c^2m)^m \qquad \textit{and} \qquad I_2 < 1.1H(0.5c^2m)^m.$$

Proof. To prove the first assertion, we use Lemma I.9 with

$$a_n = \sum_{\substack{p_1 \dots p_m = n \\ p_1, \dots, p_m \le z}} a_1(p_1) \cdots a_1(p_m)$$

and note that $|a_n| \le m! c^m$. The second estimate is obtained in a similar way.

Lemma 3.9. *Suppose that $\nu \ge 0$ is an integer, $y \ge 2$, $z \ge 1$, and $y^3 z^2 \le Y$. Then*

$$\int_T^{T+H} \left(\sigma_{y,t} - \frac{1}{2}\right)^\nu z^{\sigma_{y,t}-1/2}\, dt < \left(\frac{2}{\log y}\right)^\nu H \left(e^{(2\log z)/\log y} + 52.1\nu! \frac{\log T}{\log y} \left(\frac{\log y}{\log Y}\right)^\nu\right).$$

Proof. We divide all points t in the interval $(T, T+H]$ into two classes. The first class consists of those t for which any zero $\beta + i\gamma$ satisfying the condition

$$|t - \gamma| \le \frac{y^{3|\beta - 1/2|}}{\log y} = \delta(\varrho) = \delta$$

satisfies also the inequality

$$\left|\beta - \frac{1}{2}\right| \le \frac{1}{\log y}.$$

The remaining points form the second class. The integral I_1 over the points in the first class does not exceed

$$\left(\frac{2}{\log y}\right)^\nu z^{2/\log y} H = \left(\frac{2}{\log y}\right)^\nu H e^{(2\log z)/\log y}.$$

Precisely the same argument as in the proof of Lemma 1.9 proves that the integral I_2 over the points in the second class is at most

$$\begin{aligned} &2 \sum_{\substack{T - \sqrt{Y} < \gamma \le T + H + \sqrt{Y} \\ \beta > 0.5}} \int_{\gamma - \delta}^{\gamma + \delta} 2^\nu \left(\beta - \frac{1}{2}\right)^\nu z^{2(\beta - 1/2)}\, dt \\ &\quad = \frac{2^{\nu+2}}{\log y} \sum_{\substack{T - \sqrt{Y} < \gamma \le T + H + \sqrt{Y} \\ \beta > 0.5}} \left(\beta - \frac{1}{2}\right)^\nu (y^3 z^2)^{\beta - 1/2}. \end{aligned}$$

As shown in the proof of Lemma 2.3, the last sum is bounded above by

$$(1 + o(1)) \cdot 6.5 H (\log T) \frac{2\nu!}{(\log Y)^\nu}.$$

This implies the assertion of the lemma.

Lemma 3.10. *Suppose that k, m, and r are integers, $m \ge 0$, $r \ge 0$, $m + r = k$, and $e^2 \le y \le T^{1/(3k)}$. Let $p_1, \dots, p_m$, $q_1, \dots, q_r$ range over all primes in the interval $(1, y]$ satisfying the condition $p_1 \cdots p_m \ne q_1 \cdots q_r$. Suppose that all terms in the sequence of $a(p)$ with $p \le y$ satisfy the condition $|a(p)| \le \delta$. Then the integral*

$$I = \int_T^{T+H} \sum_{\substack{p_1, \dots, p_m \\ q_1, \dots, q_r}} \frac{a(p_1) \cdots a(p_m) \overline{a}(q_1) \cdots \overline{a}(q_r)}{(p_1 \cdots p_m\, q_1 \cdots q_r)^{0.5}} \left(\frac{q_1 \dots q_r}{p_1 \cdots p_m}\right)^{it} dt$$

is estimated as

$$|I| < (\delta y)^k.$$

Proof. Let μ and ν be integers such that $\mu > \nu \geq 1$. Then, obviously,

$$\left|\int_T^{T+H}\left(\frac{\mu}{\nu}\right)^{it} dt\right| \leq 2\left(\log\frac{\mu}{\nu}\right)^{-1} \leq \frac{2\nu}{\log 2} < 3\sqrt{\mu\nu}.$$

Evaluating the integral I termwise and applying the last inequality to μ and ν being the maximum and minimum of the numbers $p_1\cdots p_m$ and $q_1\cdots q_r$, we obtain

$$|I| < \delta^k \sum_{\substack{p_1,\dots,p_m\\ q_1,\dots,q_r}} (p_1\cdots q_r)^{-0.5}\cdot 3(p_1\cdots q_r)^{0.5} \leq 3\delta^k(\pi(y))^k < (\delta y)^k.$$

This proves the lemma.

Lemma 3.11. *Suppose that k is an integer such that $1 \leq k \leq c\log Y$, where c is a sufficiently small absolute constant, and $y = Y^{1/(8k+3)}$. Then*

$$\int_T^{T+H}\left|S(t) + \frac{1}{\pi}\sum_{m\leq y^3}\frac{\Lambda_y(m)}{m^{\sigma_{y,t}}}\frac{\sin(t\log m)}{\log m}\right|^{2k} dt < \frac{101}{\varepsilon}\left(\frac{1440}{\varepsilon}\right)^{2k} H.$$

Proof. Lemma 2.4 implies that the initial integral does not exceed the quantity

$$2^{2k-1}((a_1\log(T+H))^{2k} j_1 + a_2^{2k} j_2),$$

where

$$j_1 = \int_T^{T+H}\left(\sigma_{y,t} - \frac{1}{2}\right)^{2k} dt \qquad \text{and} \qquad j_2 = \int_T^{T+H}\left(\sigma_{y,t} - \frac{1}{2}\right)^{2k}\left|\sum_{p\leq y^3}\frac{\Lambda_y(p)}{p^{\sigma_{y,t}+it}}\right|^{2k} dt.$$

Repeating the proof of Lemma 2.5 virtually word for word and using Lemma 3.9 instead of Lemma 2.3 at the appropriate places, we successively obtain

$$j_1 < \frac{201}{\varepsilon} H\left(\frac{160k}{\varepsilon\log T}\right)^{2k} \qquad \text{and} \qquad j_2 < \frac{1}{2}(2\log y)^{2k}(\sqrt{j_3 j_5} + \sqrt{j_4 j_6}),$$

where

$$j_3 = \int_T^{T+H}\left(\sigma_{y,t} - \frac{1}{2}\right)^{4k} dt < \frac{9.5H}{\varepsilon}\left(\frac{2}{\log y}\right)^{4k},$$
$$j_4 = \int_T^{T+H}\left(\sigma_{y,t} - \frac{1}{2}\right)^{4k} y^{4k(\sigma_{y,t}-1/2)} < \frac{H}{240\varepsilon}\left(\frac{2e^2}{\log y}\right)^{4k},$$

and

$$j_5,\, j_6 = \int_T^{T+H}\left|\sum_{p\leq y^3}\frac{a_j(p)}{\sqrt{p}}\, p^{it}\right|^{4k} dt, \qquad j = 1,2, \quad a_1(p) = \frac{\Lambda_y(p)}{\log y}, \quad |a_2(p)| \leq \frac{\Lambda_y(p)(\log p)\log yp}{(\log y)^3}.$$

Estimating the integrals j_5 and j_6 by using Lemma 3.8 in the same way as we estimated the sums Σ_7 and Σ_8 in Lemma 2.5, we obtain

$$j_5 < 1.1H(18k)^{2k} \qquad \text{and} \qquad j_6 < 1.1H(18a^2k)^{2k}, \quad a = 3.446\,088\,218\ldots,$$

whence it follows that

$$\sqrt{j_3 j_5} < \frac{4H}{\sqrt{\varepsilon}}\frac{(72k)^k}{(\log y)^{2k}}, \qquad \sqrt{j_4 j_6} < \frac{H}{14\sqrt{\varepsilon}}\frac{(72e^4a^2k)^k}{(\log y)^{2k}},$$

and, finally,

$$j_1 < \frac{H}{2\sqrt{\varepsilon}}\left(4(288k)^k + \frac{1}{14}(288e^4a^2k)^k\right) < \frac{H}{25\sqrt{\varepsilon}}(288e^4a^2k)^k.$$

Thus, the initial integral does not exceed

$$\frac{1}{2}\left((4a_1^2)^k\frac{201H}{\varepsilon}\left(\frac{160k}{\varepsilon}\right)^{2k}\left(\frac{\log(T+H)}{\log T}\right)^{2k}+(4a_2^2)^k\frac{H}{25\sqrt{\varepsilon}}(288e^4a^2k)^k\right)<\frac{101}{\varepsilon}\left(\frac{1440k}{\varepsilon}\right)^{2k}H.$$

This completes the proof of the lemma.

Lemma 3.12. *Under the conditions of Lemma* 3.11*,*

$$\int_T^{T+H}\left|\sum_{m\le y^3}\frac{\Lambda_y(m)}{m^{\sigma_{y,t}}}\frac{\sin(t\log m)}{\log m}-\sum_{p\le y}\frac{\sin(t\log p)}{\sqrt{p}}\right|^{2k}dt<\frac{H}{2\varepsilon}(1552e^4k)^k.$$

Proof. Repeating the proof of Lemma 2.6 virtually word for word and using the corresponding notation, we conclude that the integral in the statement of the lemma does not exceed

$$4^{2k-1}(I_1+I_2+I_3+I_4),\qquad I_\nu=\int_T^{T+H}R_\nu(t)\,dt,$$

where

$$R_1(t)=\left|\sum_{p\le y^3}\frac{a_1(p)}{\sqrt{p}}\,p^{it}\right|,\qquad R_2(t)=\frac{1}{2}\left|\sum_{p\le y^3}\frac{a_2(p)}{p}\,p^{2it}\right|,$$

$$R_3(t)=\frac{y^{\sigma_{y,t}-1/2}}{\log y}\int_0^{+\infty}y^{-v}\left|\sum_{p\le y^3}\frac{\Lambda_y(p)\log(py)}{p^{1/2+v}}\,p^{it}\right|dv,\qquad R_4(t)=2\left(\sigma_{y,t}-\frac{1}{2}\right)\log y,$$

and, moreover,

$$|a_1(p)|\le\frac{3\log p}{\log(y^3)}\qquad\text{and}\qquad|a_2(p)|\le1$$

for $p\le y^3$.

Estimating the integrals I_1 and I_2 according to Lemma 3.8, we obtain

$$I_1<1.1H(9k)^k\qquad\text{and}\qquad I_2<1.1Hk^k.$$

Applying Hölder's inequality to $R_3(t)$ yields

$$\begin{aligned}I_3&\le\frac{1}{(\log y)^{4k-1}}\int_0^{+\infty}y^{-v}\int_T^{T+H}y^{2k(\sigma_{y,t}-1/2)}\left|\sum_{p\le y^3}\frac{\Lambda_y(p)\log(py)}{p^{1/2+v}}\,p^{it}\right|^{2k}dt\,dv\\&\le\int_T^{T+H}y^{2k(\sigma_{y,t}-1/2)}\left|\sum_{p\le y^3}\frac{a_3(p)}{\sqrt{p}}\,p^{it}\right|^{2k}dt\le\sqrt{I_5}\sqrt{I_6},\end{aligned}$$

where

$$|a_3(p)|\le\frac{\Lambda_y(p)\log(py)}{(\log y)^2}\le\frac{6\log p}{\log(y^3)},$$

$$I_5=\int_T^{T+H}y^{4k(\sigma_{y,t}-1/2)}\,dt,\qquad\text{and}\qquad I_6=\int_T^{T+H}\left|\sum_{p\le y^3}\frac{a_3(p)}{\sqrt{p}}\,p^{it}\right|^{4k}dt.$$

Estimating the integrals I_5 and I_6 by using Lemmas 3.9 and 3.8, we obtain

$$I_5<H\left(e^{8k}+52.1\frac{\log T}{\log y}\right)<\frac{2H}{\varepsilon}\,e^{8k}\qquad\text{and}\qquad I_6<1.1H(72k)^{2k},$$

whence

$$I_3<\frac{3H}{2\sqrt{\varepsilon}}(72e^4k)^k.$$

Finally, by virtue of Lemma 3.9, we have

$$I_4 < (2\log y)^{2k}\biggl(\frac{2}{\log y}\biggr)^{2k} H\biggl(1+52.1\frac{\log T}{\log y}(2k)!\biggl(\frac{\log y}{\log Y}\biggr)^{2k}\biggr)$$
$$= 16^k H\biggl(1+\frac{521}{\varepsilon}\frac{(2k)!}{(8k+3)^{2k-1}}\biggr) < 16^k\frac{95H}{\varepsilon}.$$

Adding up the obtained estimates, we arrive at

$$4^{2k-1}\sum_{\nu=1}^{4} I_\nu < \frac{H}{4\varepsilon}(16\cdot 72e^4k)^k\biggl\{1.1\varepsilon\biggl(\frac{1}{8e^4}\biggr)^k + 1.1\varepsilon\biggl(\frac{1}{144e^4}\biggr)^k + \frac{3\sqrt{\varepsilon}}{2} + 95\varepsilon\biggl(\frac{2}{9e^4k}\biggr)^k\biggr\}$$
$$< \frac{H}{4\varepsilon}(1152e^4k)^k.$$

This completes the proof of the lemma.

Lemma 3.13. *Under the conditions of Lemma* 3.11*,*

$$\int_T^{T+H}\biggl|S(t)+\frac{1}{\pi}\sum_{p\le y}\frac{\sin(t\log p)}{\sqrt{p}}\biggr|^{2k}dt < \frac{c_0}{56}(c_0k)^{2k}H.$$

Proof. Using the estimates of Lemmas 3.11 and 3.12 and Hölder's inequality, we obtain

$$\int_T^{T+H}\biggl|S(t)+\frac{1}{\pi}\sum_{p\le y}\frac{\sin(t\log p)}{\sqrt{p}}\biggr|^{2k}dt$$
$$\le 2^{2k-1}\biggl\{\int_T^{T+H}\biggl|S(t)+\frac{1}{\pi}\sum_{m\le y^3}\frac{\Lambda_y(m)}{m^{\sigma_{y,t}}}\frac{\sin(t\log m)}{\log m}\biggr|^{2k}dt$$
$$+\frac{1}{\pi^{2k}}\int_T^{T+H}\biggl|\sum_{m\le y^3}\frac{\Lambda_y(m)}{m^{\sigma_{y,t}}}\frac{\sin(t\log m)}{\log m}-\sum_{p\le y}\frac{\sin(t\log p)}{\sqrt{p}}\biggr|^{2k}dt\biggr\}$$
$$\le 2^{2k-1}H\biggl\{\frac{101}{\varepsilon}\biggl(\frac{1440k}{\varepsilon}\biggr)^{2k}+\frac{1}{2\varepsilon}\biggl(\frac{1152e^4k}{\pi^2}\biggr)^k\biggr\} < \frac{51H}{\varepsilon}\biggl(\frac{2880k}{\varepsilon}\biggr)^{2k} < \frac{c_0}{56}(c_0k)^{2k}H.$$

This proves the lemma.

Lemma 3.14. *Let k be an integer such that $1\le k\le c\log Y$, where c is a sufficiently small absolute constant, and let $h=\pi v(\log T)^{-1}$, where $0<v<\log T$ is any number. Then the integral*

$$I_{2k}=\int_T^{T+H}(S(t+2h)-S(t))^{2k}\,dt$$

is estimated as

$$I_{2k} < \exp\biggl\{\frac{v\varepsilon}{40c_0\sqrt{ke}}\biggr\}\frac{c_0}{56}(2c_0k)^{2k}H,$$

where $c_0=2880\varepsilon^{-1}$.

Proof. Setting $y=Y^{1/(8k+3)}$, we obtain

$$S(t+2h)-S(t)=-V(t)+W(t),$$

where

$$V(t)=\frac{1}{\pi}(U(t)+\overline{U}(t)),\qquad U(t)=\sum_{p\le y}\frac{a(p)}{\sqrt{p}}p^{it},\quad a(p)=p^{ih}\sin(h\log p),$$

and

$$W(t)=R(t+2h)-R(t),\qquad R(t)=S(t)+\frac{1}{\pi}\sum_{p\le y}\frac{\sin(t\log p)}{\sqrt{p}}.$$

Therefore,

$$I_{2k}=\sum_{\nu=0}^{2k}(-1)^{\nu}\binom{2k}{\nu}I(\nu),\qquad I(\nu)=\int_T^{T+H}V^{\nu}(t)W^{2k-\nu}(t)\,dt.$$

By virtue of Hölder's inequality, we have

$$|I(\nu)|\le\left(\int_T^{T+H}V^{2k}(t)\,dt\right)^{\nu/(2k)}\left(\int_T^{T+H}W^{2k}(t)\,dt\right)^{1-\nu/(2k)}=(I(2k))^{\nu/(2k)}(I(0))^{1-\nu/(2k)},$$

whence

$$I_{2k}\le\sum_{\nu=0}^{2k}\binom{2k}{\nu}|I(\nu)|\le\left\{(I(2k))^{1/(2k)}+(I(0))^{1/(2k)}\right\}^{2k}.$$

Lemma 3.10 implies

$$I(2k)=\pi^{-2k}\int_T^{T+H}(U(t)+\overline{U}(t))^{2k}\,dt=\pi^{-2k}\sum_{\nu=0}^{2k}\binom{2k}{\nu}\int_T^{T+H}U^{\nu}(t)\overline{U}^{\,2k-\nu}(t)\,dt$$

$$<\pi^{-2k}\left\{\binom{2k}{k}k!\,H\left(\sum_{p\le y}\frac{|a(p)|^2}{p}\right)^k+\sum_{\nu=0}^{2k}\binom{2k}{\nu}\left(y\max_{p\le y}|a(p)|\right)^{2k}\right\}.$$

Since

$$|a(p)|\le h\log p\le h\log y\qquad\text{and}\qquad y^{2k}=Y^{2k/(8k+3)}<Y^{1/4},$$

it follows from Lemma I.2 that

$$\sum_{p\le y}\frac{|a(p)|^2}{p}\le h^2(\log y)\sum_{p\le y}\frac{\log p}{p}<(h\log y)^2.$$

Applying the inequality

$$\frac{(2k)!}{k!}<\sqrt{2}\left(\frac{4k}{e}\right)^k,$$

we obtain

$$I(2k)\le\pi^{-2k}\left\{\sqrt{2}\left(\frac{4k}{e}\right)^k H(\log y)^{2k}+3Y^{3/8}(2h\log y)^{2k}\right\}$$

$$<\frac{3}{2}\left(\frac{2}{\pi}\sqrt{\frac{k}{e}}\,h\log y\right)^{2k}H=\frac{3}{2}\left(\frac{2}{\pi}\sqrt{\frac{k}{e}}\,\frac{\pi v}{\log T}\,\frac{0.1\varepsilon\log T}{8k+3}\right)^{2k}H\le\frac{3}{2}\left(\frac{v\varepsilon}{40\sqrt{ek}}\right)^{2k}H.$$

By virtue of Lemma 3.14, we have

$$I(0)\le 2^{2k-1}\int_T^{T+H}(R^{2k}(t+2h)+R^{2k}(t))\,dt<\frac{c_0}{56}(2c_0k)^{2k}H.$$

Therefore, returning to the estimate of I_{2k}, we obtain

$$I_{2k}\le H\left\{\left(\frac{c_0}{56}\right)^{1/(2k)}2c_0k+\left(\frac{3}{2}\right)^{1/(2k)}\frac{v\varepsilon}{40\sqrt{ek}}\right\}^{2k}$$

$$= \frac{c_0}{56}(2c_0k)^{2k}H\left\{1+\left(\frac{84}{c_0}\right)^{1/(2k)}\frac{v\varepsilon}{40c_0\sqrt{ke}}\frac{1}{2k}\right\}^{2k} \le \frac{c_0}{56}(2c_0k)^{2k}\exp\left\{\frac{v\varepsilon}{40c_0\sqrt{ke}}\right\}.$$

This completes the proof of the lemma.

Lemma 3.15. *Let h be a number satisfying the conditions of Lemma* 3.14*, and let $D(\lambda)$ be the set of those t in the interval $T \le t \le T+H$ for which*

$$|S(t+2h) - S(t)| \ge \lambda.$$

Then, for any $\lambda > 0$,

$$\operatorname{mes} D(\lambda) \le \frac{c_0e^2}{56}\exp\left\{\frac{v\varepsilon}{20\sqrt{c_0\lambda}}\right\}H\exp\left\{-\frac{\lambda}{ec_0}\right\}.$$

Proof. If $\lambda > 0.3\log T$, then $\operatorname{mes} D(\lambda) = 0$ by Lemma I.9. If $0 < \lambda \le 4ec_0$, then the required estimate follows from the trivial estimate

$$\operatorname{mes} D(\lambda) \le H \le e^{4-\lambda/(ec_0)}H < \frac{c_0e^2}{56}H\exp\left\{-\frac{\lambda}{ec_0}\right\}.$$

Thus, it suffices to consider the case where $4ec_0 < \lambda \le 0.3\log T$. We set $k = [\lambda/(2ec_0)]$, so that

$$2 \le k \le \frac{\log Y}{1980e} \le c\log Y.$$

Integrating both sides of the inequality

$$(S(t+2h) - S(t))^{2k} \ge \lambda^{2k}$$

over the set $D(\lambda)$ and applying Lemma 3.14, we obtain

$$\lambda^{2k}\cdot\operatorname{mes} D(\lambda) \le \int_T^{T+H}(S(t+2h)-S(t))^{2k}\,dt \le \frac{c_0}{56}\exp\left\{\frac{v\varepsilon}{40c_0\sqrt{ek}}\right\}(2c_0k)^{2k}H,$$

whence

$$\begin{aligned}\operatorname{mes} D(\lambda) &\le \frac{c_0}{56}\exp\left\{\frac{v\varepsilon}{40c_0\sqrt{ek}}\right\}\left(\frac{2c_0k}{e\lambda}\right)^{2k}H\\ &\le \frac{c_0}{56}\exp\left\{\frac{v\varepsilon}{40c_0\sqrt{e}}\sqrt{\frac{4ec_0}{\lambda}}\right\}e^{-2k}H \le \frac{c_0e^2}{56}\exp\left\{\frac{v\varepsilon}{20\sqrt{c_0\lambda}}\right\}H\exp\left\{-\frac{\lambda}{ec_0}\right\}.\end{aligned}$$

This completes the proof of the lemma.

Lemma 3.16. *Let h be a number satisfying the conditions of Lemma* 3.14*, and let γ be the ordinate of a zero of $\zeta(s)$ for which $T \le \gamma - h < \gamma \le T+H$. Suppose that the interval $(\gamma-h,\gamma]$ contains precisely m ordinates of zeros of $\zeta(s)$ and the multiplicity of each of these ordinates is j. Then, for any t in the interval $(\gamma-2h,\gamma-h]$,*

$$S(t+2h) - S(t) \ge mj - v.$$

Proof. Take any number t, $\gamma - 2h < t \le \gamma - h$. Let $\gamma^{(1)} < \cdots < \gamma^{(l)}$ be all different ordinates of zeros of $\zeta(s)$ in the interval $(t, t+2h]$, and let $\kappa^{(1)},\dots,\kappa^{(l)}$ be their multiplicities.

Consider an interval (a,b) contained entirely in $(t,t+2h]$ and containing no points of discontinuity of $S(t)$. Representing the difference of the values of $S(t)$ at the endpoints of this interval by means of the finite increment formula, we obtain

$$S(b) - S(a) = (b-a)S'(\xi) = -(b-a)\left(\frac{1}{2\pi}\log\frac{\xi}{2\pi} + O\left(\frac{1}{\xi^2}\right)\right),$$

where $a<\xi<b$. Since $\log\xi=\log T+O(HT^{-1})$, it follows that

$$S(b)-S(a)=-(b-a)\biggl(\frac{1}{2\pi}\log\frac{T}{2\pi}+O\biggl(\frac{H}{T}\biggr)\biggr)=-(b-a)\biggl(\frac{v}{2h}-\frac{\log 2\pi}{2\pi}+O\biggl(\frac{H}{T}\biggr)\biggr).$$

Using this the relation, we transform the identity

$$\begin{aligned}S(t+2h)-S(t)&=\{S(t+2h)-S(\gamma^{(l)}+0)\}+\sum_{r=1}^{l}(S(\gamma^{(r)}+0)-S(\gamma^{(r)}-0))\\&\quad+\sum_{r=2}^{l}(S(\gamma^{(r)}-0)-S(\gamma^{(r-1)}+0))+(S(\gamma^{(1)}-0)-S(t)).\end{aligned}$$

Note that $S(\gamma^{(r)}+0)-S(\gamma^{(r)}-0)=\kappa^{(r)}$; hence

$$\begin{aligned}&S(t+2h)-S(t)\\&\quad=\kappa^{(1)}+\cdots+\kappa^{(l)}-\biggl(\frac{v}{2h}-\frac{\log 2\pi}{2\pi}+O\biggl(\frac{H}{T}\biggr)\biggr)\\&\qquad\qquad\times\{(t+2h-\gamma^{(l)})+(\gamma^{(l)}-\gamma^{(l-1)})+\cdots+(\gamma^{(2)}-\gamma^{(1)})+(\gamma^{(1)}-t)\}\\&\quad=\kappa^{(1)}+\cdots+\kappa^{(l)}-2h\biggl(\frac{v}{2h}-\frac{\log 2\pi}{2\pi}+O\biggl(\frac{H}{T}\biggr)\biggr).\end{aligned}$$

Since $\gamma-h\ge t$ and $\gamma\le t+2h$, it follows that the interval $(\gamma-h,\gamma]$ is entirely contained in $(t,t+2h]$. Hence, by the assumption of the lemma, there are at least m numbers greater than or equal to j among $\kappa^{(1)},\dots,\kappa^{(l)}$. Therefore,

$$S(t+2h)-S(t)\ge mj-2h\biggl(\frac{v}{2h}-\frac{\log 2\pi}{2\pi}+O\biggl(\frac{H}{T}\biggr)\biggr)>mj-v.$$

This completes the proof of the lemma.

Let $N_j(T)$ denote the number of ordinates of zeros of $\zeta(s)$ in the rectangle $0<\operatorname{Re}s<1$, $0<\operatorname{Im}s\le T$ that have multiplicity j.

Lemma 3.17. *For any $j\ge ec_0+1$,*

$$N_j(T+H)-N_j(T)<0.832\frac{H}{2\pi}(\log T)\exp\biggl\{-\frac{j}{ec_0}\biggr\}.$$

Proof. We set $h=\pi v(\log T)^{-1}$, where $v=ec_0$. Let $\gamma^{(1)}$ be the maximal ordinate of multiplicity j in the interval $(T,T+H]$. We set $E_1=(\gamma^{(1)}-h,\gamma^{(1)}]$ and let $\gamma^{(2)}$ denote the maximal ordinate of multiplicity j in $(T,\gamma^{(1)}-h]$. Then, we set $E_2=(\gamma^{(2)}-h,\gamma^{(2)}]$, let $\gamma^{(3)}$ denote the maximal ordinate of multiplicity j in $(T,\gamma^{(2)}-h]$, and so on.

We continue this construction until we find an ordinate $\gamma^{(r)}$ of multiplicity j such that either $\gamma^{(r)}-h<T\le\gamma^{(r)}$ or the interval $(T,\gamma^{(r)}-h]$ contains no ordinates of multiplicity j. In each of these cases, we set $E_r=(\gamma^{(r)}-h,\gamma^{(r)}]$.

The intervals $E_1,E_2,\dots,E_r$ are pairwise disjoint and have the same length h, and their union contains all ordinates of multiplicity j from the interval $(T,T+H]$. We divide the sets $E_1,E_2,\dots,E_r$ into classes $\mathcal{E}_1,\mathcal{E}_2,\dots$ so that each class $\mathcal{E}_m$ consists of intervals containing precisely m ordinates of multiplicity j. Let k_m be the number of elements in $\mathcal{E}_m$; then

$$N_j(T+H)-N_j(T)\le 1\cdot k_1+2\cdot k_2+\cdots+m\cdot k_m+\cdots. \tag{3.24}$$

Let us estimate each number k_m from above. Suppose that an interval $E=(\gamma-h,\gamma]$ belongs to $\mathcal{E}_m$. We set $E'=E-h=(\gamma-2h,\gamma-h]$. According to Lemma 3.16, for any $t\in E'$, we have

$$S(t+2h)-S(t)\ge mj-v.$$

Let $D(\lambda)$ denote the set of points $t \in (T-2h, T+H]$ for which $|S(t+2h)-S(t)| \geq \lambda$. Since E' is contained in $(T-2h, T+H]$, we conclude that E' is entirely contained in $D(mj-v)$.

Arguing similarly for each interval E in each class $\mathcal{E}_m$, we see that all of the k_m intervals are contained in $D(mj-v)$. Since all of them have the same length h and are pairwise disjoint, it follows that their total length does not exceed the measure of the set $D(mj-v)$, i.e., $k_m h \leq \operatorname{mes} D(mj-v)$. According to Lemma 3.15, we have

$$\begin{aligned}\operatorname{mes} D(mj-v) &\leq \frac{c_0 e^2}{56}\exp\left\{\frac{v\varepsilon}{20\sqrt{c_0}}\frac{1}{\sqrt{mj-v}}\right\}H\exp\left\{-\frac{mj-v}{ec_0}\right\}\\ &\leq \frac{c_0 e^3}{56}\exp\left\{\frac{e}{20}\sqrt{c_0}\varepsilon\right\}H\exp\left\{-\frac{mj}{ec_0}\right\},\end{aligned}$$

whence

$$k_m \leq \frac{c_0 e^3}{56h}\exp\left\{6e\sqrt{\frac{\varepsilon}{5}}\right\}H\exp\left\{-\frac{mj}{ec_0}\right\} = \frac{e^2}{28}\frac{H}{2\pi}(\log T)\exp\left\{6e\sqrt{\frac{\varepsilon}{5}}\right\}\exp\left\{-\frac{mj}{ec_0}\right\}.$$

Returning to inequality (3.24), we obtain

$$\begin{aligned}N_j(T+H)-N_j(T) &< \frac{e^2}{28}\frac{H}{2\pi}(\log T)\exp\left\{6e\sqrt{\frac{\varepsilon}{5}}\right\}\sum_{m=1}^{+\infty} m\exp\left\{-\frac{mj}{ec_0}\right\}\\ &\leq \frac{e^2}{28}\exp\left\{6e\sqrt{\frac{\varepsilon}{5}}\right\}(1-e^{-j/ec_0})^{-2}\frac{H}{2\pi}(\log T)\exp\left\{-\frac{mj}{ec_0}\right\}\\ &< \frac{e^2}{28}\left(1-\frac{1}{e}\right)^{-2}\exp\left\{6e\sqrt{\frac{\varepsilon}{5}}\right\}\frac{H}{2\pi}(\log T)\exp\left\{-\frac{mj}{ec_0}\right\}.\end{aligned}$$

Using inequality $\varepsilon < 10^{-3}$, we arrive at

$$N_j(T+H)-N_j(T) < 0.832\frac{H}{2\pi}(\log T)\exp\left\{-\frac{j}{ec_0}\right\}.$$

This completes the proof of the lemma.

Given $a > 0$, by

$$K_0(a) = K_0(a; N, M) = \sum_n \kappa_n^a$$

we mean the sum of the numbers κ_n^a over all consecutive ordinates γ_n, $N < n \leq N+N$, and by

$$K_1(a) = K_1(a; N, M) = \sum_n{}' \kappa_n^a$$

we mean the sum over only different ordinates γ_n, $N < n \leq N+M$.

Thus, if relations (3.3) hold, then $K_0(a)$ contains all summands $\kappa_l^a, \ldots, \kappa_{l+p-1}^a$, while $K_1(a)$ contains only one of them, κ_l^a.

It is easy to note that the sums K_0 and K_1 thus defined are related by

$$K_0(a) = \sum_n{}' \kappa_n \cdot \kappa_n^a = K_1(a+1)$$

for any $a \geq 0$.

Lemma 3.18. *For any $a \geq 1$,*

$$K_1(a) < 2.3 c_0 (ec_0)^a \Gamma(a+1) M.$$

Proof. We set $T=\gamma_N$ and $H=\gamma_{N+M}-\gamma_N$. Note that

$$N(T+H)-N(T)=(1+o(1))\frac{H}{2\pi}(\log T)=(1+o(1))M;$$

hence, by Lemma 3.17, we have

$$\begin{aligned}K_1(a)&=\Bigg(\sum_{n\colon \kappa_n<ec_0+1}+\sum_{n\colon \kappa_n\ge ec_0+1}\Bigg)\kappa_n^a\\&\le (ec_0+1)^a(N(T+H)-N(T))+0.832\frac{H}{2\pi}(\log T)\sum_{j\ge ec_0+1}j^a\exp\left\{-\frac{j}{ec_0}\right\}\\&=(1+o(1))M\Bigg((ec_0+1)^a+0.832\sum_{j\ge ec_0+1}j^a\exp\left\{-\frac{j}{ec_0}\right\}\Bigg).\end{aligned}\tag{3.25}$$

The function $f(u)=u^a\exp\left(-u/(ec_0)\right)$ increases in the interval $1\le u\le u_a$, where $u_a=aec_0$, attains its maximum value

$$u_a^ae^{-a}=(ac_0)^a\tag{3.26}$$

at the point u_a, and then monotonically decreases. Let m be the integer for which $m<u_a\le m+1$. Estimating the terms corresponding to $j=m,m+1$ in the sum on the right-hand side of (3.25) by value (3.26) and the sum of all terms by integrals of $f(u)$, we obtain

$$\begin{aligned}\sum_{j\ge ec_0+1}j^a\exp\left\{-\frac{j}{ec_0}\right\}&\le\left(\int_0^m+\int_{m+1}^{+\infty}\right)u^a\exp\left\{-\frac{u}{ec_0}\right\}du+2(ac_0)a\\&<\int_0^{+\infty}u^a\exp\left\{-\frac{u}{ec_0}\right\}du+2(ac_0)a=(ac_0)^{a+1}\Gamma(a+1)+2(ac_0)a\\&<(ec_0)^{a+1}\Gamma(a+1)\left(1+\frac{2}{ec_0}\frac{1}{\sqrt{2\pi a}}\right)\le\left(1+\frac{1}{ec_0}\sqrt{\frac{2}{\pi}}\right)(ec_0)^{a+1}\Gamma(a+1).\end{aligned}$$

Thus,

$$\begin{aligned}K_1(a)&<(1+o(1))M\left((ec_0+1)^a+0.832\left(1+\frac{1}{ec_0}\sqrt{\frac{2}{\pi}}\right)(ec_0)^{a+1}\Gamma(a+1)\right)\\&<(ec_0)^{a+1}\Gamma(a+1)\left(\frac{b}{ec_0}+0.833\right),\end{aligned}$$

where

$$b=\left(1+\frac{1}{ec_0}\right)^a\frac{1}{\Gamma(a+1)}.$$

If $1\le a\le e$, then

$$b\le\left(1+\frac{1}{ec_0}\right)^e<\exp\left\{\frac{1}{c_0}\right\},$$

and if $a>e$, then, by virtue of Stirling's formula,

$$b<\frac{1}{\sqrt{2\pi a}}\left(\frac{e}{a}\right)^a\left(1+\frac{1}{ec_0}\right)^a<\left(\frac{e}{a}+\frac{1}{ac_0}\right)^a<\left(1+\frac{1}{ac_0}\right)^a<\exp\left\{\frac{1}{c_0}\right\}.$$

Therefore, $b(ec_0)^{-1}<10^{-3}$ and

$$K_1(a)<0.834(ec_0)^{a+1}\Gamma(a+1)<2.3c_0(ec_0)^a\Gamma(a+1).$$

This proves the lemma.

Now we have all the tools needed to prove theorems about the moments of $S(\gamma_n\pm 0)$.

Theorem 3.12. *Let k be an integer such that $1 \le k \le c_1 \sqrt[3]{\log\log N}$. Then*

$$\sum_n S^{2k}(\gamma_n \pm 0) = \frac{(2k)!}{k!}\frac{M}{(2\pi)^{2k}}(\log\log N)^k\left(1 + \frac{215\theta_1\sqrt{c_0}(c_0k)^{3/2}}{\sqrt{\log\log N}}\right),$$

$$\left|\sum_n S^{2k-1}(\gamma_n \pm 0)\right| < 28.2c_0^2\frac{(2k)!}{(k-1)!}\frac{M(\log\log N)^{k-1}}{(2\pi)^{2(k-1)}}.$$

Proof. Let us prove the first relation; for definiteness, we choose the sign "+" in $S(\gamma_n \pm 0)$. If $|\theta| \le 1$, then, by Lemma I.9 and the finite increment formula, we have

$$\left(S(\gamma_n + 0) + \frac{\theta(\log N)^2}{80N}\right)^{2k} = S^{2k}(\gamma_n + 0) + \frac{2\theta_1 k}{N}(0.15\log N)^{2k+1}.$$

This and Lemma 3.7 readily imply that the difference between the absolute values of the sums

$$\sum_n S^{2k}(\gamma_n + 0) \qquad \text{and} \qquad \sum_n \Delta_n^{2k}$$

does not exceed

$$\sum_{\nu=1}^{2k}\binom{2k}{\nu}w_\nu + \frac{2kM}{N}(0.15\log N)^{2k+1},$$

where

$$w_\nu = \sum_n |\Delta_n|^{2k-\nu}\kappa_n^\nu.$$

Using the estimate of Lemma 3.18, the assertion of Theorem 3.1, and the notation of Theorem 2.1, we obtain

$$w_\nu \le \left(\sum_n \Delta_n^{2k}\right)^{1-\nu/(2k)}\left(\sum_n \kappa_n^{2k}\right)^{\nu/(2k)} = \left(\sum_n \Delta_n^{2k}\right)^{1-\nu/(2k)}\left(\sum_n{}' \kappa_n^{2k+1}\right)^{\nu/(2k)}$$

$$\le (2.03\mathcal{L}_k)^{1-\nu/(2k)}(2.3ec_0^2(ec_0)^{2k}(2k+1)!M)^{\nu/(2k)} \le 2.03\mathcal{L}_k q^\nu,$$

where

$$q = \left\{\frac{2.3c_0(ec_0)^{2k+1}(2k+1)!M}{2.03\mathcal{L}_k}\right\}^{1/(2k)} < \frac{2\pi ec_0}{\sqrt{L}}(3.08c_0^2(2k+1)k!)^{1/(2k)} < \frac{6.08\pi e\sqrt{k}c_0^{1+1/k}}{\sqrt{L}}. \tag{3.27}$$

It is easy to verify that $q < 1/116k$ for any k. Indeed, if $1 \le k \le 100$, then

$$q < \frac{60.8\pi ec_0^2}{\sqrt{L}} < \frac{1}{116k}$$

and if $100 < k \le c_1\sqrt[3]{L}$, then

$$q \le \frac{6.08\pi e}{k}\frac{(c_0k)^{3/2}}{\sqrt{L}}c_0^{1/k-1/2} \le \frac{6.08\pi e}{k}\frac{3}{4\pi}c_0^{-0.49} < \frac{1}{116k}.$$

Therefore,

$$\sum_{\nu=1}^{2k}\binom{2k}{\nu}w_\nu < 2.03\mathcal{L}_k\sum_{\nu=1}^{2k}\binom{2k}{\nu}q^\nu = 2.03\mathcal{L}_k((1+q)^{2k}-1) < 4.06kq(1+q)^{2k}\mathcal{L}_k$$

$$< 4.06\left(1 + \frac{1}{116k}\right)^{2k}6.08\pi e\frac{(c_0k)^{3/2}}{\sqrt{L}}c_0^{1/k-1/2}\mathcal{L}_k < 214.5\sqrt{c_0}\frac{(c_0k)^{3/2}}{\sqrt{L}}\mathcal{L}_k.$$

A word-for-word repetition of the above argument proves that the absolute value of the difference between the sums

$$\sum_n S^{2k-1}(\gamma_n+0) \qquad \text{and} \qquad \sum_n \Delta_n^{2k-1}$$

does not exceed

$$\sum_{\nu=1}^{2k-1}\binom{2k-1}{\nu} w_\nu + \frac{kM}{N}(0.15\log N)^{2k}, \qquad w_\nu = \sum_n |\Delta_n|^{2k-\nu-1}\kappa_n^\nu.$$

Twice applying Hölder's inequality to w_ν, we obtain

$$w_\nu \le \left(\sum_n \Delta_n^{2k}\right)^{1-\nu+1/(2k)}\left(\sum_n \kappa_n^{2k\nu/\nu+1}\right)^{\nu+1/(2k)} \le \left(\sum_n \Delta_n^{2k}\right)^{1-\nu+1/(2k)}\left(\sum_n \kappa_n^{2k}\right)^{\nu/(2k)} M^{1/(2k)}.$$

Theorem 3.1 and Lemma 3.18 imply

$$\begin{aligned} w_\nu &\le M^{1/(2k)}(2.03\mathcal{L}_k)^{1-\nu+1/(2k)}(2.3c_0(ec_0)^{2k+1}(2k+1)!\,M)^{\nu/(2k)} \\ &\le \frac{2.03\mathcal{L}_k}{\sqrt{L}}\,2\pi\left(\frac{k!}{(2k)!}\right)^{1/(2k)} q^\nu < \frac{4.06\pi\mathcal{L}_k}{\sqrt{kL}}\frac{\sqrt{e}}{2}\,q^\nu < \frac{10.52\mathcal{L}_k}{kL}\,q^\nu, \end{aligned}$$

where q is defined by (3.27). Thus,

$$\begin{aligned} \sum_{\nu=1}^{2k-1}\binom{2k-1}{\nu} w_\nu &< \frac{10.52\mathcal{L}_k}{\sqrt{kL}}((1+q)^{2k-1}-1) < \frac{10.52\mathcal{L}_k}{\sqrt{kL}}\,2kq(1+q)^{2k} \\ &< 1092.5\frac{kc_0^{1+1/k}}{L}\left(1+\frac{1}{116k}\right)^{2k}\mathcal{L}_k < 1111.5\frac{kc_0^2}{L}\,\mathcal{L}_k < 28.16c_0^2\frac{(2k)!}{(k-1)!}\frac{ML^{k-1}}{(2\pi)^{2(k-1)}}. \end{aligned}$$

This completes the proof of the theorem.

Theorem 3.13. *Let a be any number not being an even integer and satisfying the condition $0 < a \le c_3\sqrt[3]{\log\log N}$, where $c_3 = (6c_0)^{-2}$. Then*

$$\sum_n |S(\gamma_n \pm 0)|^a = \frac{M}{\pi^a\sqrt{\pi}}\Gamma\left(\frac{a+1}{2}\right)(\log\log N)^{a/2}(1+\theta r(a)),$$

where

$$r(a) = \begin{cases} 22.2c_0^2\left(\dfrac{\log\log\log N}{\log\log N}\right)^{0.5a} & \text{if } 0 < a \le 1, \\ \dfrac{782a}{\sqrt{\log\log N}}(\sqrt{\log\log\log N} + 8.6c_0^3\sqrt{a}) & \text{if } a > 1. \end{cases}$$

Proof. Setting

$$\delta_n = \kappa_n + \frac{(\log N)^2}{80N}$$

and applying Lemma 3.7, we obtain $|S(\gamma_n \pm 0)| = |\Delta_n| + \theta\,\delta_n$. In the case $0 < a \le 1$, Theorem 3.2 and Lemma 3.18 imply

$$\begin{aligned} \sum_n |S(\gamma_n \pm 0)|^a &= \sum_n |\Delta_n|^a + \theta\left(\sum_n \kappa_n + \left(\frac{(\log N)^2}{80N}\right)^a\right) \\ &= \sum_n |\Delta_n|^a + \theta_1(2.3c_0(ec_0)^{1+a}\Gamma(a+2)+1)M = \frac{M}{\pi^a\sqrt{\pi}}\Gamma\left(\frac{a+1}{2}\right)L^{a/2}(1+\theta_2 R_a), \end{aligned}$$

where

$$R_a = 14.1\left(\frac{\log L}{L}\right)^{a/2} + 2.31e\sqrt{\pi}c_0^2\frac{\Gamma(a+2)}{\Gamma((a+1)/2)}\left\{\frac{\pi e c_0}{\sqrt{L}}\right\}^a < 22.2c_0^2\left(\frac{\log L}{L}\right)^{a/2}.$$

In the case $a > 1$, we use the argument in the proof of Theorem 2.3 and obtain

$$\sum_n |S(\gamma_n \pm 0)|^a = \sum_n |\Delta_n|^a + \theta(u + 3v + 9w)$$

for

$$u = \sum_{\nu=1}^{k}\binom{k}{\nu}u_\nu, \qquad v = \sum_{\nu=0}^{k}\binom{k}{\nu}v_\nu, \qquad \text{and} \qquad w = \sum_{\nu=0}^{k}\binom{k}{\nu}w_\nu,$$

where u_ν, v_ν, and w_ν are sums of the form

$$\tau_{\nu,c} = \sum_n |\Delta_n|^{a-\nu-c}\delta_n^{\nu+c};$$

as above, a equals $k + b$, where $k \geq 0$ is an integer and $1 \leq b \leq 2$, and c takes the values 0, 1, and b, respectively.

Next, for $1 \leq a \leq c_3\sqrt[3]{L}$, Theorem 3.2 gives the estimate

$$\sum_n |\Delta_n|^a \leq \mathcal{L}_a\left(1 + 782\left(c_3\frac{\sqrt{\log L}}{\sqrt[6]{L}} + \frac{6.6}{\pi}\sqrt{\frac{2}{e}}\,c_0^{-1.5}\right)\right) < 1.00025\mathcal{L}_a = I_a,$$

where

$$\mathcal{L}_a = \frac{M}{\pi^a\sqrt{\pi}}\Gamma\left(\frac{a+1}{2}\right)L^{a/2}.$$

By virtue of Lemma 3.18, we have

$$\sum_n \delta_n^a < 1.001\sum_n \kappa_n^a < 2.301c_0(ec_0)^{1+a}\Gamma(a+2)M = J_a.$$

Therefore,

$$\tau_{\nu,c} \leq I_a q^{\nu+c}, \qquad q = \left(\frac{J_a}{I_a}\right)^{1/a}.$$

Now, note that

$$\frac{\Gamma(a+2)}{\Gamma((a+1)/2)} = \frac{2^a(a+1)}{\sqrt{\pi}}\Gamma\left(\frac{a}{2}+1\right) \leq \sqrt{2e}\,a^{3/2}\left(\frac{2a}{e}\right)^{a/2}.$$

Therefore,

$$q^a = \frac{J_a}{I_a} < 25.85a^{3/2}\left(\pi c_0\sqrt{\frac{2ae}{L}}\right)^a, \qquad q < 190c_0^3\sqrt{\frac{a}{L}} < \frac{1}{a},$$

whence $u < I_a aeq$ and $v, w < I_a eq$, so that

$$u + 3v + 9w < I_a(a+12)eq < 6716\frac{a^{3/2}c_0^3}{\sqrt{L}}.$$

Thus,

$$\sum_n |S(\gamma_n \pm 0)|^a = \mathcal{L}_a(1 + \theta R_a),$$

where

$$R_a \le \frac{782a}{\sqrt{L}}(\sqrt{\log L} + 2.2c_0^{1.5}\sqrt{a}) + 6716\frac{a^{3/2}c_0^3}{\sqrt{L}} < \frac{782a}{\sqrt{L}}(\sqrt{\log L} + 8.6c_0^3\sqrt{a}).$$

This completes the proof of the theorem.

Theorem 3.14. *Let $\Theta(v) = \Theta_{\pm}(v)$ be the distribution function of $S(\gamma_n + 0)$ (of $S(\gamma_n - 0)$), i.e.,*

$$\Theta_{\pm}(v) = \frac{1}{M}\#\left\{n \;\middle|\; N < n \le N + M,\ S(\gamma_n \pm 0) \le \frac{v}{\pi\sqrt{2}}\sqrt{\log\log N}\right\}.$$

Then, for any u, $\Theta(u)$ can be expressed as[17]

$$\Theta(u) = G(u) + \frac{1.79\theta}{\sqrt{\log\log\log N}}.$$

Moreover, if $1.5 \le u \le \sqrt[6]{\log\log N}$, then this function satisfies the relations

$$\frac{\Theta(-u)}{G(-u)} = 1 + O(R) \qquad \text{and} \qquad \frac{1-\Theta(u)}{1-G(u)} = 1 + O(R),$$

in which

$$R = \frac{u(u^2 + \log\log\log N)}{\sqrt{\log\log N}}$$

and the constants in the big-O notations depend on ε.

The proof of this theorem is quite similar to those of Theorems 2.5 and 3.3.

Corollary 3.4. *Let $f(x)$ be a positive monotone function unboundedly increasing as $x \to +\infty$. Then*

$$\frac{1}{f(n)}\sqrt{\log\log n} < |S(\gamma_n \pm 0)| \le f(n)\sqrt{\log\log n}$$

for almost all n.

Theorem 3.15. *Let $f(x)$ be any monotonically increasing function unbounded as $x \to +\infty$ and satisfying the conditions $a \le f(x) \le b\sqrt{\log\log x}$, where a and b are absolute constants, for $N \le x \le N + M$, and let*

$$\mathcal{F} = f(N) \qquad \text{and} \qquad \delta = 0.5\exp\left\{-(ec_0)^2\frac{\log\log N}{\mathcal{F}^2}\right\}.$$

Then the interval $N \le n \le N + M$ contains at least $M\delta$ numbers n satisfying the condition

$$S(\gamma_n \pm 0) \le -\frac{1}{\mathcal{F}}\log\log N$$

and at least $M\delta$ numbers satisfying the condition

$$S(\gamma_n \pm 0) > \frac{1}{\mathcal{F}}\log\log N.$$

The proof of these assertions repeats that of Theorem 2.6 almost word for word.

[17] The remainder term in the expression for $\Theta(u)$ can be improved; see Remark 2.4.

3.15. Gram intervals with an anomalous number of ordinates of $\zeta(s)$ zeros. Let $m \geq 2$ be an integer. Consider m Gram intervals $G_{n+1}, G_{n+2}, \dots, G_{n+m}$ adjacent to each other, or, equivalently, the interval $(t_n, t_{n+m}]$. According to the Riemann–Mangoldt formula, the number of ordinates of $\zeta(s)$ zeros in this interval is

$$\begin{aligned} N(t_{n+m}+0) - N(t_n+0) &= \frac{1}{\pi}(\vartheta(t_{n+m}) - \vartheta(t_n)) + S(t_{n+m}+0) - S(t_n+0) \\ &= m + \Delta(n+m) - \Delta(n). \end{aligned} \tag{3.28}$$

Since one Gram interval contains one ordinate of a zero of $\zeta(s)$ on average, it is natural to consider m to be the mean, normal number of ordinates in the interval $(t_n, t_{n+m}]$. Therefore, the difference

$$S(t_{n+m}+0) - S(t_n+0) = \Delta(n+m) - \Delta(n) \tag{3.29}$$

equals the deviation of the number of ordinates in $(t_n, t_{n+m}]$ from the normal value. If $\Delta(n+m) - \Delta(n) \neq 0$, then such an interval is considered anomalous, i.e., containing an anomalous number of ordinates.

This section is devoted to a lower bound for the number of anomalous Gram intervals G_n. Namely, we prove that a positive proportion of Gram intervals contain no ordinates, and, at the same time, a positive proportion of Gram intervals contain at least two ordinates of zeros of $\zeta(s)$ (Theorem 3.18).

This result strengthens Trudgian's theorem [11] that a positive proportion of Gram intervals are anomalous. It should, however, be mentioned that an analogue of Theorem 3.18 for a long summation interval of the type $1 \leq n \leq N$ is given without proof in Fujii's paper [52].

Briefly, the idea underlying the argument used below is as follows. Theorem 3.18 means that, in a positive proportion of cases, we have

$$\Delta(n+1) - \Delta(n) = -1,$$

and in a positive proportion of cases, we have

$$\Delta(n+1) - \Delta(n) \geq 1.$$

These facts can be derived, e.g., from an estimate of the form

$$\sum_n |\Delta(n+1) - \Delta(n)|^k = \sum_n |S(t_{n+1}+0) - S(t_n+0)|^k \geq cM, \tag{3.30}$$

where $c > 0$ is a constant, possibly depending on k. However, the straightforward application of Selberg's formulas (Lemmas 2.4–2.7) to study differences (3.29) with $m = 1$ does not help. Because of the closeness of the points t_{n+1} and t_n, the estimate of the presumed remainder term in (3.30) which these formulas yield is rough, and, as a consequence, an upper bound for sum (3.30) is insufficient to prove Theorem 3.18.

However, the replacement in this argument of t_{n+1} by t_{n+m}, where m is a very large but fixed number, leads to an instructive relation similar to (3.29) for the sum, which makes it possible to obtain lower bounds of correct order for this sum (Theorem 3.16 and Corollary 3.5).

Yet another simple consideration based on Hölder's inequality allows us to pass from the moments of $\Delta(n+m) - \Delta(n)$ to sums (3.29) and obtain lower bounds of correct order for these sums (Theorem 3.17).

An analogue of Theorem 3.17 for the case of a long summation interval and a parameter m increasing with N is contained in Fujii's paper [52]. The reason why we give its proof here is that our proof of Theorem 3.18 essentially uses the fact that m can be taken constant (although very large).

To prove Theorem 3.16, we need one more auxiliary relation.

Lemma 3.19. *Suppose that $0 < h_0 < 0.5$ is a sufficiently small constant, $0 < h < h_0$, $h \log x > 2$, and*

$$V(x;h) = \sum_{p \leq x} \frac{1}{p} \sin^2\left(\frac{h}{2} \log p\right).$$

Then

$$V(x;h) = \frac{1}{2} \log(h \log x) + 1.05\theta.$$

Proof. Setting $y = e^{\lambda/h}$, where $1 < \lambda < 2$, we obtain

$$V(x;h) = \left(\sum_{p\le y} + \sum_{y<p\le x}\right)\frac{1}{p}\sin^2\left(\frac{h}{2}\log p\right)$$
$$= \sum_{p\le y}\frac{1}{p}\sin^2\left(\frac{h}{2}\log p\right) + \frac{1}{2}\sum_{y<p\le x}\frac{1-\cos(h\log p)}{p} = V_1 + \frac{1}{2}(V_2 - V_3).$$

Lemma 1.5 implies

$$V_1 \le \left(\frac{h}{2}\right)^2 \sum_{p\le y}\frac{(\log p)^2}{p} \le \left(\frac{h}{2}\right)^2 (\log y)\sum_{p\le y}\frac{\log p}{p} < \left(\frac{h}{2}\log y\right)^2 = \frac{\lambda^2}{4}$$

and

$$V_2 = \left(\sum_{p\le x} - \sum_{p\le y}\right)\frac{1}{p} = \log(h\log x) - \log\lambda + 2\theta\left(\frac{h}{\lambda}\right)^2.$$

To estimate V_3, we partition the range of variation of p by points of the form $a = 2^k y$, $k = 0, 1, 2, \dots$, into intervals of the form $a < p \le b$, where $y \le a < x$ and $b \le 2a$. We have

$$V_3 = \operatorname{Re}\sum_{y<p\le x}\frac{p^{ih}}{p} = \operatorname{Re}\sum_a V_3(a), \qquad V_3(a) = \sum_{a<p\le b}\frac{p^{ih}}{p}.$$

Setting

$$\mathbb{C}(u) = \sum_{a<p\le u}\frac{1}{p} = \log\log u - \log\log a + \frac{2\theta}{\log^2 a}$$

and applying the Abel transform to $V_3(a)$, we obtain

$$V_3(a) = \mathbb{C}(b)b^{ih} - \int_a^b \mathbb{C}(u)\,du^{ih} = \int_a^b \frac{u^{ih}\,du}{u\log u} + \frac{3\theta_1}{\log^2 a},$$

whence

$$V_3 = \operatorname{Re}(j) + 3\theta_2\sum_{k\ge 0}\frac{1}{(k\log 2 + \log y)^2}, \qquad j = \int_y^x \frac{u^{ih}\,du}{u\log u}.$$

Integration by parts yields

$$j = \frac{1}{ih}\left(\frac{x^{ih}}{\log x} - \frac{y^{ih}}{\log y} - \int_y^x u^{ih}\,d\frac{1}{\log u}\right) \qquad \text{and} \qquad |j| \le \frac{2}{h\log y} = \frac{2}{\lambda}.$$

Finally, we obtain

$$|V_3| \le \frac{2}{\lambda} + \frac{9}{\log y} = \frac{2+9h}{\lambda} \qquad \text{and} \qquad V(x;h) = \frac{1}{2}\log(h\log x) + v(x;h),$$

where

$$|v(x;h)| \le \frac{\lambda^2}{4} + \frac{1}{\lambda} + \frac{1}{2}\log\lambda + \frac{9h}{\lambda} + \frac{h^2}{\lambda^2}.$$

Setting $\lambda = 1.5$, we arrive at the assertion of the lemma.

Theorem 3.16. *Let k and m be integers such that $k \ge 1$ and $k\varepsilon^{-1}\exp\{(4c_0k)^3\} < m \le c\log N$, where c is a sufficiently small absolute constant. Then*

$$\sum_n (\Delta(n+m) - \Delta(n))^{2k} = \frac{(2k)!}{k!}\frac{M}{(2\pi^2)^k}\left(\log\frac{m\varepsilon}{k}\right)^k\left(1 + \frac{5.4\theta(c_0k)^{3/2}}{\sqrt{\log(m\varepsilon/k)}}\right).$$

Proof. As above, we set

$$x = X^{1/(8k+3)}, \qquad W(t) = -\frac{1}{\pi}\sum_{p\le x}\frac{\sin(t\log p)}{\sqrt{p}}, \qquad \text{and} \qquad R(t) = S(t) - W(t),$$

so that

$$\Delta(n) = R(t_n + 0) + W(t_n). \tag{3.31}$$

Lemma 1.1 implies that, for n and m under consideration,

$$t_{n+m} - t_n = h + \varepsilon_n, \qquad h = \frac{\pi m}{\vartheta'(t_N)}, \qquad |\varepsilon_n| \le \frac{2\pi m M}{N(\log N)^2},$$

whence we obtain

$$W(t_{n+m}) = W(t_n + h) + \frac{2\pi\theta_1 m M\sqrt{x}}{N(\log N)^2}$$

and

$$W(t_{n+m}) - W(t_n) = -\frac{1}{\pi}(U(t_n) + \overline{U}(t_n)) + \frac{2\pi\theta_1 m M\sqrt{x}}{N(\log N)^2},$$

where

$$U(t) = \sum_{p\le x}\frac{a(p)}{\sqrt{p}}p^{it}, \qquad a(p) = p^{ih/2}\sin\left(\frac{h}{2}\log p\right).$$

According to Lemma 2.1, we have

$$\begin{aligned} W &= \sum_n (W(t_{n+m}) - W(t_n))^{2k} = \pi^{-2k}\sum_{\nu=0}^{2k}\binom{2k}{\nu}\sum_n U^\nu(t_n)\overline{U}^{\,2k-\nu}(t_n) + \theta_2 N^{-1/3} \\ &= \pi^{-2k}\binom{2k}{k}M\mathfrak{S} + \theta_3\sqrt{X}, \end{aligned}$$

where

$$\mathfrak{S} = k!(\mathfrak{S}_1^k + \theta_4 k^2\mathfrak{S}_1^{k-2}\mathfrak{S}_2), \qquad \mathfrak{S}_1 = \sum_{p\le x}\frac{1}{p}\sin^2\left(\frac{h}{2}\log p\right), \qquad \mathfrak{S}_2 \le \sum_{p\le x}\frac{1}{p^2} < \frac{1}{2}.$$

Using Lemma 3.19 and the relations

$$h\log x = \frac{\pi m\log x}{\vartheta'(t_N)} = \frac{\pi m\varepsilon}{10(8k+3)}\frac{\log N}{\vartheta'(t_N)} = (1+o(1))\frac{\pi m\varepsilon}{40k}\left(1+\frac{3}{8k}\right)^{-1},$$

we obtain

$$\mathfrak{S}_1 = \frac{1}{2}\log\frac{\pi m\log x}{40k} - \frac{1}{2}\log\left(1+\frac{3}{8k}\right) + 1.05\theta_5 + o(1) = \frac{\Lambda}{2} + 2.52\theta_6,$$

where $\Lambda = \log(m\varepsilon/k)$. It follows from the assumptions on k and m that $\Lambda \ge (4c_0k)^3$. Therefore,

$$\mathfrak{S}_1^k = \left(\frac{\Lambda}{2}\right)^k\left(1+\frac{5.04\theta_6}{\Lambda}\right)^k = \left(\frac{\Lambda}{2}\right)^k\left(1+\frac{5.05\theta_6 k}{\Lambda}\right), \qquad \mathfrak{S}_1^{k-2} < 1.001\left(\frac{\Lambda}{2}\right)^k,$$

and, finally,

$$S = k!\left(\frac{\Lambda}{2}\right)^k\left(1+\frac{5.06\theta_8 k}{\Lambda}\right).$$

Thus,

$$W = \frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}\biggl(1 + \frac{5.07\theta_9 k}{L}\biggr).$$

In particular, for the integers k under consideration, we have

$$W < 1.001\frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}. \tag{3.32}$$

Let V denote the sum in the statement of the lemma. Then it follows from relation (3.31) that

$$V = W + \sum_{\nu=1}^{2k}(-1)^\nu\binom{2k}{\nu}V_\nu,$$

where

$$V_\nu = \sum_n (W(t_{n+m}) - W(t_n))^{2k-\nu}(R(t_{n+m}+0) - R(t_n+0))^\nu.$$

Using Hölder's inequality, Lemma 2.7, and inequality (3.32), we obtain

$$\begin{aligned}|V_\nu| &\le V^{1-\nu/(2k)}\biggl\{\sum_n (R(t_{n+m}+0) - R(t_n+0))^{2k}\biggr\}^{\nu/(2k)}\\ &\le \biggl\{1.001\frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}\biggr\}^{1-\nu/(2k)}\biggl\{\frac{c_0}{56}(2c_0k)^{2k}M\biggr\}^{\nu/(2k)} < 1.001\frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}\,q^\nu,\end{aligned}$$

where

$$q = \biggl(\frac{c_0}{56}\biggr)^{1/(2k)}\pi c_0\sqrt{\frac{2ke}{\Lambda}} \le \frac{\pi}{2}\sqrt{\frac{e}{7}}\,c_0^{3/2}\sqrt{\frac{k}{\Lambda}}.$$

Note that

$$q \le \frac{1}{k}\frac{\pi}{2}\sqrt{\frac{e}{7}}\frac{(c_0k)^{3/2}}{\sqrt{\Lambda}} < \frac{1}{8k}.$$

Therefore, the sum over ν in (3.32) is estimated as

$$\begin{aligned}\biggl|\sum_{\nu=1}^{2k}(-1)^\nu\binom{2k}{\nu}V_\nu\biggr| &< 1.001\frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}\{(1+q)^{2k}-1\} < 1.001\cdot 2kq(1+q)^{2k-1}\frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}\\ &< 1.001\cdot 2k\frac{\pi}{2}\sqrt{\frac{e}{7}}\,c_0^{3/2}\sqrt{\frac{k}{L}}\biggl(1+\frac{1}{8k}\biggr)^{2k}\frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}\\ &< 1.001\frac{\pi e^{3/4}}{\sqrt{7}}\frac{(c_0k)^{3/2}}{\sqrt{\Lambda}}\frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k} < 2.12\frac{(c_0k)^{3/2}}{\sqrt{\Lambda}}\frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}.\end{aligned}$$

Returning to the initial sum, we obtain

$$V = \frac{(2k)!}{k!}\frac{M\Lambda^k}{(2\pi^2)^k}(1+\theta\delta), \qquad \delta = \frac{5.07k}{\Lambda} + 2.12\frac{(c_0k)^{3/2}}{\sqrt{\Lambda}} < 2.2\frac{(c_0k)^{3/2}}{\sqrt{\Lambda}}.$$

This completes the proof of the theorem.

Corollary 3.5. *Let $m = [\mu]+1$, where $\mu = \varepsilon^{-1}\exp\{(4c_0)^3\}$. Then*

$$\sum_n (\Delta(n+m) - \Delta(n))^2 > 2c_0^3M.$$

Theorem 3.17. *The inequality*

$$\sum_n |\Delta(n+1) - \Delta(n)| > K_1 M$$

holds, where

$$K_1 = \frac{c_0^2 \varepsilon^3}{4} \exp\{-192 c_0^3\}.$$

Proof. As above, we set $r(n) = \Delta(n) - \Delta(n-1)$ and denote the sum of the kth powers of $|r(n)|$ by V_k. Let $m = [\mu] + 1$, where $\mu = \varepsilon^{-1} \exp\{(4c_0)^3\}$. The application of Cauchy's inequality to the identity

$$\Delta(n+m) - \Delta(n) = r(n+1) + r(n+2) + \cdots + r(n+m)$$

yields

$$(\Delta(n+m) - \Delta(n))^2 \le m \sum_{\nu=1}^{m} r^2(n+\nu).$$

Summing both sides of this relation over n, we obtain

$$\begin{aligned}\sum_n (\Delta(n+m) - \Delta(n))^2 &\le m \sum_{\nu=1}^{m} \sum_n r^2(n+\nu) \\ &\le m \sum_{\nu=1}^{m} \sum_{N < n \le N+M+m} r^2(n) \le m^2 \sum_{N < n \le N+M+m} r^2(n).\end{aligned}$$

From the corollary of Theorem 3.16 and the estimate of Lemma 3.1 it follows that

$$\sum_{N < n \le N+M+m} r^2(n) \ge \frac{1}{m^2} \sum_n (\Delta(n+m) - \Delta(n))^2 \ge \frac{2c_0^3}{m^2} M > 1.5 c_0^3 \varepsilon^2 \exp\{-128 c_0^3\} M,$$

so that

$$V_2 = \sum_n r^2(n) > 1.5 c_0^3 \varepsilon^2 \exp\{-128 c_0^3\} M - m(0.3 \log N)^2 > c_0^3 \varepsilon^2 \exp\{-128 c_0^3\} M = K_2 M.$$

On the other hand, applying Hölder's inequality to V_2, we obtain

$$V_2 = \sum_n |r(n)|^{2/3} \cdot |r(n)|^{4/3} \le \left(\sum_n |r(n)| \right)^{2/3} \left(\sum_n r^4(n) \right)^{1/3} = V_1^{2/3} V_4^{1/3},$$

whence $V_1 \ge V_2^{3/2} V_4^{-1/2}$. Using the estimate of V_2 found above and the inequality in Theorem 3.5 with $k = 2$, we obtain

$$V_4 \le K_4 M, \qquad \text{where} \qquad K_4 = \frac{256}{17} c_0^5,$$

and

$$V_1 = \sum_n |r(n)| \ge \frac{(K_2 M)^{3/2}}{(K_1 M)^{1/2}} = K_1 M,$$

where

$$K_1 = K_2^{3/2} K_4^{-1/2} > \frac{c_0^2 \varepsilon^3}{4} \exp\{-192 c_0^3\}.$$

This proves the theorem.

Let $\nu_k(N)$ denote the number of Gram intervals G_n, $1 \le n \le N$, containing precisely k ordinates of zeros of $\zeta(s)$. The following theorem is valid.

Theorem 3.18. *There exist positive constants $\varkappa_1$ and $\varkappa_2$ such that at least $\varkappa_1 M$ of the Gram intervals G_n, $N < n \le N + M$, contain no ordinates of zeros of the Riemann zeta-function and at least $\varkappa_2 M$ of such intervals contain two or more ordinates of zeros of $\zeta(s)$. In other words,*

$$\nu_0(N+M) - \nu_0(N) \ge \varkappa_1 M \qquad \text{and} \qquad \sum_{k\ge 2}(\nu_k(N+M) - \nu_k(N)) \ge \varkappa_2 M.$$

Proof. It follows from Lemma 3.3 that the number M_1 of empty Gram intervals coincides with the number of those n for which the quantity

$$r(n) = \Delta(n) - \Delta(n-1) = S(t_n + 0) - S(t_{n-1} + 0)$$

is negative, and the number M_2 of Gram intervals containing more than one ordinate coincides with the number of those n for which $r(n)$ is strictly positive.

Using the relation

$$\frac{1}{2}(|r(n)| - r(n)) = \begin{cases} 1 & \text{if } r(n) < 0, \\ 0 & \text{if } r(n) > 0 \end{cases}$$

and the estimate of Theorem 3.17, we obtain

$$\begin{aligned} M_1 &= \sum_n \frac{1}{2}(|r(n)| - r(n)) = \frac{1}{2}\sum_n |r(n)| - \frac{1}{2}\sum_n (\Delta(n) - \Delta(n-1)) \\ &= \frac{1}{2}\sum_n |r(n)| - \frac{1}{2}(\Delta(N+M) - \Delta(N)) > 0.5K_1 M - 0.15\log N > \varkappa_1 M, \qquad \varkappa_1 = 0.4\,K_1. \end{aligned}$$

Further, M_2 coincides with the number on nonzero terms in the sum

$$W = \sum_n \frac{1}{2}(|r(n)| + r(n)).$$

Applying Cauchy's inequality and the estimate of Lemma 3.3 to W yields

$$W \le \left(M_2 \sum_n r^2(n)\right)^{1/2} \le \left(M_2 \cdot \frac{4c_0^3}{17} M\right)^{1/2}.$$

On the other hand, $W > \varkappa_1 M$ by virtue of Theorem 3.17. Thus,

$$M_2 \ge \frac{17}{4}\frac{\varkappa_1^2 M^2}{c_0^3 M} > 4c_0^{-3}\varkappa_1^2 M = \varkappa_2 M.$$

This completes the proof of the theorem.

Remark 3.3. The constants $\varkappa_1$ and $\varkappa_2$ in Theorem 3.18 are very small. Thus, even for $\varepsilon = 10^{-3}$, they do not exceed $\exp\{-4.5 \cdot 10^{21}\}$ and $\exp\{-9 \cdot 10^{21}\}$, respectively. At the same time, an approximate calculation of zeros of $\zeta(s)$ suggests that $\varkappa_1 > 0.1$ and $\varkappa_2 > 0.1$. In this connection, it would be interesting to obtain an analogue of Theorem 3.18 with more or less reasonable values of these constants (say, of order 0.001–0.01).

3.16. A new equivalent of the "almost Riemann hypothesis." This section considers the question of what exactly Selberg meant by the quantities γ_n in the results and conjectures concerning Δ_n in [12]: whether these are the positive ordinates of zeros of the zeta-function that lie on the critical line (that is, $\mathfrak{c}_1, \mathfrak{c}_2, \dots, \mathfrak{c}_n, \dots$ in our notation) or the ordinates of all complex zeros $\zeta(s)$ in the upper half-plane.

This question, although looking artificial at first sight, is of fundamental importance for the following reasons.

At the beginning of his talk cited above Selberg mentioned that he used $\beta + i\gamma$ to denote the zeros of $\zeta(s)$ lying in the critical strip above the real axis (that is, $\gamma_1, \gamma_2, \dots, \gamma_n, \dots$ in our notation). However, defining the quantities Δ_n, Selberg cited Titchmarsh's result [9] on the unboundedness of the fractions

$$\tau_n = \frac{\mathfrak{c}_n - t_n}{t_{n+1} - t_n}$$

and mentioned that this result is directly related to the unboundedness of Δ_n.

If a zero $\mathfrak{c}_n$ of the function $\zeta(1/2+it)$ satisfies the condition

$$t_{m-1} < \mathfrak{c}_n \le t_m, \tag{3.33}$$

then it easily follows (e.g., by Lemma 3.6) that

$$\frac{\mathfrak{c}_n - t_n}{t_{n+1} - t_n} = m - n + O(1).$$

Therefore, the unboundedness of the Titchmarsh fractions implies that of the differences $m-n$ defined by (3.33). But these differences may differ from the quantities Δ_n defined in the introduction. This may happen if the zeta-function has zeros outside the critical line.

Thus, there are two mutually exclusive possibilities:

(1) Selberg used Δ_n to denote the differences defined by (3.33) and proved relations (7) and (8) for them;

(2) Selberg quoted Titchmarsh's result not quite correctly, implicitly replacing $\mathfrak{c}_n$ by γ_n in the definition of the fractions τ_n.

We are to determine which of these possibilities did indeed occur.

Thus, the correctness of the first alternative would devalue most results of the present paper, because, in this case, the relations proved in Sections 3.1–3.3 are not Selberg's formulas and their corollaries. As a result, one of the tasks of this paper, the rigorous substantiation of Selberg's formulas, remains unaccomplished.

On the contrary, the correctness of the second alternative would mean, e.g., that our results fill the lacuna that has appeared because the major part of Selberg's work on Gram's law has not been published.

Below we provide evidence in favor of the second alternative. It essentially reduces to the following reasoning.

Assuming that the first alternative is correct, we shall derive an assertion equivalent to the "almost Riemann hypothesis" from Selberg's formulas; according to this hypothesis, almost all complex zeros of $\zeta(s)$ lie on the critical line:

$$\lim_{T\to+\infty} \frac{N_0(T)}{N(T)} = 1$$

(here and in what follows, $N_0(T)$ denotes the number of zeros of $\zeta(s)$ on the critical line whose ordinates are positive and do not exceed T). Without doubt, such an equivalent must have been known to Selberg. Having proved it, he would not fail to mention such a significant result in some form.[18] Since neither Selberg nor anybody else have made similar announcements, it is natural to conclude that the first alternative is incorrect.[19]

To avoid confusion in notation, in the case where inequalities (3.33) hold, we denote the difference $m-n$ by D_n. Certainly, if the Riemann hypothesis is true, then $\mathfrak{c}_n = \gamma_n$ and $D_n = \Delta_n$ for any n.

Theorem 3.19. *The almost Riemann hypothesis is true if and only if*

$$\sum_{n\le N} |D_n| = o(N^2) \tag{3.34}$$

as $N \to +\infty$.

[18] Obviously, such a result would be a sensation. In particular, its proof would bring the fierce activity of improving the constant κ in Selberg's inequality $N_0(T) \ge \kappa \cdot N(T)$, which persists up to now, to naught.

[19] See also Remark 4.1 in Section 2 of this paper.

Proof. First, note that

$$N_0(\mathfrak{c}_n + 0) = n + O(\log n). \tag{3.35}$$

Indeed, let κ denote the multiplicity of a zero $1/2 + i\mathfrak{c}_n$; then, by Lemma I.9, we have

$$\kappa \le S(\mathfrak{c}_n + 0) - S(\mathfrak{c}_n - 0) < 0.3 \log \mathfrak{c}_n.$$

Thus,

$$N_0(\mathfrak{c}_n + 0) = n + \theta_1 \kappa = n + 0.3\theta_2 \log \mathfrak{c}_n. \tag{3.36}$$

Setting $t = \mathfrak{c}_n + 0$ in the inequalities $0.4N(t) < N_0(t) \le N(t)$, which follow from Lemma I.11 and the definition of $N_0(t)$, and taking into account relations (3.36), we obtain

$$(1 + o(1))\frac{\mathfrak{c}_n}{5\pi} \log \frac{\mathfrak{c}_n}{2\pi} < n \le (1 + o(1))\frac{\mathfrak{c}_n}{2\pi} \log \frac{\mathfrak{c}_n}{2\pi},$$

whence it follows that $\log \mathfrak{c}_n = \log n + O(\log \log n)$. Thus, we arrive at relation (3.35).

Now, suppose that the almost Riemann hypothesis is true. Then (3.35) gives

$$N(\mathfrak{c}_n + 0) = (1 + o(1))N_0(\mathfrak{c}_n + 0) = (1 + o(1))(n + O(\log n)) = n + o(n).$$

On the other hand, it follows from (3.33) that

$$m - 1 + S(t_{m-1} + 0) < N(\mathfrak{c}_n + 0) \le m + S(t_m + 0),$$

whence

$$D_n = m - n = o(n) + O(\log m) = o(n) \tag{3.37}$$

and, therefore,

$$\sum_{n \le N} |D_n| = o(N^2).$$

Conversely, suppose that condition (3.34) holds. Noting that $N(\mathfrak{c}_n + 0) \ge n + \varepsilon(n)$, where $\varepsilon(n)$ is the number of zeros of $\zeta(s)$ satisfying the conditions $0 < \operatorname{Im} s \le \mathfrak{c}_n$ and $\operatorname{Re} s \ne 1/2$, we obtain

$$n + \varepsilon(n) \le N(t_m + 0) = m + S(t_m + 0),$$

whence $0 \le \varepsilon(n) \le |D_n| + |S(t_m + 0)|$. Summing over $n \le 2N$ and applying Cauchy's inequality, we arrive at

$$\sum_{n \le 2N} \varepsilon(n) \le \sum_{n \le 2N} |D_n| + \sum_{\substack{n \le 2N \\ m = m(n)}} |S(t_m + 0)| \le \sqrt{2N}\sqrt{W} + o(N^2),$$

where

$$W = \sum_{\substack{n \le 2N \\ m = m(n)}} S^2(t_m + 0).$$

Let μ be the maximum value of $m(n)$ over $n \le 2N$. Then $t_{\mu-1} < \mathfrak{c}_{2N} \le t_\mu$, whence

$$N(\mathfrak{c}_{2N}) \ge N(t_{\mu-1}) = \mu + O(\log \mu).$$

On the other hand, Lemma I.11with $t = \mathfrak{c}_{2N}$ implies

$$N(\mathfrak{c}_{2N}) \le \left(\frac{5}{2} - 10^{-3}\right) N_0(\mathfrak{c}_{2N}) = \left(\frac{5}{2} - 10^{-3}\right)(2N + O(\log N));$$

thus, $\mu < 5N$. Changing the order of summation in the sum W, we obtain

$$W \le \sum_{l \le 5N} S^2(t_l + 0) \sum_{\substack{n \le 2N \\ m(n) = l}} 1.$$

If l is fixed, then the number of those n for which $n \le 2N$ and $m(n) = l$ does not exceed the number of all ordinates of $\zeta(s)$ zeros in the interval $(t_{l-1}, t_l]$, i.e., the quantity

$$N(t_l + 0) - N(t_{l-1} + 0) = 1 + S(t_l + 0) - S(t_{l-1} + 0) = 1 + r(l).$$

Therefore,

$$W \le \sum_{l \le 5N} S^2(t_l + 0)(1 + r(l)).$$

Using the relation in Theorems 2.1 together with the second inequality in Theorem 3.6, we successively obtain

$$W \le \frac{5N}{2\pi^2} \log\log N + O(N\sqrt{\log\log N}) < \frac{1}{3} N \log\log N,$$

$$\sum_{n \le 2N} \varepsilon(n) < N\sqrt{\log\log N} + o(N^2) = o(N^2).$$

Now, noting that

$$\varepsilon(N+1) + \varepsilon(N+2) + \cdots + \varepsilon(2N) \ge N\varepsilon(N),$$

we obtain

$$N\varepsilon(N) = o(N^2), \qquad \varepsilon(N) = o(N).$$

If the number t is sufficiently large, then, for N determined by the inequalities $\mathfrak{c}_{N-1} < t \le \mathfrak{c}_N$, the relation proved above implies

$$N(t) \ge N - 1 \qquad \text{and} \qquad N(t) - N_0(t) \le \varepsilon(N) = o(N) = o(N(t)),$$

as required. This completes the proof of the theorem.

Corollary 3.6. *The estimate*

$$\sum_{n \le N} |D_n|^k = o(N^{k+1})$$

as $N \to +\infty$ holds for at least one fixed $k \ge 1$ if and only if the almost Riemann hypothesis is true.

The proof of this assertion is similar to that of Theorem 3.19 with the only difference that sufficiency is proved by using the inequalities

$$N\varepsilon(N) \le \sum_{n \le 2N} |D_n| + N\sqrt{\log\log N} \le (2N)^{1-1/k} \Bigg(\sum_{n \le 2N} |D_n|^k \Bigg)^{1/k} + N\sqrt{\log\log N}.$$

This assertion shows, in particular, that if Selberg's formulas (7) and (8) remained valid under the replacement of Δ_n by D_n, then they would imply the almost Riemann hypothesis.

Corollary 3.7. *The relation $D_n = o(n)$ as $n \to +\infty$ is equivalent to the almost Riemann hypothesis.*

Proof. Necessity follows from (3.37) and sufficiency, from Corollary 3.6 with $k = 1$.

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